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# W1 Lesson 1

## Introduction to Probability



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# Introduction to probability

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## **What is Probability?**

# Introduction to Probability



Find the probability that a child picked at random plays soccer.



# Introduction to Probability



Find the probability that a child picked at random plays soccer.

**The probability that a child picked at random plays soccer.**

$P(\text{soccer})$

# Introduction to Probability



Find the probability that a child picked at random plays soccer.

$$P(\text{soccer}) = \frac{\text{soccer}}{\text{total}} = \frac{\text{Event}}{\text{Sample space}} = \frac{3}{10} = 0.3$$

# Introduction to Probability: Venn Diagram



Sample Space

Event

# Introduction to Probability: Coin Example 1



## Experiment

Probability of landing on heads

$P(\text{heads})$

# Introduction to Probability: Coin Example 1



50% 50%

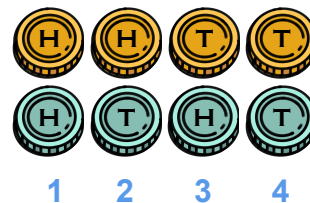
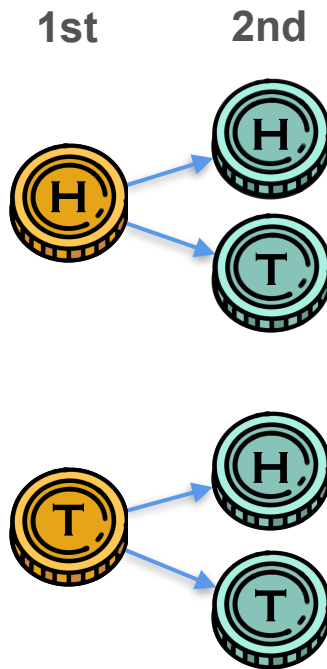
$$P(\text{heads}) = \frac{\text{H}}{\text{H} \quad \text{T}} = \frac{1}{2} = 0.5$$

# Introduction to Probability: Coin Example 2



50% 50%

What is the probability of  
landing on heads twice?



# Introduction to Probability: Coin Example 2



50% 50%



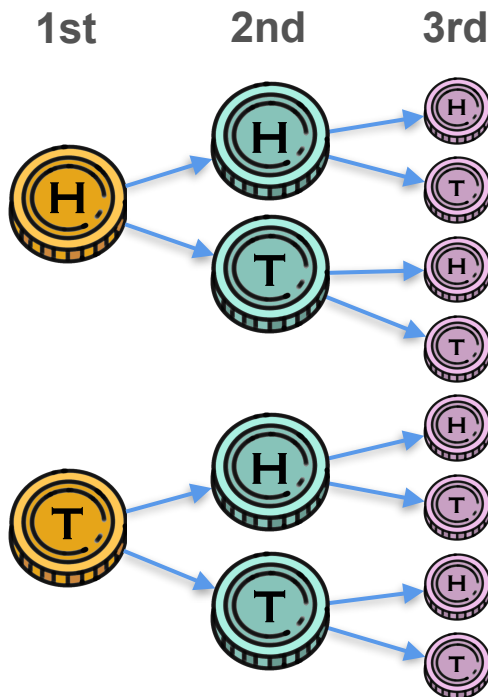
$$P(HH) = \text{—————} = \frac{1}{4} = 0.25$$

# Introduction to Probability: Coin Example 3



50% 50%

What is the probability of  
landing on heads 3 times?



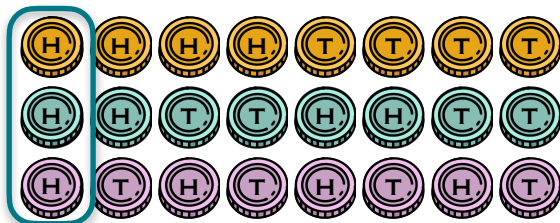
1   2   3   4   5   6   7   8



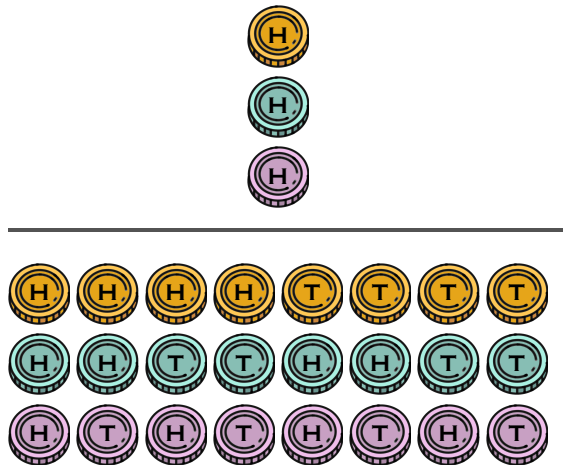
# Introduction to Probability: Coin Example 3



50% 50%



$P(HHH) =$



$$= \frac{1}{8} = 0.125$$



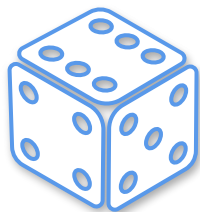
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# Introduction to Probability

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## **What is Probability? - Dice Example**

# Introduction to Probability: Dice Example 1

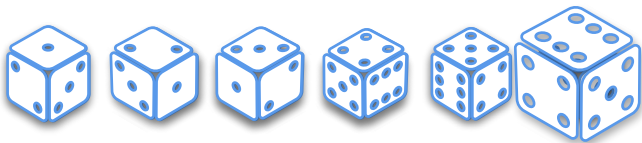


What is the probability of obtaining 6?

# Introduction to Probability: Dice Example 1

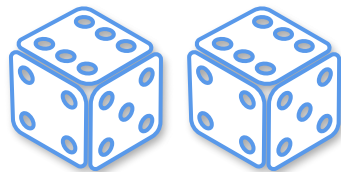


What is the probability of obtaining 6?



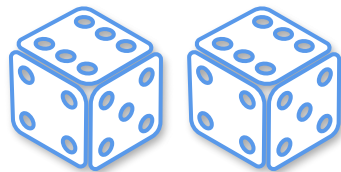
$$P(6) = \text{_____} = \frac{1}{6}$$

# Introduction to Probability: Dice Example 2



What is the probability of obtaining 6,6?

# Introduction to Probability: Dice Example 2



What is the probability of obtaining 6,6?

A vertical column of six blue-outlined dice icons is positioned to the left of the table. Each icon represents a different row of the table, showing the first die's value for that row (1 through 6).

|     |     |     |     |     |            |
|-----|-----|-----|-----|-----|------------|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6        |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6        |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6        |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6        |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6        |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | <b>6,6</b> |

$$P(6,6) = \frac{\text{Number of outcomes where both dice are 6}}{\text{Total number of outcomes}} = \frac{1}{36}$$

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |



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# Introduction to probability

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## **Complement of Probability**

# Complement of Probability



30%

What is the probability of a child NOT playing soccer?



# Complement of Probability

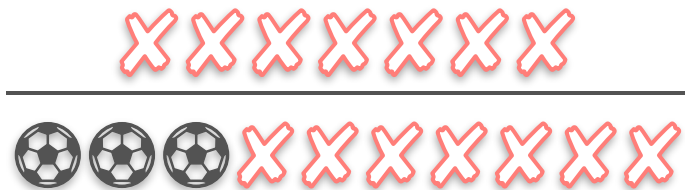


30%

What is the probability of a child NOT playing soccer?

$$P(\text{not soccer}) = \frac{\text{not soccer}}{\text{total}} = \frac{\text{X X X X X X X X}}{\text{⚽ ⚽ ⚽ X X X X X X X X}} = \frac{7}{10} = 0.7$$

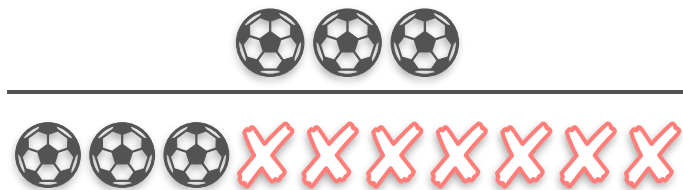
# Complement of Probability



$P(\text{not soccer})$

0.7

+



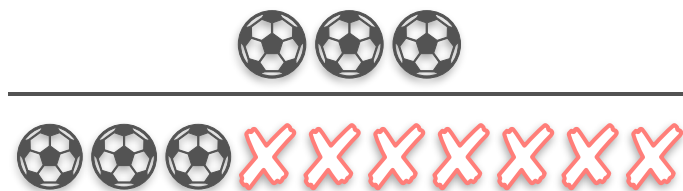
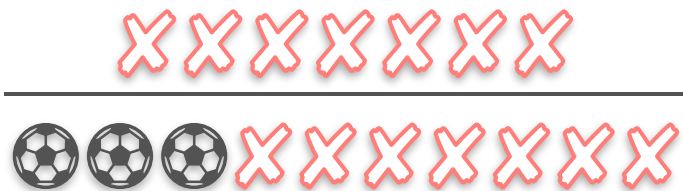
$P(\text{soccer})$

0.3

=

1

# Complement of Probability



$P(\text{not soccer})$

$P(\text{soccer})$

0.7

=

1

-

0.3

Complement Rule

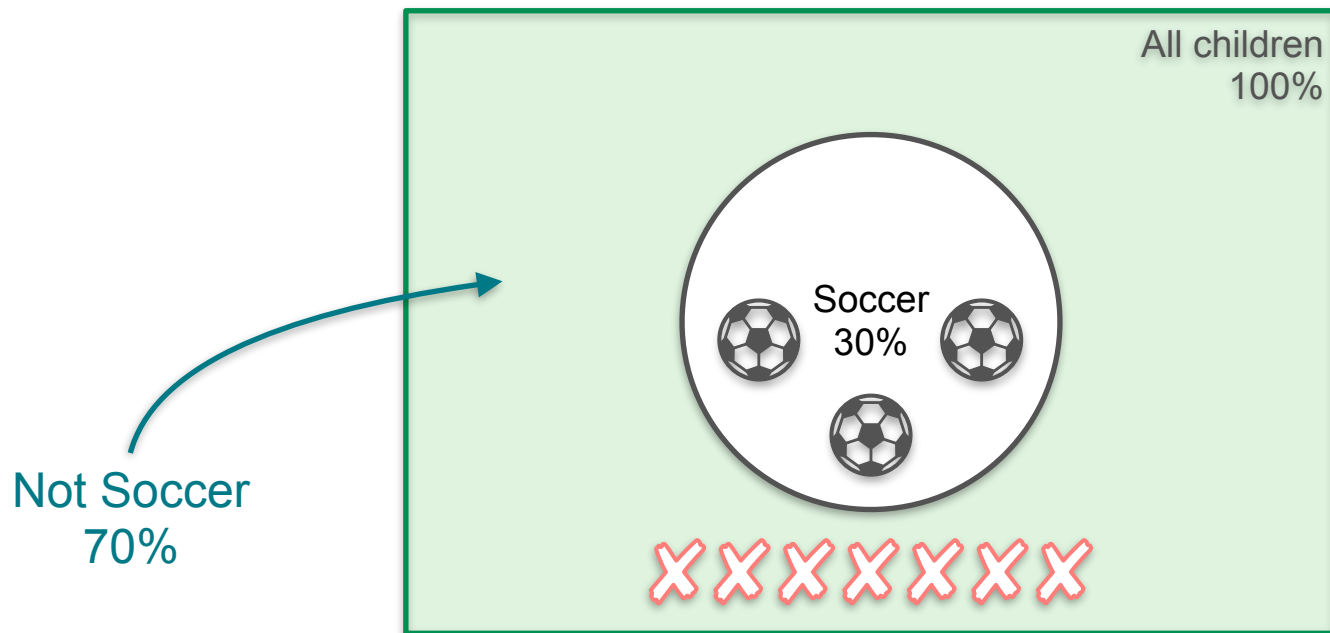
# Complement of Probability

$$P(\text{not soccer}) = 1 - P(\text{soccer})$$

$$P(A') = 1 - P(A)$$

Complement Rule

# Complement of Probability: Venn Diagram



# Complement of Probability: Coin Example 1



What is the probability of not landing on heads 3 times?

# Complement of Probability: Coin Example 1



What is the probability of not landing on heads 3 times?

$$P(\text{not } HHH) = 1 - P(HHH)$$

$$P(HHH) = \frac{\begin{array}{c} \text{H} \\ \text{H} \\ \text{H} \end{array}}{\begin{array}{cccccccc} \text{H} & \text{H} & \text{H} & \text{H} & \text{T} & \text{T} & \text{T} & \text{T} \\ \text{H} & \text{H} & \text{T} & \text{T} & \text{H} & \text{H} & \text{T} & \text{T} \\ \text{H} & \text{T} & \text{H} & \text{T} & \text{H} & \text{T} & \text{H} & \text{T} \end{array}}$$

# Complement of Probability: Coin Example 1



What is the probability of not landing on heads 3 times?

$$P(\text{not } HHH) = 1 - P(HHH)$$

$$P(\text{not } HHH) = 1 - \frac{1}{8}$$
A diagram illustrating the sample space for three coins. It consists of three rows of coins. The top row has three coins, all showing 'H' (heads). The middle row has eight coins, with the first three showing 'H' and the remaining five showing 'T' (tails). The bottom row has eight coins, with the first three showing 'H' and the remaining five showing 'T'. The coins are colored yellow, teal, and purple from left to right in each row.



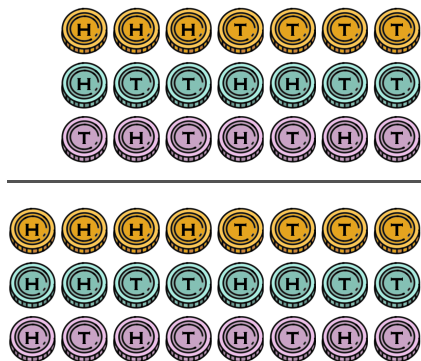
# Complement of Probability: Coin Example 1



What is the probability of not landing on heads 3 times?

$$P(\text{not } HHH) = 1 - P(HHH)$$

$$\begin{aligned} P(\text{not } HHH) &= 1 - \frac{1}{8} \\ &= \frac{7}{8} \end{aligned}$$



# Complement of Probability: Dice Example 1



What is the probability of obtaining anything other than 6?

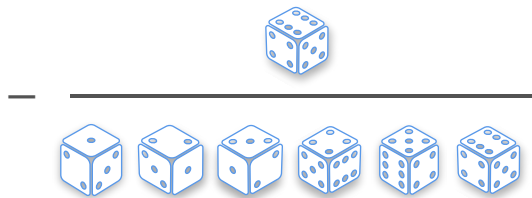
# Complement of Probability: Dice Example 1



What is the probability of obtaining anything other than 6?



$P(\text{not } 6) =$



$=$

$$= \frac{5}{6}$$



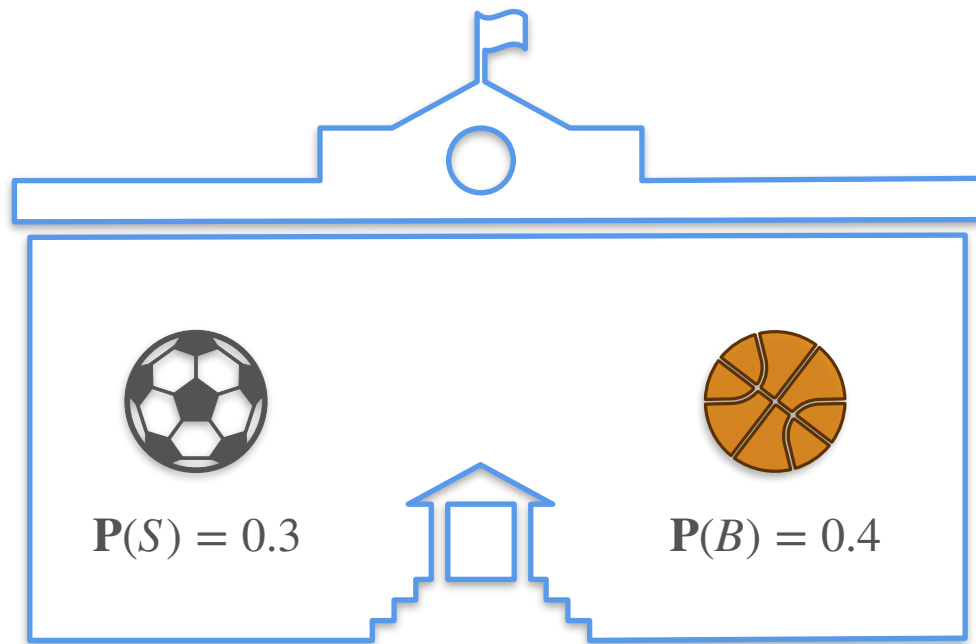
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# Introduction to probability

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## **Sum of Probabilities (Disjoint Events)**

# Sum of Probabilities: Quiz 1

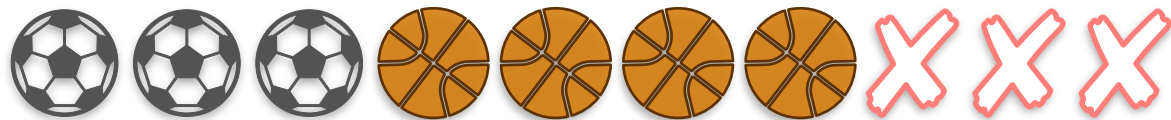


At a school, kids can only play one sport.

What is the probability that a kid plays soccer or basketball?

Hint: What if there were only 10 kids?

# Sum of Probabilities: Quiz 1 Solution



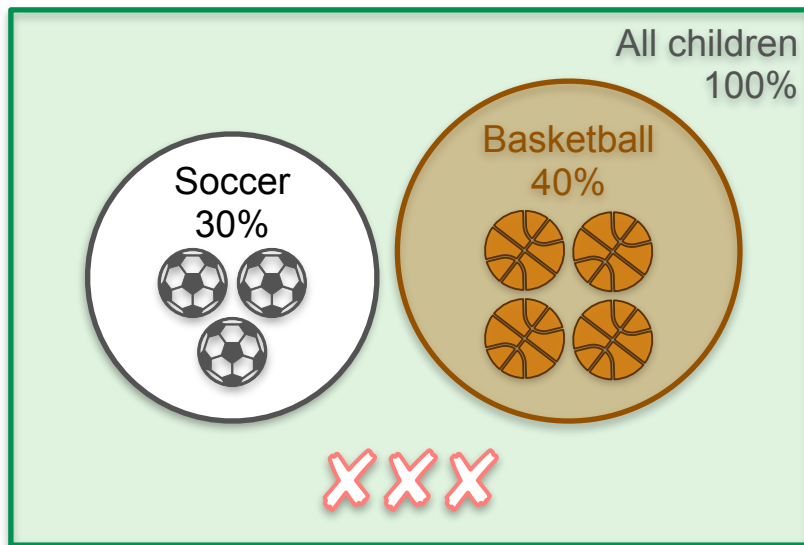
30% 40%

70%

$$P(\text{soccer or basketball}) = \frac{\text{soccer or basketball}}{\text{total}} = \frac{3 + 4}{10} = 0.7$$

$$P(\text{soccer or basketball}) = P(\text{soccer}) + P(\text{basketball})$$

# Sum of Probabilities: Quiz 1 Solution



$$P(\text{soccer} \cup \text{basketball}) = P(\text{soccer}) + P(\text{basketball})$$

$$P(A \cup B) = P(A) + P(B)$$

# Sum of Probabilities: Dice Example 1



What is the probability of obtaining an even number or a 5?



# Sum of Probabilities: Dice Example 1

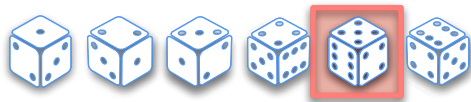


What is the probability of obtaining an even number or a 5?

*A*



*B*

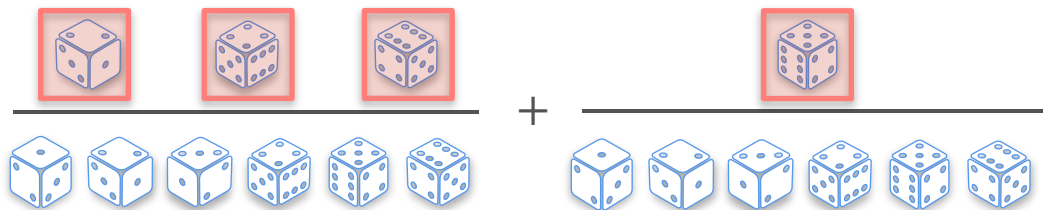


\_\_\_\_\_ + \_\_\_\_\_

# Sum of Probabilities: Dice Example 1

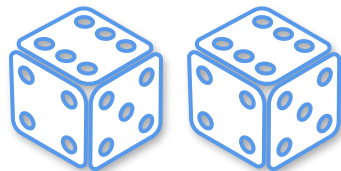


What is the probability of obtaining an even number or a 5?



$$P(\text{even number or } 5) = P(\text{even number}) + P(5) = \frac{3}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$$

# Sum of Probabilities: Dice Example 2



What is the probability of obtaining a sum of 7 or a sum of 10?

# Sum of Probabilities: Dice Example 2

*A*

sum of 7



|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

*B*

sum of 10

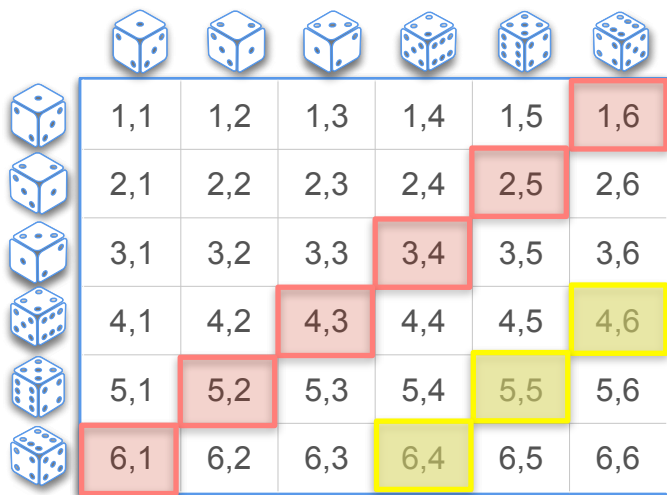


|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

# Sum of Probabilities: Dice Example 2

$A$  or  $B$

sum of 7 or sum of 10



|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

# Sum of Probabilities: Dice Example 2



|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

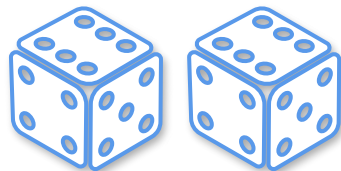
P(sum of 7 or sum of 10)

$$= \boxed{P(\text{sum of 7})} + \boxed{P(\text{sum of 10})}$$

$$= \frac{6}{36} + \frac{3}{36}$$

$$= \frac{9}{36} = \frac{1}{4}$$

# Sum of Probabilities: Dice Example 3



What is the probability of obtaining a difference of 2 or a difference of 1?

# Sum of Probabilities: Dice Example 3

*A*

diff = 2



|                                                                                   |     |     |     |     |     |     |
|-----------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|
|  | 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
|  | 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
|  | 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
|  | 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
|  | 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
|  | 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

*B*

diff = 1



|                                                                                     |     |     |     |     |     |     |
|-------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|
|  | 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
|  | 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
|  | 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
|  | 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
|  | 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
|  | 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |



# Sum of Probabilities: Dice Example 3

$A$  or  $B$

diff = 2 or diff = 1

|  |     |     |     |     |     |     |
|--|-----|-----|-----|-----|-----|-----|
|  |     |     |     |     |     |     |
|  | 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
|  | 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
|  | 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
|  | 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
|  | 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
|  | 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

# Sum of Probabilities: Dice Example 3

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

$P(\text{diff} = 2 \text{ or } \text{diff} = 1)$

$$= \boxed{P(\text{diff} = 2)} + \boxed{P(\text{diff} = 1)}$$

$$= \frac{8}{36} + \frac{10}{36}$$

$$= \frac{18}{36}$$



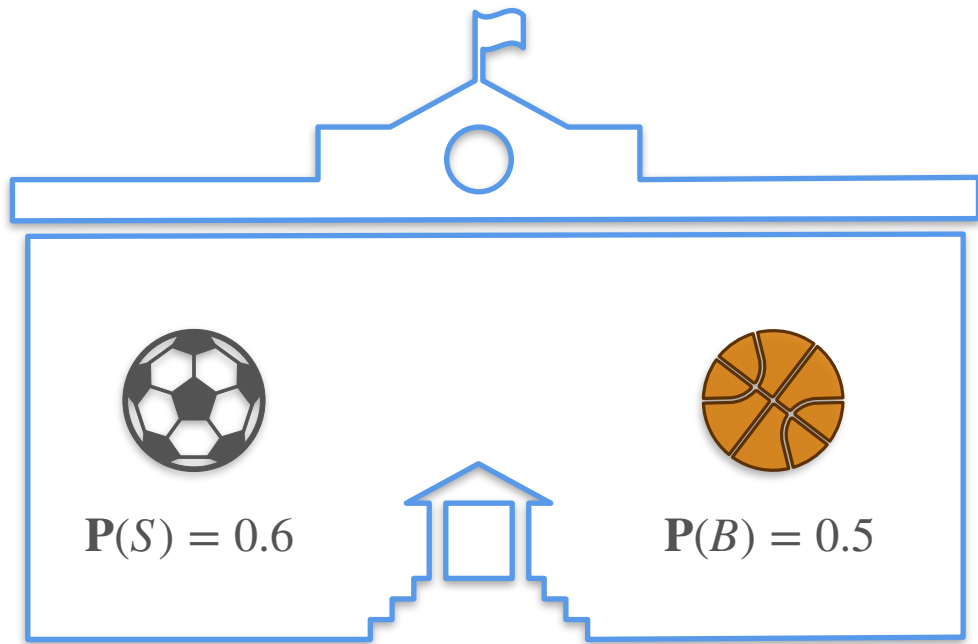
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# Introduction to probability

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## **Sum of Probabilities (Joint Events)**

# Sum of Probabilities (Joint Events): Quiz 1

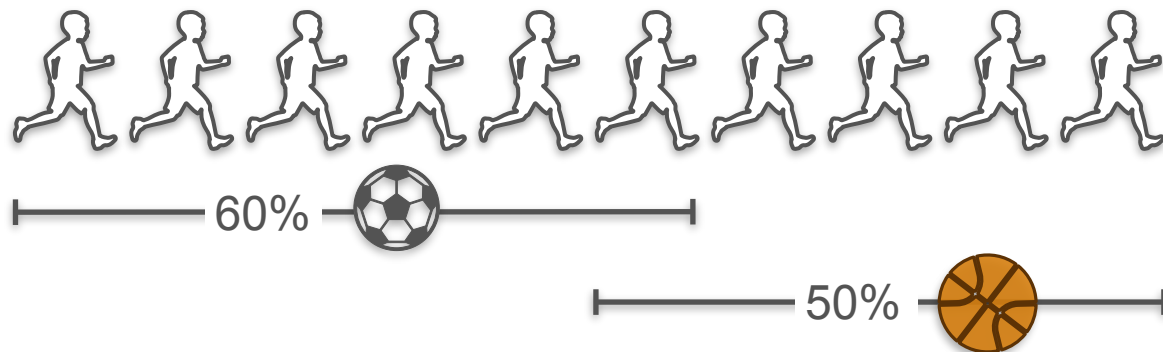


At a school, kids can play as many sports as they want.

What is the probability that a kid plays soccer or basketball?

Hint: What if there were only 10 kids?

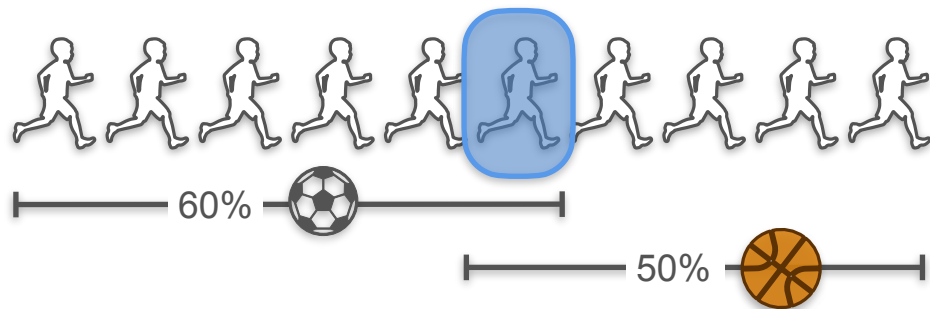
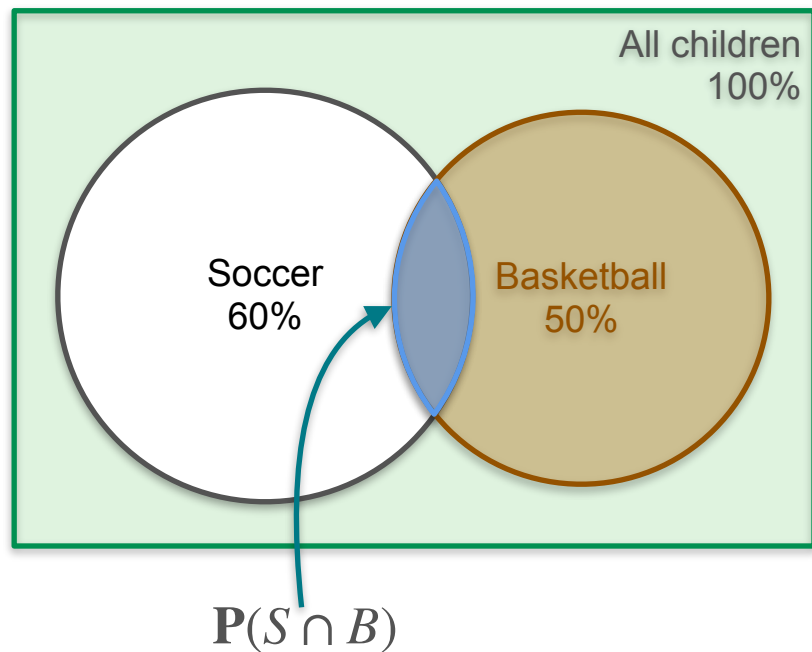
# Sum of Probabilities (Joint Events): Quiz 1 Solution



$$P(\text{soccer} \cup \text{basketball}) = ?$$

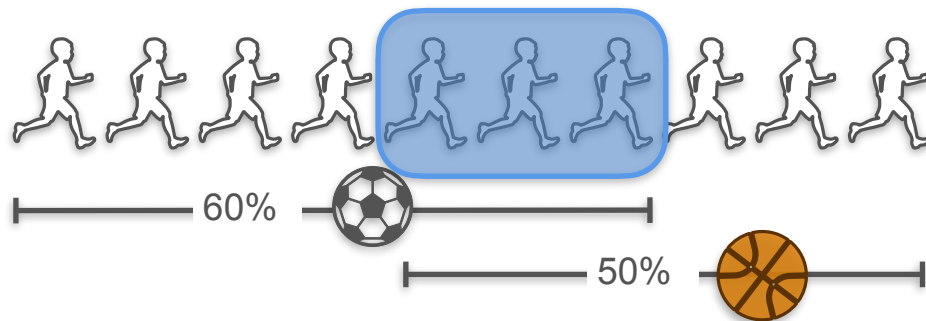
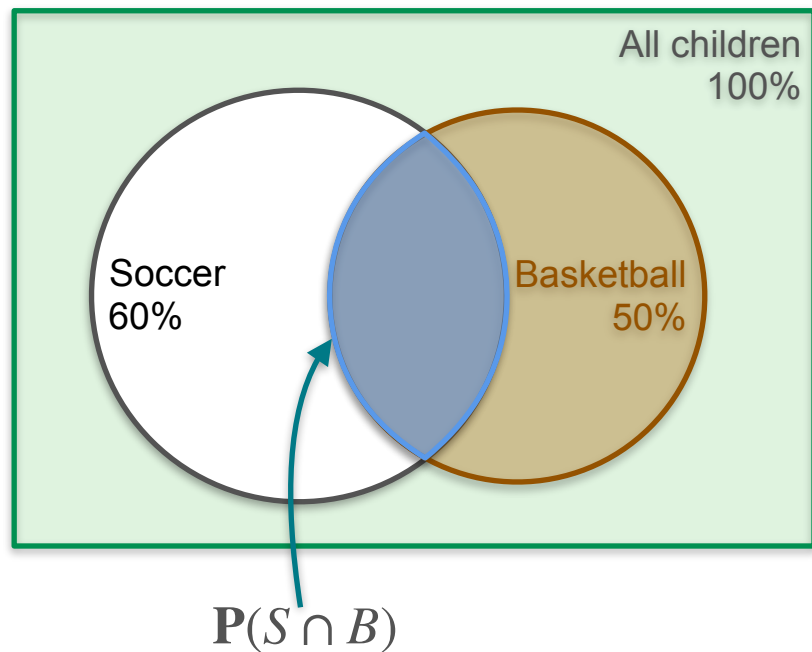
We don't know how many children play multiple sports

# Sum of Probabilities (Joint Events): Quiz 1 Solution



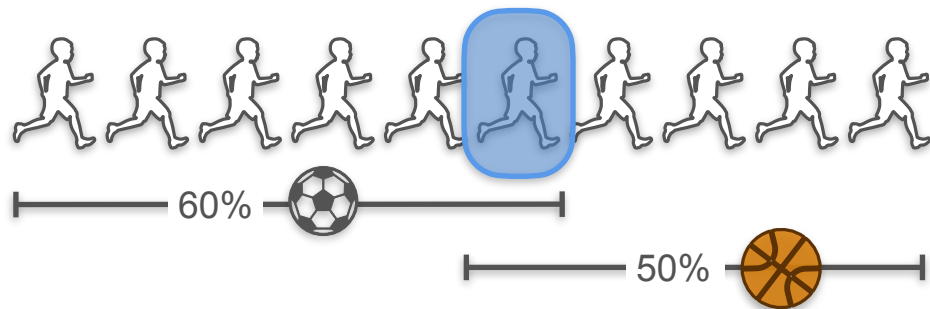
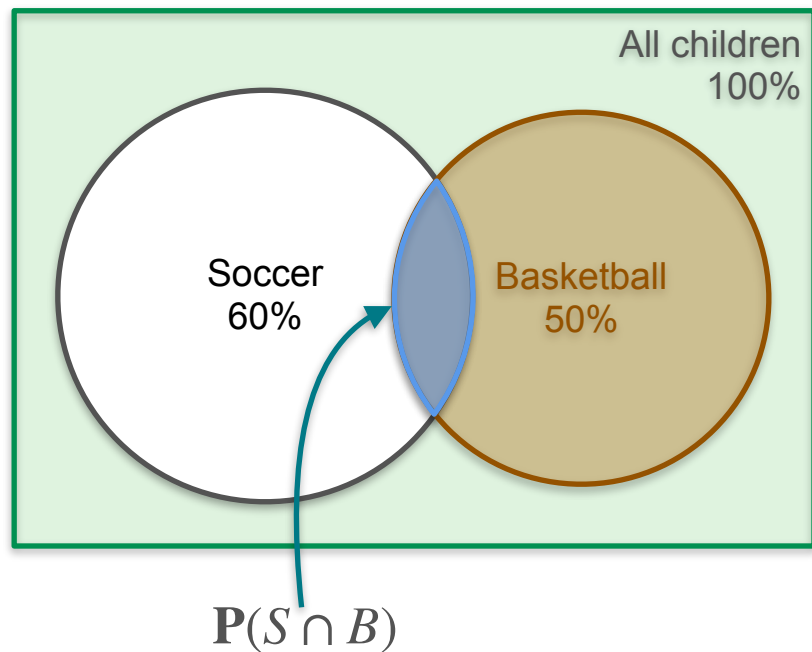
$$P(S \cup B) = P(S) + P(B)$$

# Sum of Probabilities (Joint Events): Quiz 1 Solution



$$P(S \cup B) = P(S) + P(B)$$

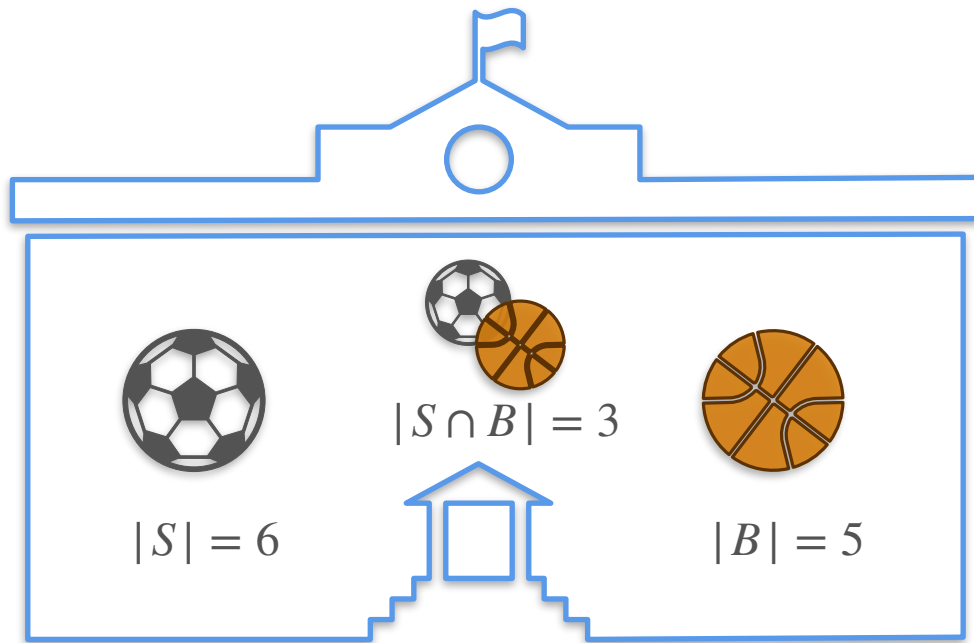
# Sum of Probabilities (Joint Events): Quiz 1 Solution



$$P(S \cup B) = P(S) + P(B)$$



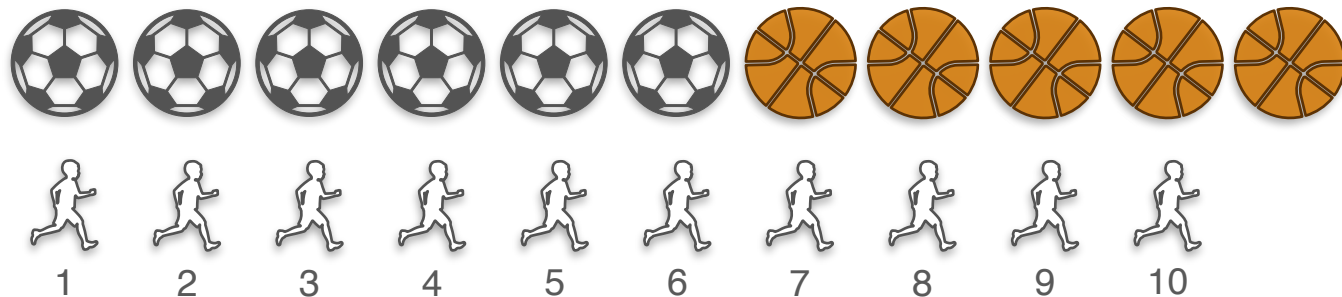
# Sum of Probabilities (Joint Events): Quiz 2



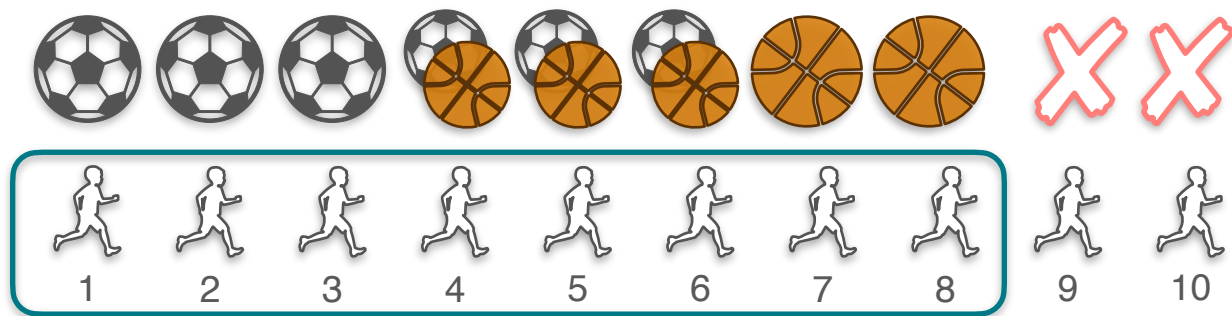
How many kids play soccer or basketball?

Hint: What if there were only 10 kids?

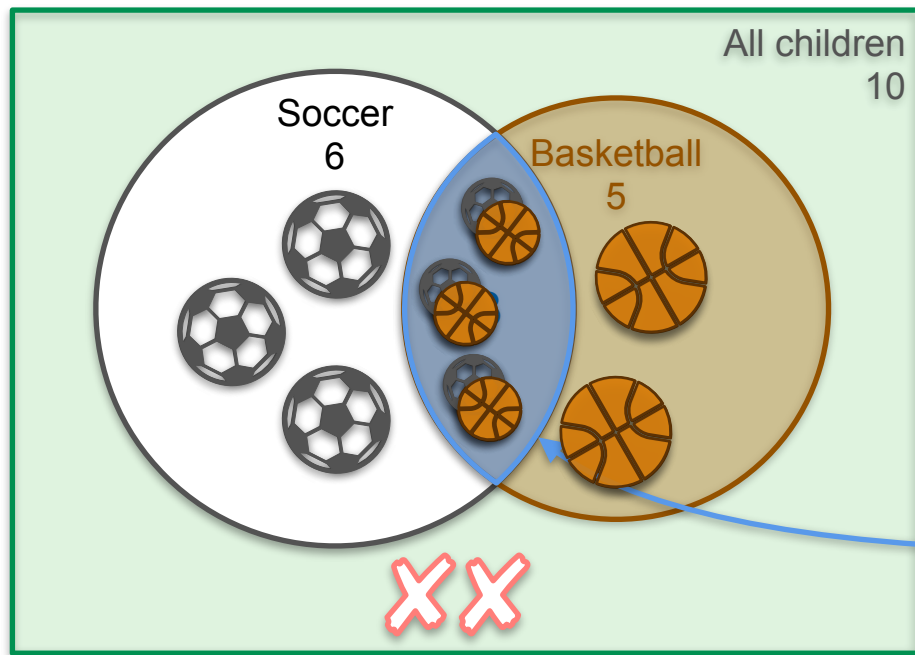
# Sum of Probabilities (Joint Events): Quiz 2 Solution



# Sum of Probabilities (Joint Events): Quiz 2 Solution



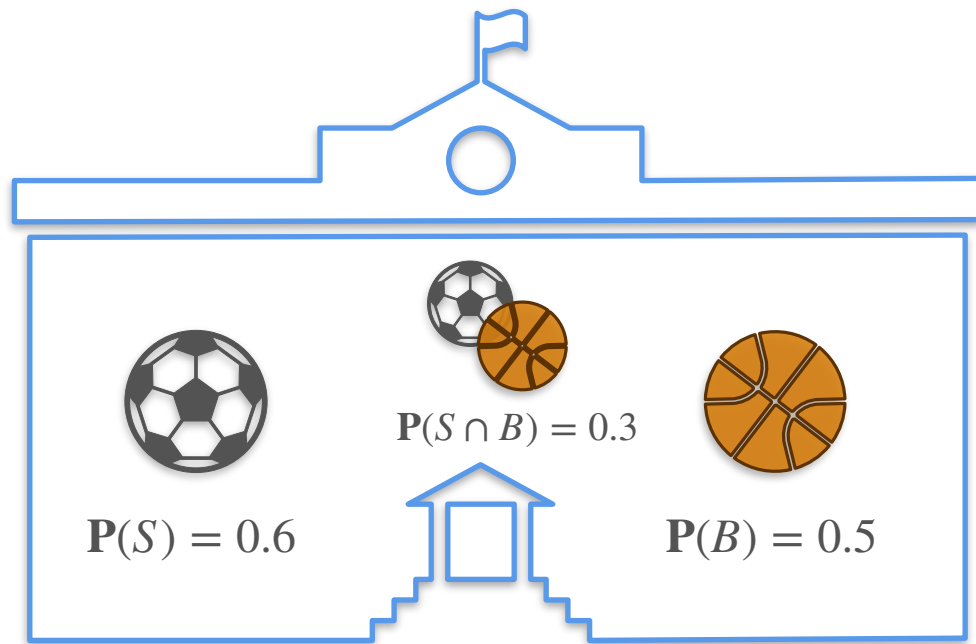
# Sum of Probabilities (Joint Events): Venn Diagram



$$\begin{aligned} |S \cup B| &= |S| + |B| - |S \cap B| \\ &= 6 + 5 - 3 \\ &= 8 \end{aligned}$$

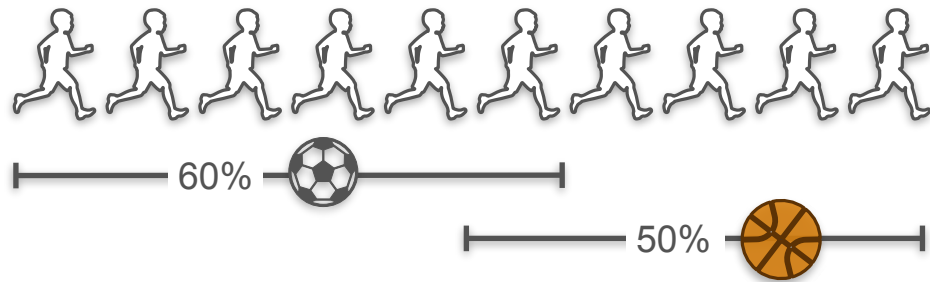
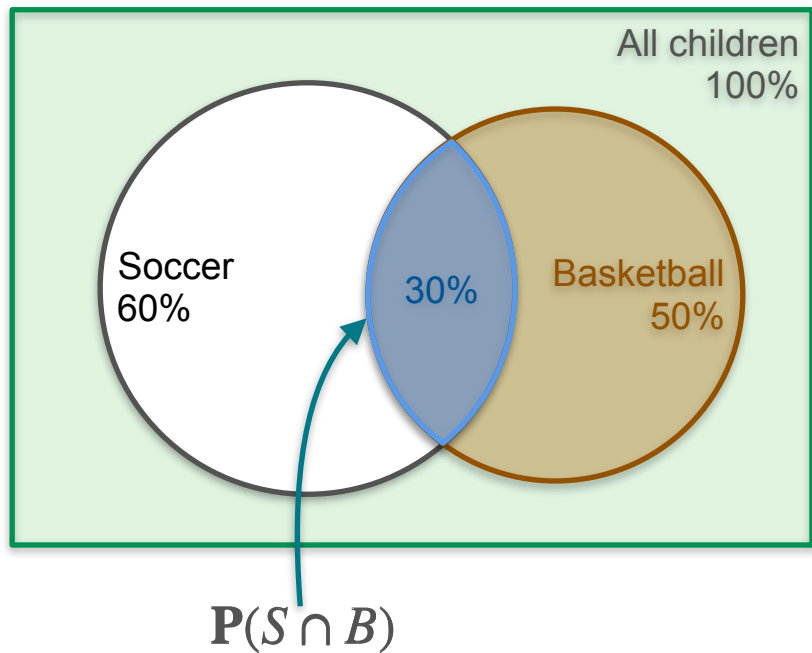
Soccer and Basketball

# Sum of Probabilities (Joint Events): Quiz 3



What is the probability that a child plays soccer or basketball?

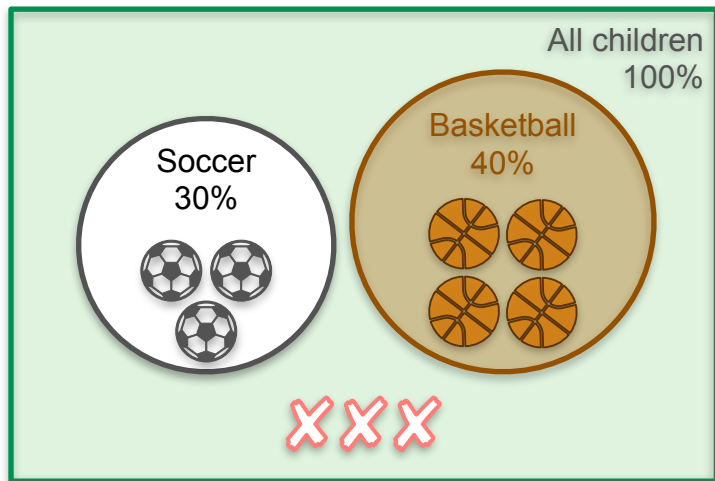
# Sum of Probabilities (Joint Events): Quiz 3 Solution



$$\begin{aligned} P(S \cup B) &= P(S) + P(B) - P(S \cap B) \\ &= 0.6 + 0.5 - 0.3 \\ &= 0.8 \end{aligned}$$

# Disjoint Events vs Joint Events

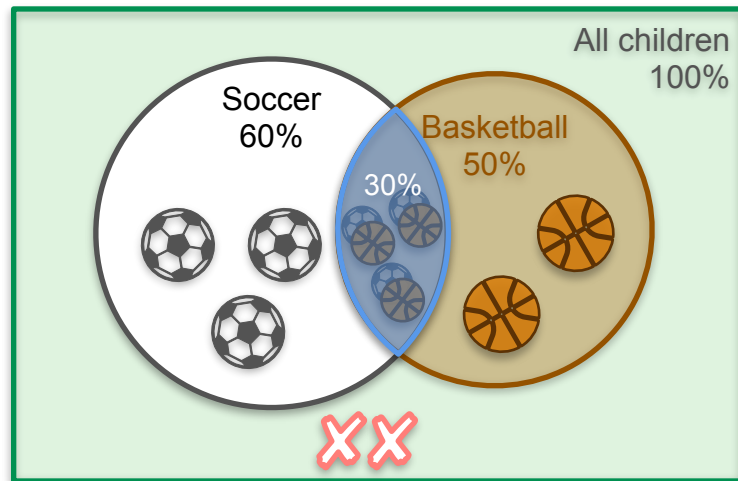
Disjoint



Mutually exclusive

$$\mathbf{P}(S \cup B) = \mathbf{P}(S) + \mathbf{P}(B)$$

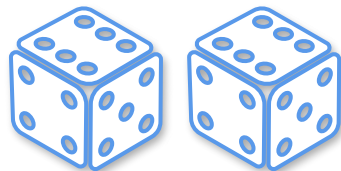
Joint



Non-mutually exclusive

$$\mathbf{P}(S \cup B) = \mathbf{P}(S) + \mathbf{P}(B) - \mathbf{P}(S \cap B)$$

# Sum of Probabilities (Joint Events): Dice Example 1



What is the probability of obtaining a sum of 7 or a difference of 1?



# Sum of Probabilities (Joint Events): Dice Example 1

*A*

sum = 7



|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

*B*

diff = 1



|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

# Sum of Probabilities (Joint Events): Dice Example 1

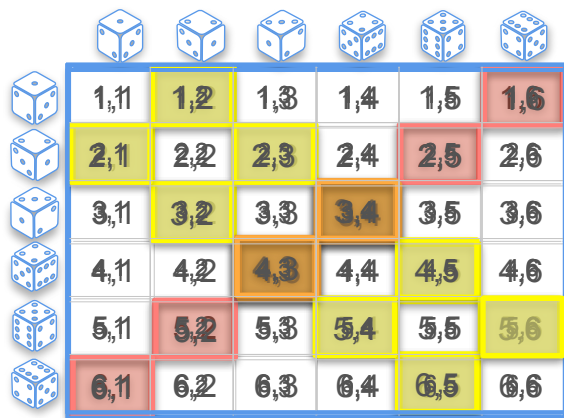
$A$  or  $B$

sum = 7 or diff = 1

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

sum = 7 and diff = 1

# Sum of Probabilities (Joint Events): Dice Example 1



A 6x6 grid representing all possible outcomes of two dice. The first die is on the left, and the second die is on top. The grid is color-coded: yellow for sum=7, red for difference=1, and orange for both conditions. The orange cell (3,4) is the intersection of the two sets.

|   | 1   | 2   | 3   | 4   | 5   | 6   |
|---|-----|-----|-----|-----|-----|-----|
| 1 | 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2 | 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3 | 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4 | 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5 | 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6 | 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

$P(\text{sum} = 7 \text{ or } \text{diff} = 1)$

$$= \boxed{P(\text{sum} = 7)} + \boxed{P(\text{diff} = 1)} - \boxed{P(\text{sum} = 7 \cap \text{diff} = 1)}$$

$$= \frac{6}{36} + \frac{10}{36} - \frac{2}{36}$$
$$= \frac{14}{36}$$



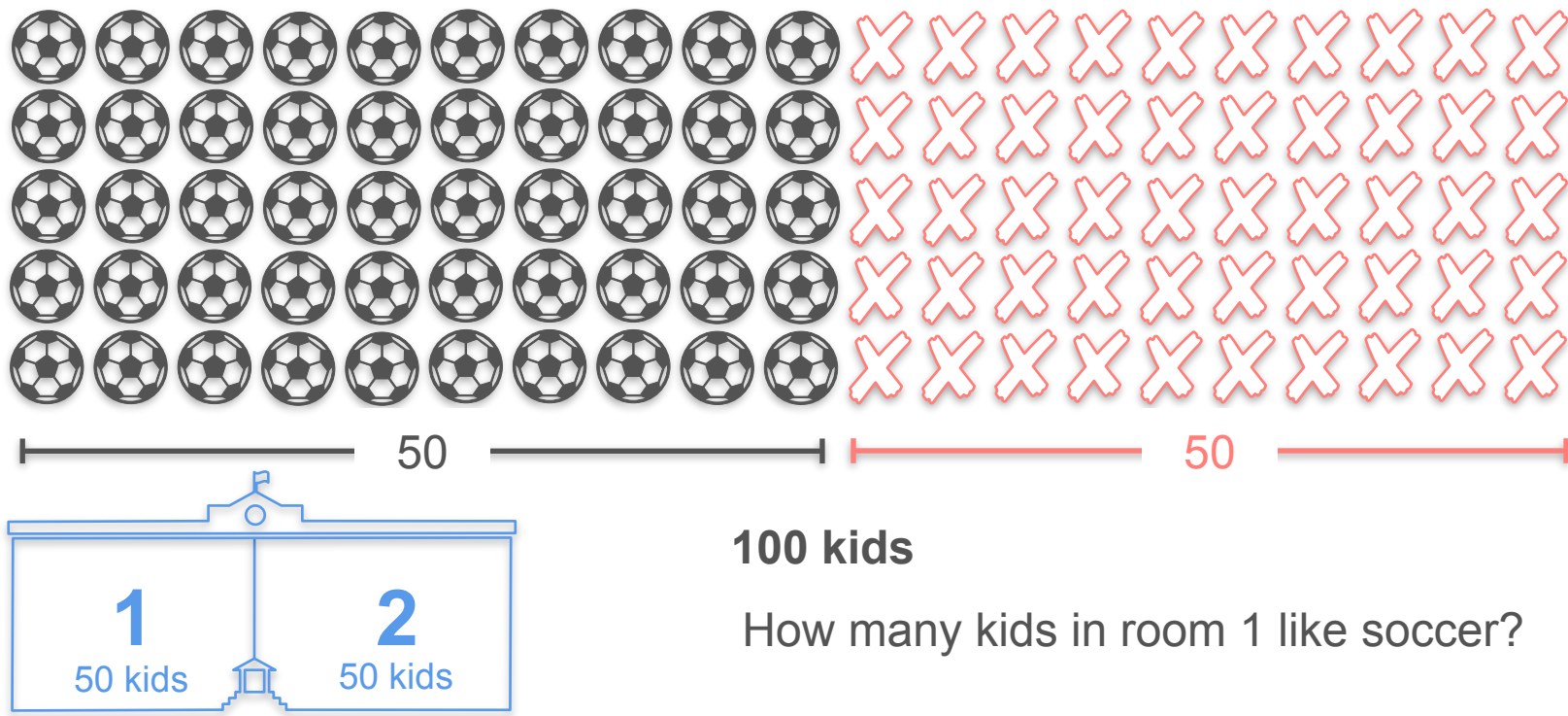
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# Introduction to probability

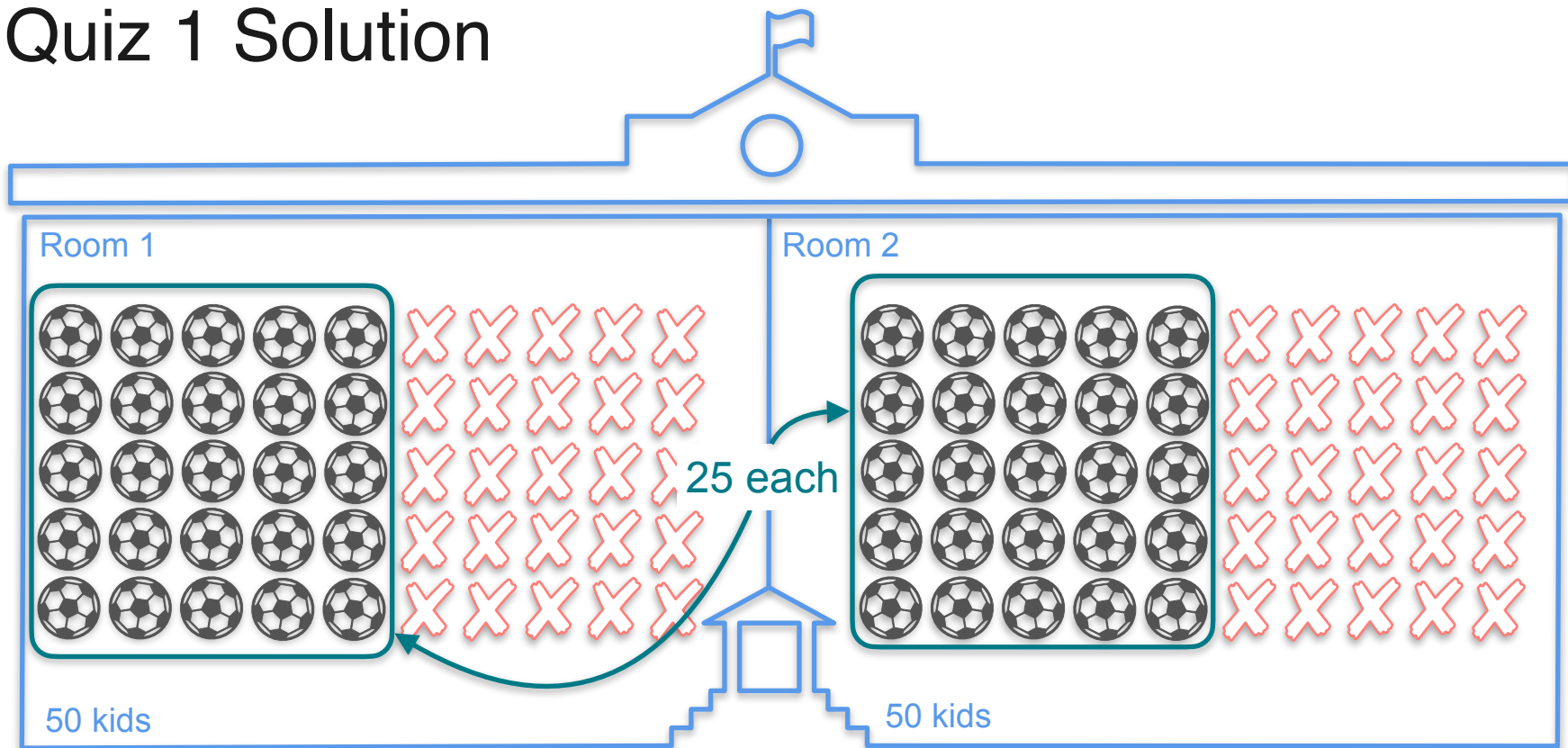
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## Independence

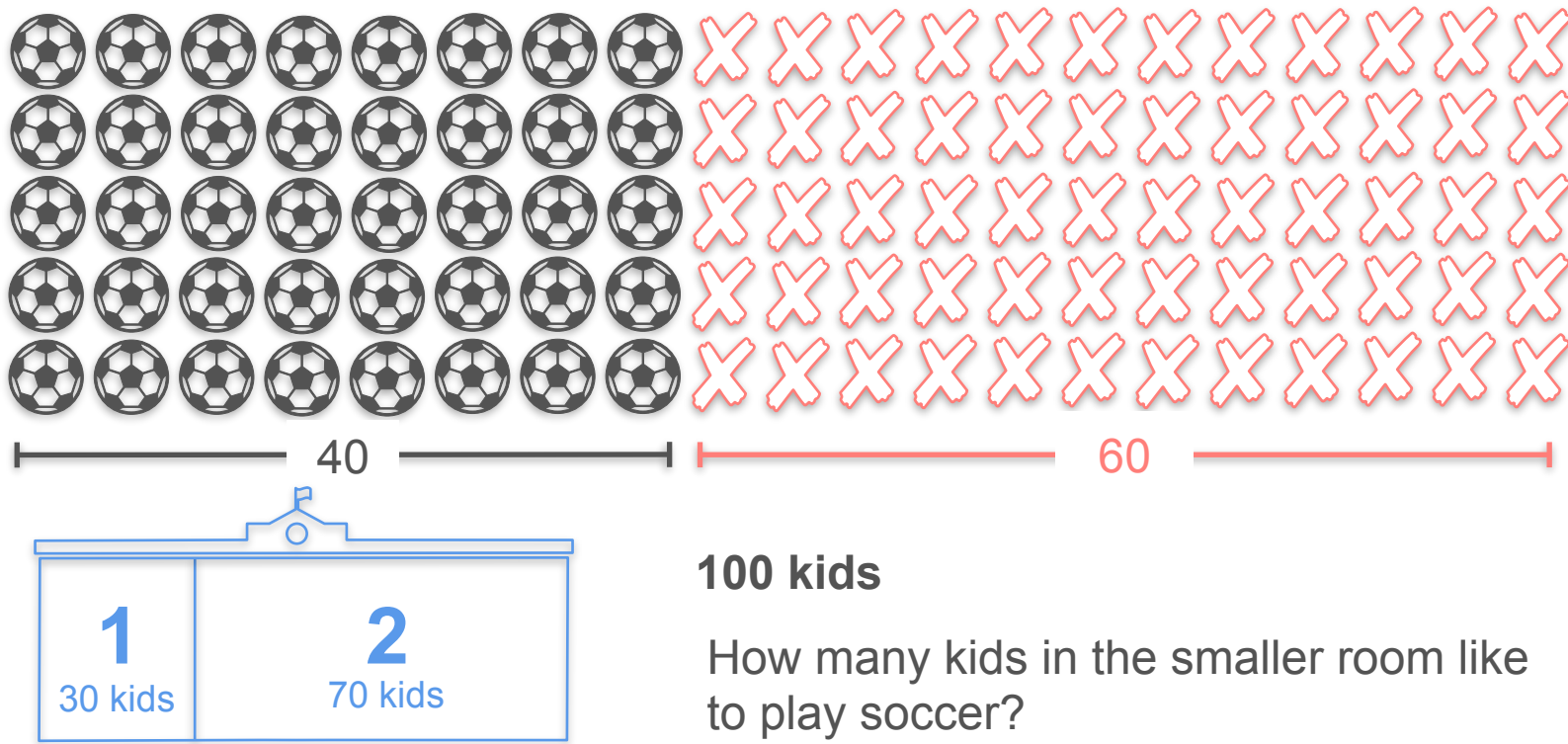
# Independence: Quiz 1



# Quiz 1 Solution



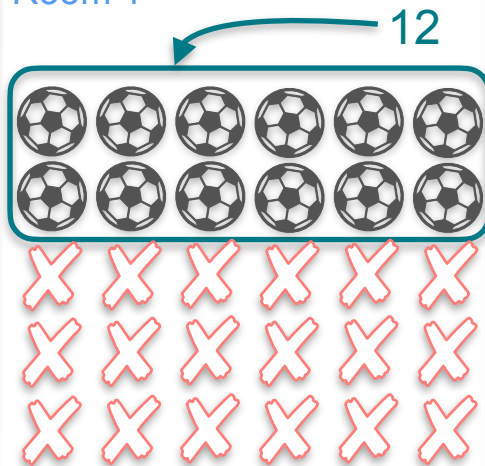
# Independence: Quiz 2



# Quiz 2 Solution

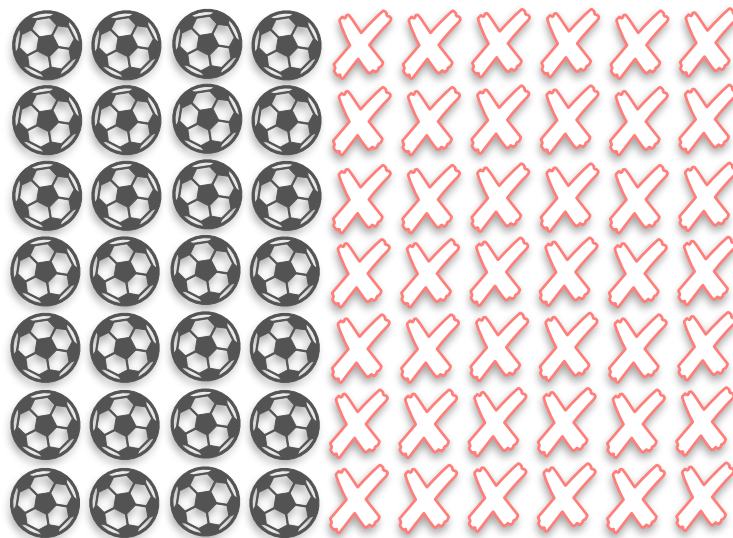


Room 1



30 kids

Room 2



70 kids



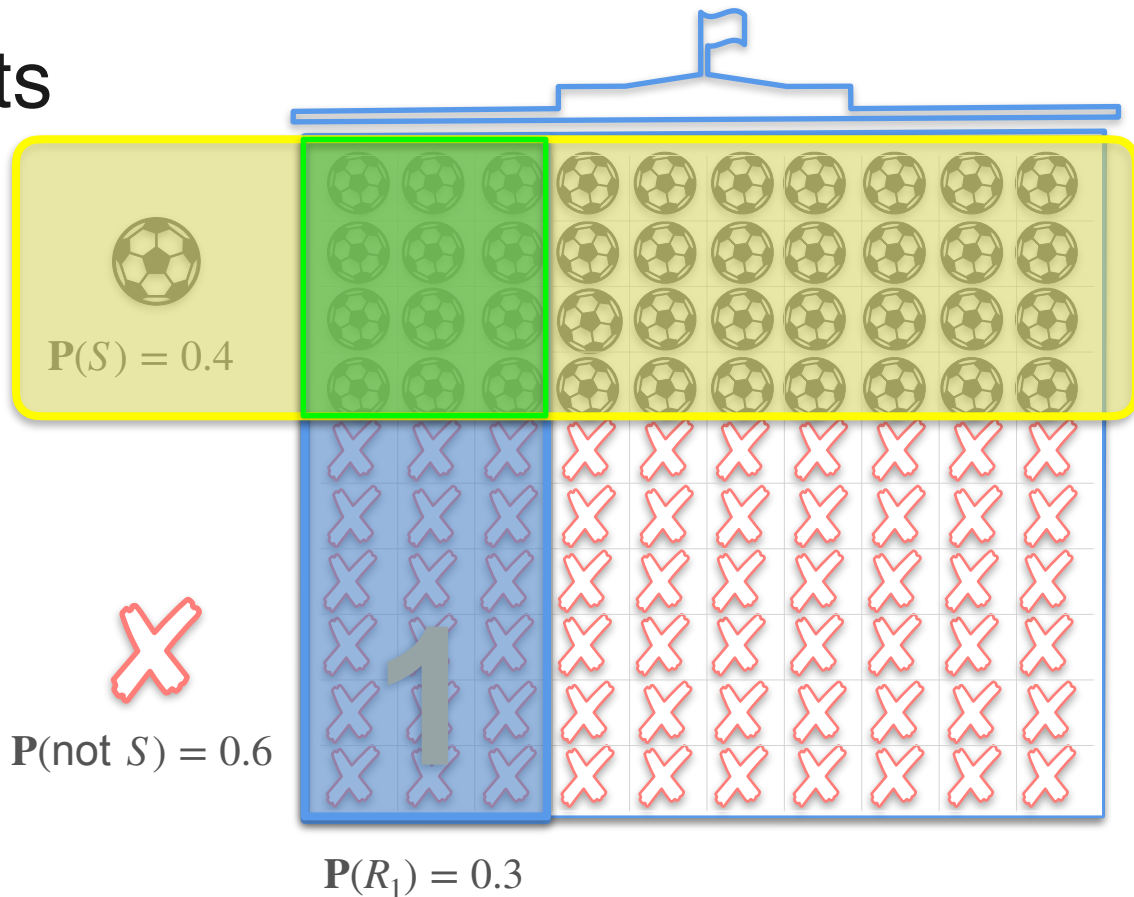
# Independent Events

$P(\text{Soccer and Room 1})$

$$P(S \cap R_1) = P(S) \bullet P(R_1)$$

$$= 0.4 \bullet 0.3$$

$$= 0.12$$



# Product Rule (for Independent Events)

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$$

# Independent Events: Coin Example 1

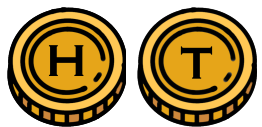


50% 50%

What is the probability of landing on heads five times?



# Independent Events: Coin Example 1



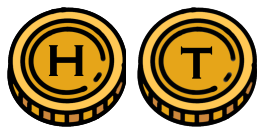
50% 50%

What is the probability of landing on heads five times?

$$P(5 \text{ heads}) = \frac{\text{H}}{\text{H} \text{ T}} \cdot \frac{\text{H}}{\text{H} \text{ T}} \cdot \frac{\text{H}}{\text{H} \text{ T}} \cdot \frac{\text{H}}{\text{H} \text{ T}} \cdot \frac{\text{H}}{\text{H} \text{ T}}$$

$\frac{1}{2} \quad \cdot \quad \frac{1}{2} \quad \cdot \quad \frac{1}{2} \quad \cdot \quad \frac{1}{2} \quad \cdot \quad \frac{1}{2}$

# Independent Events: Coin Example 2



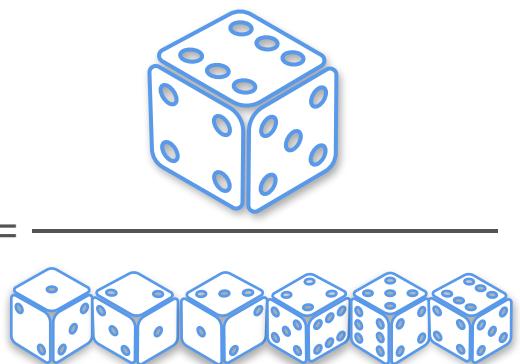
50% 50%

What is the probability of landing 5n heads five times?

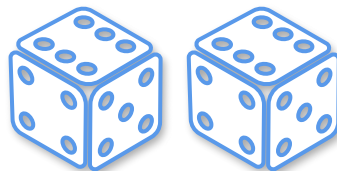
$$P(5 \text{ heads}) = \left( \frac{\text{H}}{\text{H} \text{ T}} \right)$$

$$\left( \frac{1}{2} \right)^5 = \frac{1}{32}$$

# Independent Events: Dice Example 1


$$P(6) = \frac{1}{6}$$

# Independent Events: Dice Example 1

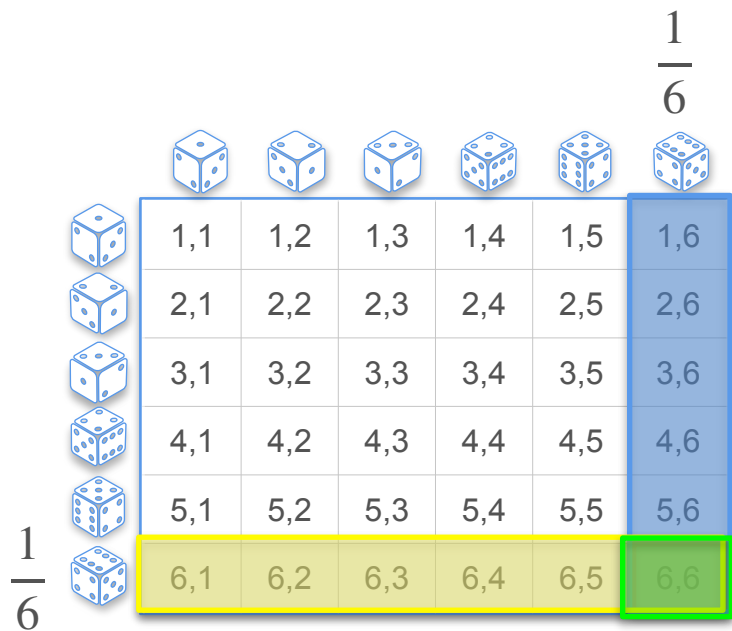


2 dice

|                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|                                                                                   |  |  |  |  |  |  |
|  | 1,1                                                                               | 1,2                                                                               | 1,3                                                                               | 1,4                                                                               | 1,5                                                                               | 1,6                                                                               |
|  | 2,1                                                                               | 2,2                                                                               | 2,3                                                                               | 2,4                                                                               | 2,5                                                                               | 2,6                                                                               |
|  | 3,1                                                                               | 3,2                                                                               | 3,3                                                                               | 3,4                                                                               | 3,5                                                                               | 3,6                                                                               |
|  | 4,1                                                                               | 4,2                                                                               | 4,3                                                                               | 4,4                                                                               | 4,5                                                                               | 4,6                                                                               |
|  | 5,1                                                                               | 5,2                                                                               | 5,3                                                                               | 5,4                                                                               | 5,5                                                                               | 5,6                                                                               |
|  | 6,1                                                                               | 6,2                                                                               | 6,3                                                                               | 6,4                                                                               | 6,5                                                                               | 6,6                                                                               |

$$P(6,6) = \frac{1}{36}$$

# Independent Events: Dice Example 1



A 6x6 grid of dice outcomes. The top row is highlighted in blue, and the bottom row is highlighted in yellow. The cell (6,6) is highlighted in green.

|                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|                                                                                   |  |  |  |  |  |  |
|  | 1,1                                                                               | 1,2                                                                               | 1,3                                                                               | 1,4                                                                               | 1,5                                                                               | 1,6                                                                               |
|  | 2,1                                                                               | 2,2                                                                               | 2,3                                                                               | 2,4                                                                               | 2,5                                                                               | 2,6                                                                               |
|  | 3,1                                                                               | 3,2                                                                               | 3,3                                                                               | 3,4                                                                               | 3,5                                                                               | 3,6                                                                               |
|  | 4,1                                                                               | 4,2                                                                               | 4,3                                                                               | 4,4                                                                               | 4,5                                                                               | 4,6                                                                               |
|  | 5,1                                                                               | 5,2                                                                               | 5,3                                                                               | 5,4                                                                               | 5,5                                                                               | 5,6                                                                               |
|  | 6,1                                                                               | 6,2                                                                               | 6,3                                                                               | 6,4                                                                               | 6,5                                                                               | 6,6                                                                               |

$\frac{1}{6}$

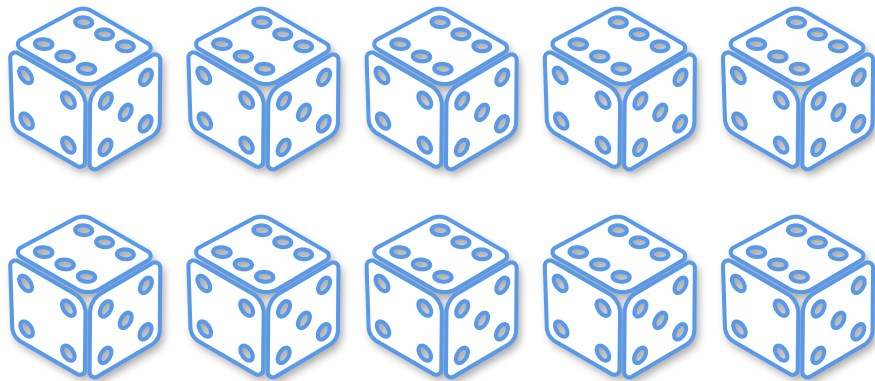
2 dice

$$P(6,6) = \frac{\text{1 die showing 6}}{\text{6 possible outcomes}} \cdot \frac{\text{1 die showing 6}}{\text{6 possible outcomes}}$$
$$= \left(\frac{1}{6}\right) \cdot \left(\frac{1}{6}\right) = \frac{1}{36}$$



# Independent Events: Dice Example 2

What is the probability of getting 10 sixes?



# Independent Events: Dice Example 2

What is the probability of getting 10 sixes?

$$\mathbf{P(10\ sixes)} = \left( \frac{\img alt="A single blue die showing a six" data-bbox="475 455 525 545}}{\img alt="Ten blue dice arranged in a 2x5 grid, all showing a six" data-bbox="460 575 540 665}} \right)^{10}$$
$$= \left( \frac{1}{6} \right)^{10}$$



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# Introduction to probability

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## **Birthday problem**

# Quiz

- You have 30 friends at a party. What do you think is more likely:
  - That there exist two people with the same birthday
  - That no two of them have the same birthday
- (Assume the year has 365 days, nobody has a birthday on Feb 29).

# Quiz

- Answer: It's more likely that 2 people have the same birthday.
- In fact, the probability of no two people having the same birthday is around 0.3.

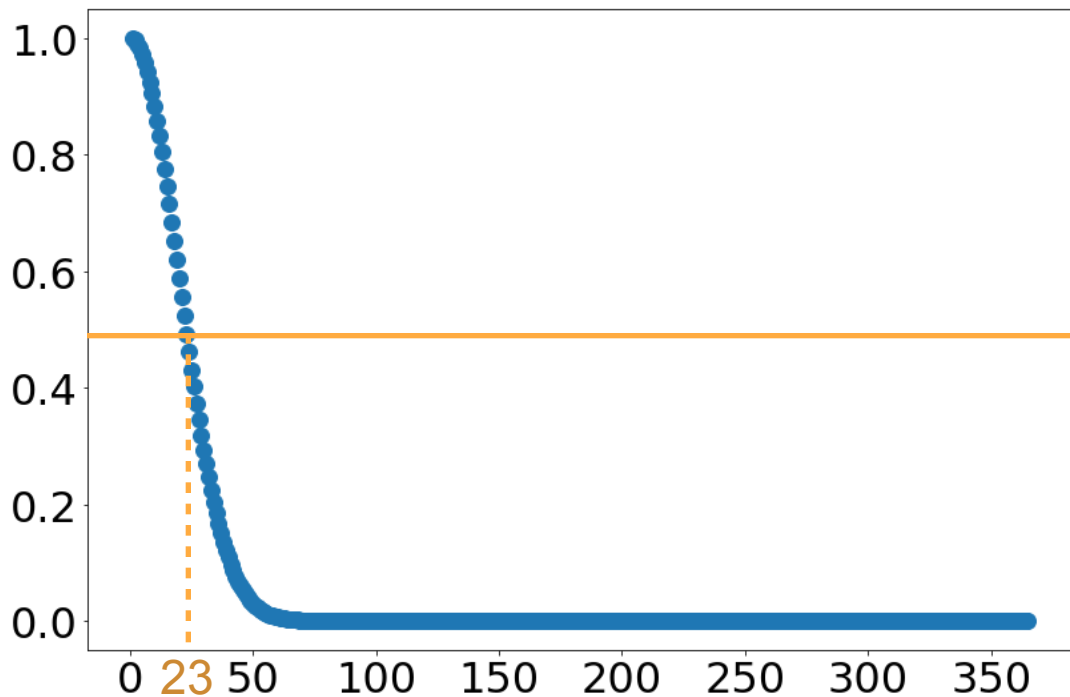
# Probability That Everyone Has a Different Birthday



$$\frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdot \frac{362}{365} \cdot \frac{361}{365} \cdot \frac{360}{365} \cdot \frac{359}{365} \cdot \frac{358}{365} \cdot \frac{357}{365} = 0.905$$

# Probability That no Two People Have the Same Birthday

1 person: 1  
2 people: 0.997  
3 people: 0.992  
4 people: 0.984  
5 people: 0.973  
10 people: 0.883  
20 people: 0.589  
**23 people: 0.493**  
30 people: 0.294  
50 people: 0.030  
100 people: 0.0000003  
365 people: 0





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# Introduction to Probability

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## **Conditional probability**



# Conditional Probability: Coin Example 1



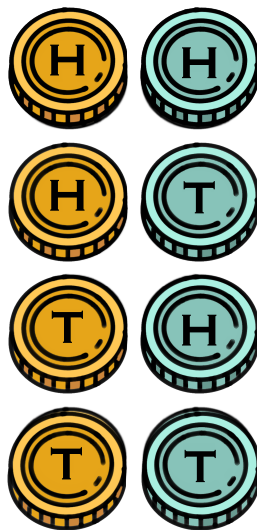
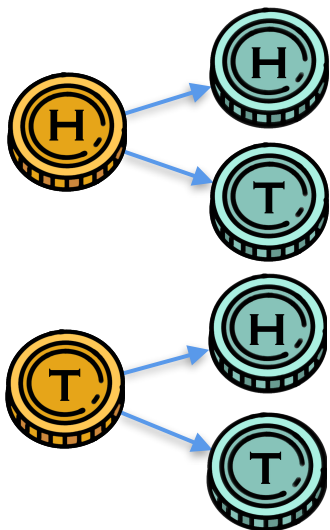
50% 50%

What is the probability of landing on heads twice?

1st

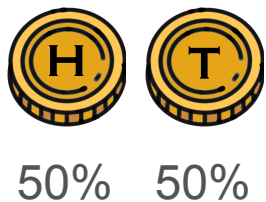
2nd

**GIVEN that the first one is heads**



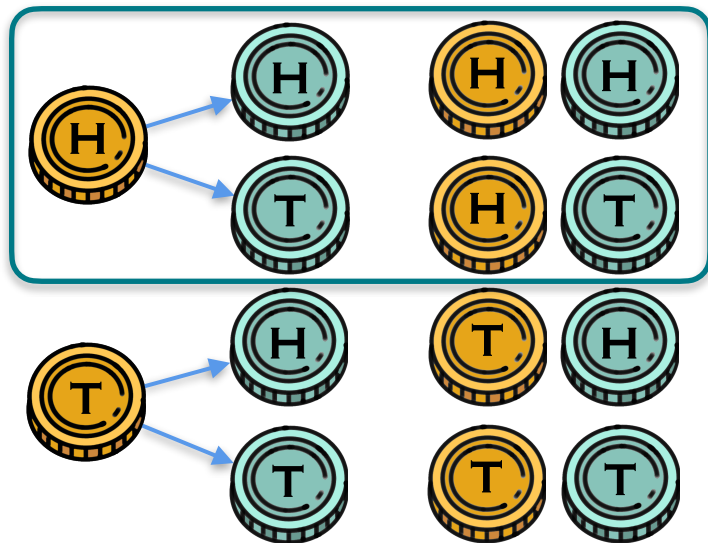
$$P(HH) = \frac{\text{Number of outcomes where both are heads}}{\text{Total number of outcomes}} = \frac{1}{4}$$

# Conditional Probability: Coin Example 1



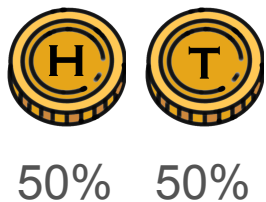
What is the probability of landing on heads twice?

1st 2nd **GIVEN that the first one is heads**



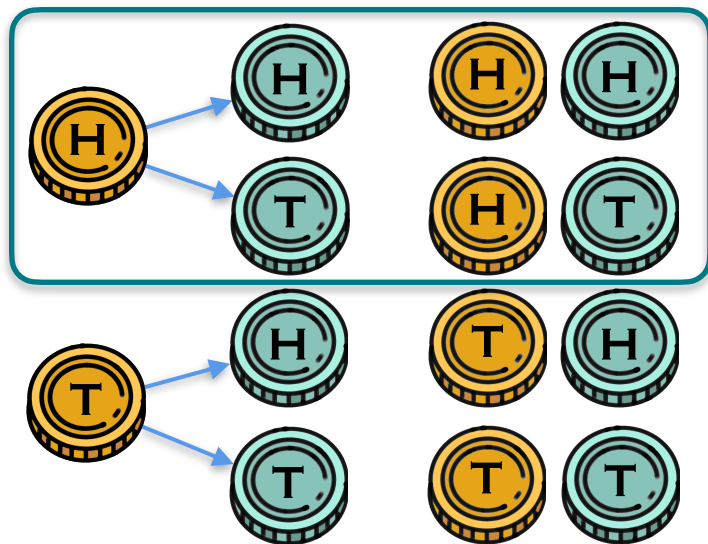
$$P(HH) = \frac{\text{Number of outcomes where both are heads}}{\text{Total number of outcomes}} = \frac{1}{4}$$

# Conditional Probability: Coin Example 1



What is the probability of landing on heads twice?

1st 2nd **GIVEN that the first one is heads**



$$P(HH \mid 1st \text{ is } H) = \frac{\text{Number of outcomes where both are heads}}{\text{Number of outcomes where the first is heads}} = \frac{1}{2}$$

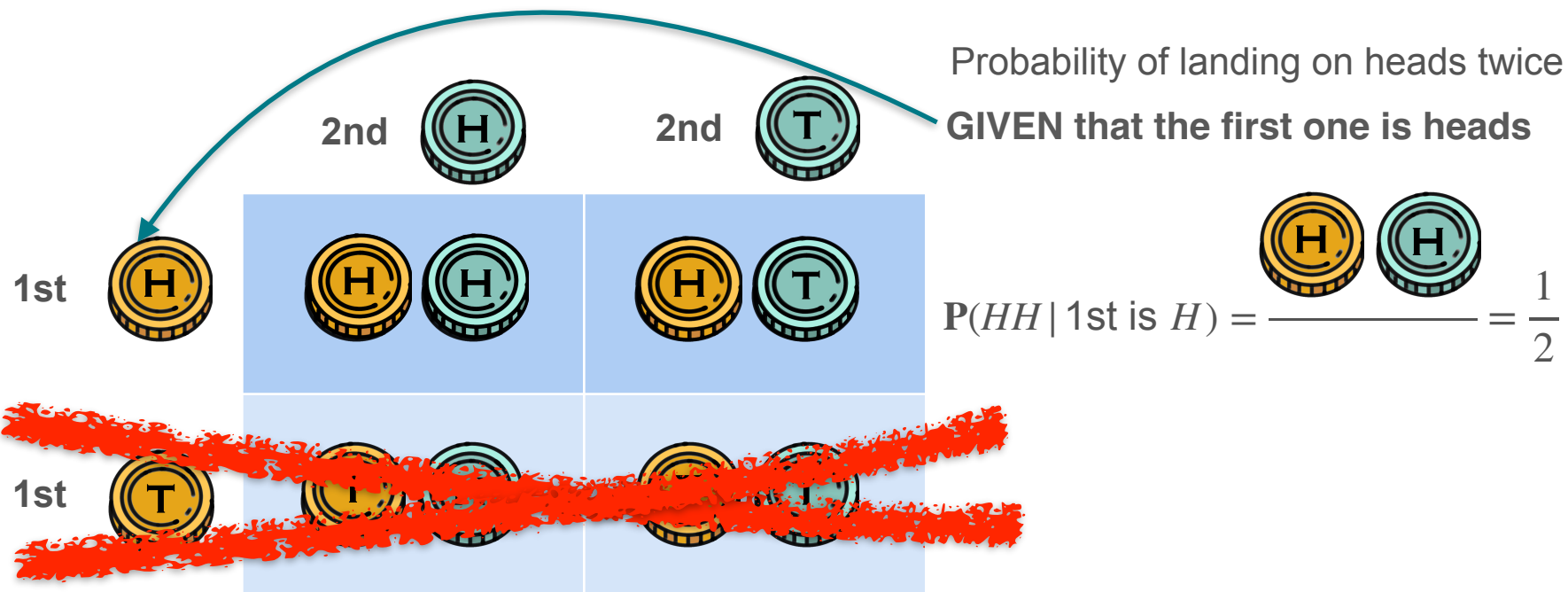
# Conditional Probability: Coin Example 1

What is the probability of landing on heads twice?

**GIVEN** that the first one is heads


$$\mathbf{P}(HH \mid \text{1st is } H)$$

# Conditional Probability: Coin Example 1



# Conditional Probability: Coin Example 2



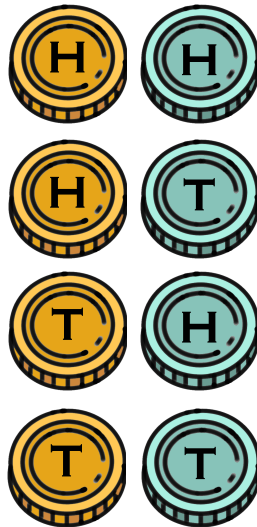
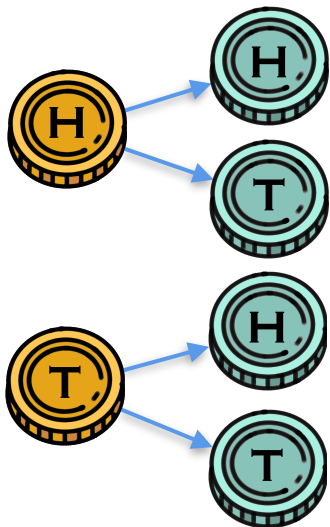
50% 50%

What is the probability of landing on heads twice?

1st

2nd

**GIVEN** that the first one is tails



$$P(HH) = \frac{\text{Number of outcomes where both are heads}}{\text{Number of outcomes where the first is tails}} = \frac{1}{4}$$

# Conditional Probability: Coin Example 2



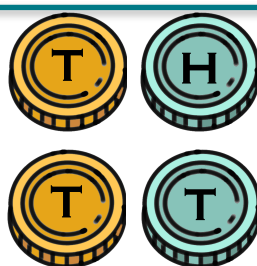
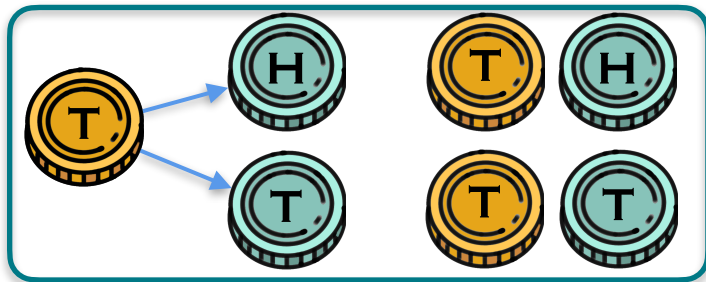
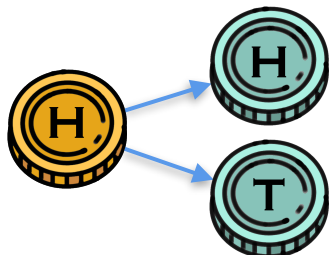
50% 50%

What is the probability of landing on heads twice?

1st

2nd

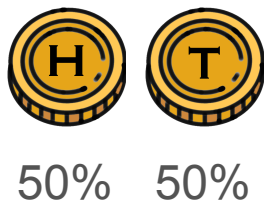
**GIVEN that the first one is tails**



$$P(HH) = \frac{\text{Number of outcomes where both are heads}}{\text{Total number of outcomes}} = \frac{1}{4}$$



# Conditional Probability: Coin Example 2

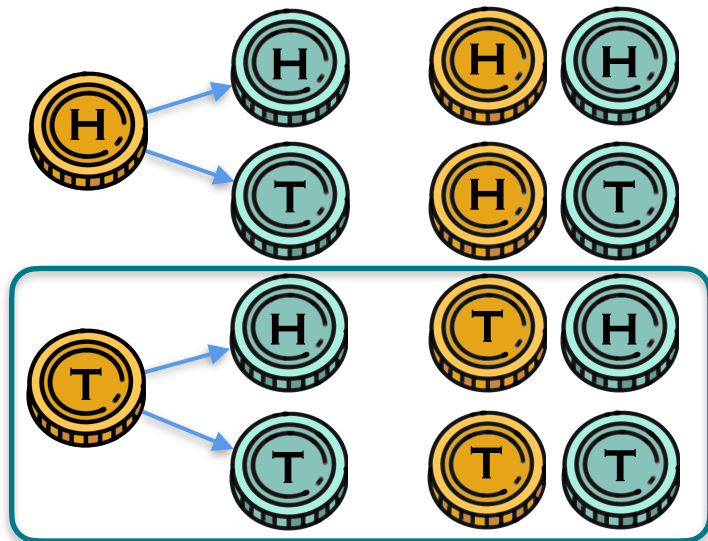


What is the probability of landing on heads twice?

1st

2nd

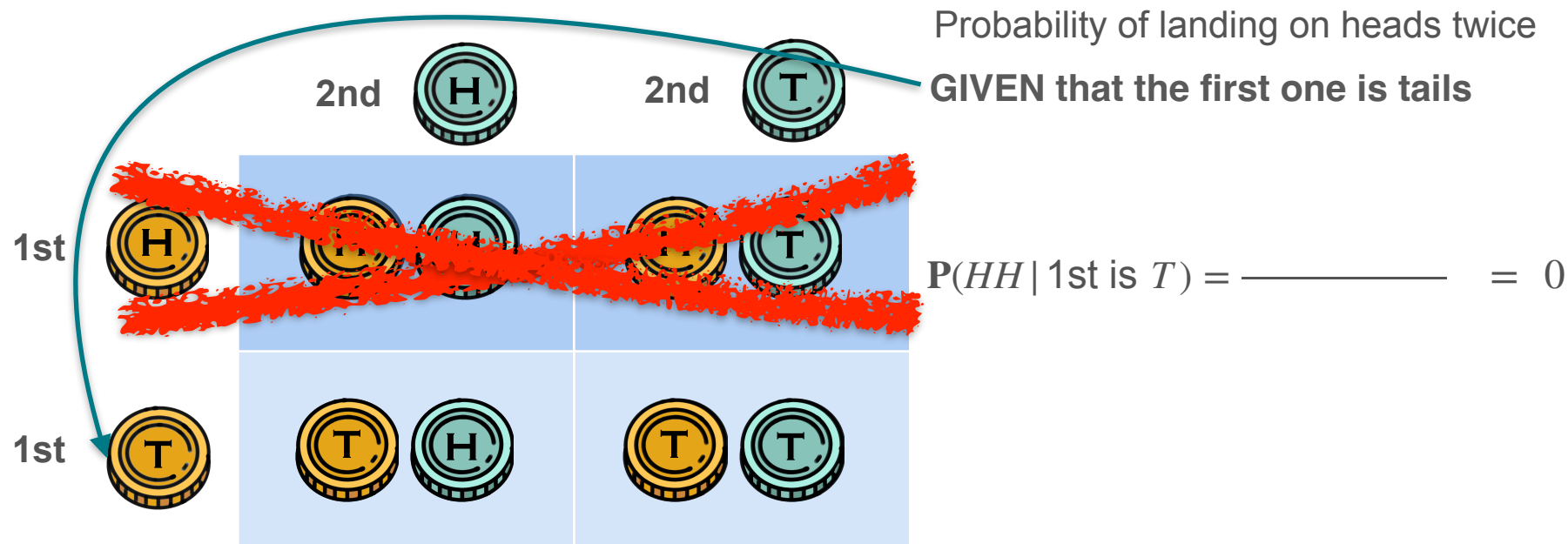
**GIVEN that the first one is tails**



$$P(HH | 1st \text{ is } T) = \frac{0}{\text{4 outcomes}} = 0$$




# Conditional Probability: Coin Example 2

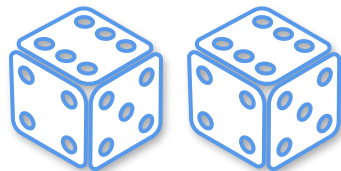


# Product Rule (for Independent Events)

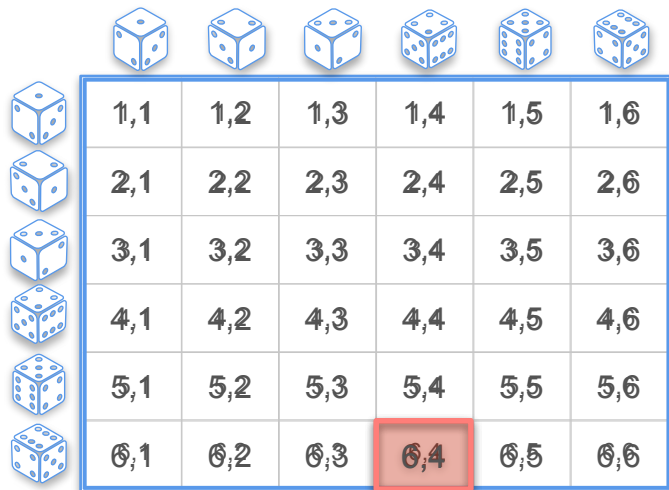
When A and B independent

$$\mathbf{P(A \cap B) = P(A) \cdot P(B)}$$

# Conditional Probability: Dice Example 3



What is the probability that  
the first is 6 **AND** the sum = 10?

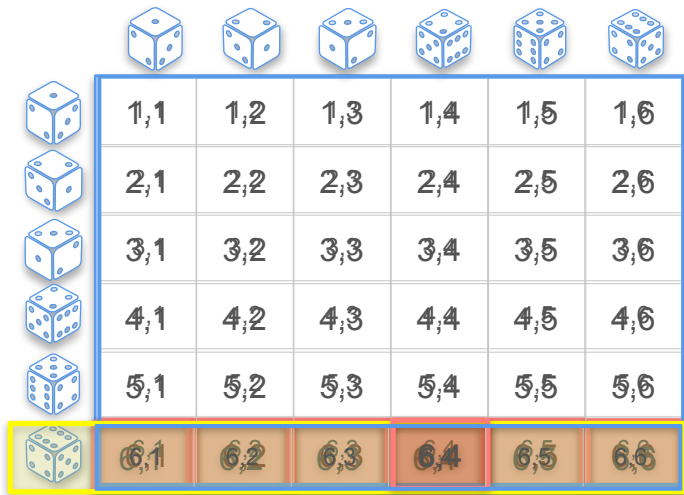
Six blue-outlined dice are arranged in a vertical column to the left of the table. Each die shows a different face: 1, 2, 3, 4, 5, and 6 from top to bottom.

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

$$P(\text{1st is 6} \cap \text{sum} = 10) = \underline{\hspace{2cm}}$$

$$= \frac{1}{36}$$

# Conditional Probability: Dice Example 3



|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

$P(\text{1st is 6} \cap \text{sum} = 10) =$

$P(\text{1st is 6}) \quad \bullet \quad P(\text{sum} = 10 \mid \text{1st 6})$

\_\_\_\_\_

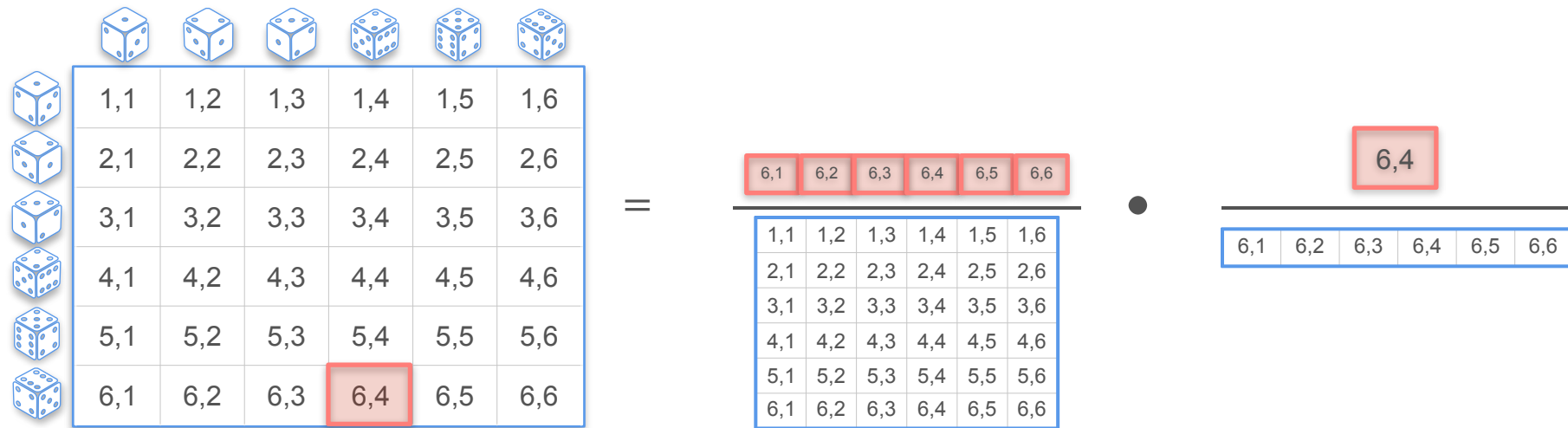
$$\frac{6}{36}$$

•

$$\frac{1}{6}$$

$$= \frac{1}{36}$$

# Conditional Probability: Dice Example 3



|  |     |     |     |     |     |     |
|--|-----|-----|-----|-----|-----|-----|
|  |     |     |     |     |     |     |
|  | 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
|  | 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
|  | 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
|  | 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
|  | 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
|  | 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

=

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |
|-----|-----|-----|-----|-----|-----|

•

|     |
|-----|
| 6,4 |
|-----|

•

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |
|-----|-----|-----|-----|-----|-----|

$$\mathbf{P(1st\ is\ 6 \cap\ sum = 10) = P(1st\ is\ 6) \cdot P(sum = 10 \mid 1st\ 6)}$$

$$\mathbf{P(A \cap B) = P(A) \cdot P(B \mid A)}$$

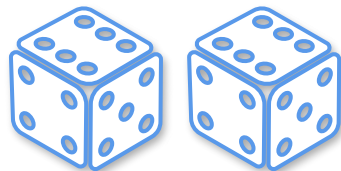
# The General Product Rule

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B | A)$$

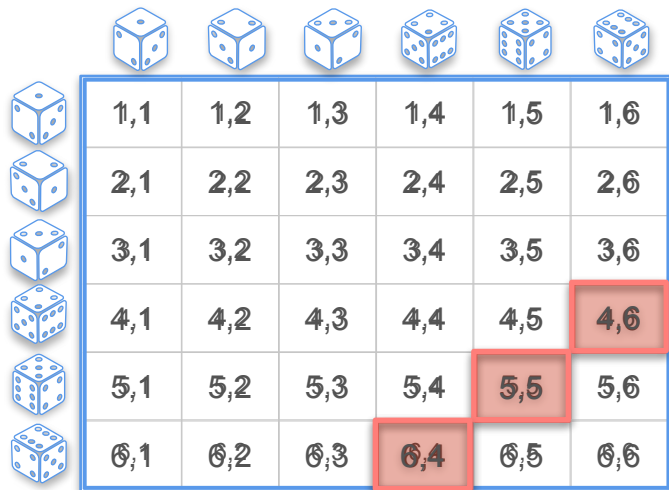
When independent

$$\mathbf{P}(B | A) = \mathbf{P}(B)$$

# Conditional Probability: Dice Example 1



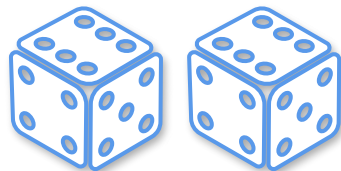
What is the probability that the sum is 10?

A vertical column of six blue-outlined dice icons is positioned to the left of the table, each corresponding to a row.

|                                                                                  |  |  |  |  |  |  |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  | 1,1                                                                               | 1,2                                                                               | 1,3                                                                               | 1,4                                                                               | 1,5                                                                               | 1,6                                                                               |
|  | 2,1                                                                               | 2,2                                                                               | 2,3                                                                               | 2,4                                                                               | 2,5                                                                               | 2,6                                                                               |
|  | 3,1                                                                               | 3,2                                                                               | 3,3                                                                               | 3,4                                                                               | 3,5                                                                               | 3,6                                                                               |
|  | 4,1                                                                               | 4,2                                                                               | 4,3                                                                               | 4,4                                                                               | 4,5                                                                               | 4,6                                                                               |
|  | 5,1                                                                               | 5,2                                                                               | 5,3                                                                               | 5,4                                                                               | 5,5                                                                               | 5,6                                                                               |
|  | 6,1                                                                               | 6,2                                                                               | 6,3                                                                               | 6,4                                                                               | 6,5                                                                               | 6,6                                                                               |

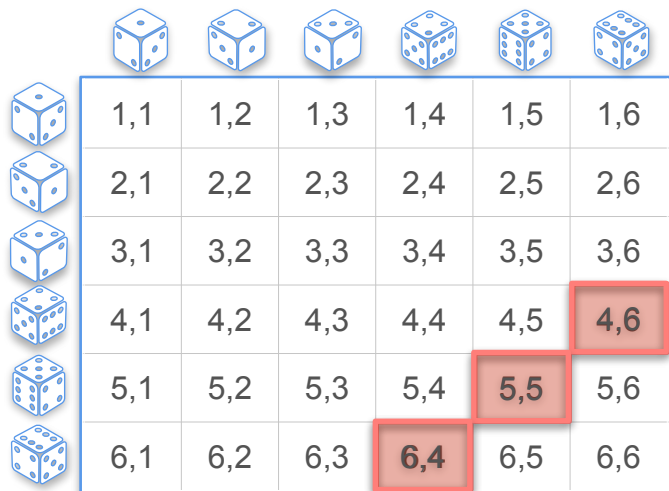
$$P(\text{sum} = 10) = \frac{3}{36}$$

# Conditional Probability: Dice Example 1



What is the probability that the sum is 10?

**GIVEN** that the first one is 6

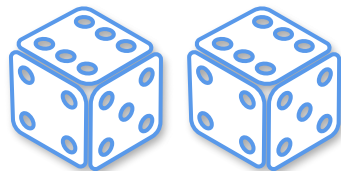
Six small blue-outlined dice icons are arranged vertically to the left of the table, each corresponding to a row of the table.

|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

$P(\text{sum} = 10 \mid \text{1st is 6}) = \underline{\hspace{2cm}}$

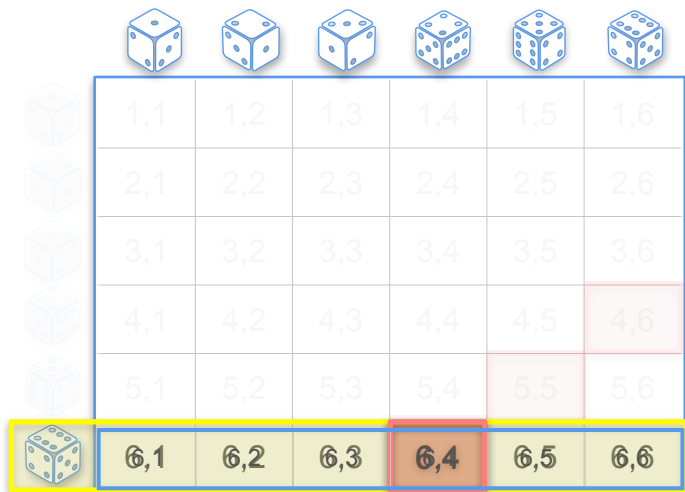


# Conditional Probability: Dice Example 1



What is the probability that the sum is 10?

**GIVEN** that the first one is 6

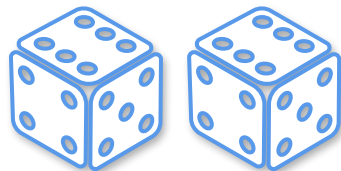


|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

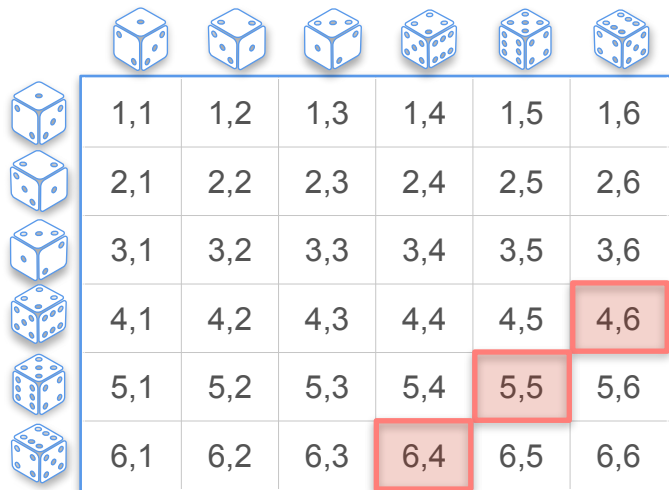
$P(\text{sum} = 10 \mid \text{1st is 6}) = \underline{\hspace{2cm}}$

$$= \frac{1}{6}$$

# Conditional Probability: Dice Example 2



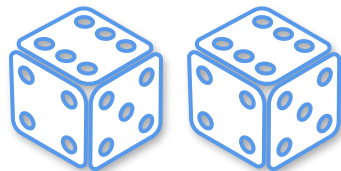
What is the probability that the sum is 10?



|                                                                                  |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|                                                                                  |  |  |  |  |  |  |
|  | 1,1                                                                               | 1,2                                                                               | 1,3                                                                               | 1,4                                                                               | 1,5                                                                               | 1,6                                                                               |
|  | 2,1                                                                               | 2,2                                                                               | 2,3                                                                               | 2,4                                                                               | 2,5                                                                               | 2,6                                                                               |
|  | 3,1                                                                               | 3,2                                                                               | 3,3                                                                               | 3,4                                                                               | 3,5                                                                               | 3,6                                                                               |
|  | 4,1                                                                               | 4,2                                                                               | 4,3                                                                               | 4,4                                                                               | 4,5                                                                               | 4,6                                                                               |
|  | 5,1                                                                               | 5,2                                                                               | 5,3                                                                               | 5,4                                                                               | 5,5                                                                               | 5,6                                                                               |
|  | 6,1                                                                               | 6,2                                                                               | 6,3                                                                               | 6,4                                                                               | 6,5                                                                               | 6,6                                                                               |

$$P(\text{sum} = 10) = \frac{\begin{array}{c} \boxed{4,6} \\ \boxed{5,5} \\ \boxed{6,4} \end{array}}{\begin{array}{|c|c|c|c|c|c|} \hline 1,1 & 1,2 & 1,3 & 1,4 & 1,5 & 1,6 \\ \hline 2,1 & 2,2 & 2,3 & 2,4 & 2,5 & 2,6 \\ \hline 3,1 & 3,2 & 3,3 & 3,4 & 3,5 & 3,6 \\ \hline 4,1 & 4,2 & 4,3 & 4,4 & 4,5 & 4,6 \\ \hline 5,1 & 5,2 & 5,3 & 5,4 & 5,5 & 5,6 \\ \hline 6,1 & 6,2 & 6,3 & 6,4 & 6,5 & 6,6 \\ \hline \end{array}} = \frac{3}{36}$$

# Conditional Probability: Dice Example 2



What is the probability that the sum is 10?

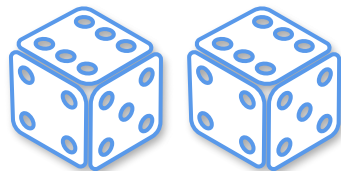
**GIVEN** that the first one is 1



|  |  |  |  |  |  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1,1                                                                               | 1,2                                                                               | 1,3                                                                               | 1,4                                                                               | 1,5                                                                               | 1,6                                                                               |
| 2,1                                                                               | 2,2                                                                               | 2,3                                                                               | 2,4                                                                               | 2,5                                                                               | 2,6                                                                               |
| 3,1                                                                               | 3,2                                                                               | 3,3                                                                               | 3,4                                                                               | 3,5                                                                               | 3,6                                                                               |
| 4,1                                                                               | 4,2                                                                               | 4,3                                                                               | 4,4                                                                               | 4,5                                                                               | 4,6                                                                               |
| 5,1                                                                               | 5,2                                                                               | 5,3                                                                               | 5,4                                                                               | 5,5                                                                               | 5,6                                                                               |
| 6,1                                                                               | 6,2                                                                               | 6,3                                                                               | 6,4                                                                               | 6,5                                                                               | 6,6                                                                               |

$P(\text{sum} = 10 \mid \text{1st is } 1) = \underline{\hspace{2cm}}$

# Conditional Probability: Dice Example 2



What is the probability that the sum is 10?

**GIVEN** that the first one is 1

|  |  |  |  |  |  |     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----|
|   | 1,1                                                                               | 1,2                                                                               | 1,3                                                                               | 1,4                                                                               | 1,5                                                                               | 1,6 |
|   | 2,1                                                                               | 2,2                                                                               | 2,3                                                                               | 2,4                                                                               | 2,5                                                                               | 2,6 |
|   | 3,1                                                                               | 3,2                                                                               | 3,3                                                                               | 3,4                                                                               | 3,5                                                                               | 3,6 |
|   | 4,1                                                                               | 4,2                                                                               | 4,3                                                                               | 4,4                                                                               | 4,5                                                                               | 4,6 |
|   | 5,1                                                                               | 5,2                                                                               | 5,3                                                                               | 5,4                                                                               | 5,5                                                                               | 5,6 |
|   | 6,1                                                                               | 6,2                                                                               | 6,3                                                                               | 6,4                                                                               | 6,5                                                                               | 6,6 |

$$P(\text{sum} = 10 \mid \text{1st is } 1) = \frac{0}{\quad}$$

$$= 0$$



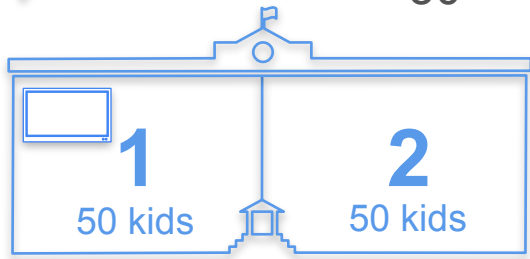
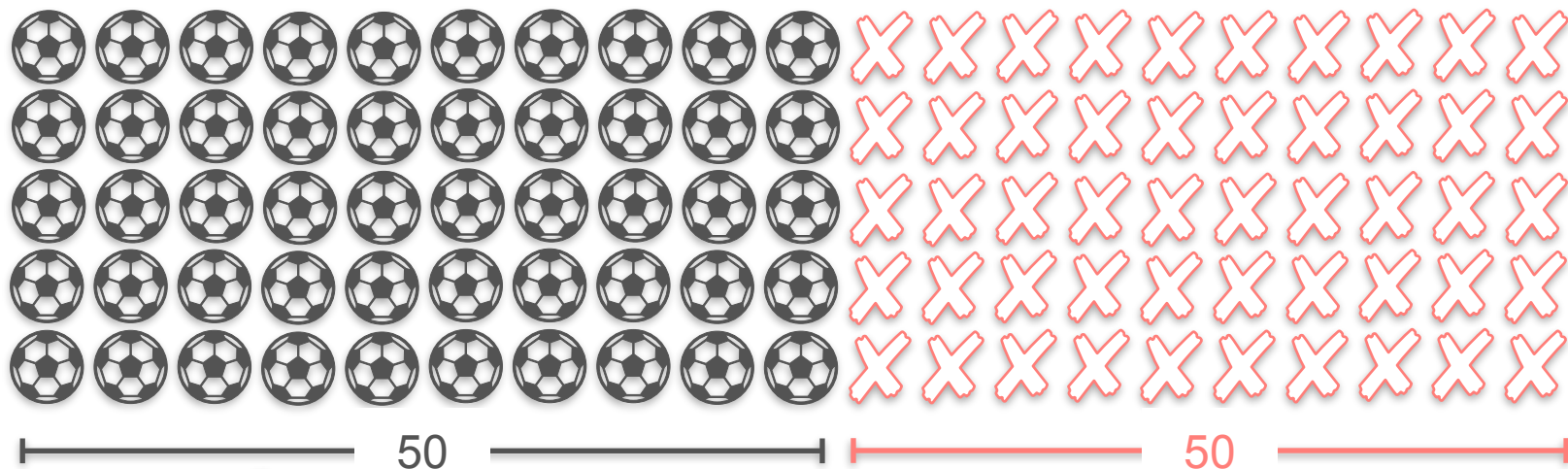
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# Introduction to Probability

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## **Conditional probability - Part 2**

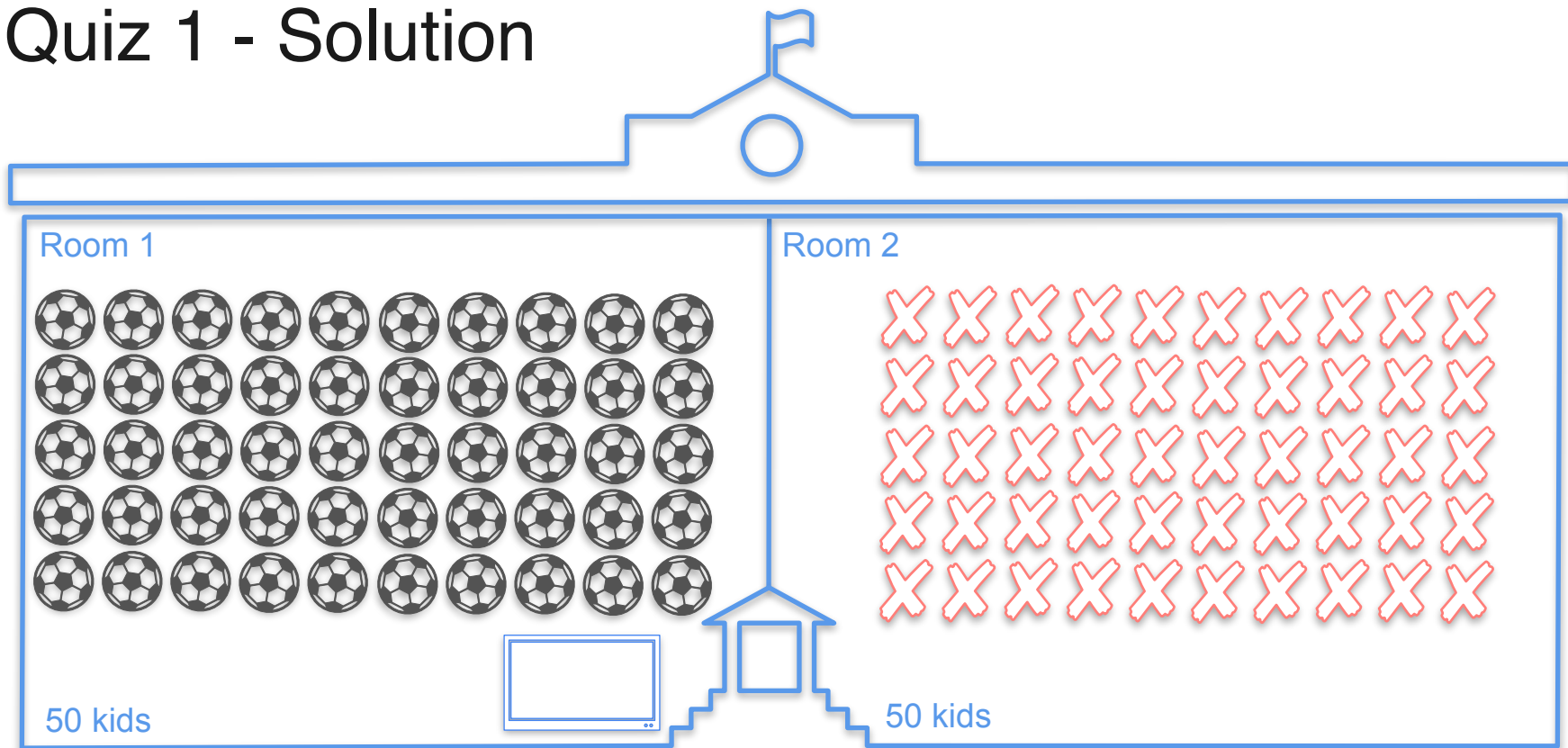
# Quiz 1



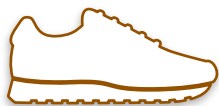
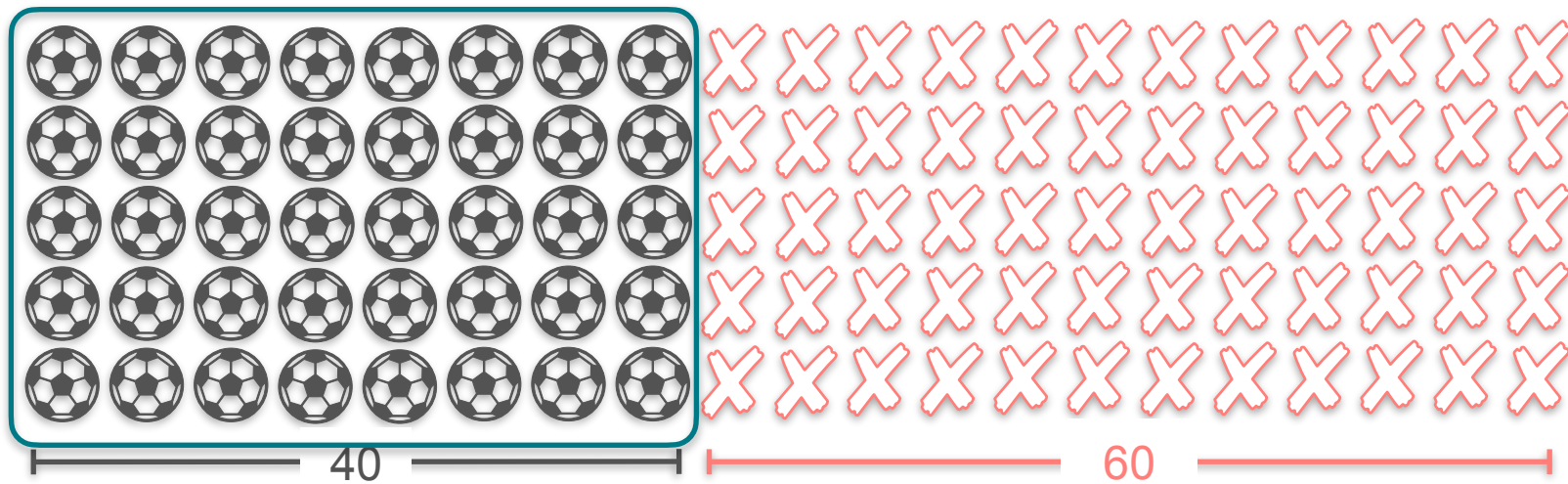
**100 kids**

How many kids in room 1 like soccer?

# Quiz 1 - Solution



## Quiz 2



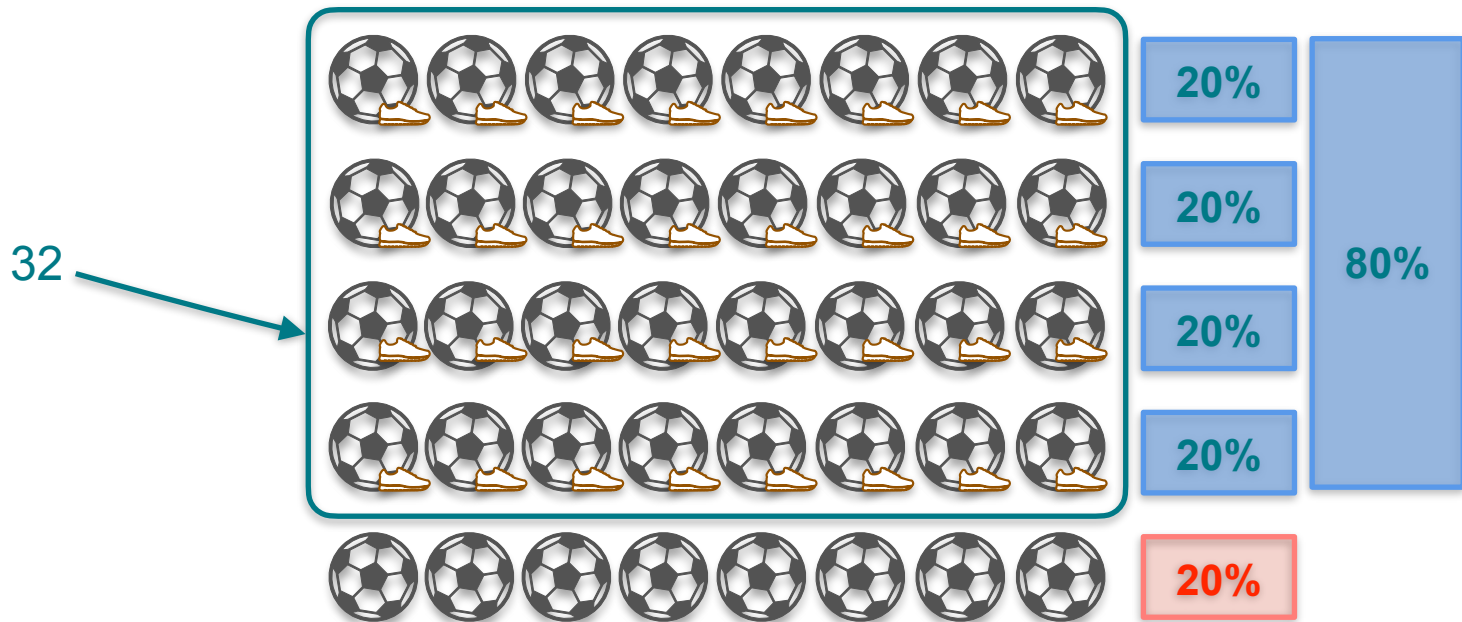
80%

**100 kids**

How many kids play soccer and wear running shoes



# Quiz 2 - Solution



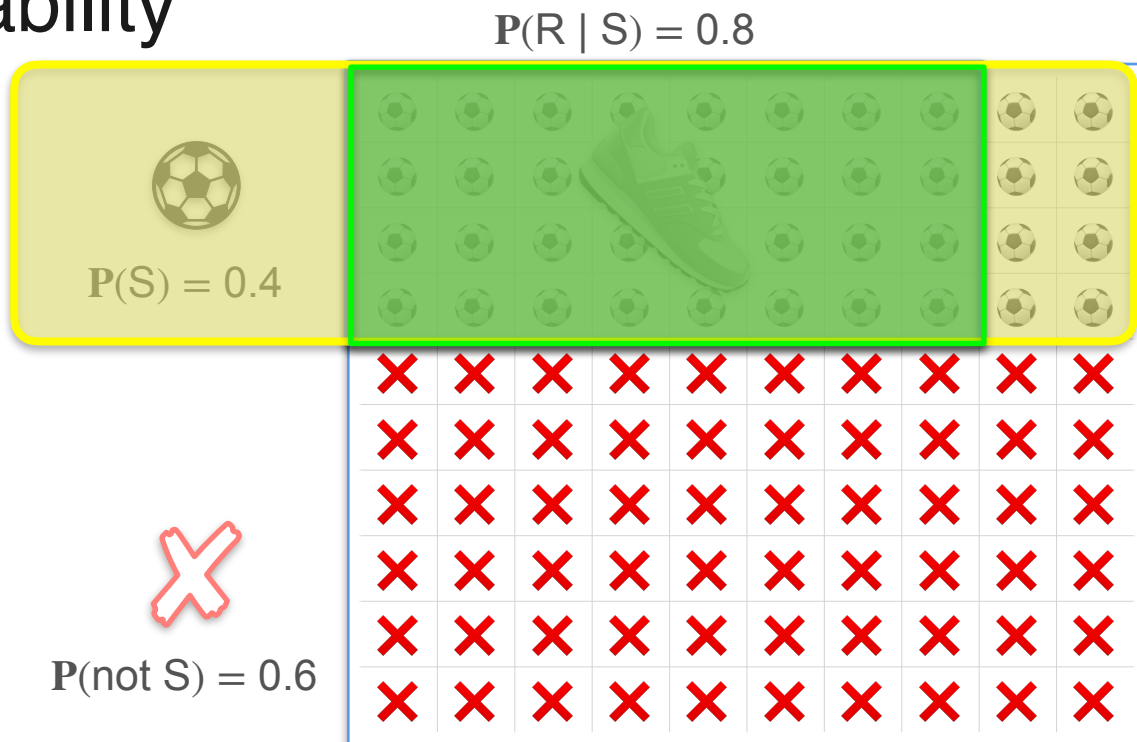
# Conditional Probability

$P(\text{Soccer and Running shoes})$

$$P(S \cap R) = P(S) \bullet P(R|S)$$

$$= 0.4 \bullet 0.8$$

$$= 0.32$$



# Conditional Probability

$P(\text{Soccer and Running shoes})$

$$P(S \cap R) = 0.32$$

$P(\text{not Soccer and Running shoes})$

$$P(\text{not } S \cap R) = P(\text{not } S) \bullet P(R | \text{not } S)$$

$$= 0.6 \bullet 0.5$$

$$= 0.3$$



$$P(S) = 0.4$$



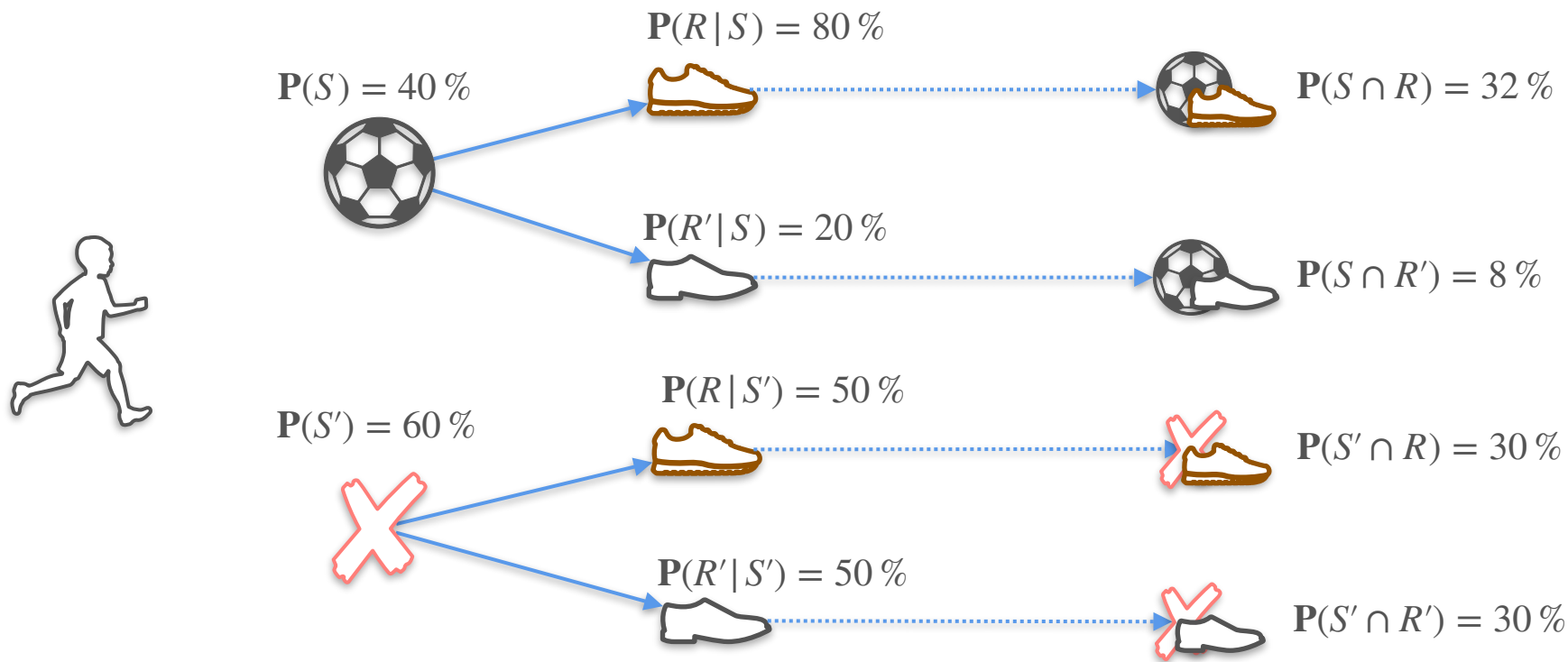
$$P(\text{not } S) = 0.6$$

$$P(R | S) = 0.8$$



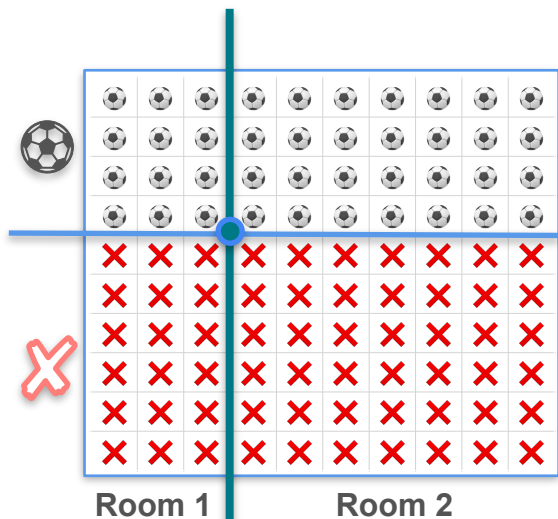
$$P(R | \text{not } S) = 0.5$$

# Conditional Probability

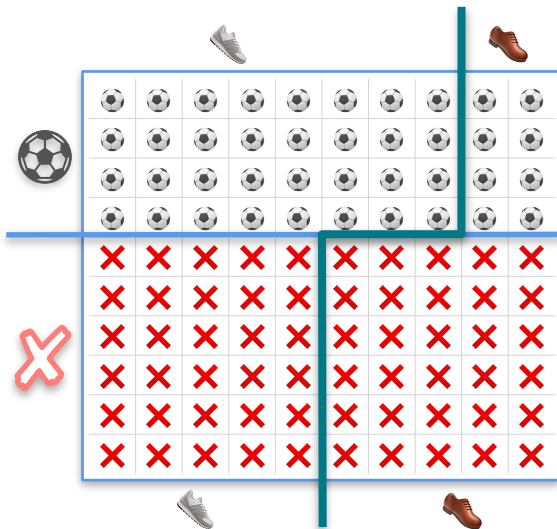


# Independent vs Dependent Events

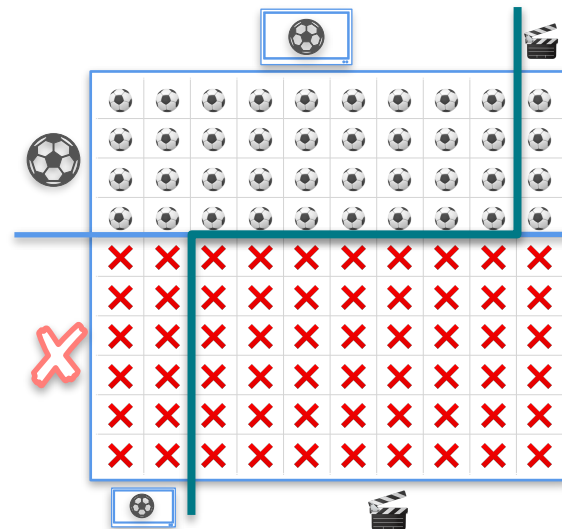
Independent



Dependent



Dependent





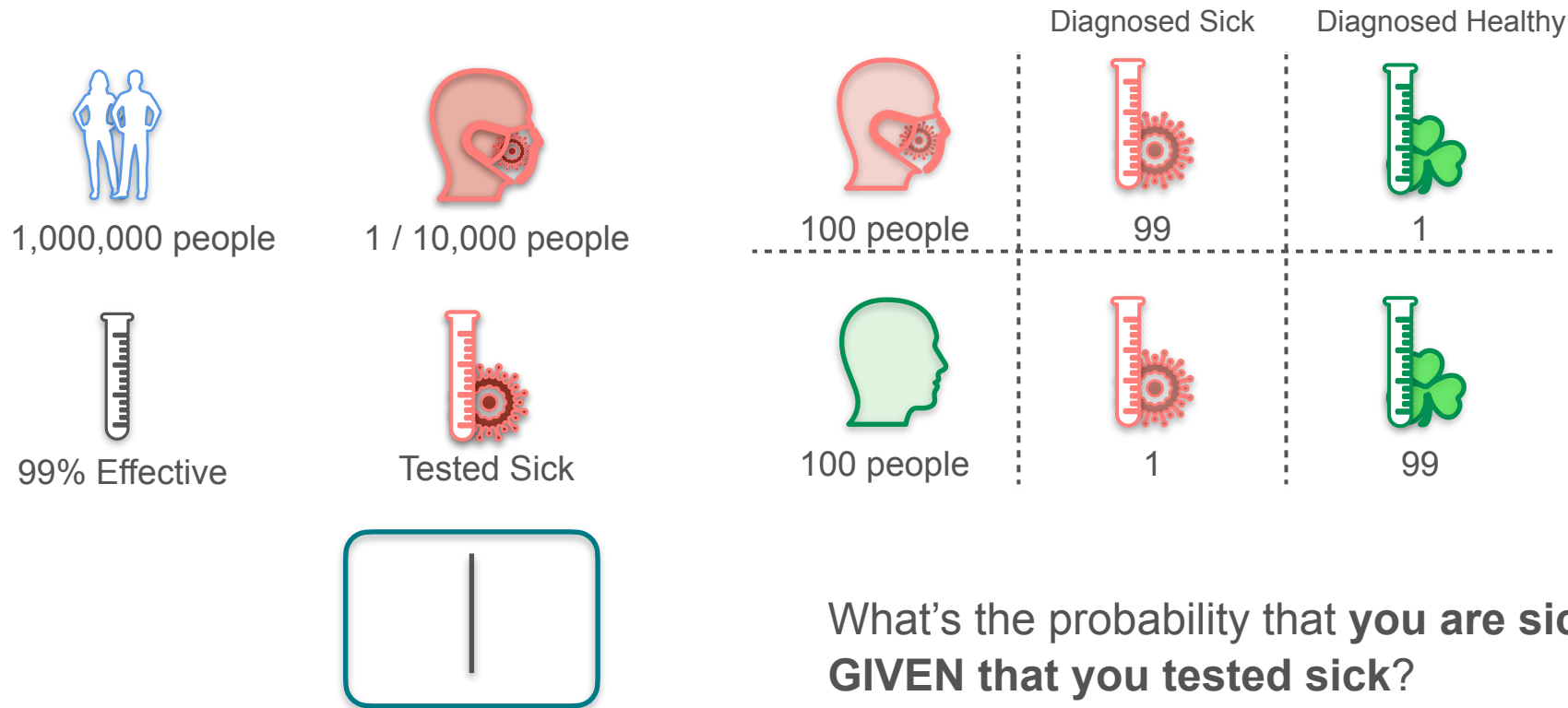
DeepLearning.AI

# Introduction to probability

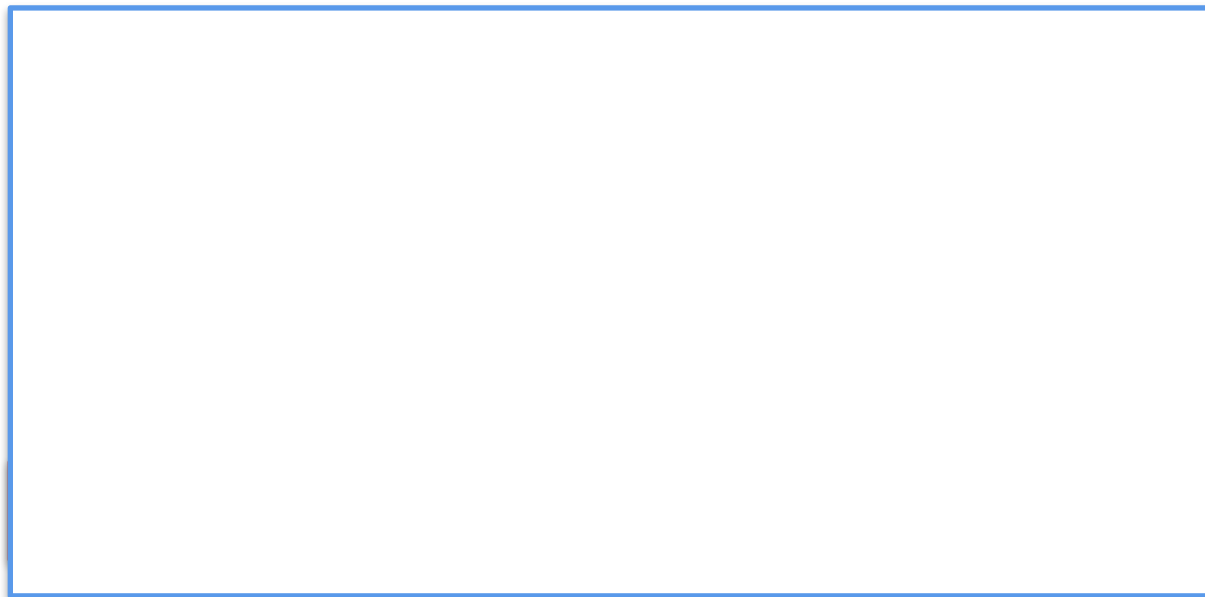
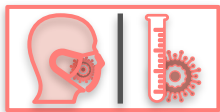
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## **Bayes Theorem - Intuition**

# Bayes Theorem: Intuition



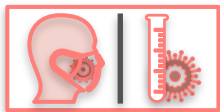
# Bayes Theorem: Intuition



1,000,000 people



# Bayes Theorem: Intuition



Healthy



999,900

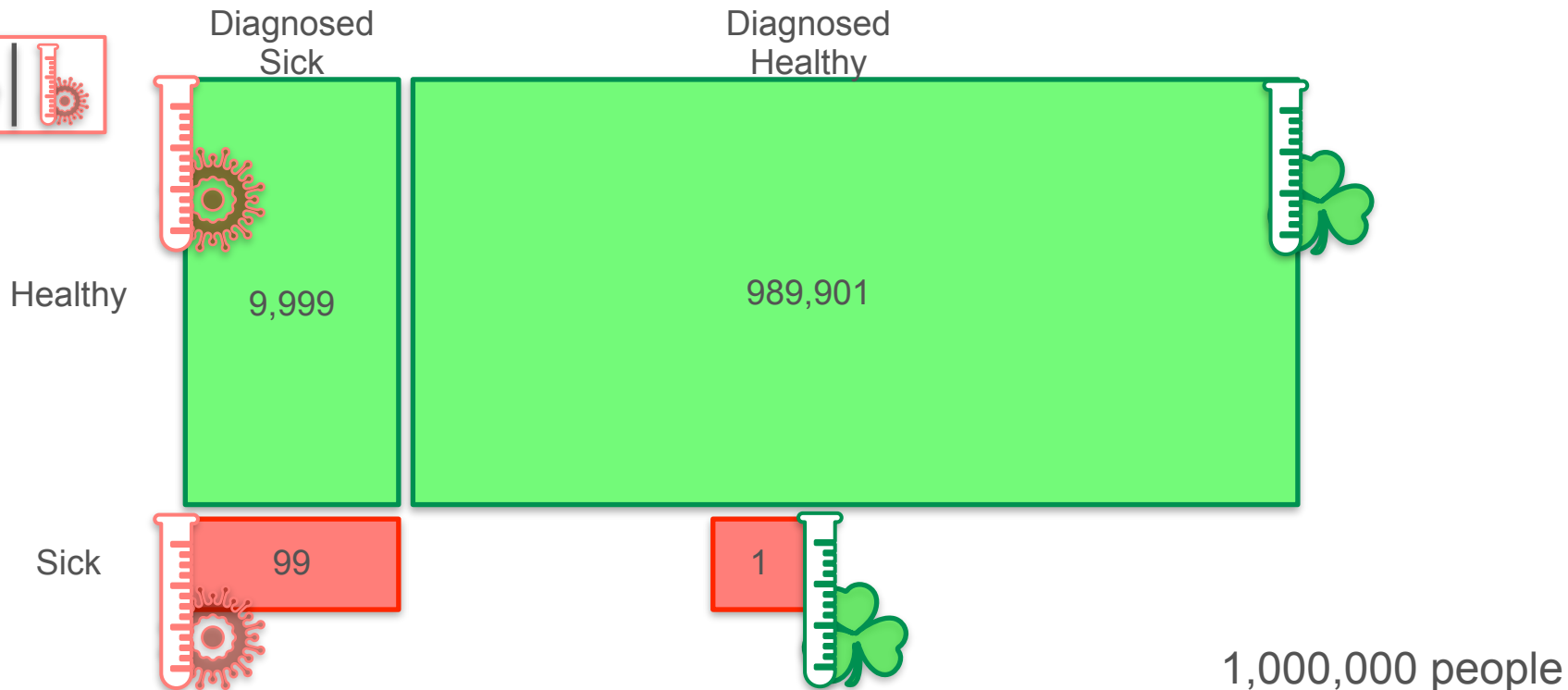
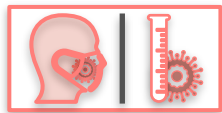
Sick



100

1,000,000 people

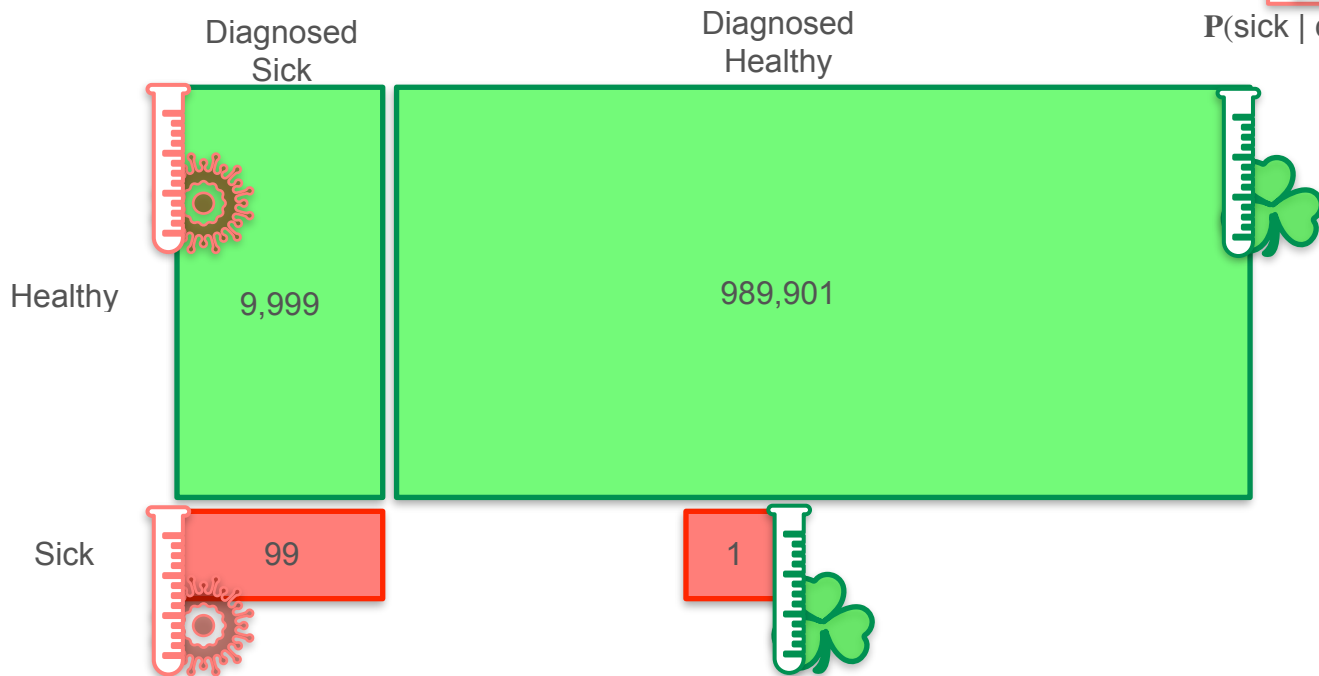
# Bayes Theorem: Intuition



# Bayes Theorem: Intuition



$P(\text{sick} \mid \text{diagnosed sick}) = \text{_____}$

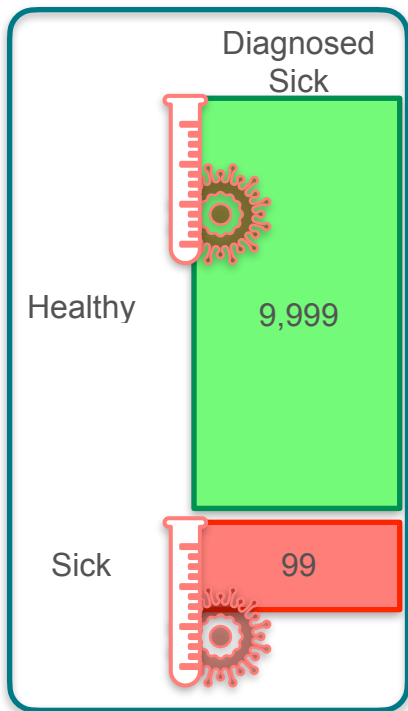


1,000,000 people

# Bayes Theorem: Intuition

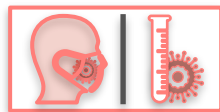
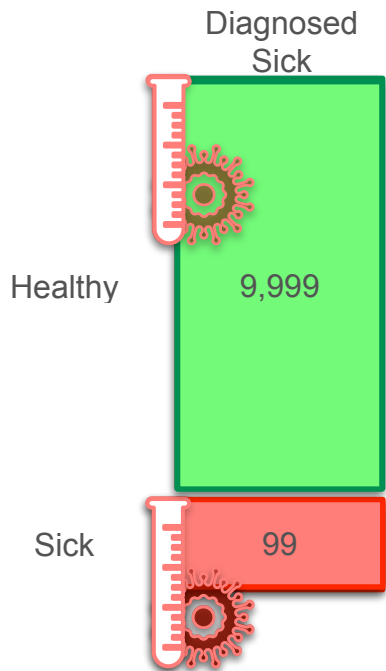


$P(\text{sick} \mid \text{diagnosed sick}) = \text{_____}$

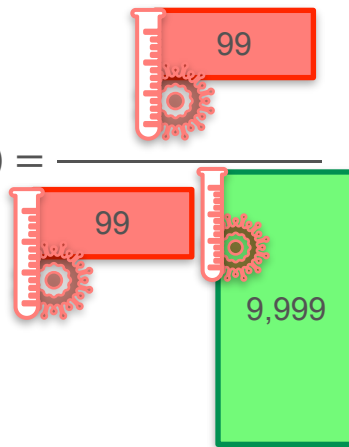


1,000,000 people

# Bayes Theorem: Intuition



$P(\text{sick} \mid \text{diagnosed sick}) =$

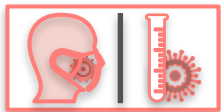


$$= \frac{99}{99 + 9999}$$

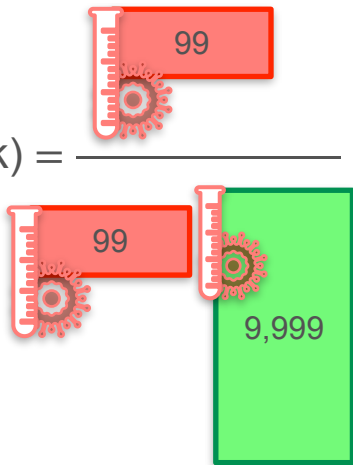
$$= \frac{99}{10098} = 0.0098$$

$$= \frac{\text{sick and diagnosed sick}}{\text{sick and diagnosed sick} + \text{healthy and diagnosed sick}}$$

# Bayes Theorem: Intuition



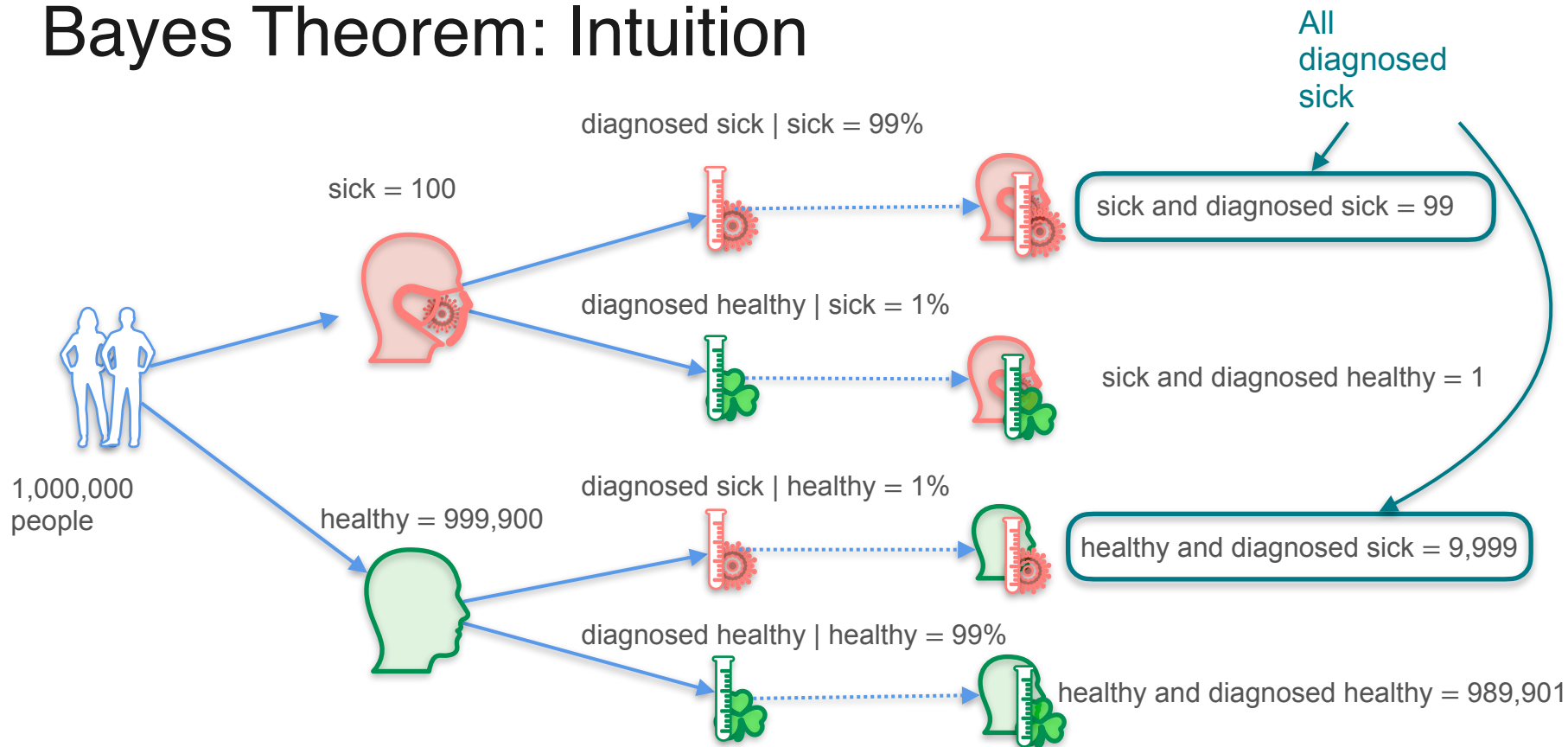
$$P(\text{sick} \mid \text{diagnosed sick}) = \frac{\quad}{\quad}$$



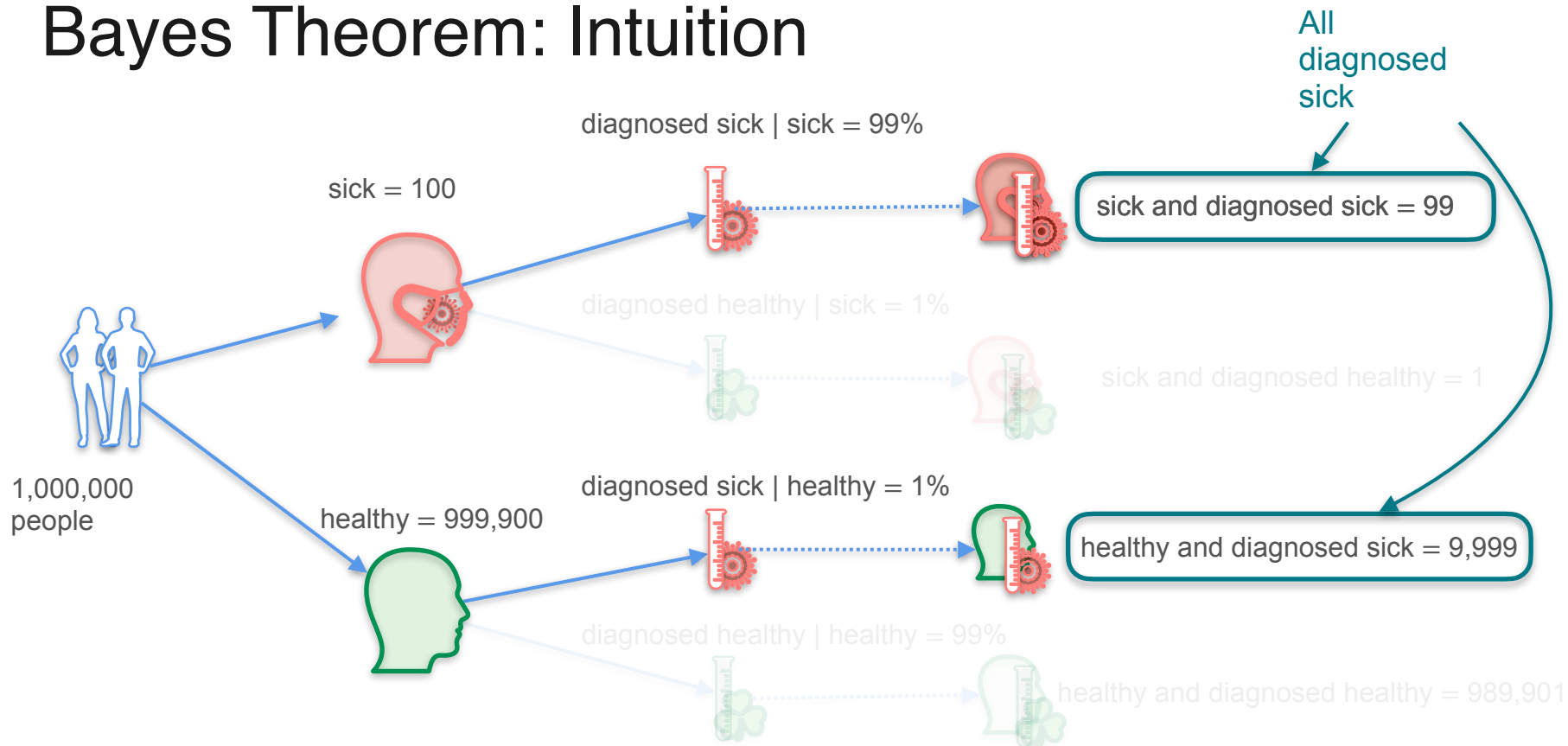
$$= \frac{\text{sick and diagnosed sick}}{\text{sick and diagnosed sick} + \text{healthy and diagnosed sick}}$$

$$= \frac{\text{sick and diagnosed sick}}{\text{everyone diagnosed sick}}$$

# Bayes Theorem: Intuition



# Bayes Theorem: Intuition





# Bayes Theorem: Intuition

$$P(\text{sick} \mid \text{diagnosed sick}) = \frac{\text{red head icon}}{\text{red head icon} + \text{green head icon}}$$

$$P(\text{sick} \mid \text{diagnosed sick}) = \frac{\text{sick and diagnosed sick} = 99}{\text{healthy and diagnosed sick} = 9,999 + \text{sick and diagnosed sick} = 99}$$

$$P(\text{sick} \mid \text{diagnosed sick}) = \frac{99}{10098} = 0.0098$$



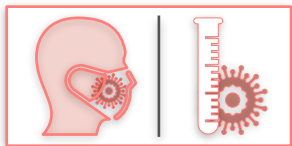
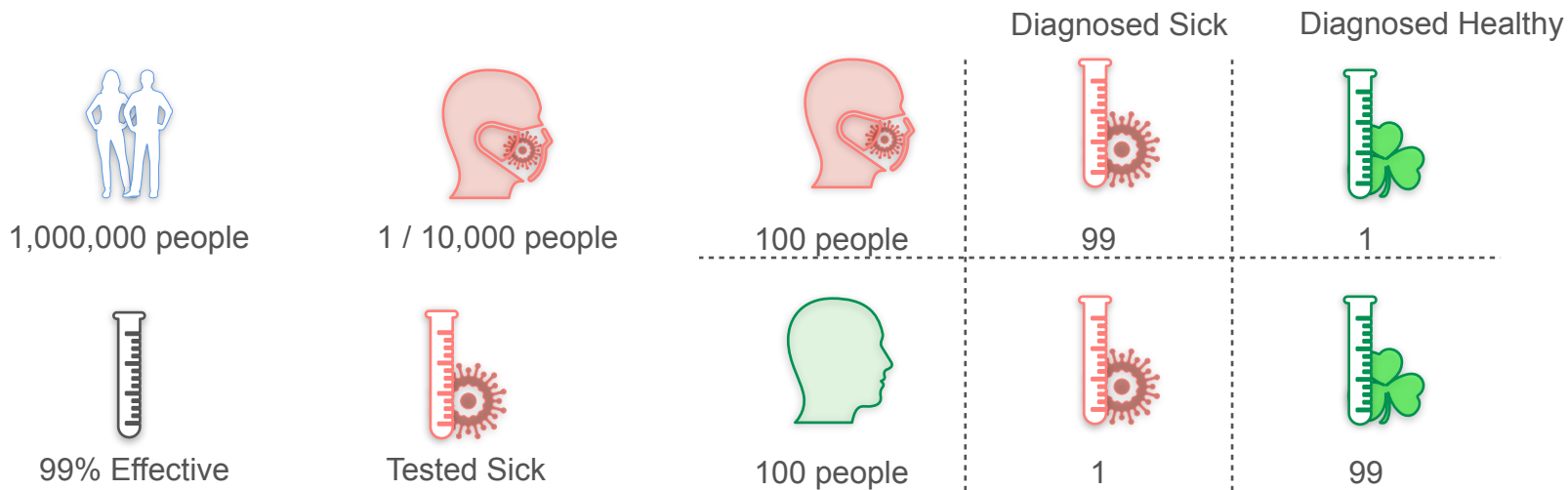
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# Introduction to probability

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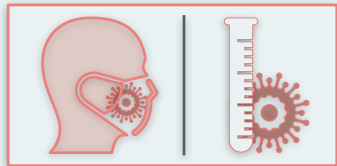
## **Bayes Theorem - Mathematical Formula**

# Bayes Theorem: Formula



The probability that **you are sick**  
**GIVEN** that you tested sick

# Bayes Theorem: Formula



$$P(\text{sick} \mid \text{diagnosed sick}) = ?$$

$$P(\text{sick}) = 0.01\%$$

$$P(\text{not sick}) = 99.99\%$$

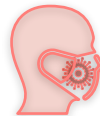
$$P(\text{diagnosed sick} \mid \text{sick}) = 99\%$$

$$P(\text{diagnosed sick} \mid \text{not sick}) = 1\%$$

The probability that **you are sick**  
**GIVEN** that you tested sick



1,000,000

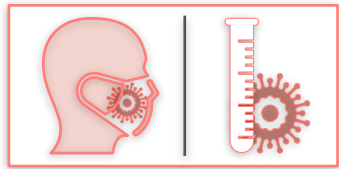


1 / 10,000



99% Effective

# Bayes Theorem: Formula



$A$ : sick

$B$ : diagnosed sick

$$P(\text{sick} \mid \text{diagnosed sick}) = ?$$

$$P(A \mid B) = ?$$

$$P(\text{sick}) = 0.01\%$$

$$P(\text{not sick}) = 99.99\%$$

$$P(\text{diagnosed sick} \mid \text{sick}) = 99\%$$

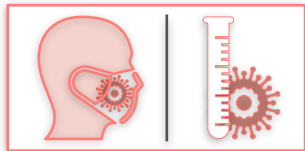
$$P(\text{diagnosed sick} \mid \text{not sick}) = 1\%$$

**From Conditional Probability**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \mid B) = \frac{P(\text{sick and diagnosed sick})}{P(\text{diagnosed sick})}$$

# Bayes Theorem: Formula



$A$ : sick

$B$ : diagnosed sick

$P(\text{sick} \mid \text{diagnosed sick}) = ?$

$P(\text{sick}) = 0.01\%$

$P(\text{not sick}) = 99.99\%$

$P(\text{diagnosed sick} \mid \text{sick}) = 99\%$

$P(\text{diagnosed sick} \mid \text{not sick}) = 1\%$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

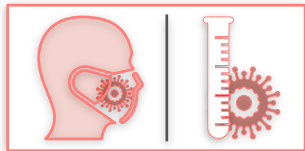
$$P(A \mid B) = \frac{P(\text{sick and diagnosed sick})}{P(\text{diagnosed sick})}$$

$P(\text{sick and diagnosed sick}) = ?$

$P(\text{diagnosed sick}) = ?$

**BAYES THEOREM FORMULA CAN HELP**

# Bayes Theorem: Formula



$A$ : sick

$B$ : diagnosed sick

$P(\text{sick} \mid \text{diagnosed sick}) = ?$

$P(\text{sick}) = 0.01\%$

$P(\text{not sick}) = 99.99\%$

$P(\text{diagnosed sick} \mid \text{sick}) = 99\%$

$P(\text{diagnosed sick} \mid \text{not sick}) = 1\%$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

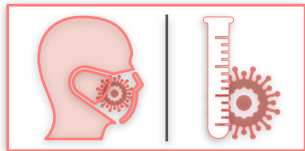
$$P(A \mid B) = \frac{P(\text{sick and diagnosed sick}) = ?}{P(\text{diagnosed sick}) = ?}$$

**From Conditional Probability**

$$P(A \cap B) = P(A) \cdot P(B \mid A)$$

$$P(\text{sick and diagnosed sick}) = P(\text{sick}) \cdot P(\text{diagnosed sick} \mid \text{sick})$$

# Bayes Theorem: Formula



$A$ : sick

$B$ : diagnosed sick

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A | B) = \frac{P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick})}{P(\text{diagnosed sick})} = ?$$

$$P(\text{sick} | \text{diagnosed sick}) = ?$$

$$P(\text{sick}) = 0.01\%$$

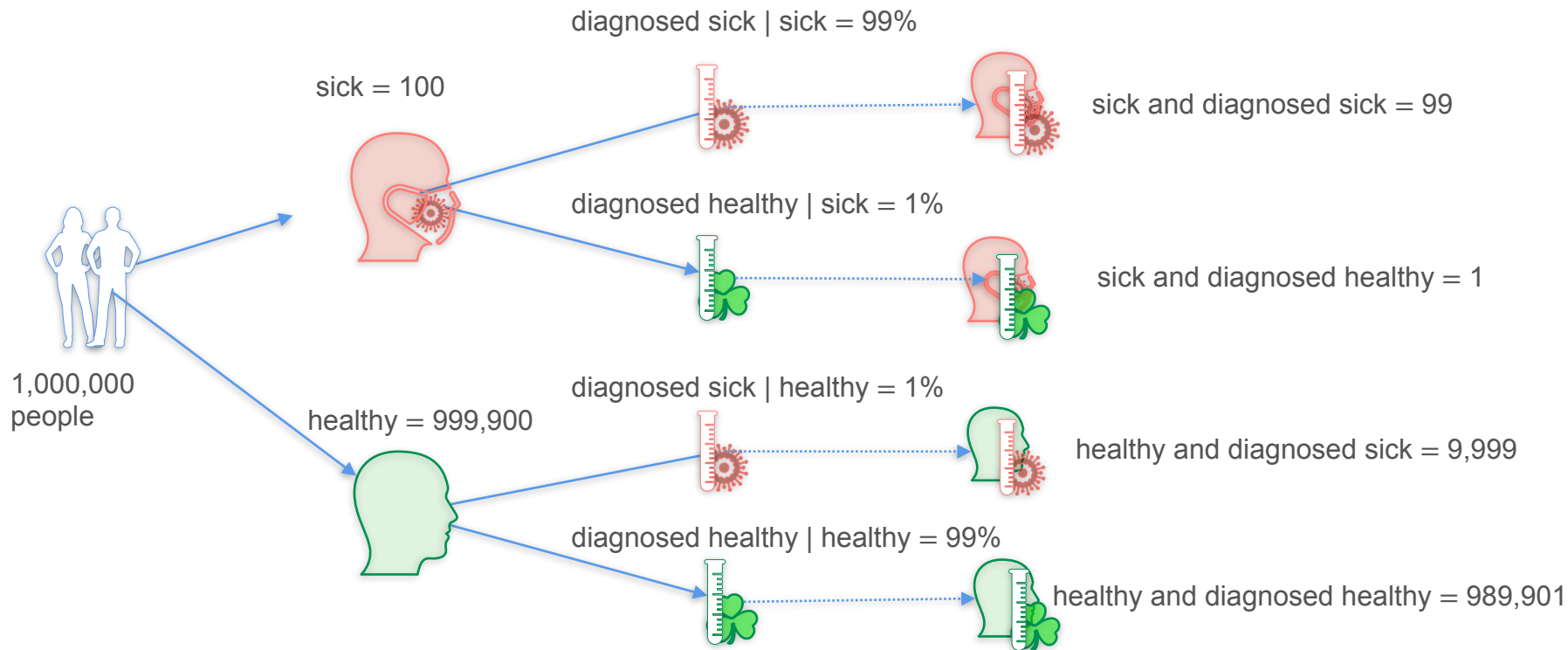
$$P(\text{not sick}) = 99.99\%$$

$$P(\text{diagnosed sick} | \text{sick}) = 99\%$$

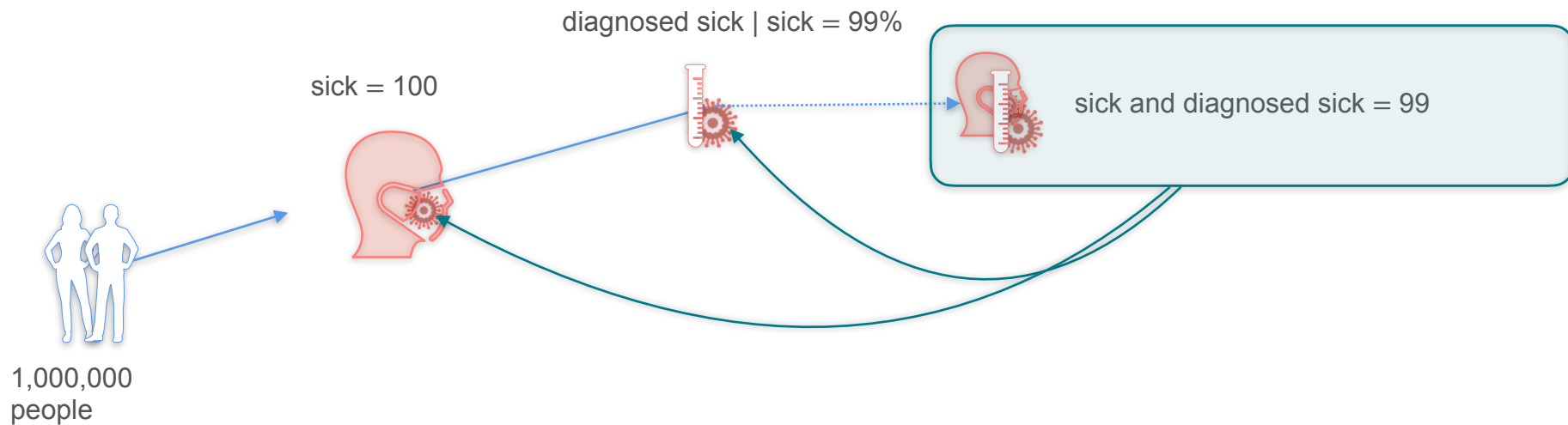
$$P(\text{diagnosed sick} | \text{not sick}) = 1\%$$



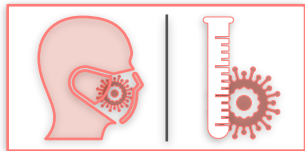
# Bayes Theorem: Formula



# Bayes Theorem: Formula



# Bayes Theorem: Formula



$A$ : sick

$B$ : diagnosed sick

$P(\text{sick} \mid \text{diagnosed sick}) = ?$

$P(\text{sick}) = 0.01\%$

$P(\text{not sick}) = 99.99\%$

$P(\text{diagnosed sick} \mid \text{sick}) = 99\%$

$P(\text{diagnosed sick} \mid \text{not sick}) = 1\%$

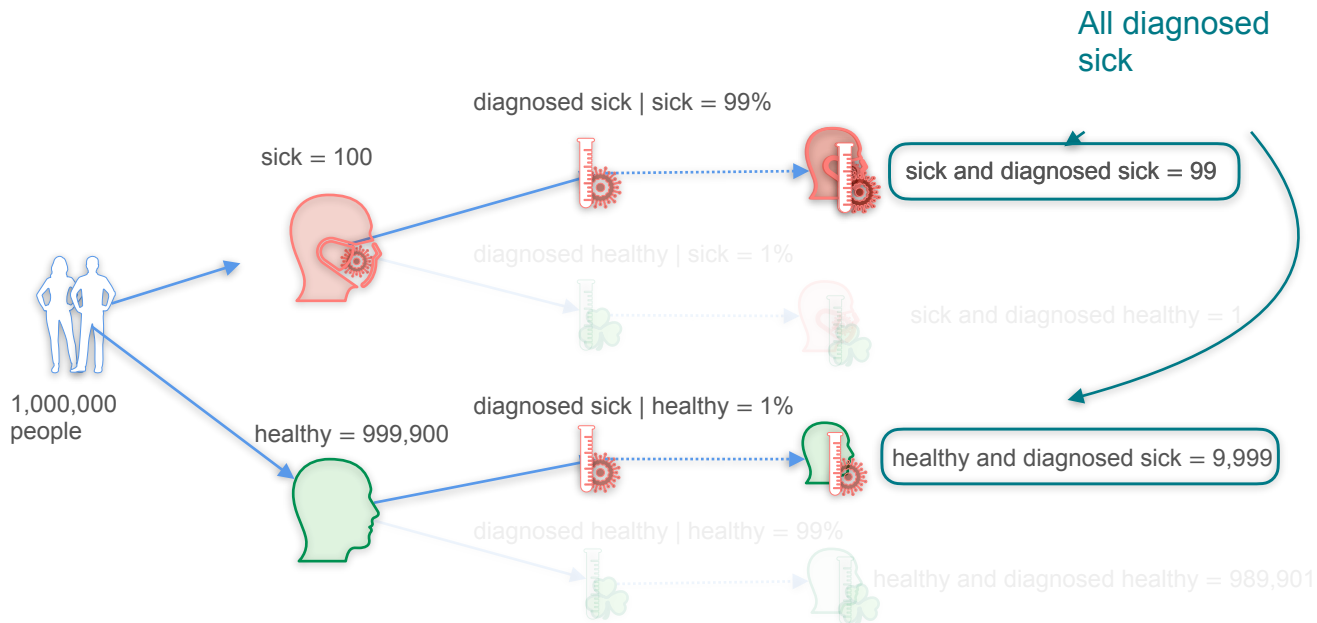
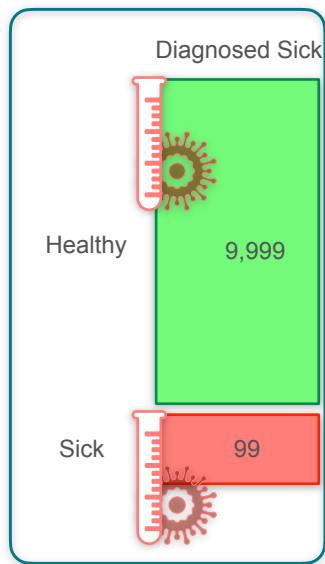
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \mid B) = \frac{P(\text{sick}) \cdot P(\text{diagnosed sick} \mid \text{sick})}{P(\text{diagnosed sick}) = ?}$$

$P(\text{diagnosed sick}) =$

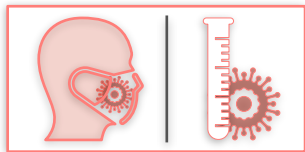
# Bayes Theorem: Formula

$P(\text{diagnosed sick}) =$



$$P(\text{sick} \cap \text{diagnosed sick}) + P(\text{not sick} \cap \text{diagnosed sick})$$

# Bayes Theorem: Formula



$A$ : sick

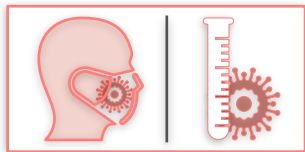
$B$ : diagnosed sick

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A | B) = \frac{P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick})}{P(\text{diagnosed sick}) = ?}$$

$$P(\text{diagnosed sick}) = P(\text{sick} \cap \text{diagnosed sick}) + P(\text{not sick} \cap \text{diagnosed sick})$$

# Bayes Theorem: Formula



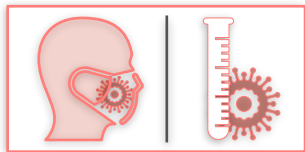
$A$ : sick

$B$ : diagnosed sick

$$\mathbf{P}(A | B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

$$\mathbf{P}(A | B) = \frac{\mathbf{P}(\text{sick}) \bullet \mathbf{P}(\text{diagnosed sick} | \text{sick})}{\mathbf{P}(\text{sick} \cap \text{diagnosed sick}) + \mathbf{P}(\text{not sick} \cap \text{diagnosed sick})}$$

# Bayes Theorem: Formula



$A$ : sick

$B$ : diagnosed sick

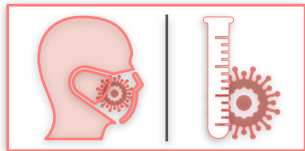
$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A | B) = \frac{P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick})}{P(\text{sick} \cap \text{diagnosed sick}) + P(\text{not sick} \cap \text{diagnosed sick})}$$

$$\begin{aligned} P(\text{sick} \cap \text{diagnosed sick}) &= P(A \cap B) \\ &= P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick}) \end{aligned}$$

$$\begin{aligned} P(\text{not sick} \cap \text{diagnosed sick}) &= P(A' \cap B) \\ &= P(\text{not sick}) \cdot P(\text{diagnosed sick} | \text{not sick}) \end{aligned}$$

# Bayes Theorem: Formula



$A$ : sick

$B$ : diagnosed sick

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad ?$$

$$P(A | B) = \frac{P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick})}{P(\text{sick}) \cdot P(\text{diagnosed sick} | \text{sick}) + P(\text{not sick}) \cdot P(\text{diagnosed sick} | \text{not sick})}$$

$$P(\text{sick}) = 0.01\%$$

$$P(\text{not sick}) = 99.99\%$$

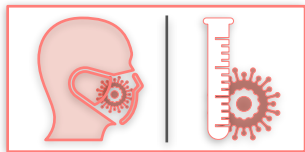
$$P(\text{diagnosed sick} | \text{sick}) = 99\%$$

$$P(\text{diagnosed sick} | \text{not sick}) = 1\%$$

**BAYES THEOREM  
FORMULA**



# Bayes Theorem: Formula



$A$ : sick

$B$ : diagnosed sick

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A | B) = \frac{P(A) \cdot P(B | A)}{P(A) \cdot P(B | A) + P(A') \cdot P(B | A')}$$

$$P(A) = 0.01\%$$

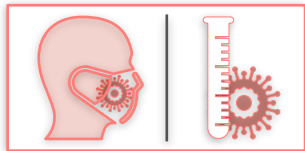
$$P(A') = 99.99\%$$

$$P(B | A) = 99\%$$

$$P(B | A') = 1\%$$

**BAYES THEOREM  
FORMULA**

# Bayes Theorem: Formula



$A$ : sick

$B$ : diagnosed sick

$$\mathbf{P}(A | B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

$$\mathbf{P}(A | B) = \frac{\mathbf{P}(A) \cdot \mathbf{P}(B | A)}{\mathbf{P}(A) \cdot \mathbf{P}(B | A) + \mathbf{P}(A') \cdot \mathbf{P}(B | A')}$$

$$\mathbf{P}(A) = 0.01\%$$

$$\mathbf{P}(A') = 99.99\%$$

$$\mathbf{P}(B | A) = 99\%$$

$$\mathbf{P}(B | A') = 1\%$$

$$\mathbf{P}(A | B) = \frac{0.0001 \times 0.99}{(0.0001 \times 0.99) + (0.9999 \times 0.01)}$$

$$\mathbf{P}(A | B) = 0.0098$$

**BAYES THEOREM  
FORMULA**



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# Introduction to probability

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## **Bayes Theorem - Spam Example**

# Bayes Theorem: Spam Example



# Bayes Theorem: Spam Example



20 spam

80 not spam (ham)



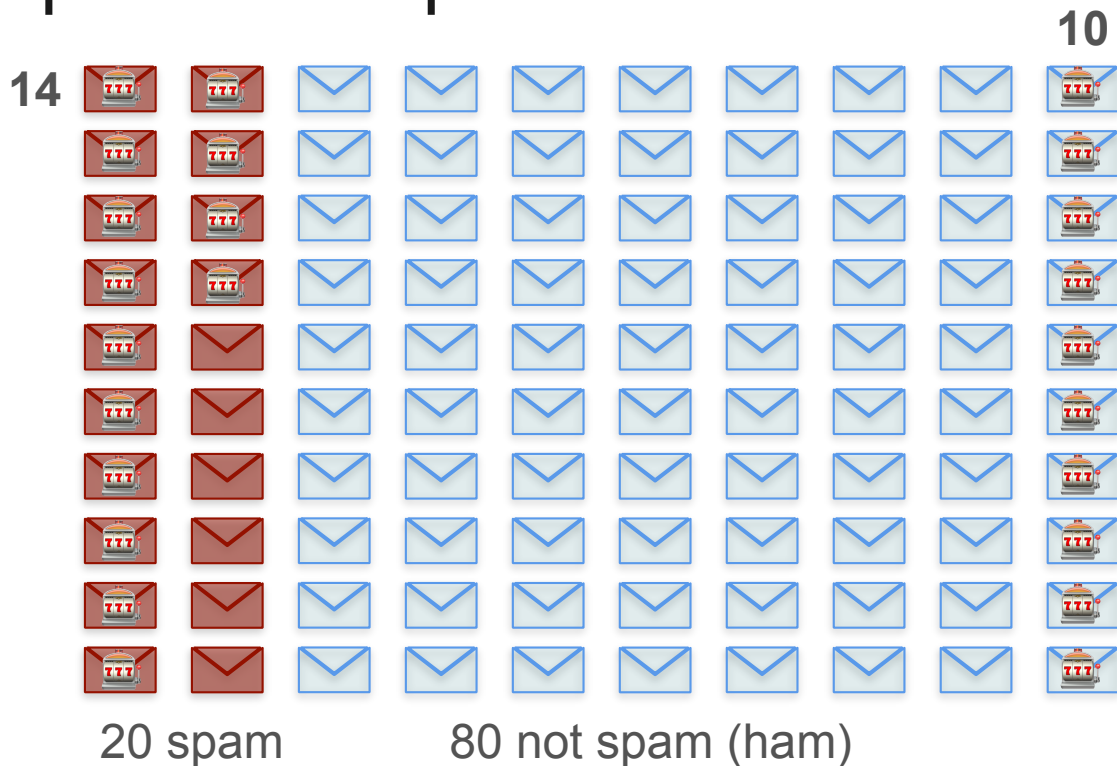
“lottery”

# Bayes Theorem: Spam Example



What is the probability that an email containing lottery is a spam?

$$P(\text{spam} \mid \text{lottery})$$



# Bayes Theorem: Spam Example (Intuition Solution)



$P(\text{spam} \mid \text{lottery})$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$



14

24 emails  
containing lottery

$$P(\text{spam} \mid \text{lottery}) = \frac{\text{spam and lottery}}{\text{all lottery}}$$

$$= \frac{14}{24} \\ = \frac{7}{12} = 0.583$$



10

# Bayes Theorem: Spam Example (Formula Solution)

$P(\text{spam} \mid \text{lottery})$

$$P(A \mid B) = \frac{P(A) \cdot P(B \mid A)}{P(A) \cdot P(B \mid A) + P(A') \cdot P(B \mid A')}$$

$A$ : Email is spam       $B$ : Email contains lottery

$$P(\text{spam} \mid \text{lottery}) = \frac{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam})}{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam}) + P(\text{not spam}) \cdot P(\text{lottery} \mid \text{not spam})}$$



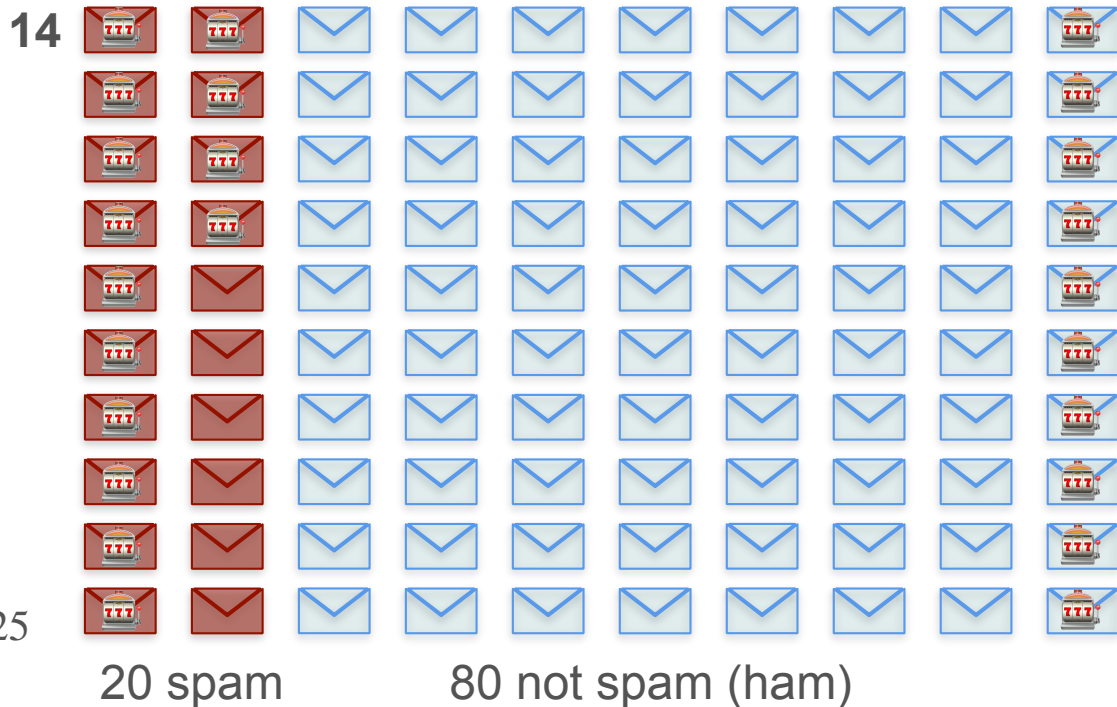
# Bayes Theorem: Spam Example (Formula Solution)

$$P(\text{spam}) = \frac{20}{100} = 0.2$$

$$P(\text{not spam}) = \frac{80}{100} = 0.8$$

$$P(\text{lottery} \mid \text{spam}) = \frac{14}{20} = 0.7$$

$$P(\text{lottery} \mid \text{not spam}) = \frac{10}{80} = 0.125$$



# Bayes Theorem: Spam Example (Formula Solution)

$$P(\text{spam}) = 0.2$$

$$P(\text{lottery} \mid \text{spam}) = 0.7$$

$$P(\text{not spam}) = 0.8$$

$$P(\text{lottery} \mid \text{not spam}) = 0.125$$

$$P(\text{spam} \mid \text{lottery}) = \frac{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam})}{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam}) + P(\text{not spam}) \cdot P(\text{lottery} \mid \text{not spam})}$$

$$P(\text{spam} \mid \text{lottery}) = \frac{0.2 \times 0.7}{(0.2 \times 0.7) + (0.8 \times 0.125)} = 0.583$$



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# Introduction to probability

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## **Bayes Theorem - Prior and Posterior**

# Bayes Theorem

PRIOR

$$\mathbf{P}(A)$$

EVENT

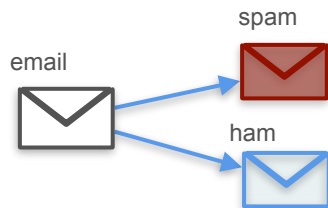
$$E$$

POSTERIOR

$$\mathbf{P}(A | E)$$

# Prior and Posterior

## PRIOR

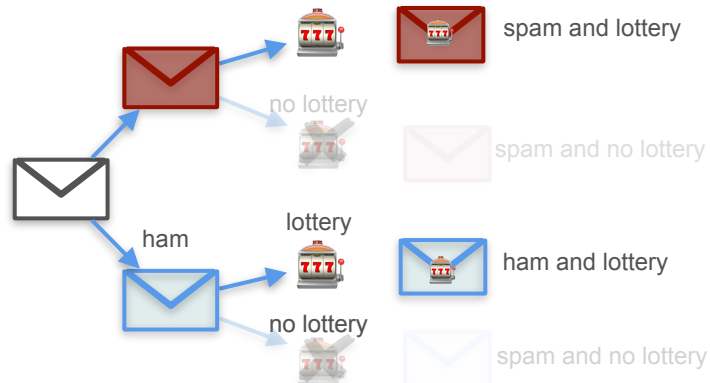


$$P(\text{spam}) = \frac{\text{red envelope icon}}{\text{red envelope icon} + \text{blue envelope icon}}$$

## EVENT

  
Email contains lottery

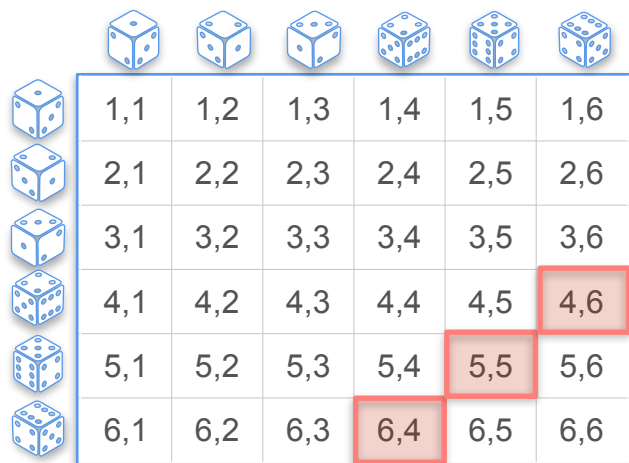
## POSTERIOR



$$P(\text{spam} \mid \text{lottery}) = \frac{\text{red envelope with lottery icon}}{\text{red envelope with lottery icon} + \text{blue envelope with lottery icon}}$$

# Prior and Posterior

PRIOR



|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

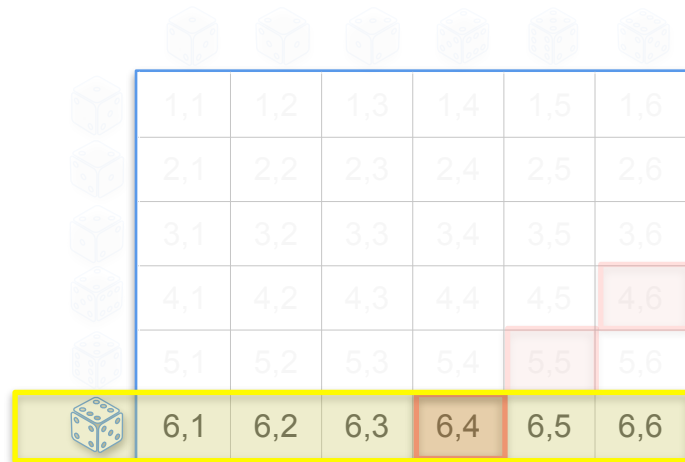
$$P(\text{sum} = 10) = \frac{3}{36}$$

EVENT



1st dice is 6

POSTERIOR

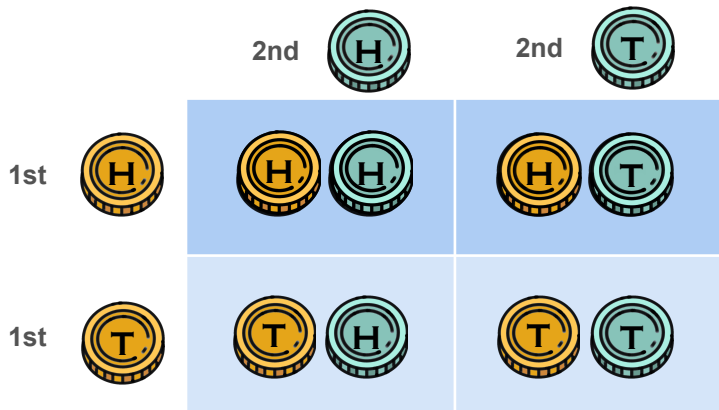


|     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|
| 1,1 | 1,2 | 1,3 | 1,4 | 1,5 | 1,6 |
| 2,1 | 2,2 | 2,3 | 2,4 | 2,5 | 2,6 |
| 3,1 | 3,2 | 3,3 | 3,4 | 3,5 | 3,6 |
| 4,1 | 4,2 | 4,3 | 4,4 | 4,5 | 4,6 |
| 5,1 | 5,2 | 5,3 | 5,4 | 5,5 | 5,6 |
| 6,1 | 6,2 | 6,3 | 6,4 | 6,5 | 6,6 |

$$P(\text{sum} = 10 \mid \text{1st is 6}) = \frac{1}{6}$$

# Prior and Posterior

## PRIOR



$$P(HH) = \frac{1}{4}$$

## EVENT



## POSTERIOR



$$P(HH | 1st \text{ is } H) = \frac{1}{2}$$



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# Introduction to probability

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## **Bayes Theorem - The Naive Bayes Model**



# What About 2 Events?

PRIOR

$$P(\text{spam}) = \frac{\text{Red Envelope}}{\text{Red Envelope} + \text{Blue Envelope}}$$

EVENT



Email contains 'lottery'



Email contains 'winning'



Email contains 'lottery' and 'winning'

POSTERIOR

$$P(\text{spam} \mid \text{lottery}) = \frac{\text{Red Envelope with Lottery Ticket}}{\text{Red Envelope with Lottery Ticket} + \text{Blue Envelope with Lottery Ticket}}$$

$$P(\text{spam} \mid \text{winning}) = \frac{\text{Red Envelope with Winning Ticket}}{\text{Red Envelope with Winning Ticket} + \text{Blue Envelope with Winning Ticket}}$$

?

# What About 2 Events?

## EVENT



Email contains 'lottery' and 'winning'

## POSTERIOR

$$P(\text{spam} \mid \text{lottery \& winning}) = \frac{\text{[Red envelope with lottery ticket and winning ticket]}}{\text{[Red envelope with lottery ticket and winning ticket]} + \text{[Blue envelope with lottery ticket and winning ticket]}}$$

$$\frac{\text{\# Spam emails with 'lottery' and 'winning'}}{\text{\# Spam emails}}$$

$$P(\text{spam} \mid \text{lottery \& winning}) = \frac{P(\text{spam}) \cdot \text{P(lottery \& winning} \mid \text{spam)}}{P(\text{spam}) \cdot P(\text{lottery \& winning} \mid \text{spam}) + P(\text{ham}) \cdot P(\text{lottery \& winning} \mid \text{ham})}$$

A red arrow points from the fraction  $\frac{\text{\# Spam emails with 'lottery' and 'winning'}}{\text{\# Spam emails}}$  to the term  $P(\text{lottery \& winning} \mid \text{spam})$  in the denominator of the equation above. A red question mark is placed above the term  $P(\text{lottery \& winning} \mid \text{spam})$ .

# What About More Than 2 Events?

EVENT

POSTERIOR

Email contains  $w_1, w_2, \dots, w_{100}$

~~$\frac{\text{\# Spam emails with } w_1, \dots, w_{100}}{\text{\# Spam emails}}$~~  <sup>0</sup>

$$\mathbf{P}(\text{spam} \mid w_1, \dots, w_{100}) = \frac{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(w_1, \dots, w_{100} \mid \text{spam})}{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(w_1, \dots, w_{100} \mid \text{spam}) + \mathbf{P}(\text{ham}) \cdot \mathbf{P}(w_1, \dots, w_{100} \mid \text{ham})}$$

<sup>?</sup>

# Is There a Quicker Way To Estimate the Probability?

## Naive assumption



The appearances of 'lottery' and 'winning' are independent

$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(\text{spam} \mid \text{lottery \& winning}) = \frac{P(\text{spam}) \cdot P(\text{lottery \& winning} \mid \text{spam})}{P(\text{spam}) \cdot P(\text{lottery \& winning} \mid \text{spam}) + P(\text{ham}) \cdot P(\text{lottery \& winning} \mid \text{ham})}$$

# Is There a Quicker Way To Estimate the Probability?

## Naive assumption



The appearances of 'lottery' and 'winning' are independent

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$$

$$\mathbf{P}(\text{spam} \mid \text{lottery \& winning}) = \frac{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(\text{lottery} \mid \text{spam}) \cdot \mathbf{P}(\text{winning} \mid \text{spam})}{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(\text{lottery} \mid \text{spam}) \cdot \mathbf{P}(\text{winning} \mid \text{spam}) + \mathbf{P}(\text{ham}) \cdot \mathbf{P}(\text{lottery} \mid \text{ham}) \cdot \mathbf{P}(\text{winning} \mid \text{ham})}$$

# Is There a Quicker Way To Estimate the Probability?

## Naive assumption

The appearances of the words  $w_1, w_2, \dots, w_n$  are independent

$$\mathbf{P}(\text{spam} \mid w_1, \dots, w_n) = \frac{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(w_1 \mid \text{spam}) \cdots \mathbf{P}(w_n \mid \text{spam})}{\mathbf{P}(\text{spam}) \cdot \mathbf{P}(w_1 \mid \text{spam}) \cdots \mathbf{P}(w_n \mid \text{spam}) + \mathbf{P}(\text{ham}) \cdot \mathbf{P}(w_1 \mid \text{ham}) \cdots \mathbf{P}(w_n \mid \text{ham})}$$

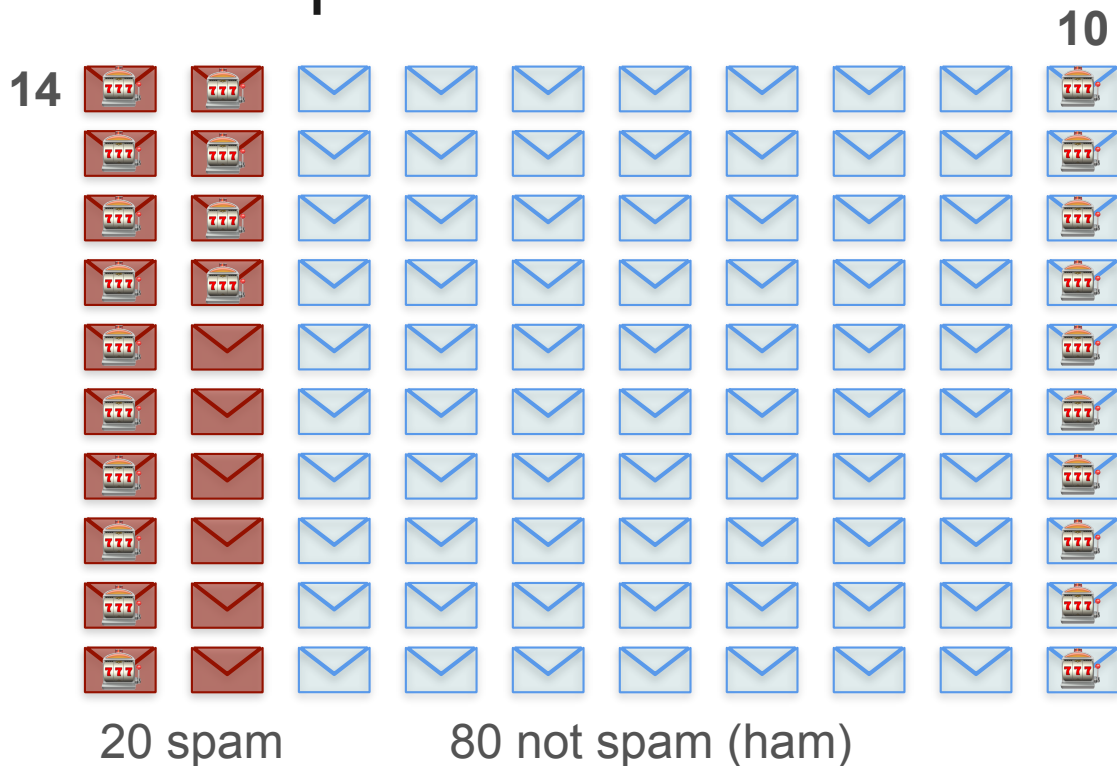
# Naive Bayes: Spam Example

$$P(\text{spam}) = \frac{20}{100} = 0.2$$

$$P(\text{ham}) = \frac{80}{100} = 0.8$$

$$P(\text{lottery} \mid \text{spam}) = \frac{14}{20} = 0.7$$

$$P(\text{lottery} \mid \text{ham}) = \frac{10}{80} = 0.125$$



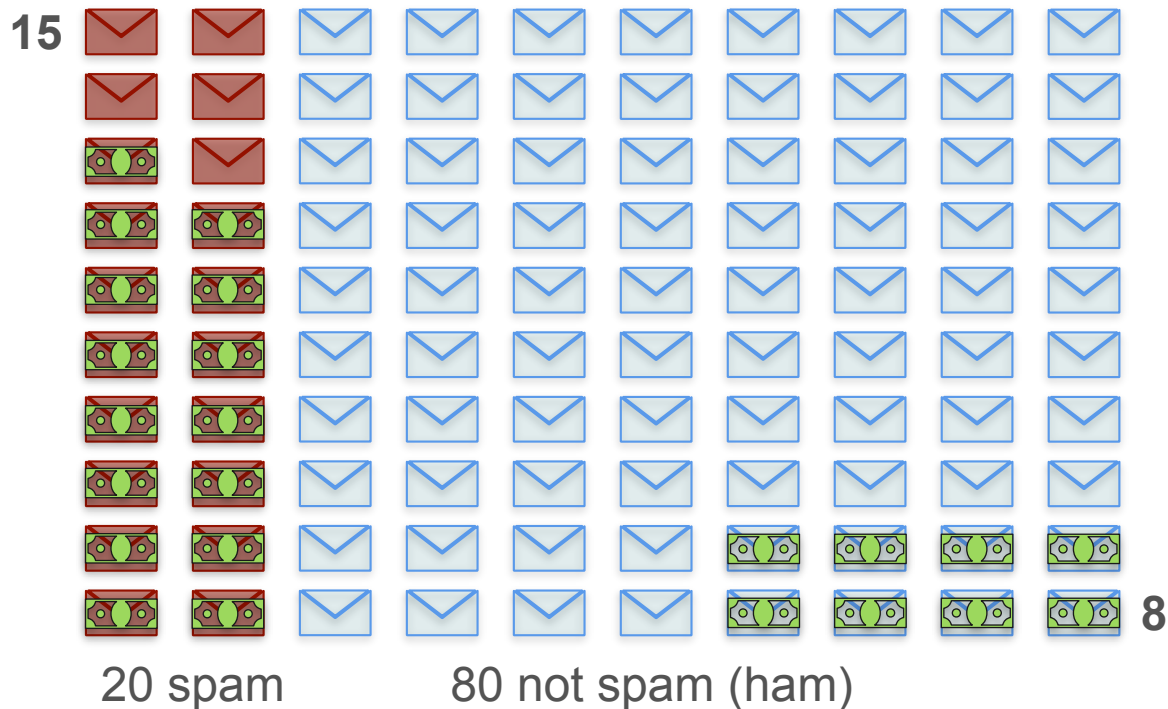
# Naive Bayes: Spam Example

$$P(\text{spam}) = \frac{20}{100} = 0.2$$

$$P(\text{ham}) = \frac{80}{100} = 0.8$$

$$P(\text{winning} \mid \text{spam}) = \frac{15}{20} = 0.75$$

$$P(\text{winning} \mid \text{ham}) = \frac{8}{80} = 0.1$$





# Naive Bayes: Spam Example

$$P(\text{spam}) = 0.2$$

$$P(\text{lottery} \mid \text{spam}) = 0.7$$

$$P(\text{winning} \mid \text{spam}) = 0.75$$

$$P(\text{not spam}) = 0.8$$

$$P(\text{lottery} \mid \text{ham}) = 0.125$$

$$P(\text{winning} \mid \text{ham}) = 0.1$$

$$P(\text{spam} \mid \text{lottery}) = \frac{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam}) \cdot P(\text{winning} \mid \text{spam})}{P(\text{spam}) \cdot P(\text{lottery} \mid \text{spam}) \cdot P(\text{winning} \mid \text{spam}) + P(\text{ham}) \cdot P(\text{lottery} \mid \text{ham}) \cdot P(\text{winning} \mid \text{ham})}$$

$$P(\text{spam} \mid \text{lottery \& winning}) = \frac{0.2 \times 0.7 \times 0.75}{(0.2 \times 0.7 \times 0.75) + (0.8 \times 0.125 \times 0.1)} = 0.913$$



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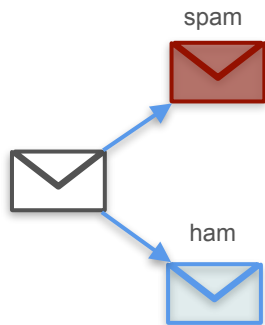
# Introduction to probability

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## **Probability in Machine Learning**

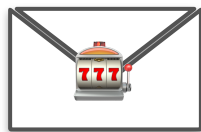
# Bayes Theorem

## PRIOR

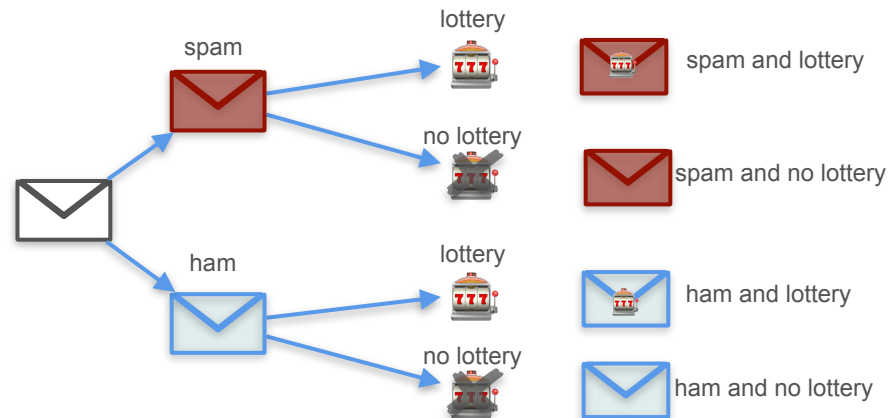


$$P(\text{spam}) = \frac{\text{red envelope icon}}{\text{red envelope icon} + \text{blue envelope icon}}$$

## EVENT



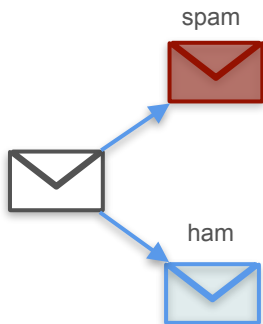
## POSTERIOR



$$P(\text{spam} \mid \text{lottery}) =$$

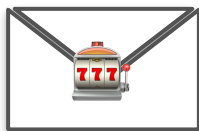
# Bayes Theorem

## PRIOR

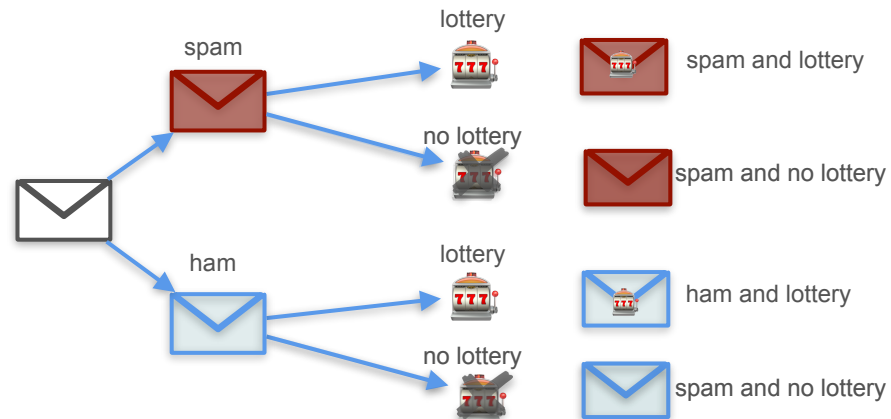


$$P(\text{spam}) = \frac{\text{spam icon}}{\text{spam icon} + \text{ham icon}}$$

## EVENT



## POSTERIOR



$$P(\text{spam} \mid \text{lottery}) = \frac{\text{spam and lottery icon}}{\text{spam and lottery icon} + \text{spam and no lottery icon}}$$

# Example Problem

## Image recognition

- What is the probability that there is a cat in the image
- $\mathbf{P}(\text{cat} \mid \text{image}) = \mathbf{P}(\text{cat} \mid \text{pixel}_1, \text{pixel}_2, \dots, \text{pixel}_n)$



# Example Problem

## Classification

| Patient 3 |                |        |
|-----------|----------------|--------|
| Patient 2 |                |        |
| Patient 1 |                |        |
| A         | Age            | 29     |
| G         | Gender         | Female |
| H         | Height         | 169 cm |
| W         | Weight         | 62 kg  |
| S         | Smoker         | No     |
| ...       | ...            | ...    |
| H         | Heart rate     | 63     |
| B         | Blood pressure | 120 90 |

- Is this patient healthy?
- Calculate  $\mathbf{P}(\text{healthy} \mid \text{symptoms and history})$

# Example Problem

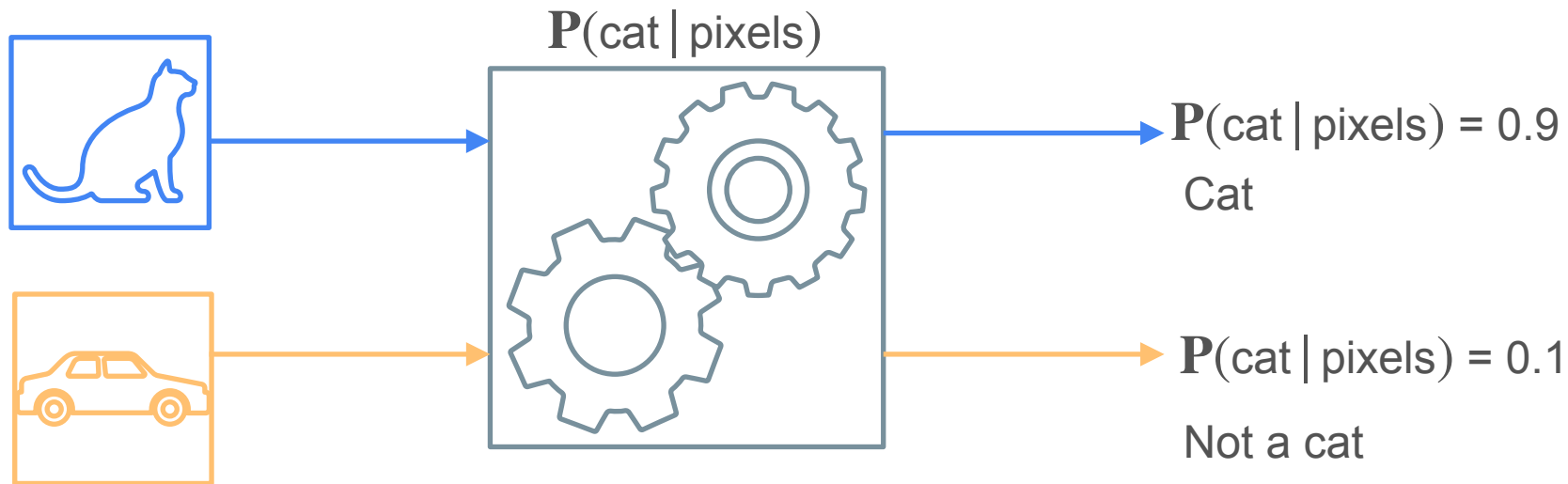
Sentiment analysis

the first cold shower  
even the monkey seems to want  
a little coat of straw

Matsuo Bashō

- Is this a happy sentence?
- Calculate  $P(\text{happy} | \text{words in the sentence})$

# Example Solution





# Example Problem: Generative Models

## Face generation

- Generate a group of pixels such that the resulting image looks like a human face.
- Goal: generate images such that  $\mathbf{P}(\text{face} \mid \text{pixels})$  is high.



Image generated by a StyleGAN

# W1 Lesson 2

# Probability Distributions



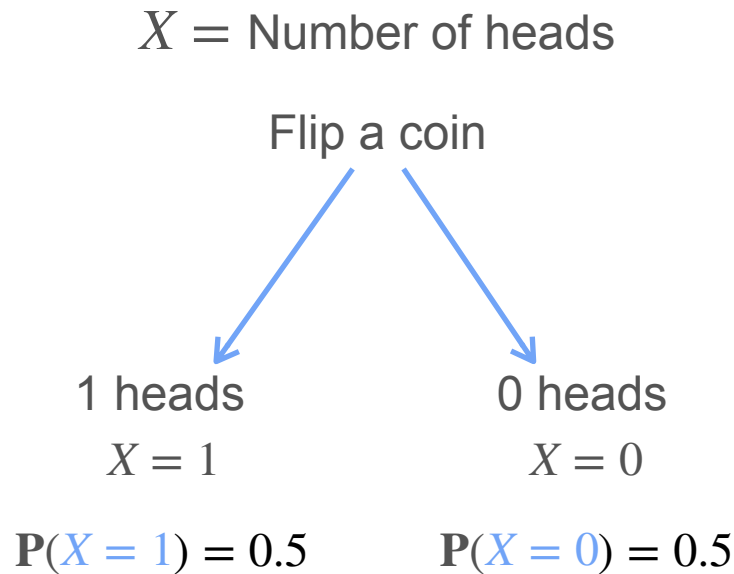
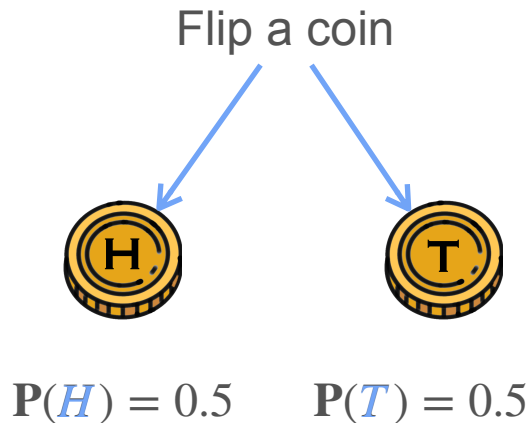
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# Probability Distributions

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## Random Variables

# From Events to Random Variables



$X$  is a random variable

# From Events to Random Variables

$X$  = Number of heads in 10 coin tosses

$$P(H) = 0.5$$



$$X = 10$$

$$0.5^{10}$$

$$P(X = 0)?$$



$$X = 9$$

$$0.5^9 0.5$$

$$P(X = 1)?$$

...



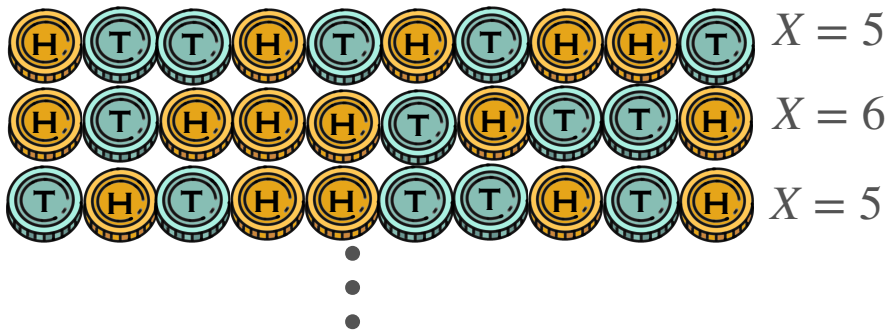
$$X = 9$$

$$0.5^9 0.5$$

$$P(X = 10)?$$

# From Events to Random Variables

$X$  = Number of heads in 10 coin tosses



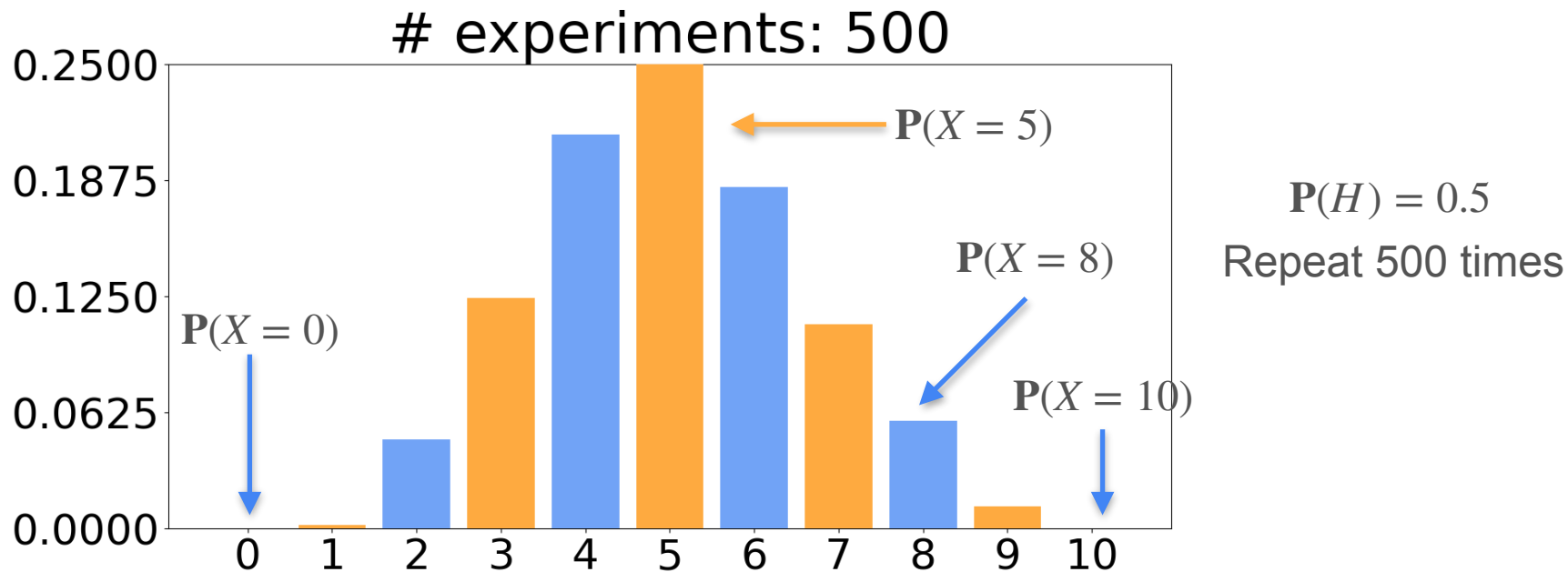
Possible outcomes:

|   |   |    |
|---|---|----|
| 0 | 4 | 8  |
| 1 | 5 | 9  |
| 2 | 6 | 10 |
| 3 | 7 |    |

$$P(H) = 0.5$$

Repeat 500 times

# Flipping a Fair Coin 500 Times



# Why Random Variables?

- Random variables allow you to model the whole experiment at once



$X$  = Number of heads



$X$  = Number of 1's



$X$  = Number of sick patients

?

$$P(X = 1) = 0.5$$

$$P(X = -7) = 0.2$$

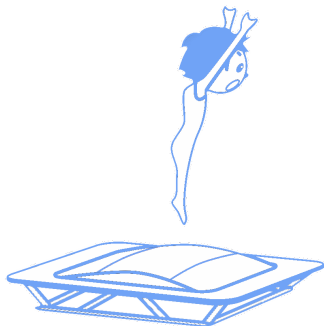
$$P(X = 3.14159) = 0.3$$



# Other Random Variables



Wait time until the  
next bus arrives



Height of an  
gymnast's jump



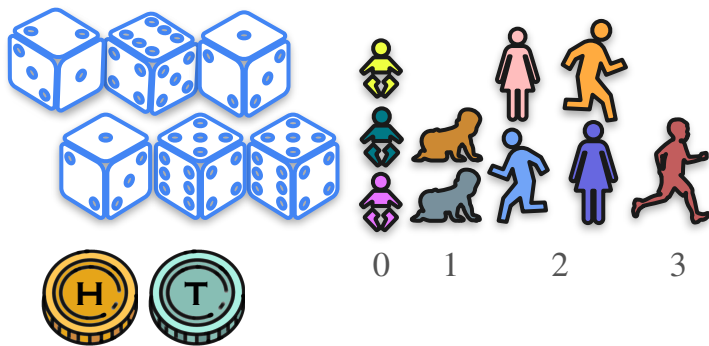
Number of  
defective products  
in a shipment



mm. of rain in  
November

# Discrete and Continuous Random Variables

## Discrete random variables



~~Finite number of values~~

(Could be infinite too)

Can take only a **countable**  
number of values

## Continuous random variables



Infinite number of values

Takes values on an interval

# Random Variable Vs. Deterministic Variable

## Deterministic

$$x = 2, f(x) = x^2$$

Fixed outcome

## Random

$X$  = number of defective items in  
a shipment

Uncertain outcome



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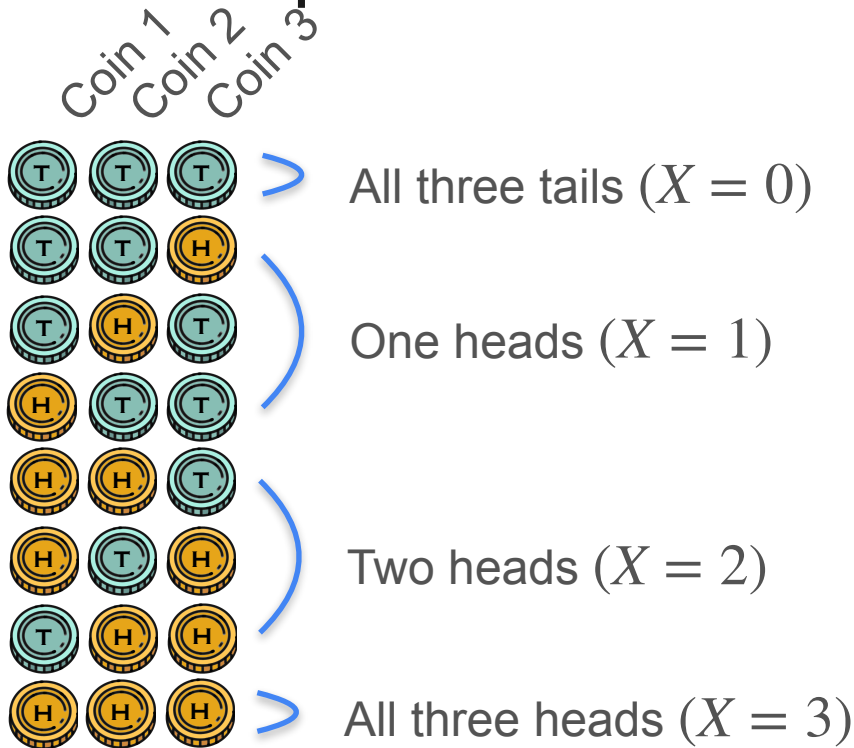
# Probability Distributions

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## **Probability Distributions (Discrete)**

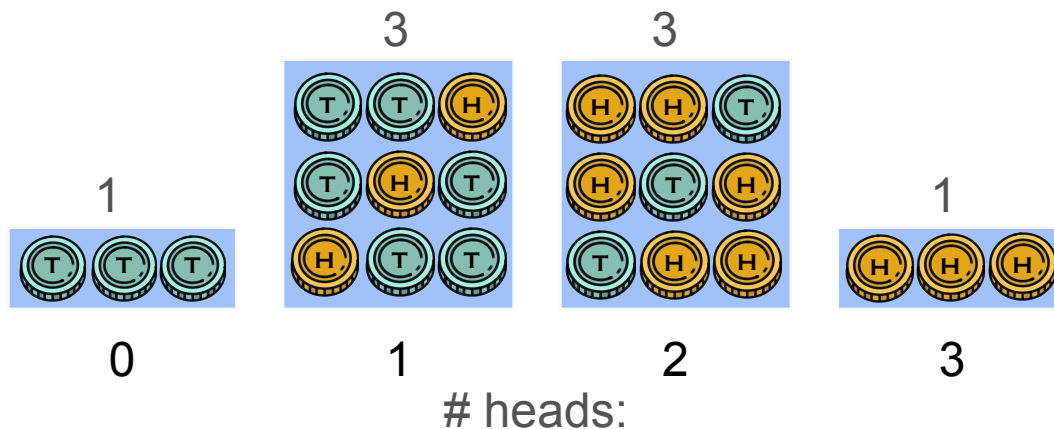
# Discrete Distributions: Flip Three Coins

$X_1$ : number of heads in 3 coin tosses



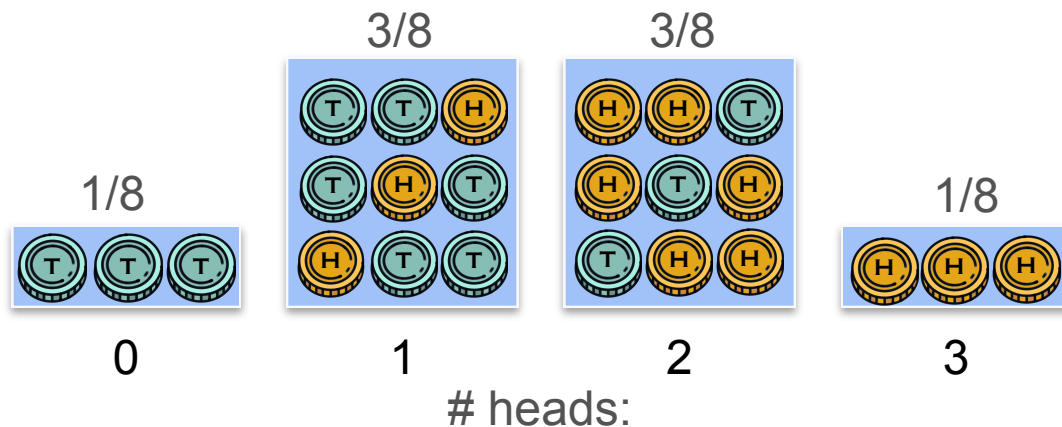
# Discrete Distributions: Flip Three Coins

$X_1$ : number of heads in 3  
coin tosses



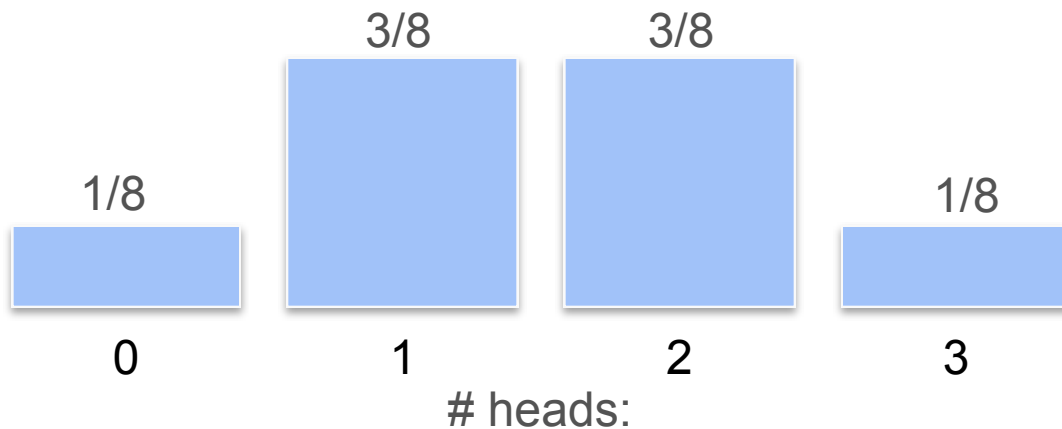
# Discrete Distributions: Flip Three Coins

$X_1$ : number of heads in 3 coin tosses



# Discrete Distributions: Flip Three Coins

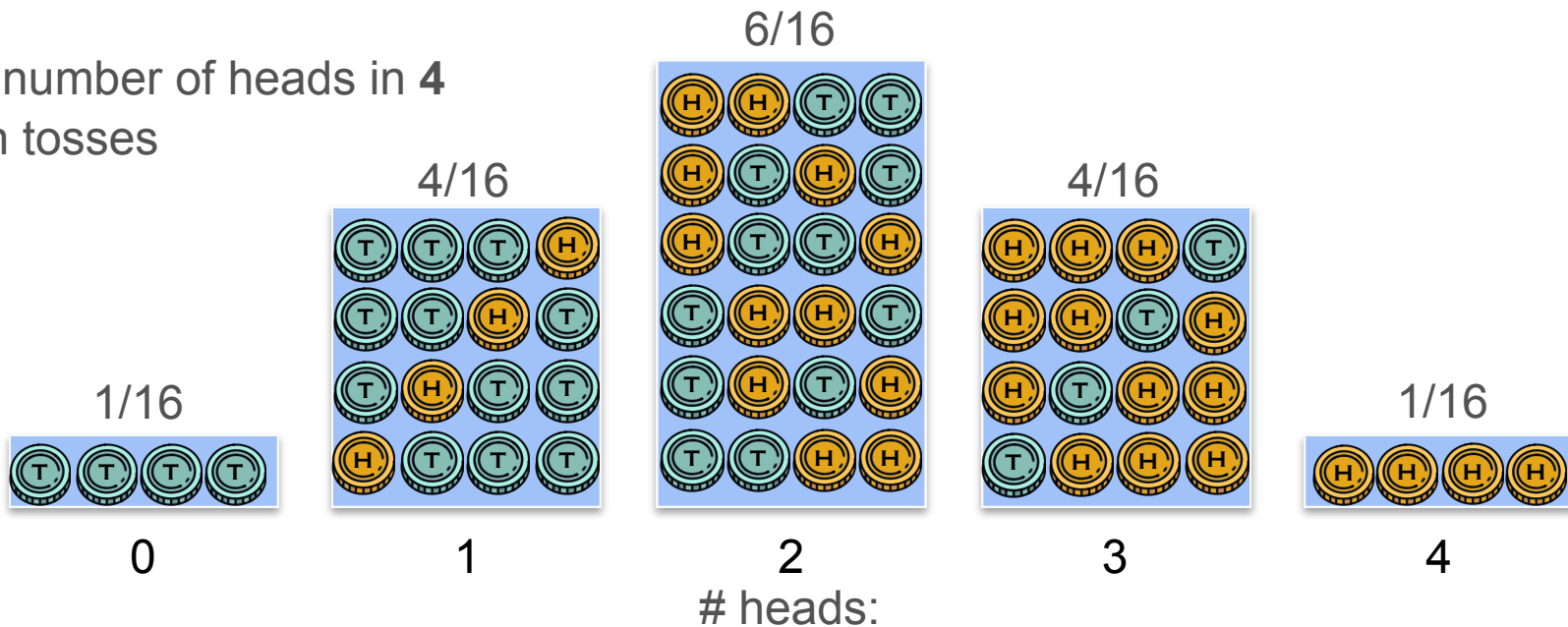
$X_1$ : number of heads in 3  
coin tosses





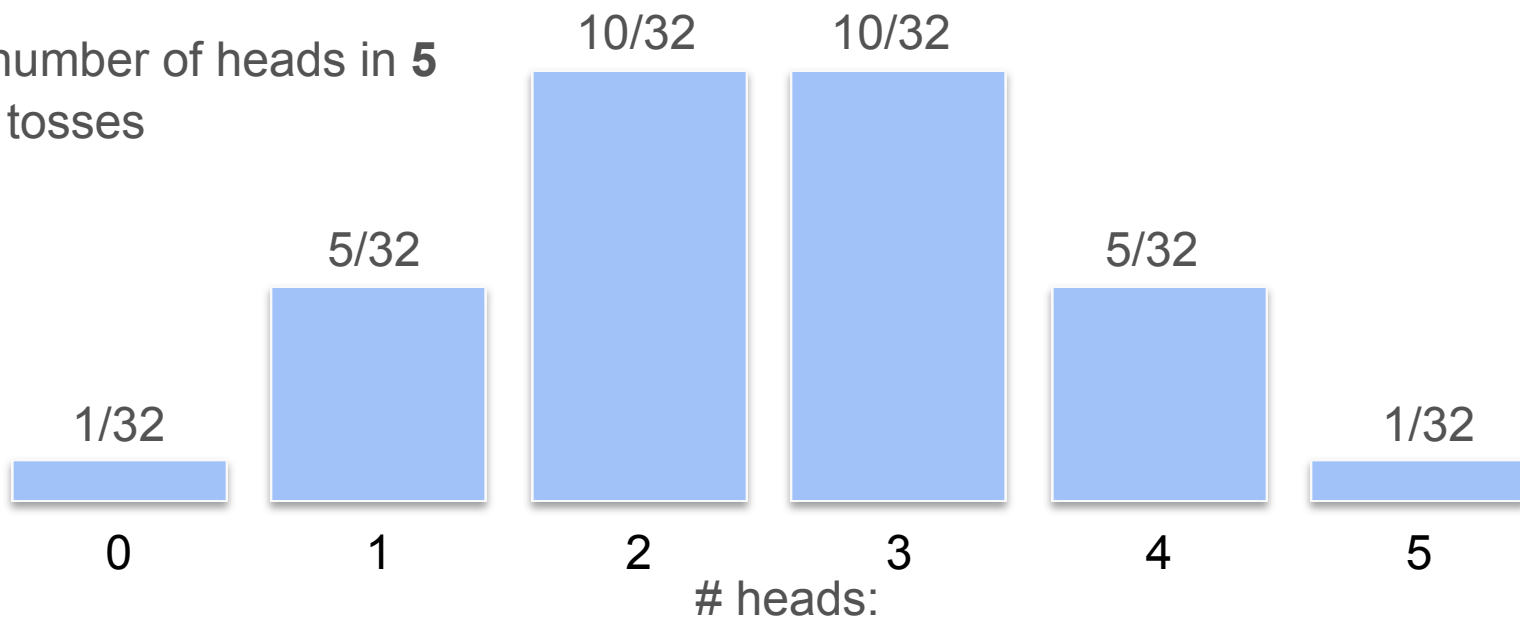
# Discrete Distributions: Flip Four Coins

$X_2$ : number of heads in 4 coin tosses



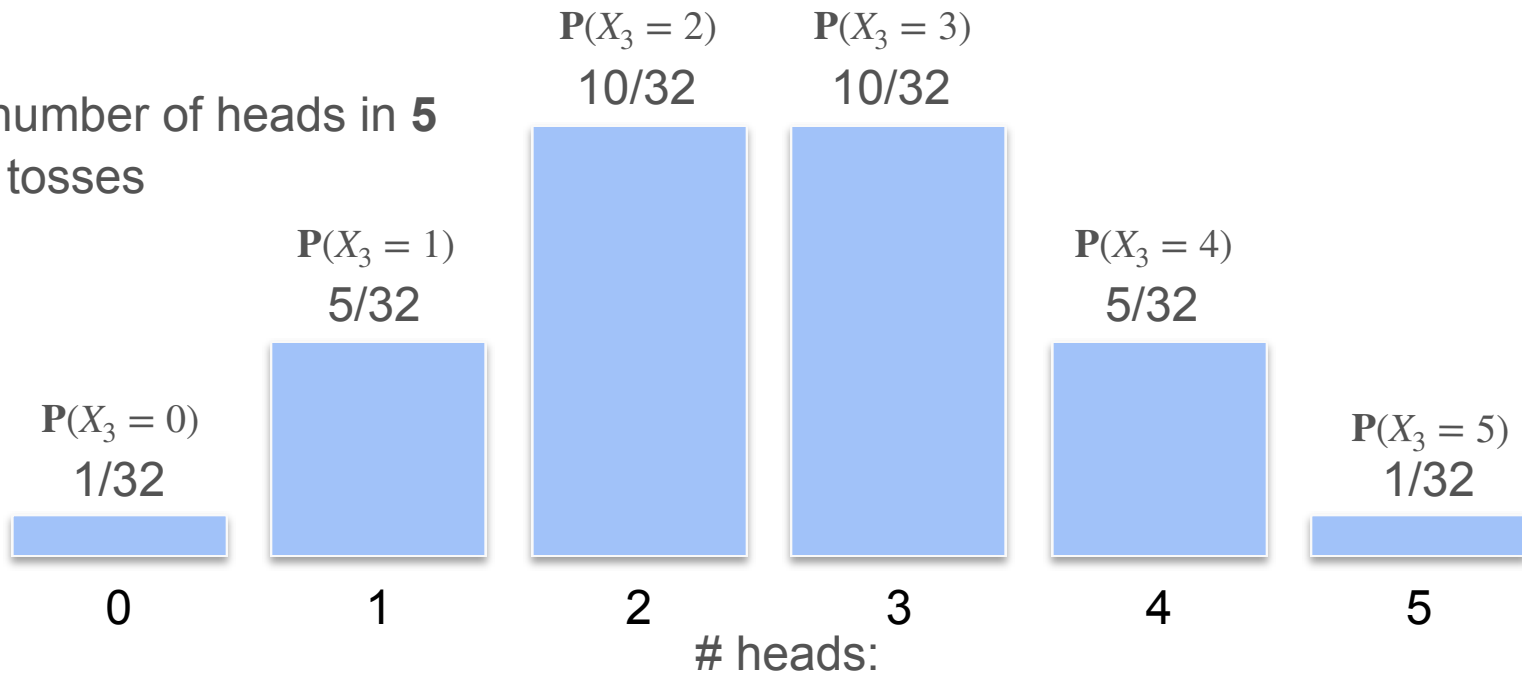
# Discrete Distributions: Flip Five Coins

$X_3$ : number of heads in **5**  
coin tosses



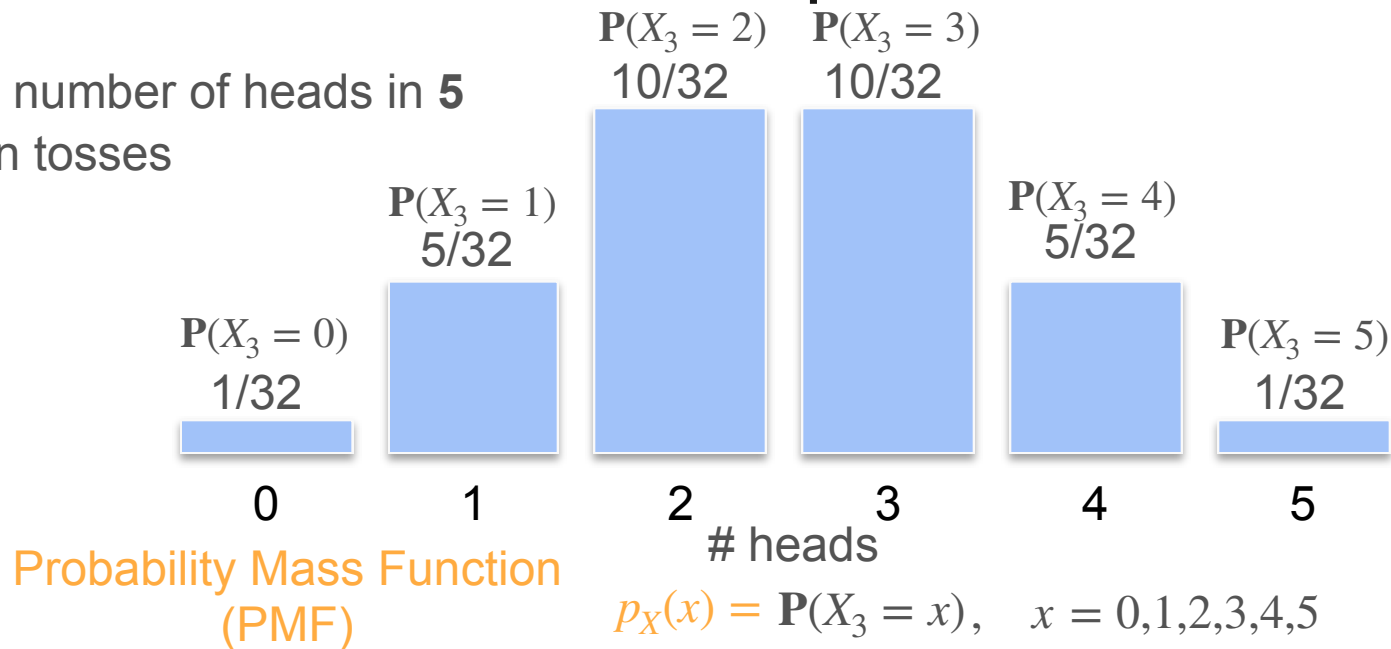
# Discrete Distributions: Flip Five Coins

$X_3$ : number of heads in 5 coin tosses



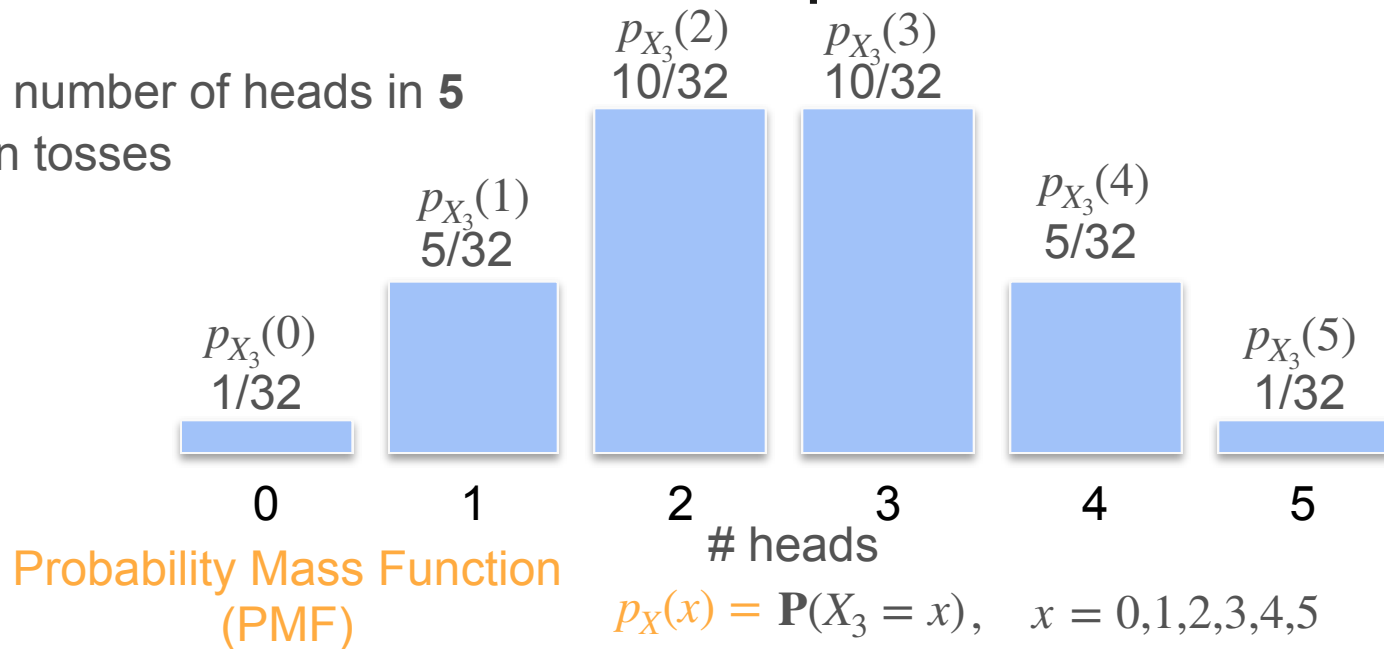
# Discrete Distributions: Flip Five Coins

$X_3$ : number of heads in 5 coin tosses



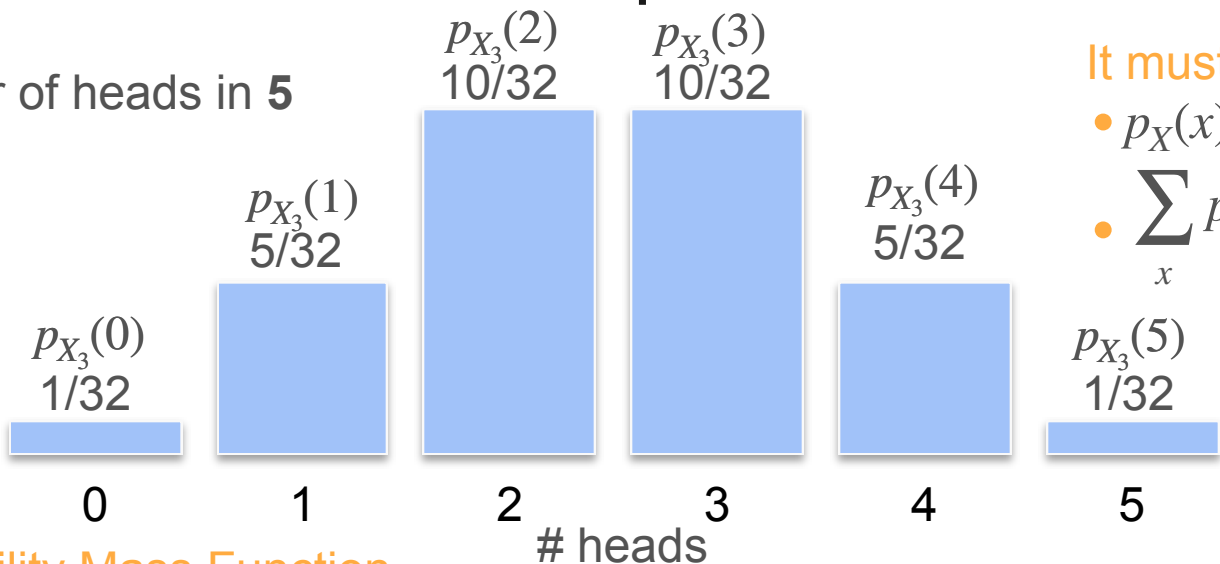
# Discrete Distributions: Flip Five Coins

$X_3$ : number of heads in 5 coin tosses



# Discrete Distributions: Flip Five Coins

$X_3$ : number of heads in 5 coin tosses



Probability Mass Function (PMF)

It must satisfy:

- $p_X(x) \geq 0$
- $\sum_x p_X(x) = 1$

$$p_X(x) = \mathbf{P}(X_4 = x), \quad x = 0, 1, 2, 3, 4, 5$$

$$\begin{aligned} &+ \quad + \quad + \quad + \quad + \quad = \frac{1}{32} + \frac{5}{32} + \frac{10}{32} + \frac{10}{32} + \frac{5}{32} + \frac{1}{32} = 1 \end{aligned}$$

# Can You See a Pattern?

$X_1, X_2, X_3, X_4$  are very similar

They all represent **number of heads in  $n$  experiments**

The way the probability distributes along the possible outcomes seems to have a similar pattern

Could there be a **single model** to represent all this variables?



**Binomial distribution**



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# Probability Distributions

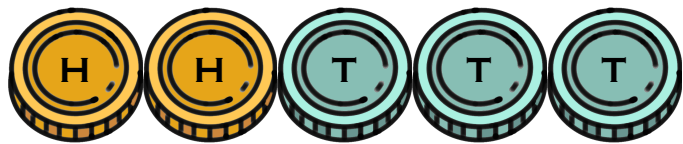
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## **Binomial Distribution**

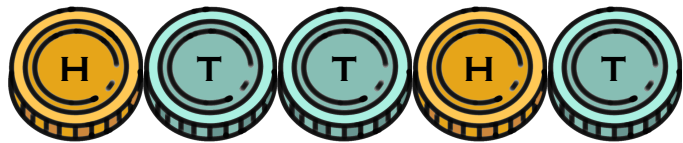


# Binomial Distribution: Example

What is the probability that if I flip 5 coins, 2 of them land in heads?



$$0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 = \frac{1}{32}$$



$$0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 = \frac{1}{32}$$

# Binomial Distribution: Example

What is the probability that if I flip 5 coins, 2 of them land in heads?



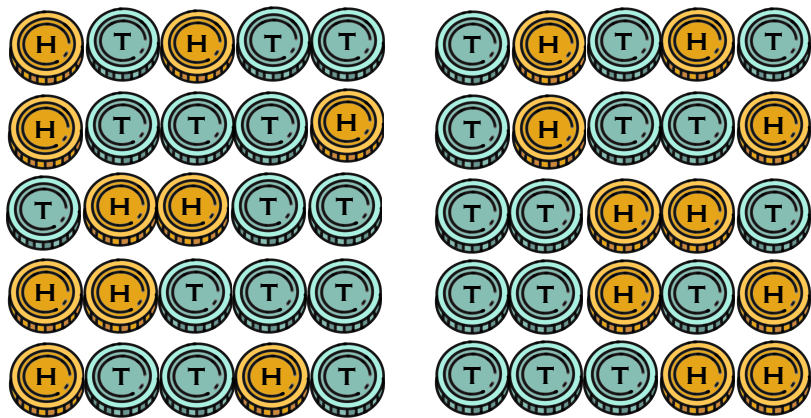
Number of ways  
you can order 5  
coins

$$10 = \frac{5!}{2!}$$

Number  
of H

# Binomial Distribution: Example

What is the probability that if I flip 5 coins, 2 of them land in heads?



$$10 = \frac{5!}{2!(5-2)!}$$

Number of ways you can order 5 coins

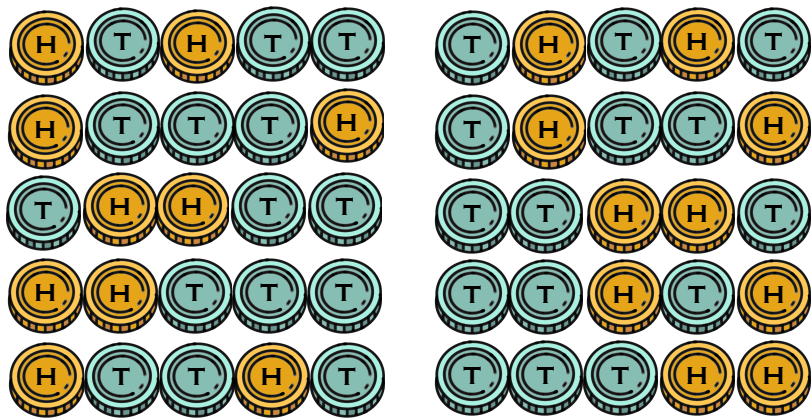
Number of H

Number of T



# Binomial Distribution: Example

What is the probability that if I flip 5 coins, 2 of them land in heads?



$$10 = \frac{5!}{2!(5-2)!} = \binom{5}{2}$$

Binomial coefficient

Number of ways you can get 2  
heads in 5 coin tosses

# Binomial Distribution: Binomial Coefficient

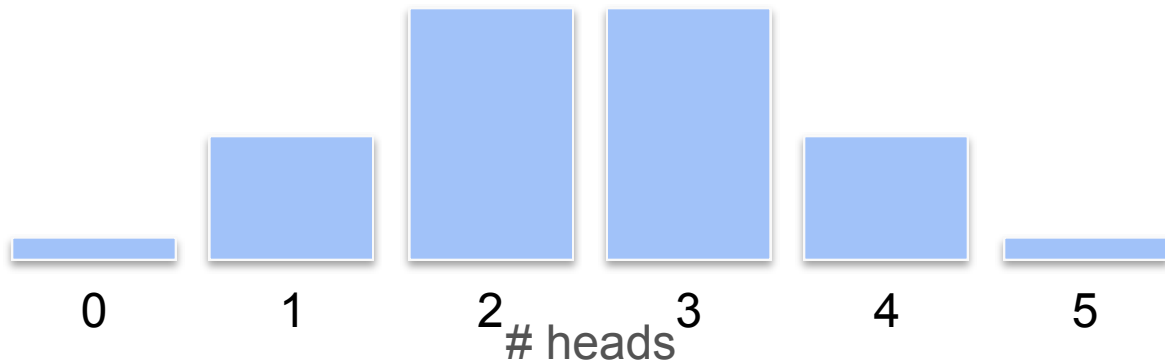
In general:

$\binom{n}{k}$  counts all the combinations for landing  $k$  heads in  $n$  coin tosses

Property:

$$\binom{n}{k} = \binom{n}{n-k}$$

The PMF with a fair coin is symmetrical

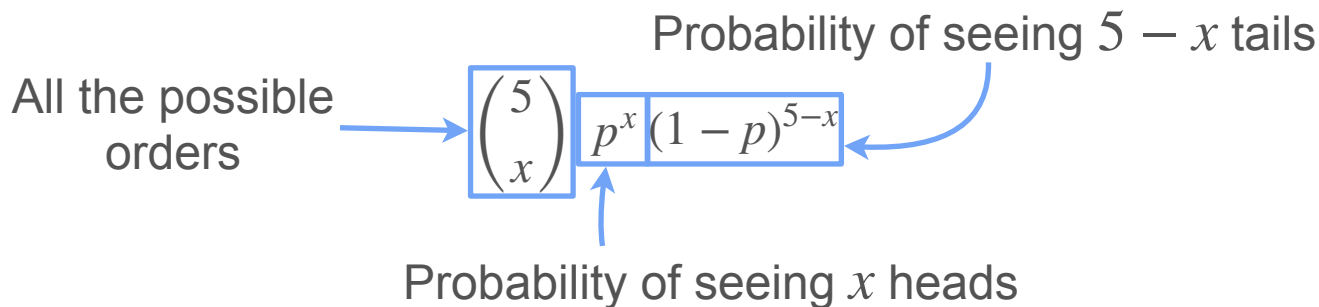


# Binomial Distribution: Binomial Coefficient

General PMF for  $X$  : number of heads in 5 coin tosses?

Your coin has  $\mathbf{P}(H) = p$

Event:  $X = x$ :  $x$  heads in 5 tosses



# Binomial Distribution

General PMF for  $X$  : number of heads in 5 coin tosses?

Your coin has  $\mathbf{P}(H) = p$

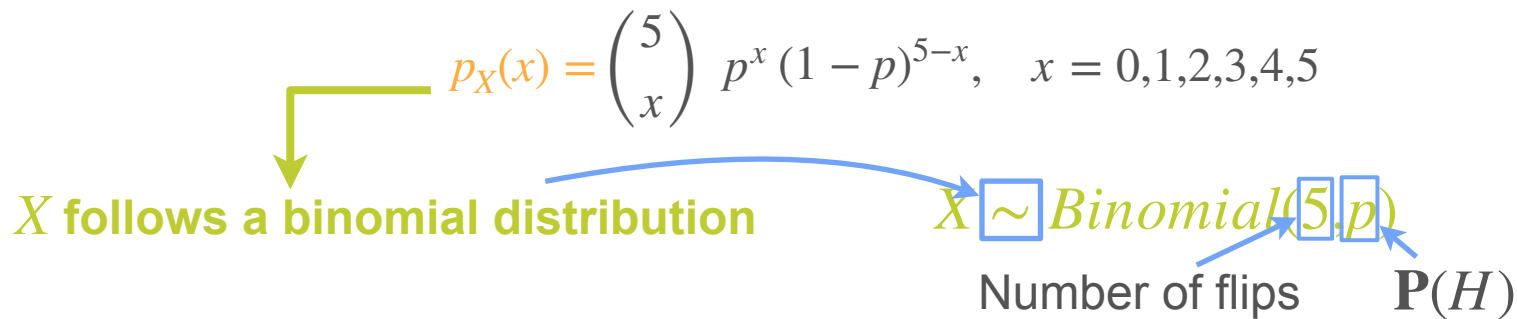
Event:  $X = x$ :  $x$  heads in 5 tosses

$$p_X(x) = \binom{5}{x} p^x (1-p)^{5-x}, \quad x = 0, 1, 2, 3, 4, 5$$

$X$  follows a binomial distribution

$X \sim \text{Binomial}(5, p)$

Number of flips       $\mathbf{P}(H)$

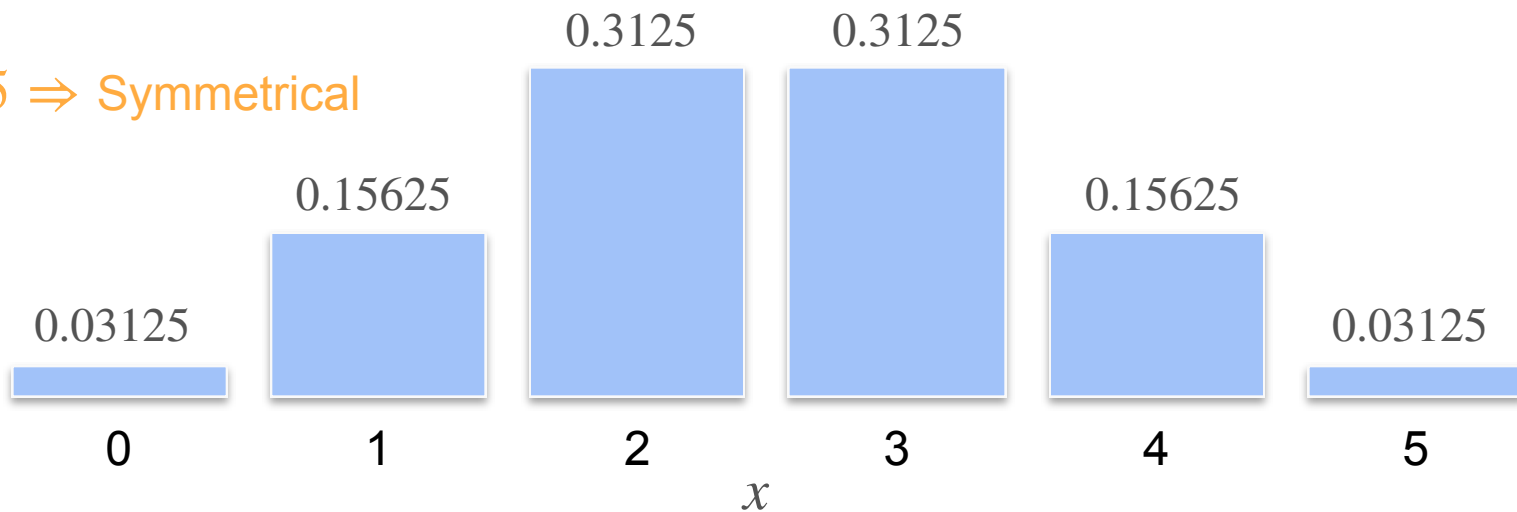


# Binomial Distribution

$$\begin{matrix} n = 5 \\ p = 0.5 \end{matrix}$$

$$p_X(x) = \mathbf{P}(X = x) = \binom{5}{k} 0.5^k 0.5^{5-k}$$

$p = 0.5 \Rightarrow$  Symmetrical

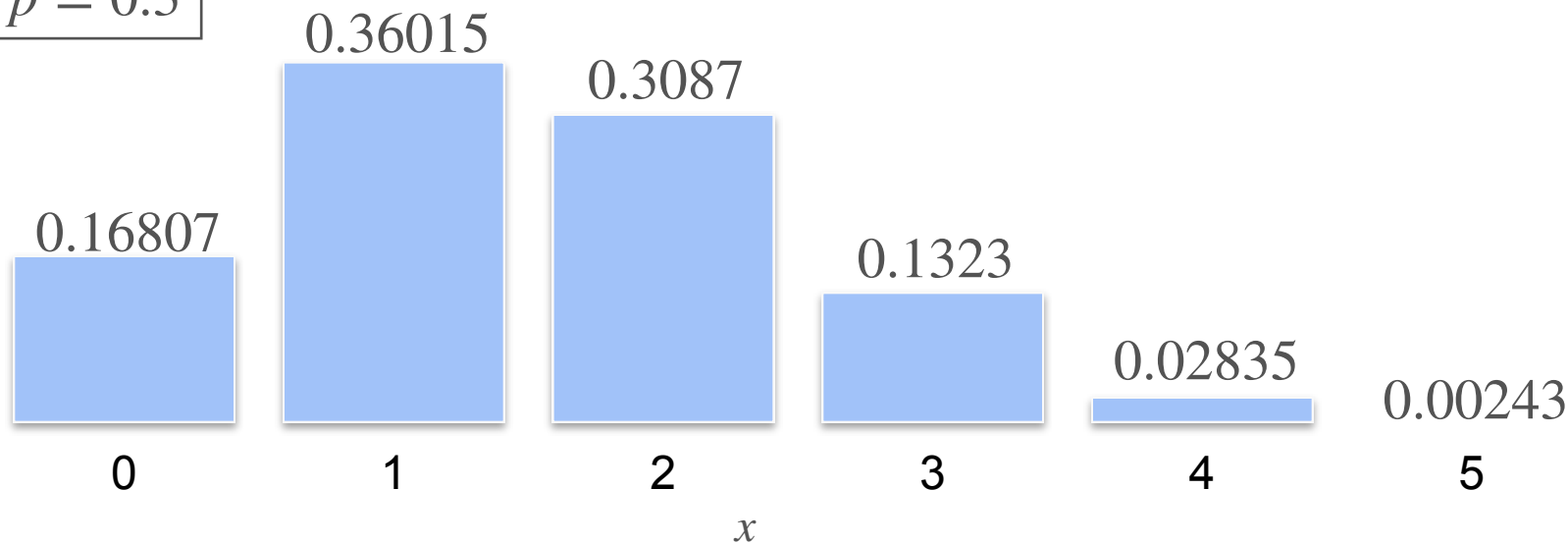




# Binomial Distribution

$$\begin{aligned} n &= 5 \\ p &= 0.3 \end{aligned}$$

$$p_X(x) = \mathbf{P}(X = x) = \binom{5}{k} 0.3^k 0.7^{5-k}$$



# Binomial Distribution

General PMF for  $X$  : number of heads in 5 coin tosses?

Your coin has  $\mathbf{P}(H) = p$

Event:  $X = x$ :  $x$  heads in 5 tosses

$$p_X(x) = \binom{5}{x} p^x (1-p)^{5-x}, \quad x = 0, 1, 2, 3, 4, 5$$

$$X \sim \textit{Binomial}(5, p)$$

# Binomial Distribution

General PMF for  $X$  : number of heads in  $n$  coin tosses?

Your coin has  $\mathbf{P}(H) = p$

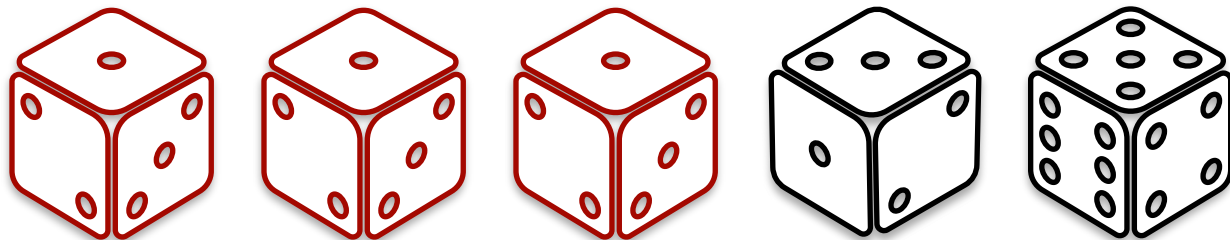
Event:  $X = x$ :  $x$  heads in  $n$  tosses

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

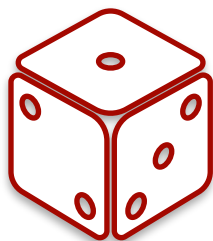
$X \sim \text{Binomial}(n, p)$       $n$  and  $p$  are called the **parameters** of the binomial distribution

# Binomial Distribution: Quiz

What is the probability of getting three ones when rolling a dice five times (no matter on which dice)?



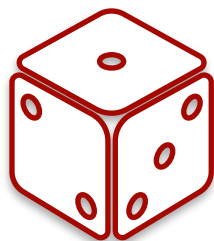
# Binomial Distribution: Dice Is a Biased Coin!



one  
 $p = \frac{1}{6}$



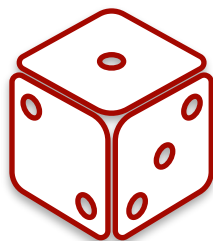
heads  
 $p = \frac{1}{6}$



one  
 $p = \frac{1}{6}$



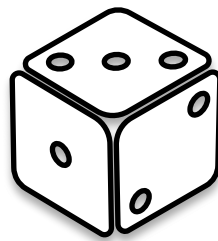
heads  
 $p = \frac{1}{6}$



one  
 $p = \frac{1}{6}$



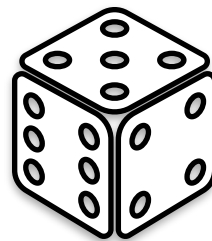
heads  
 $p = \frac{1}{6}$



not one  
 $p = \frac{5}{6}$



not heads  
 $p = \frac{5}{6}$



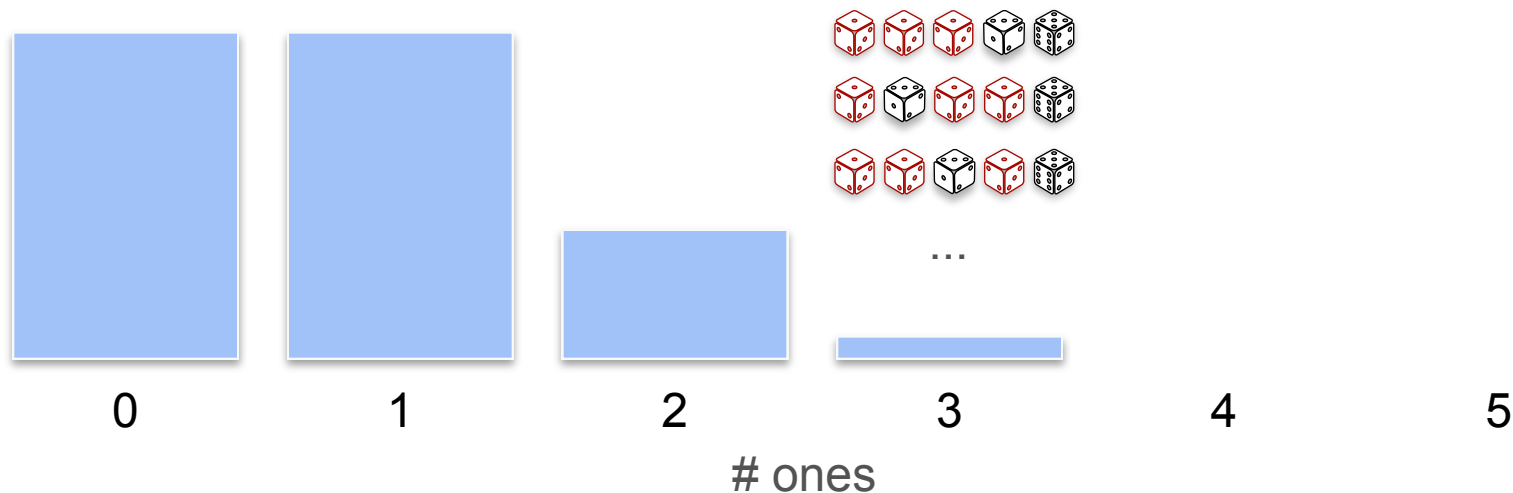
not one  
 $p = \frac{5}{6}$



not heads  
 $p = \frac{5}{6}$

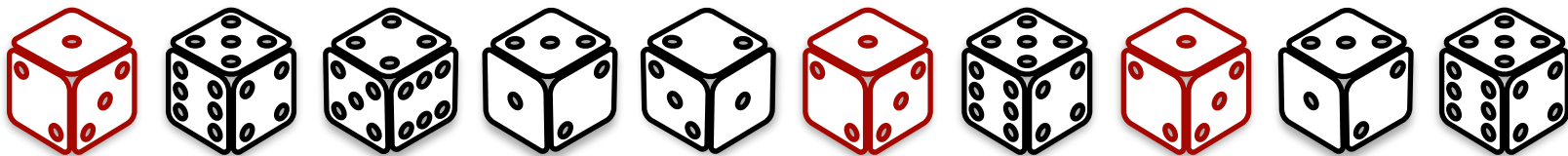
# Binomial Distribution: Dice Is a Biased Coin!

$$n = 5$$
$$p = 0.1666$$



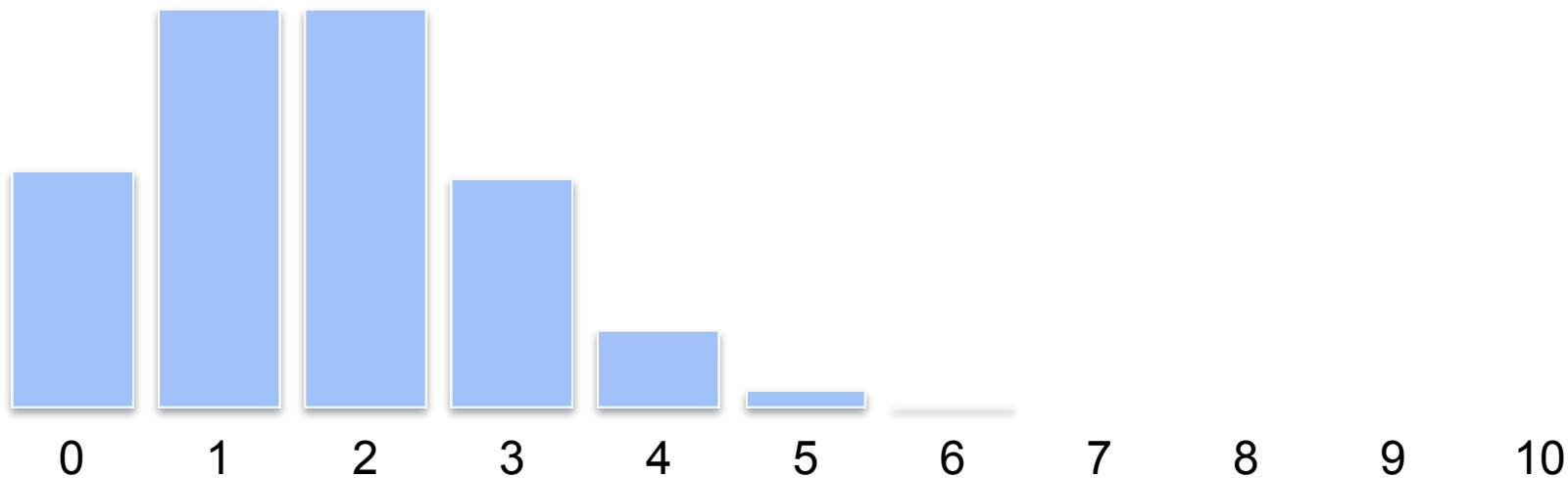
# Binomial Distribution: Quiz

- Quiz: What are the parameters for the following binomial distribution:
  - I roll 10 dice
  - I want to record the number of times I obtain the number 1



# Binomial Distribution: Quiz

$$n = 10$$
$$p = 0.1666$$







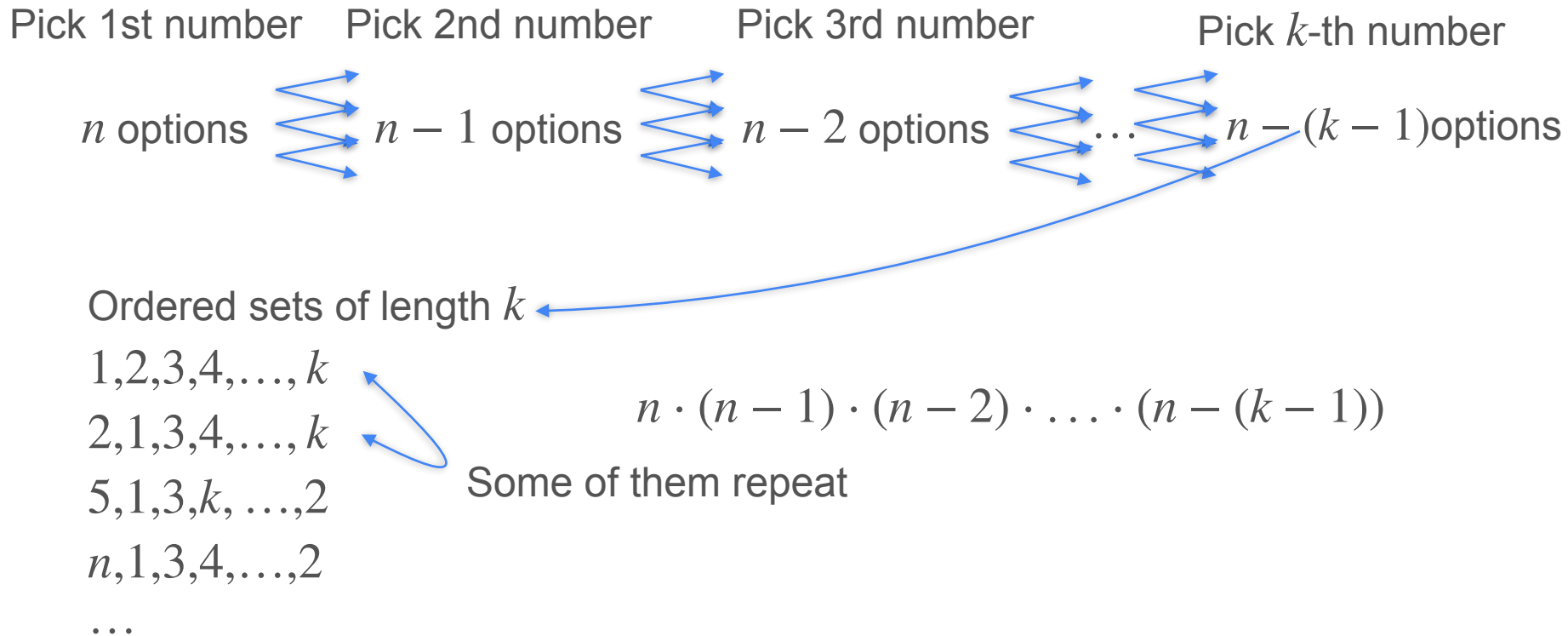
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# Probability Distributions

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**(Optional)  
Binomial Coefficient**

# Binomial Coefficient



# Binomial Coefficient

|         | Pick 1st  | Pick 2nd  | Pick 3rd  | Pick 4th |
|---------|-----------|-----------|-----------|----------|
|         | 4 options | 3 options | 2 options | 1 option |
| 1,2,3,4 |           |           |           |          |
| 1,2,4,3 |           |           |           |          |
| 1,3,2,4 |           |           |           |          |
| 1,3,4,2 |           |           |           |          |
| ...     |           |           |           |          |
| 4,3,2,1 |           |           |           |          |

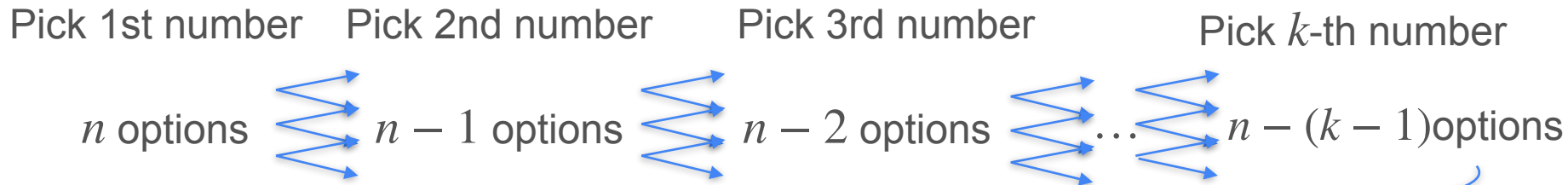
$$4 \cdot 3 \cdot 2 \cdot 1 = 4!$$

For five numbers:

$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5!$$

General solution:  $k!$

# Binomial Coefficient



Ordered sets of length  $k$

→ Unordered sets of length  $k$

1, 2, 3, 4, ...,  $k$

2, 1, 3, 4, ...,  $k$

5, 1, 3,  $k$ , ..., 2

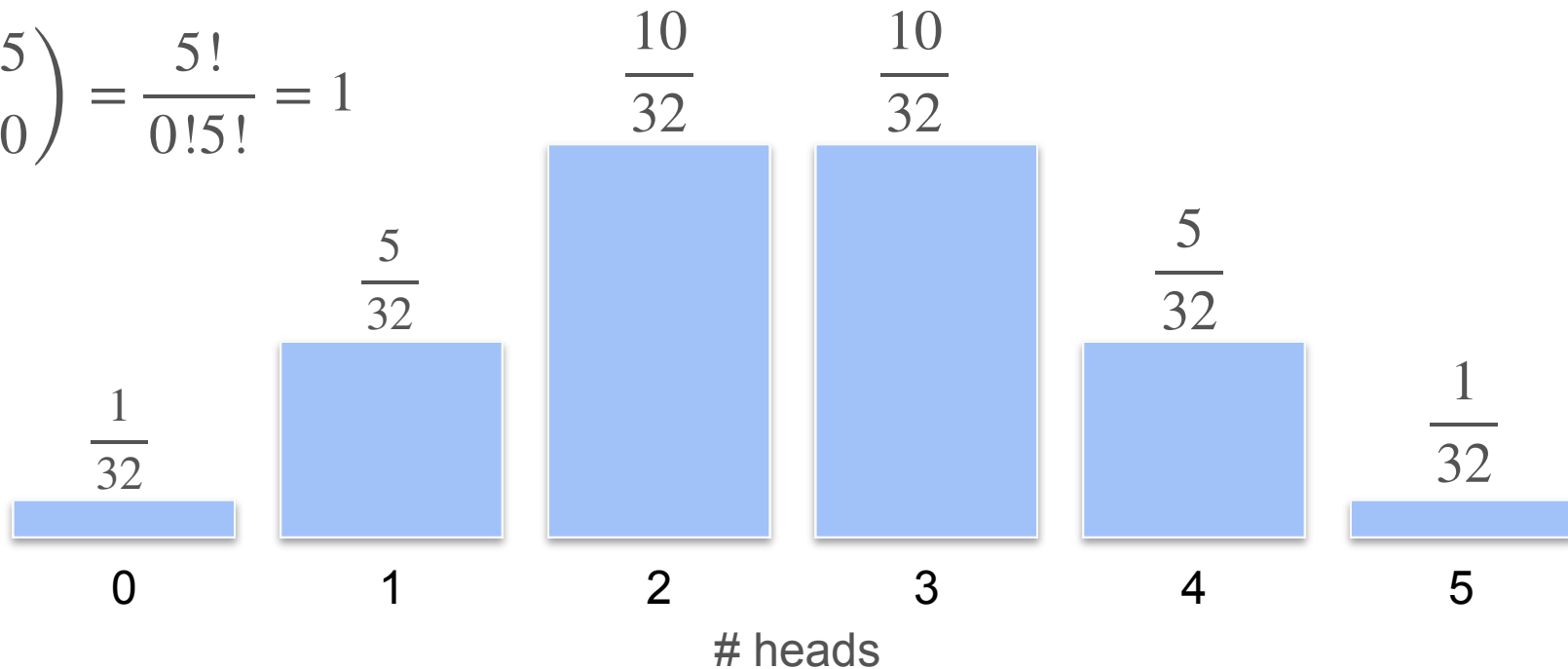
$n$ , 1, 3, 4, ..., 2

...

$$\frac{n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot (n - (k - 1))}{k!} = \frac{n!}{(n - k)! k!} = \binom{n}{k}$$

# Binomial Distribution: Fair Coins

$$\binom{5}{0} = \frac{5!}{0!5!} = 1$$



# Binomial Distribution: Fair Coins



50 %

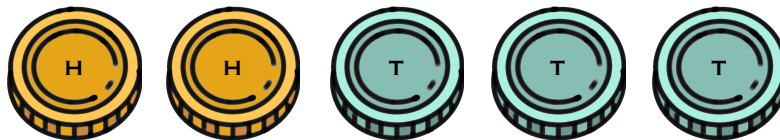


50 %



$0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5$

$$= \frac{1}{32}$$



$0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5$

$$= \frac{1}{32}$$

# Binomial Distribution: Biased Coins



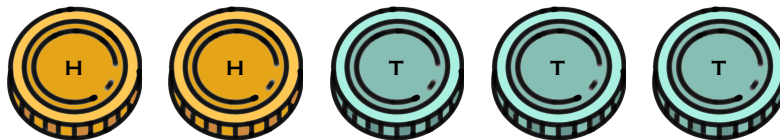
30 %



70 %



$$0.3 \cdot 0.3 \cdot 0.3 \cdot 0.3 \cdot 0.3 = 0.00243$$



$$0.3 \cdot 0.3 \cdot 0.7 \cdot 0.7 \cdot 0.7 = 0.01323$$

# Binomial Distribution: Biased Coins

|                                                                                  |                                                                   |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------|
|  | $0.3 \cdot 0.3 \cdot 0.3 \cdot 0.3 \cdot 0.3 = 0.3^5 \cdot 0.7^0$ |
|  | $0.3 \cdot 0.3 \cdot 0.3 \cdot 0.3 \cdot 0.7 = 0.3^4 \cdot 0.7^1$ |
|  | $0.3 \cdot 0.3 \cdot 0.3 \cdot 0.7 \cdot 0.7 = 0.3^3 \cdot 0.7^2$ |
|  | $0.3 \cdot 0.3 \cdot 0.7 \cdot 0.7 \cdot 0.7 = 0.3^2 \cdot 0.7^3$ |
|  | $0.3 \cdot 0.7 \cdot 0.7 \cdot 0.7 \cdot 0.7 = 0.3^1 \cdot 0.7^4$ |
|  | $0.7 \cdot 0.7 \cdot 0.7 \cdot 0.7 \cdot 0.7 = 0.3^0 \cdot 0.7^5$ |

$$= 0.3^k \cdot 0.7^{n-k}$$



# Binomial Distribution: Biased Coins



$$0.3^5 \cdot 0.7^0$$



$$0.3^4 \cdot 0.7^1$$



$$0.3^3 \cdot 0.7^2$$



$$0.3^2 \cdot 0.7^3$$



$$0.3^1 \cdot 0.7^4$$



$$0.3^0 \cdot 0.7^5$$

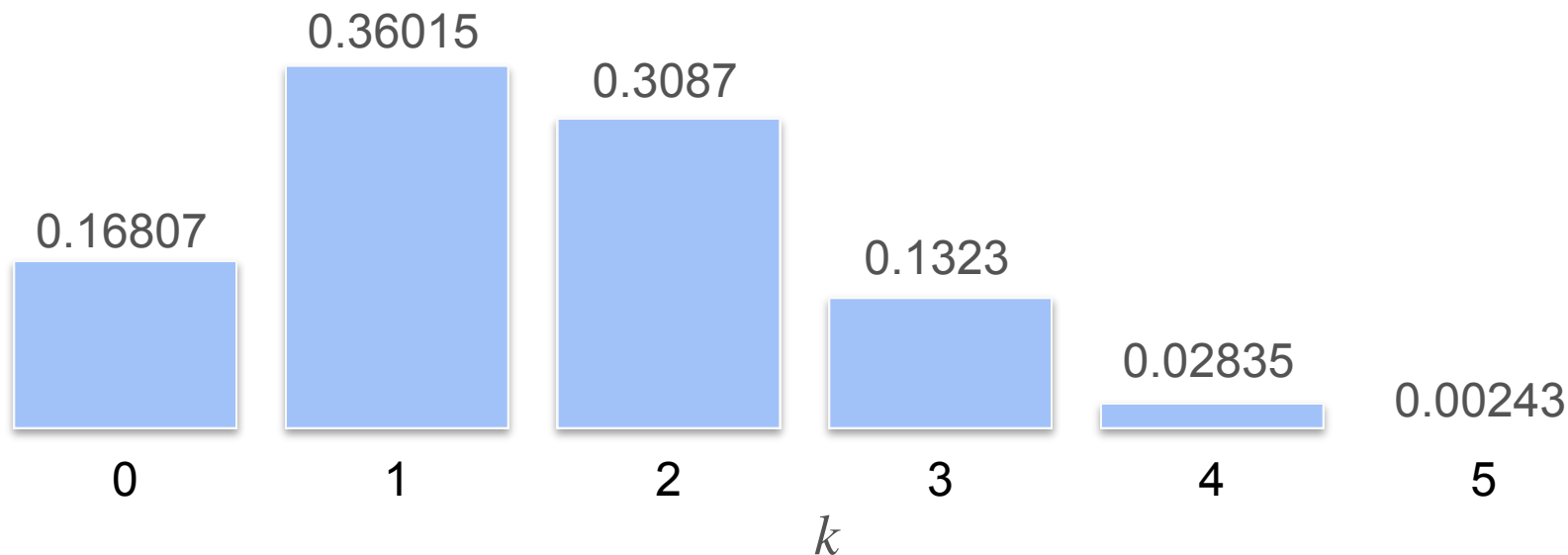
$$= 0.3^k \cdot 0.7^{n-k} \rightarrow \binom{n}{k} 0.3^k \cdot 0.7^{n-k}$$

Account for all  
possible orders of  
heads and tails

# Binomial Distribution: Biased Coins

$$= \binom{n}{k} 0.3^k \cdot 0.7^{n-k}$$

$$\begin{aligned} n &= 5 \\ p &= 0.3 \end{aligned}$$





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# Probability Distributions

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## **Bernoulli Distribution**

# Bernoulli Distribution

$X$  = Number of heads

Flip a coin



$$P(X = 1) = 0.5$$

Success



$$P(X = 0) = 0.5$$

Failure

$X$  = Number of 1's

Throw a dice



$$P(X = 1) = \frac{1}{6}$$

Success

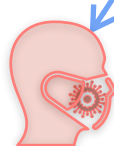


$$P(X = 0) = \frac{5}{6}$$

Failure

$X$  = Number of sick patients

Flip a coin



$$P(X = 1) = p$$

Success



$$P(X = 0) = 1 - p$$

Failure

$X \sim \text{Bernoulli}(p)$  ←  $p$  is the parameter of the Bernoulli distribution



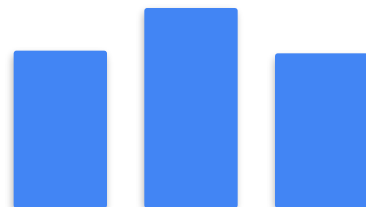
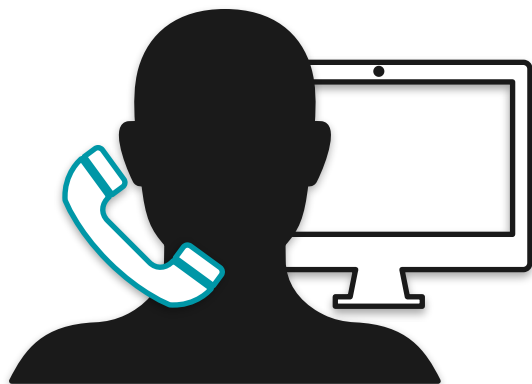
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# Probability Distributions

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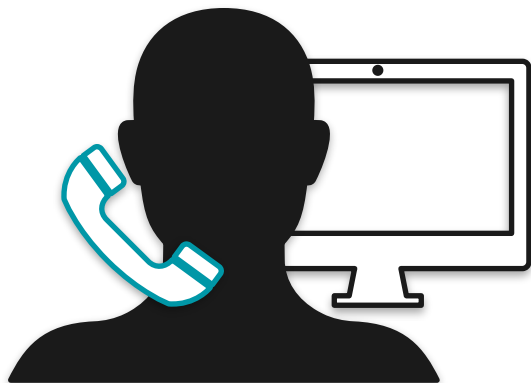
## **Probability Distributions (Continuous)**

# From Discrete to Continuous Distributions



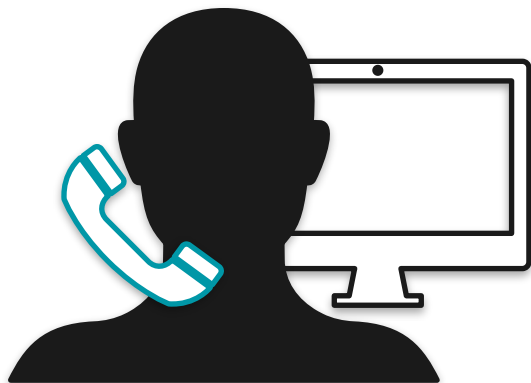
Waiting time:      1      2      3      (min)

# From Discrete to Continuous Distributions



Waiting time: 1 1.01 2 3 (min)

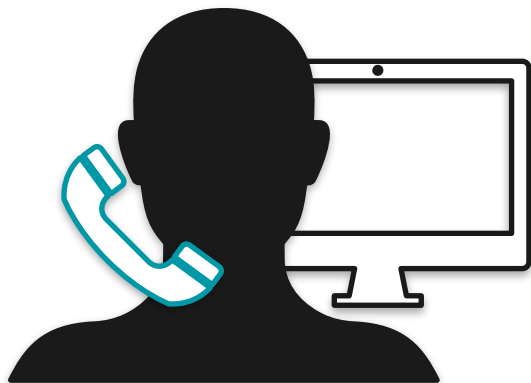
# From Discrete to Continuous Distributions



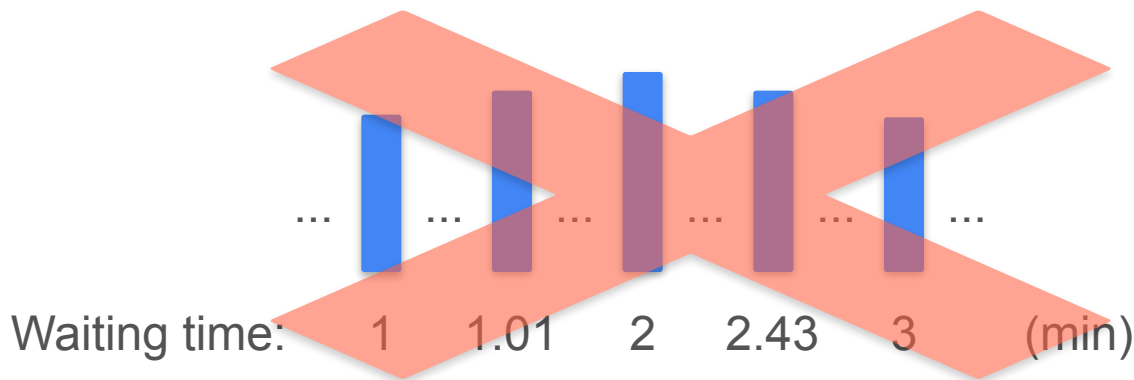
Waiting time:      1      1.01      2      2.43      3      (min)



# From Discrete to Continuous Distributions



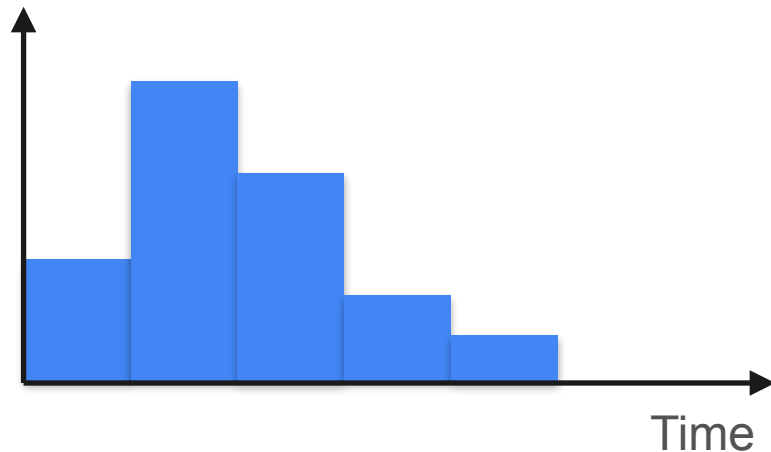
This is a continuous distribution!



What is the probability that you will wait EXACTLY one minute for the call?

Answer: ZERO

# From Discrete to Continuous Distributions



$P(\text{between 0 and 1 mins})$

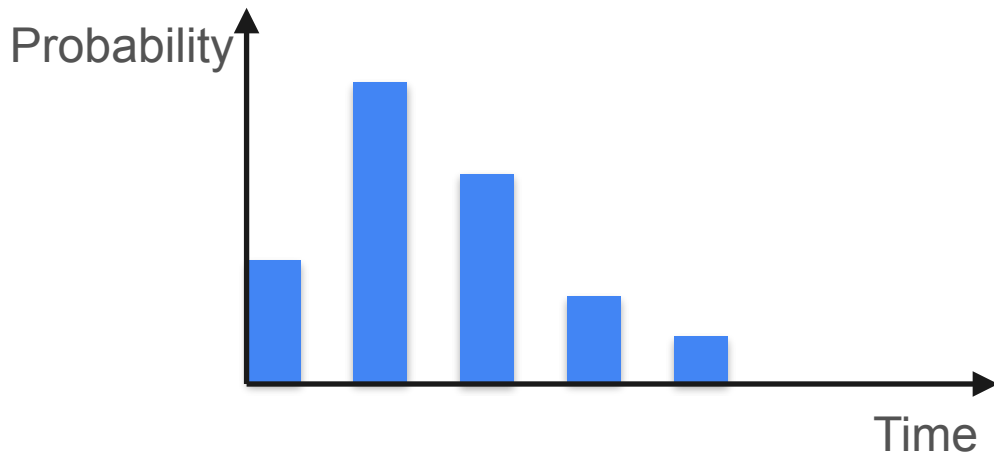
$P(\text{between 1 and 2 mins})$

$P(\text{between 2 and 3 mins})$

$P(\text{between 3 and 4 mins})$

$P(\text{between 4 and 5 mins})$

# From Discrete to Continuous Distributions



$P(\text{between 0 and 1 mins})$

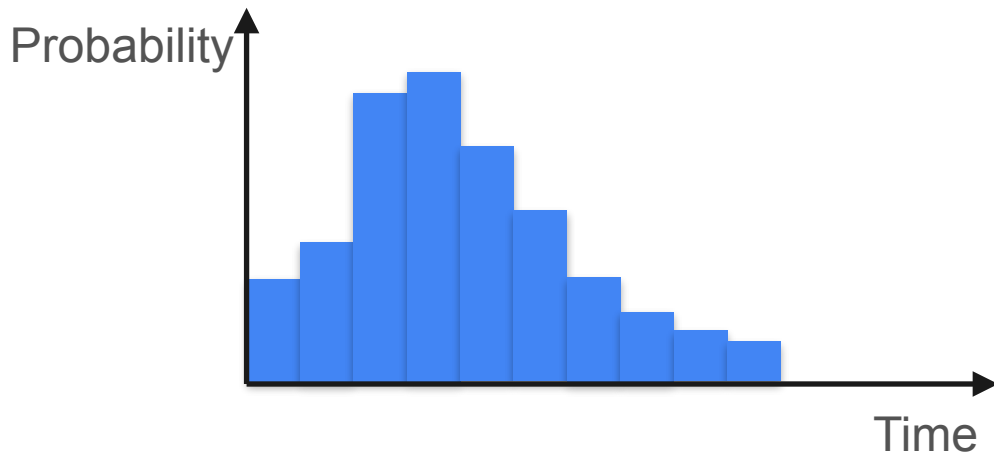
$P(\text{between 1 and 2 mins})$

$P(\text{between 2 and 3 mins})$

$P(\text{between 3 and 4 mins})$

$P(\text{between 4 and 5 mins})$

# From Discrete to Continuous Distributions



$P(\text{between 0 and 0.5 mins})$

$P(\text{between 0.5 and 1 mins})$

$P(\text{between 1 and 1.5 mins})$

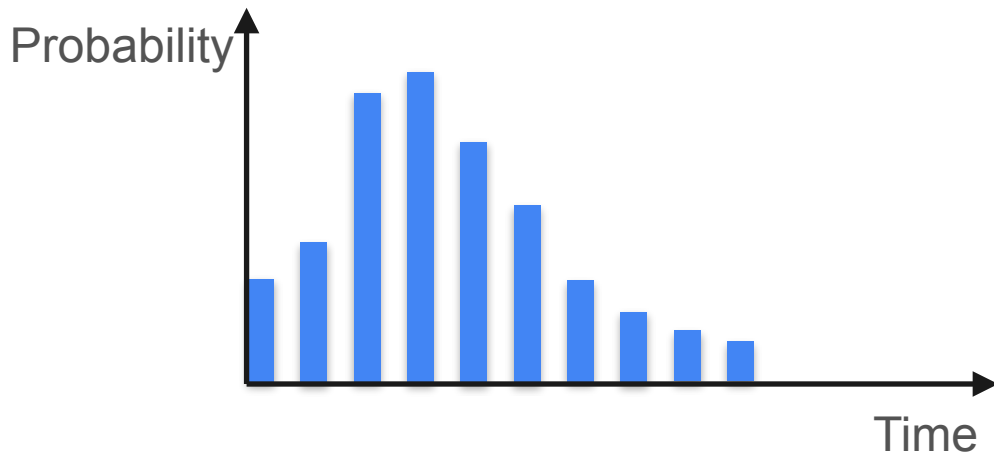
$\vdots$

$P(\text{between 3.5 and 4 mins})$

$P(\text{between 4 and 4.5 mins})$

$P(\text{between 4.5 and 5 mins})$

# From Discrete to Continuous Distributions



$P(\text{between 0 and 0.5 mins})$

$P(\text{between 0.5 and 1 mins})$

$P(\text{between 1 and 1.5 mins})$

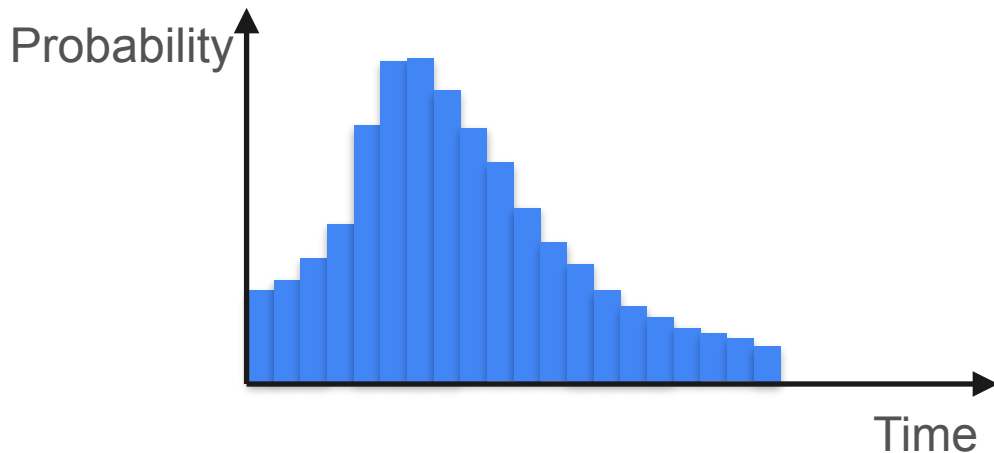
$\vdots$

$P(\text{between 3.5 and 4 mins})$

$P(\text{between 4 and 4.5 mins})$

$P(\text{between 4.5 and 5 mins})$

# From Discrete to Continuous Distributions



$P(\text{between } 0 \text{ and } 0.25 \text{ mins})$

$P(\text{between } 0.25 \text{ and } 0.5 \text{ mins})$

$P(\text{between } 0.5 \text{ and } 0.75 \text{ mins})$

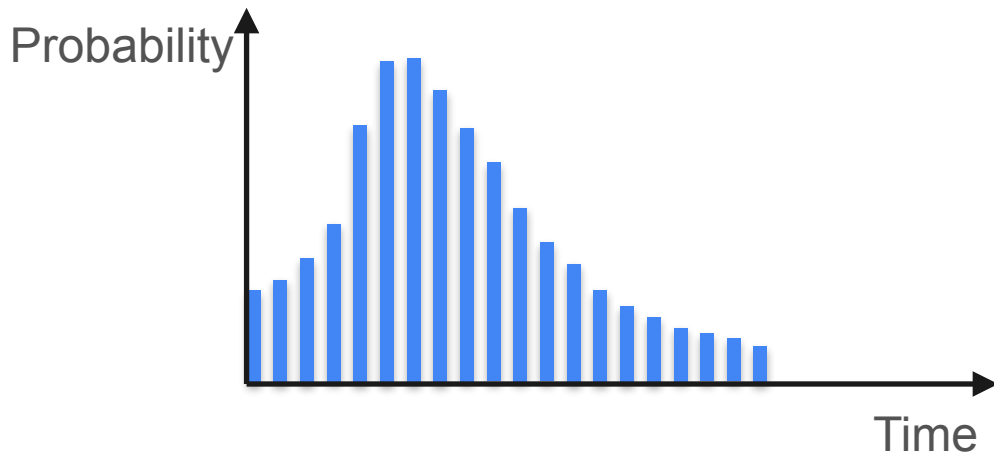
$\vdots$

$P(\text{between } 4.25 \text{ and } 4.5 \text{ mins})$

$P(\text{between } 4.5 \text{ and } 4.75 \text{ mins})$

$P(\text{between } 4.75 \text{ and } 5 \text{ mins})$

# From Discrete to Continuous Distributions



$P(\text{between 0 and 0.25 mins})$

$P(\text{between 0.25 and 0.5 mins})$

$P(\text{between 0.5 and 0.75 mins})$

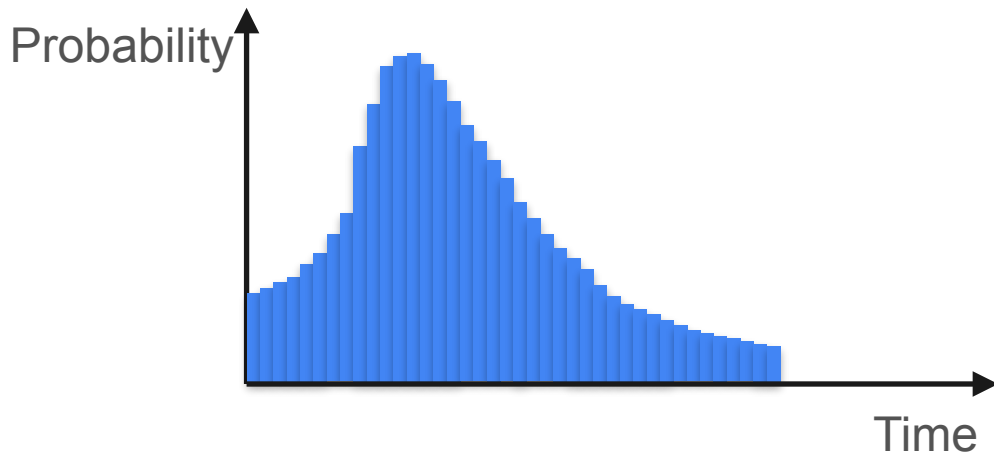
$\vdots$

$P(\text{between 4.25 and 4.5 mins})$

$P(\text{between 4.5 and 4.75 mins})$

$P(\text{between 4.75 and 5 mins})$

# From Discrete to Continuous Distributions



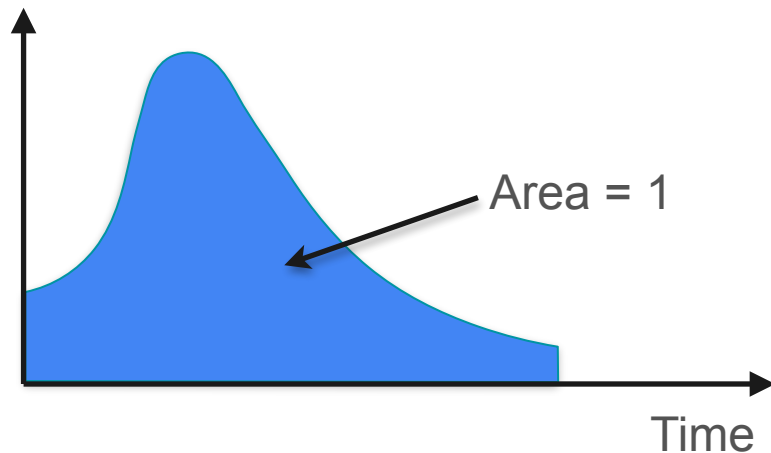
$P(\text{between } 0 \text{ and } 0.125 \text{ mins})$

$\vdots$

$P(\text{between } 4.875 \text{ and } 5 \text{ mins})$



# From Discrete to Continuous Distributions



- Discrete:
  - Sum of heights equals 1
- Continuous:
  - Area under the curve equals 1



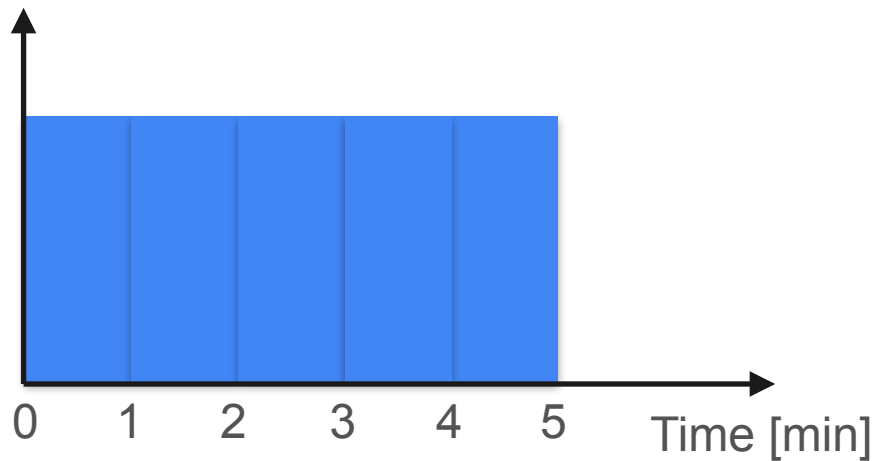
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# Probability Distributions

---

## **Probability density function**

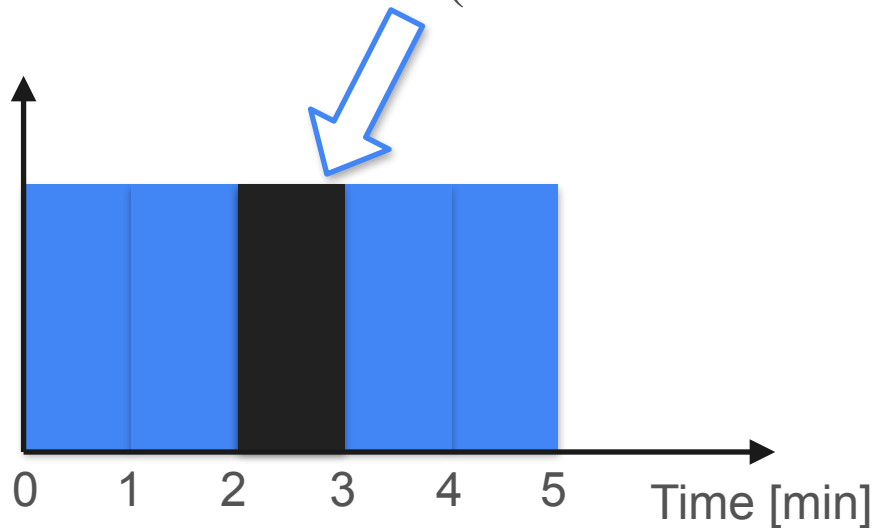
# Probability Density Function



# Probability Density Function



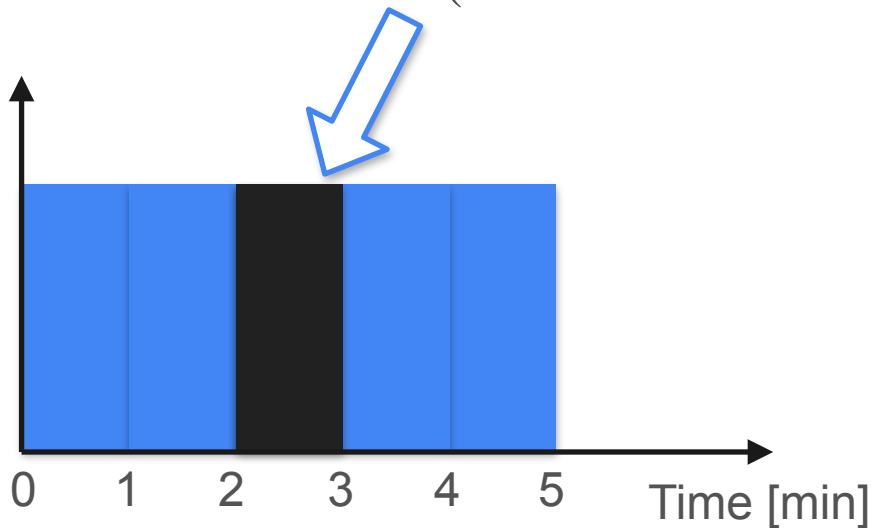
$P(\text{between 2 and 3 min}) = ?$



# Probability Density Function



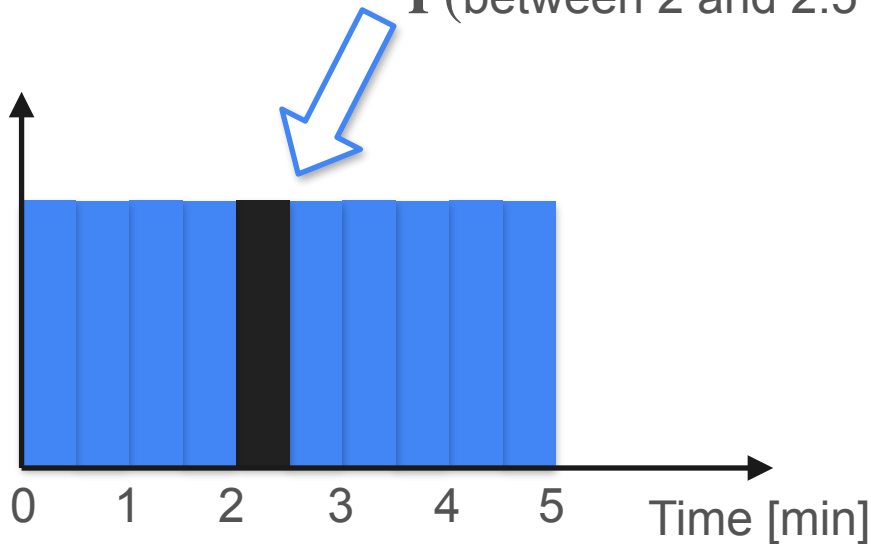
$$P(\text{between 2 and 3 min}) = 1/5$$



# Probability Density Function



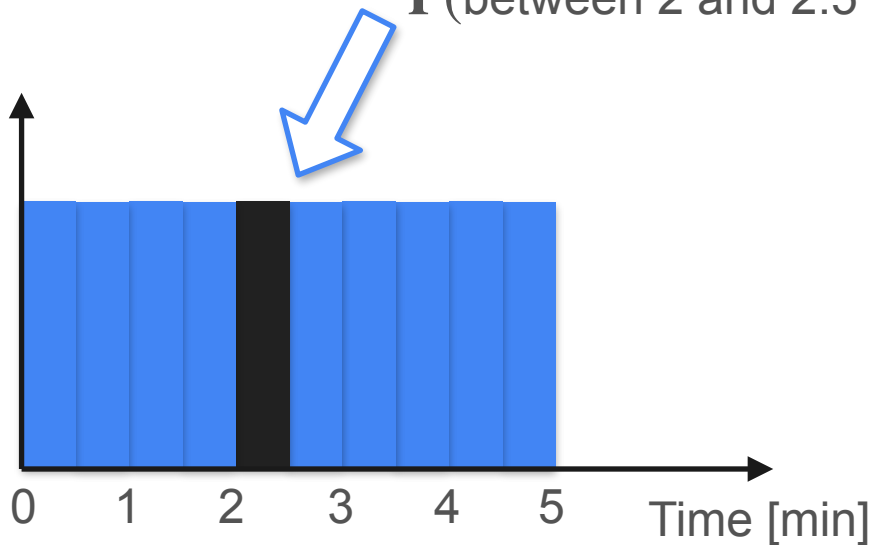
$P(\text{between 2 and 2.5 min}) = ?$



# Probability Density Function



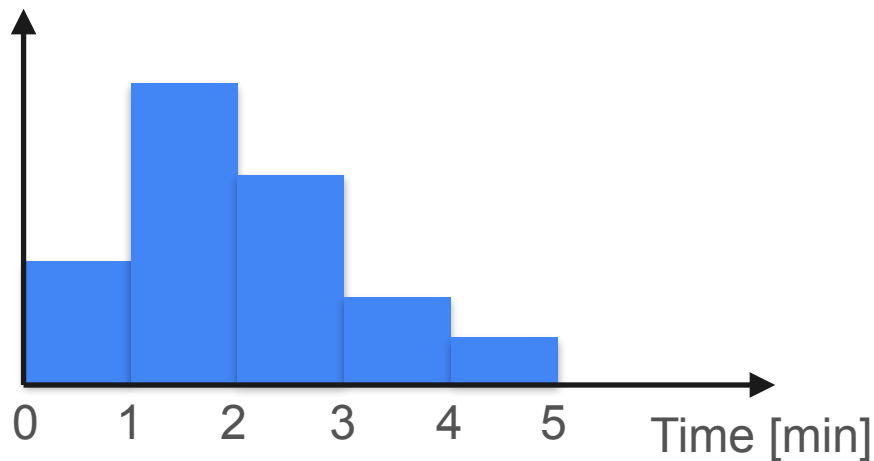
$$P(\text{between 2 and 2.5 min}) = 0.1$$



# Probability Density Function



$P(\text{between 1 and 2 min})$

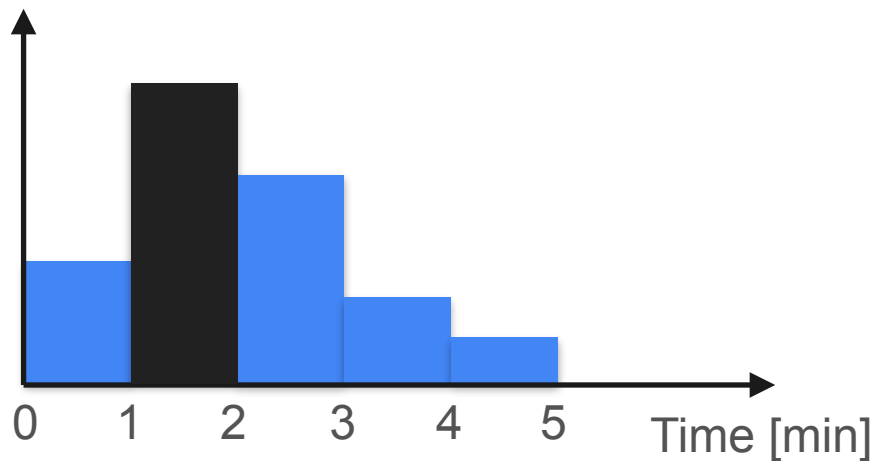




# Probability Density Function



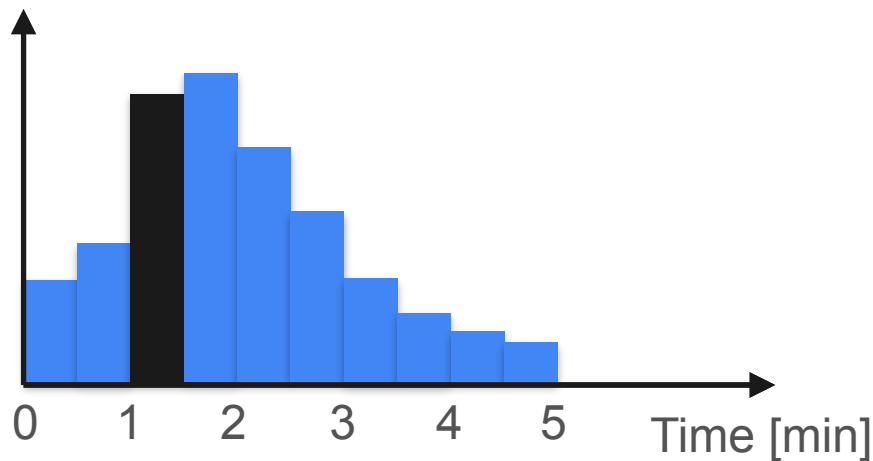
$P(\text{between 1 and 2 min})$



# Probability Density Function



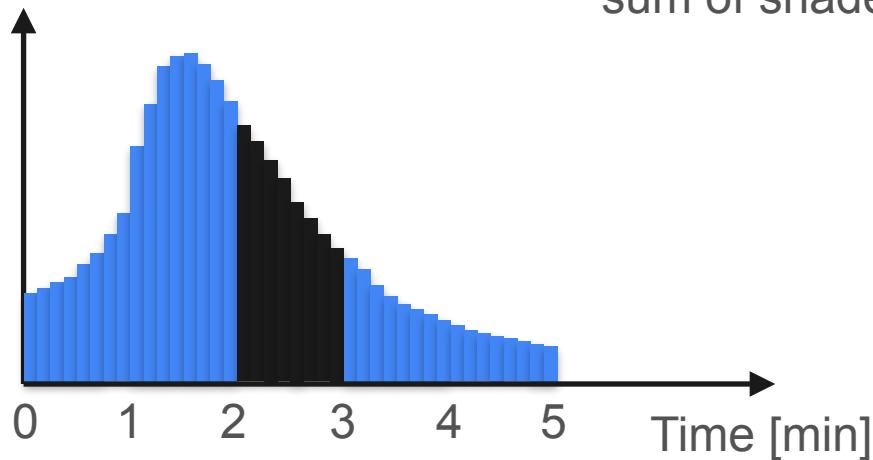
$P(\text{between 1 and 1:30})$



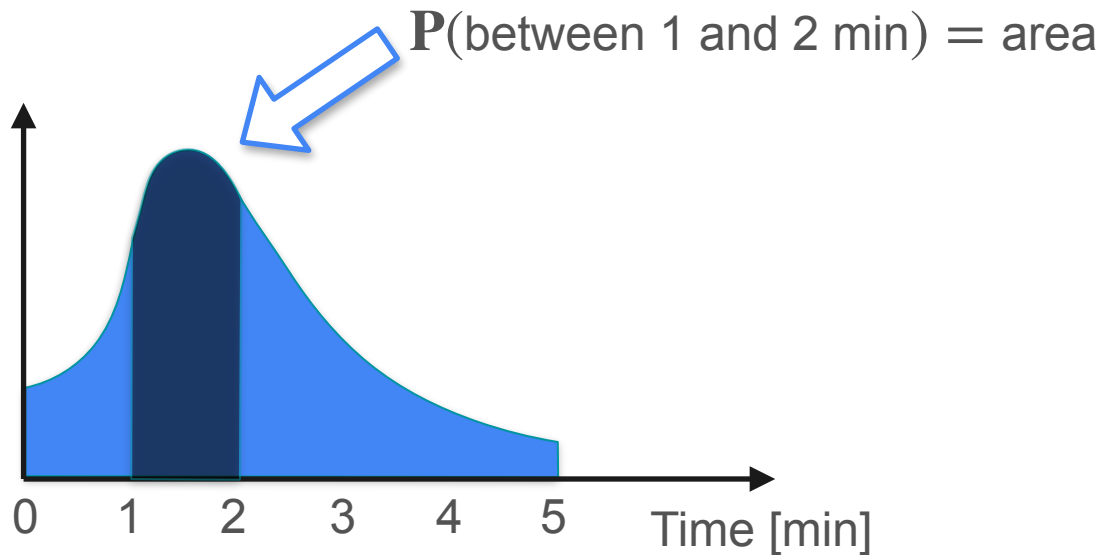
# Probability Density Function



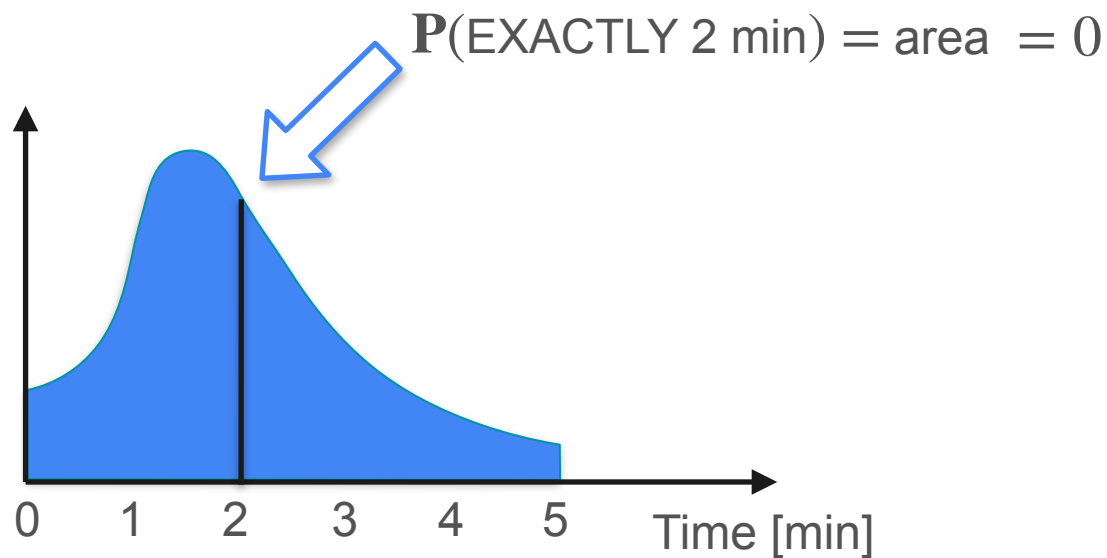
$P(\text{between 2 and 3 min}) =$   
sum of shaded areas



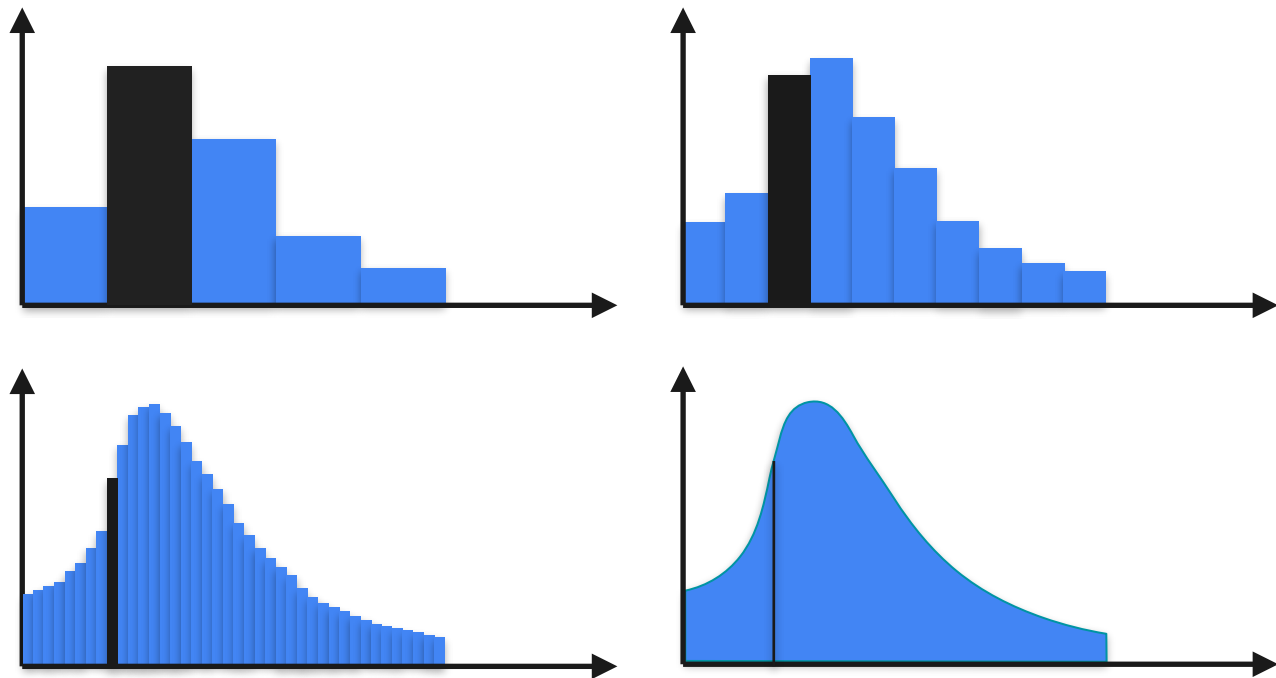
# Probability Density Function



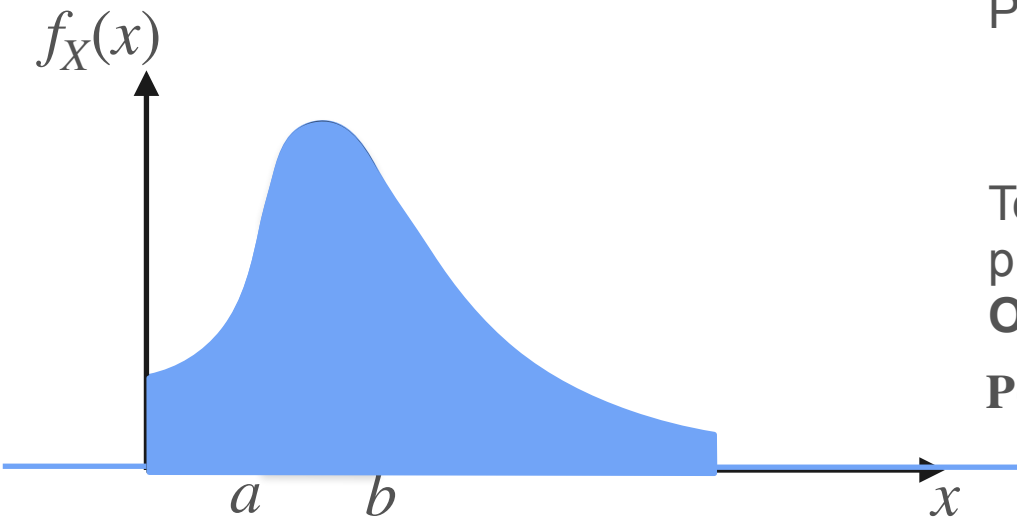
# Probability Density Function



# Probability Density Function



# Probability Density Function: Formal Definition



Probability Density Function (PDF)

$$f_X(x)$$

Tells you the rate you accumulate probability around each point.

**Only defined for continuous variables!**

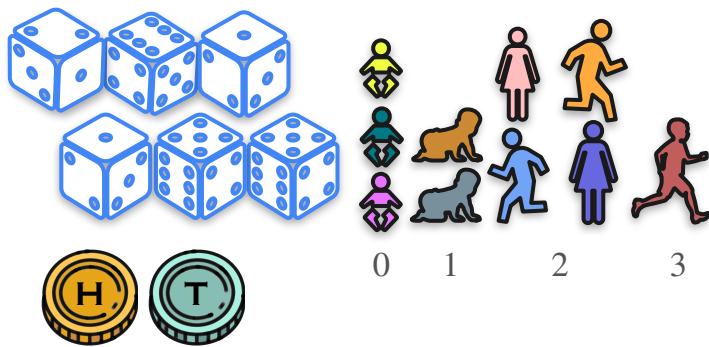
$P(a < X < b) = \text{area under } f_X(x)$

$f_X(x)$  needs to satisfy:

- It is defined for all numbers
- $f_X(x) \geq 0$
- Area under  $f_X(x) = 1$

# Discrete and Continuous Random Variables

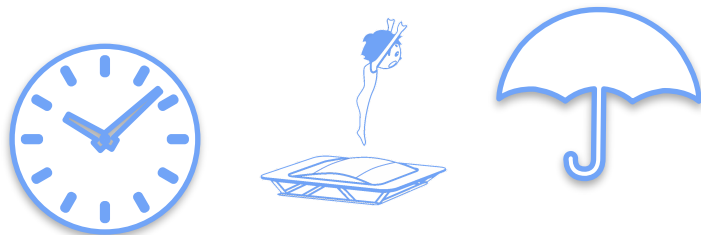
## Discrete random variables



Can take only a **finite** or at most countable number of values

$$\text{PMF: } p_X(x) = \mathbf{P}(X = x)$$

## Continuous random variables



Takes values on an interval (infinite possibilities!)

$$\begin{aligned} \text{PDF: } f_X(x) \\ \mathbf{P}(X = x) = 0 \quad \forall x \end{aligned}$$





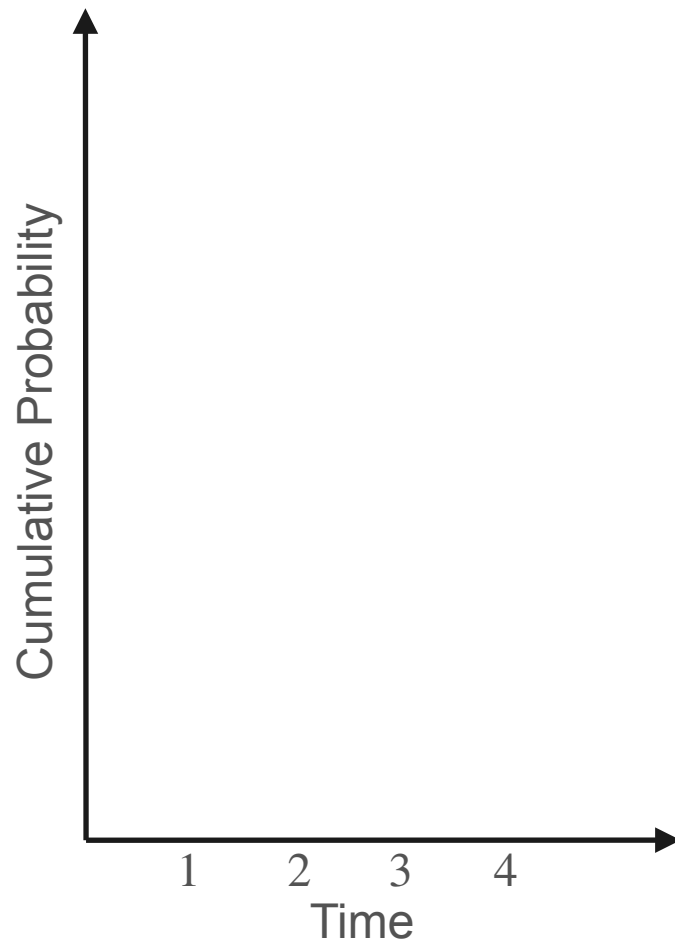
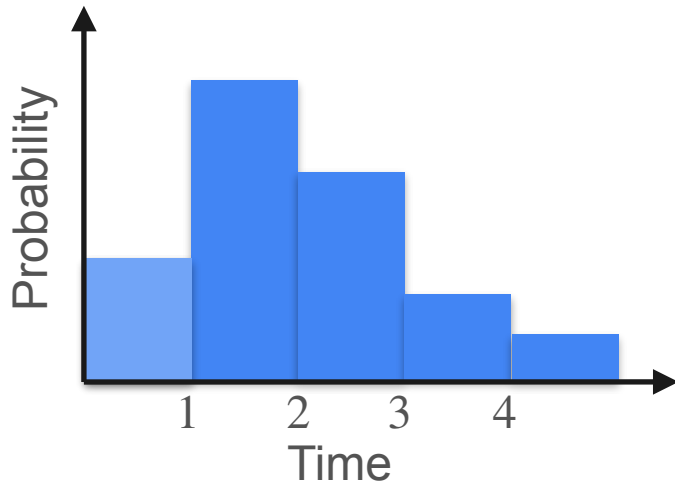
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# Probability Distributions

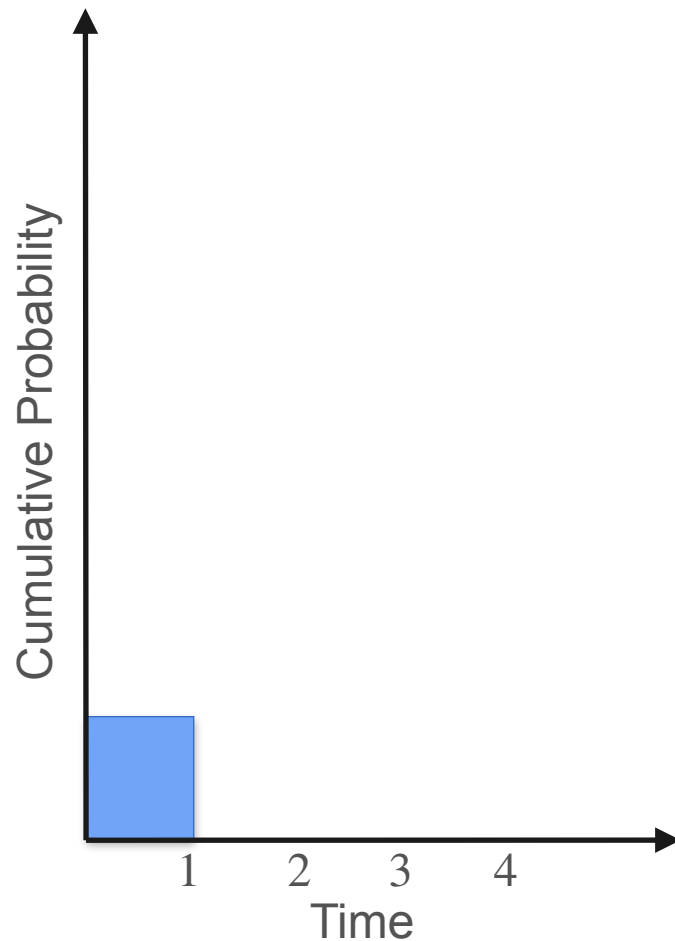
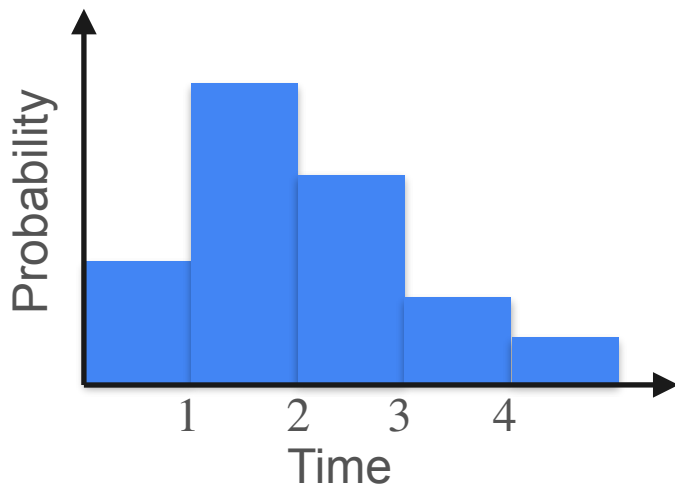
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## **Cumulative Distribution Function**

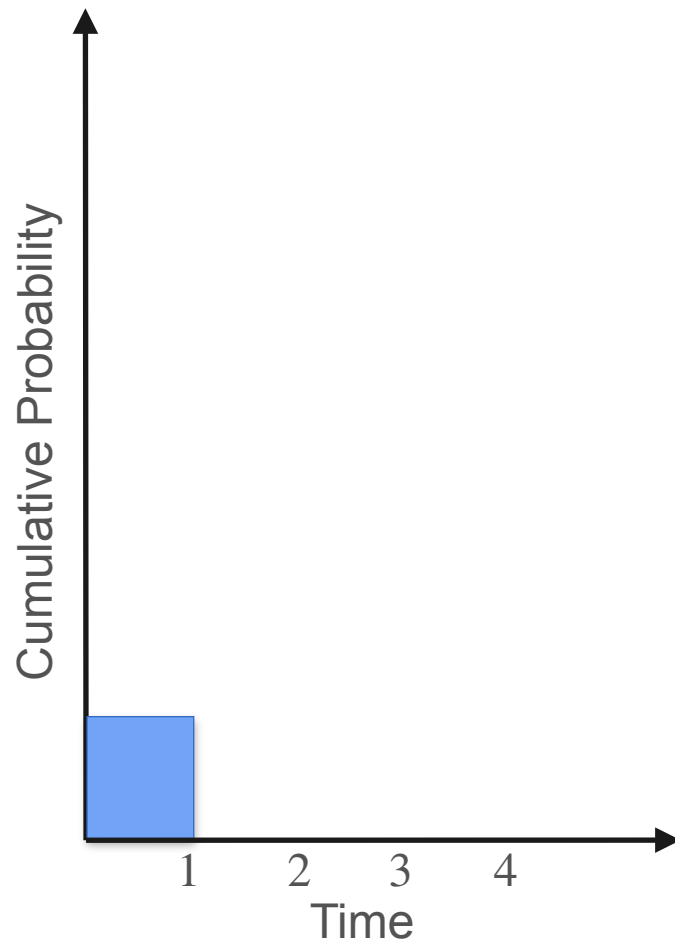
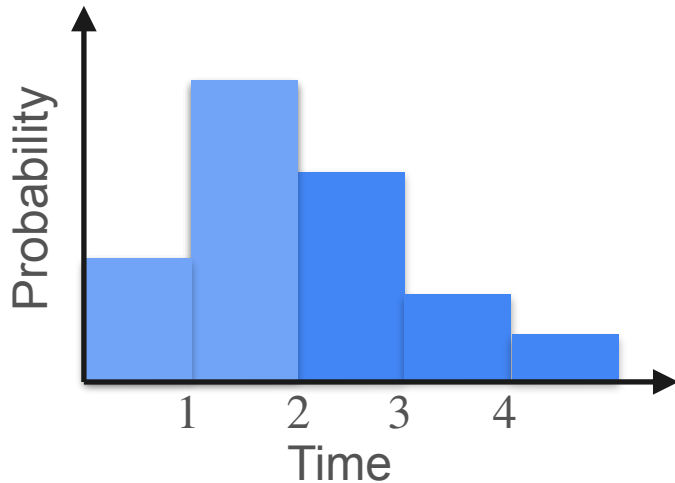
# Cumulative Distribution



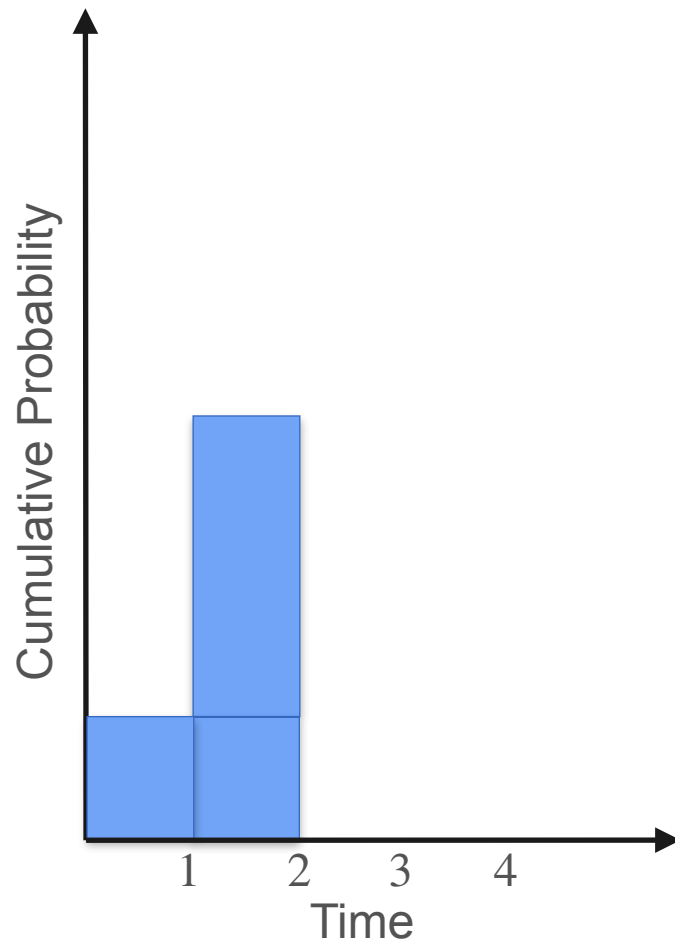
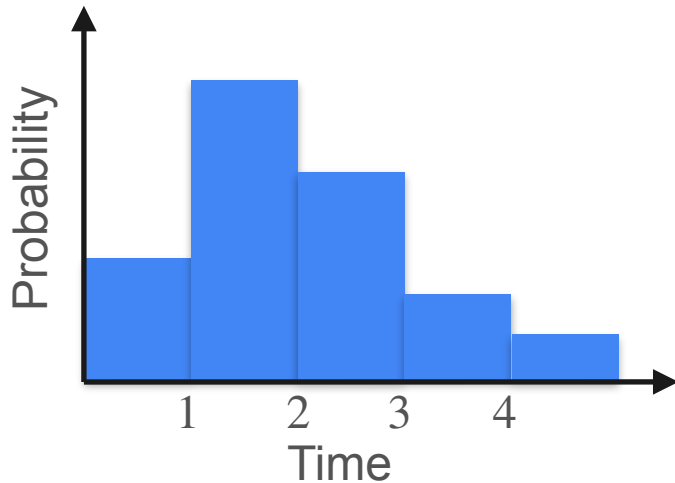
# Cumulative Distribution



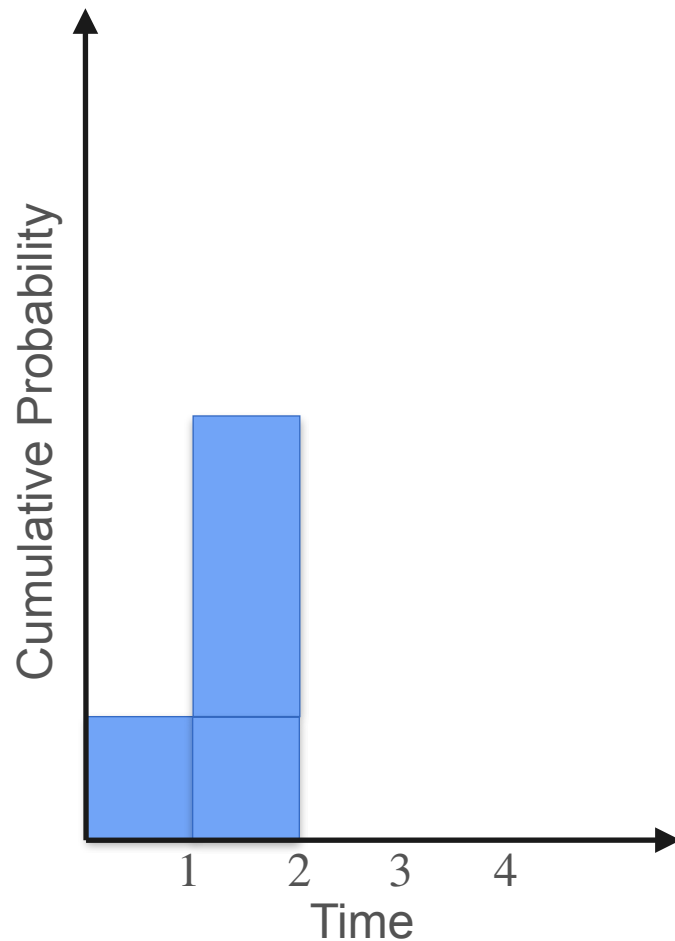
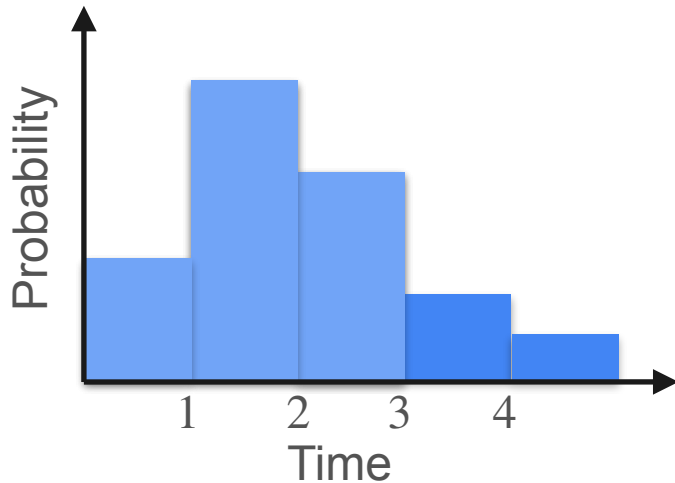
# Cumulative Distribution



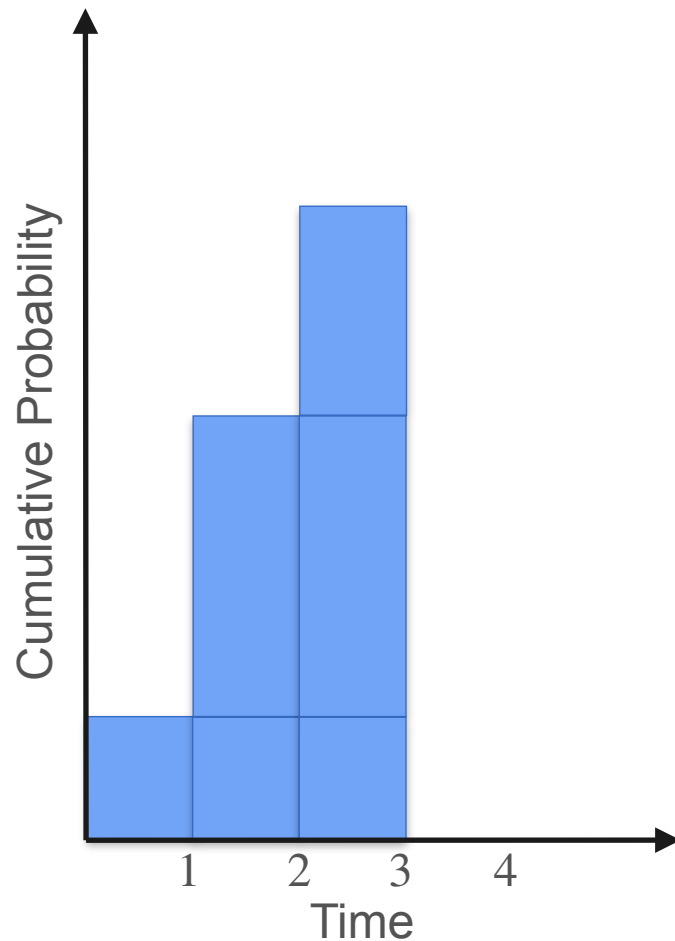
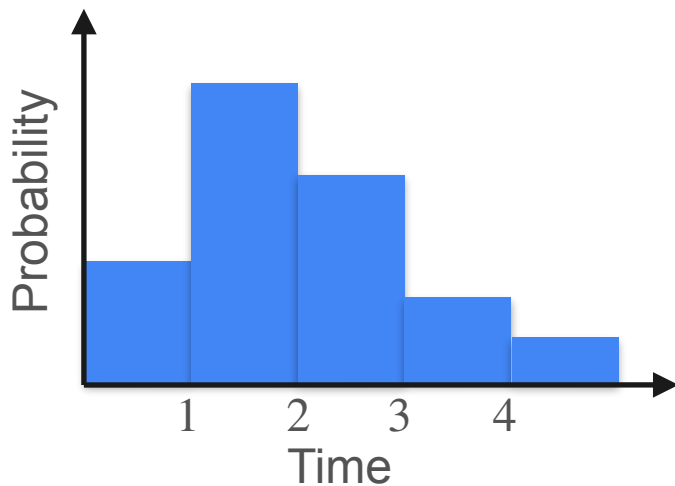
# Cumulative Distribution



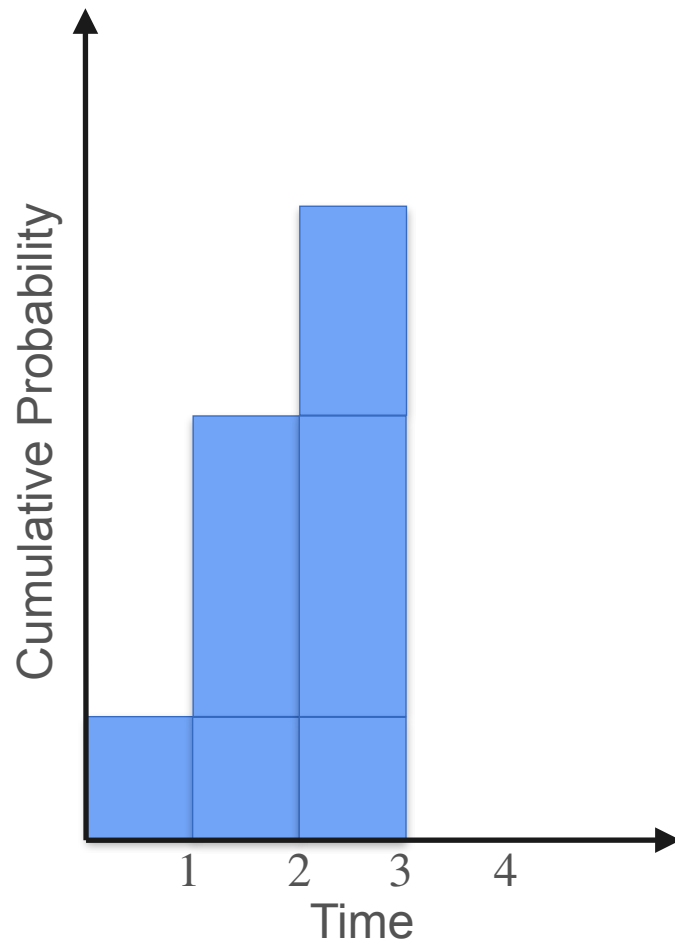
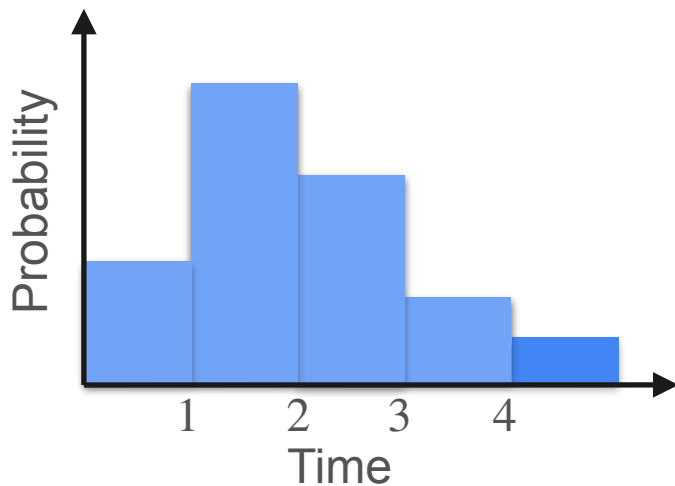
# Cumulative Distribution



# Cumulative Distribution

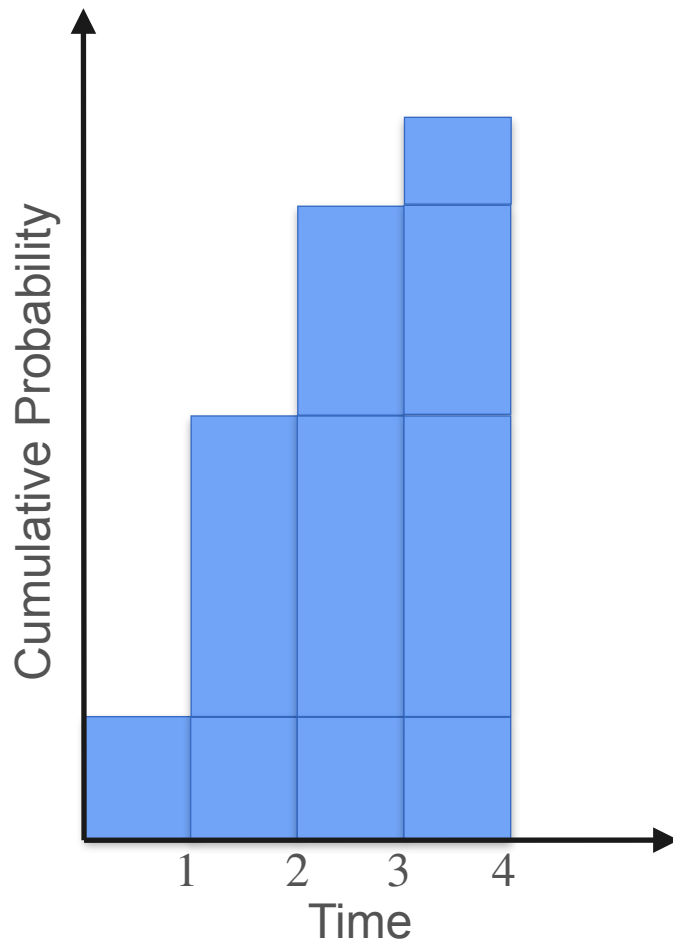
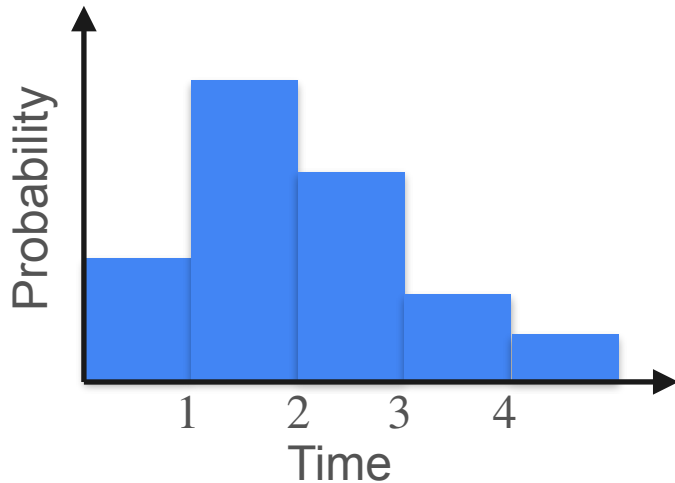


# Cumulative Distribution

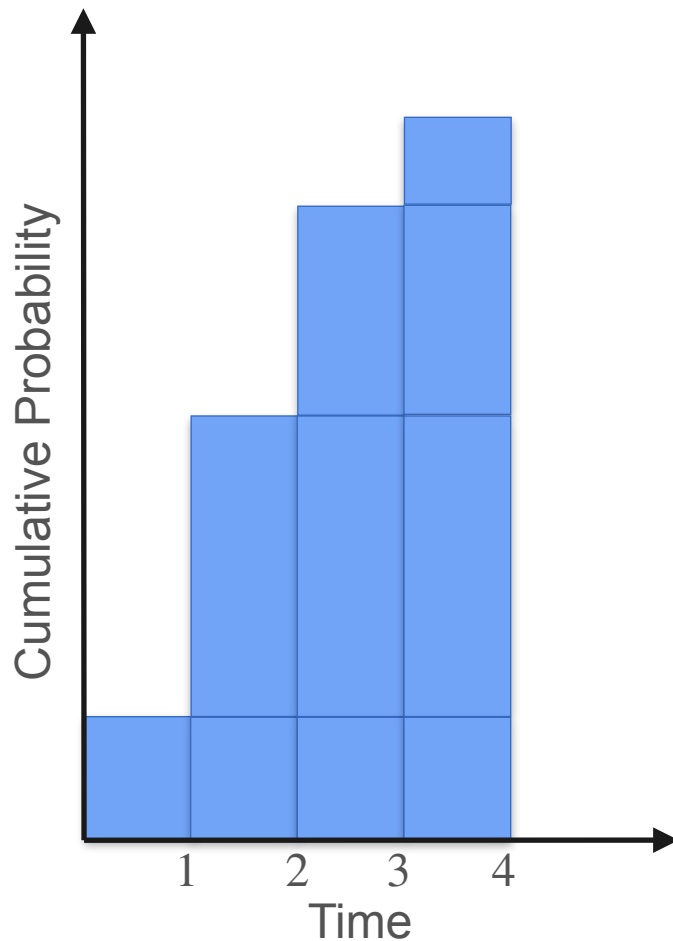
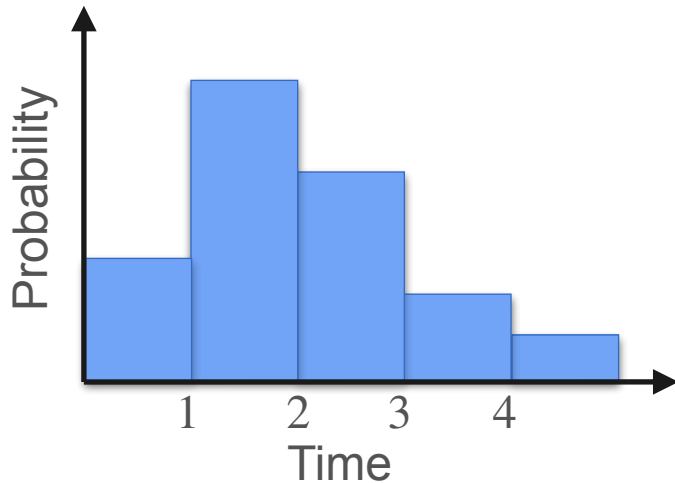




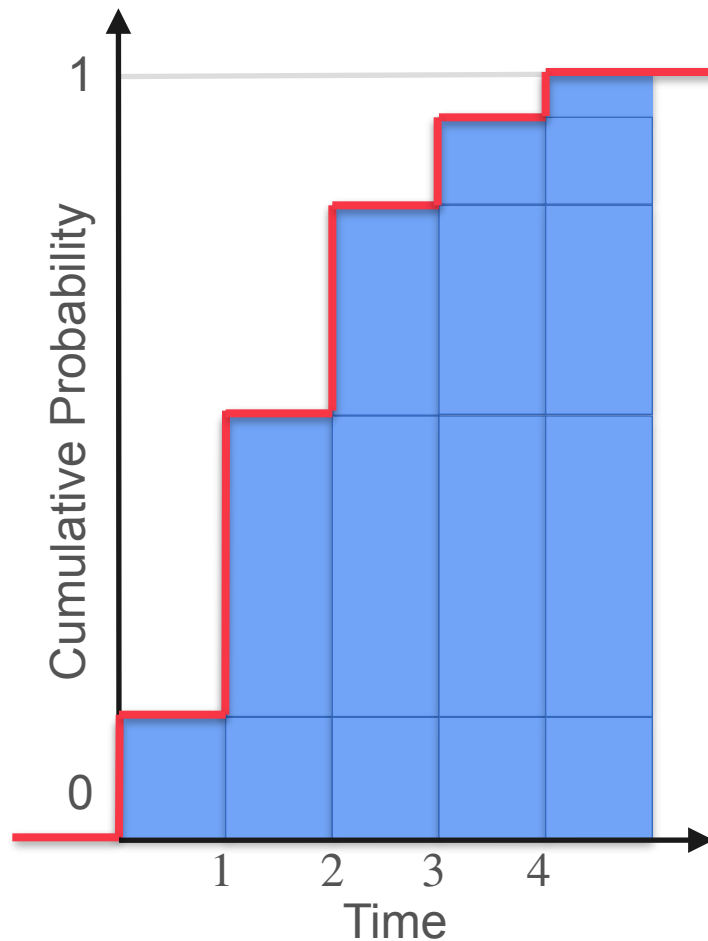
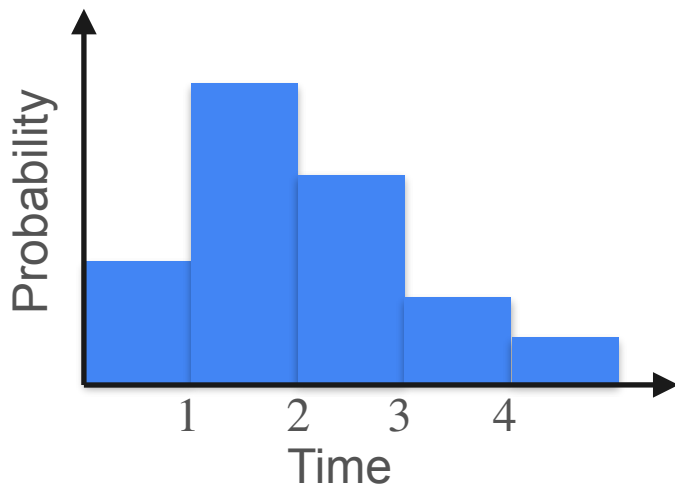
# Cumulative Distribution



# Cumulative Distribution

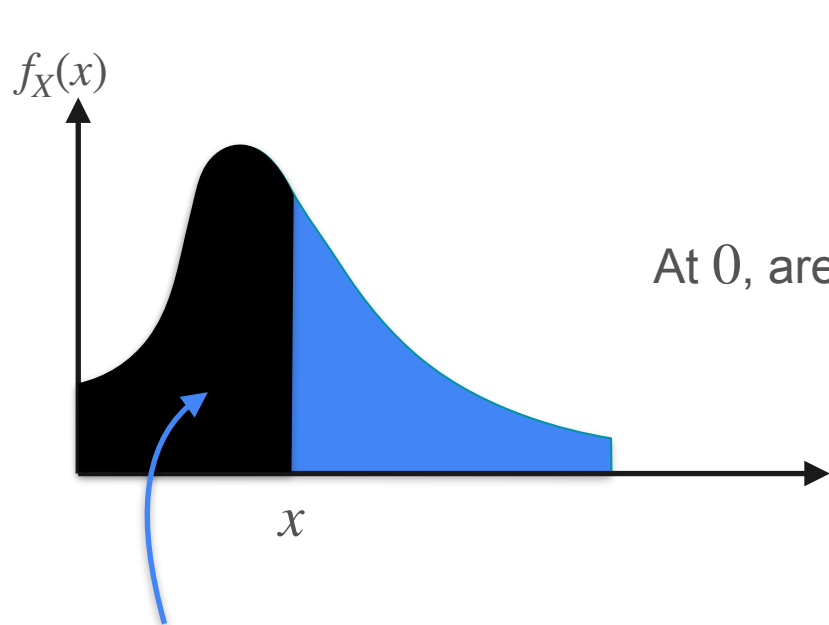


# Cumulative Distribution

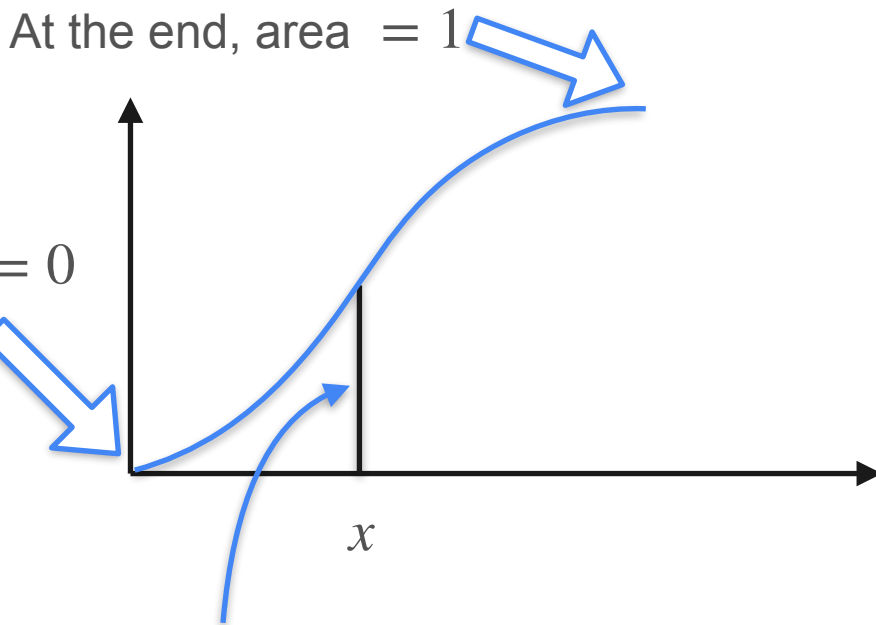


# Cumulative Distribution

CDF: Cumulative distribution function



At 0, area = 0



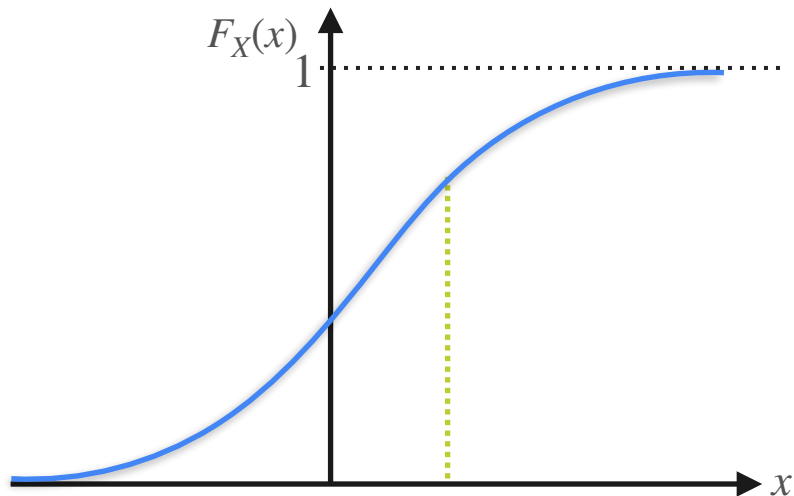
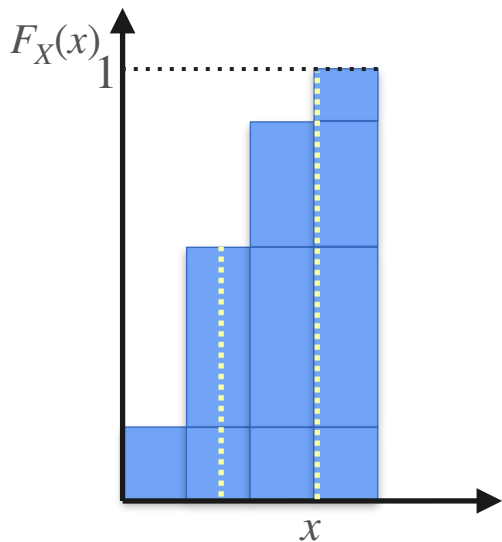
$P(\text{less than or equal to 2 minutes}) = 0.5$

$P(\text{less than or equal to 2 minutes}) = 0.5$

# Cumulative Distribution Function: Formal Definition

The CDF shows how much probability the variable has accumulated until a certain value

That means that  $F_X(x) = \mathbf{P}(X \leq x)$  ← It is defined for every real number

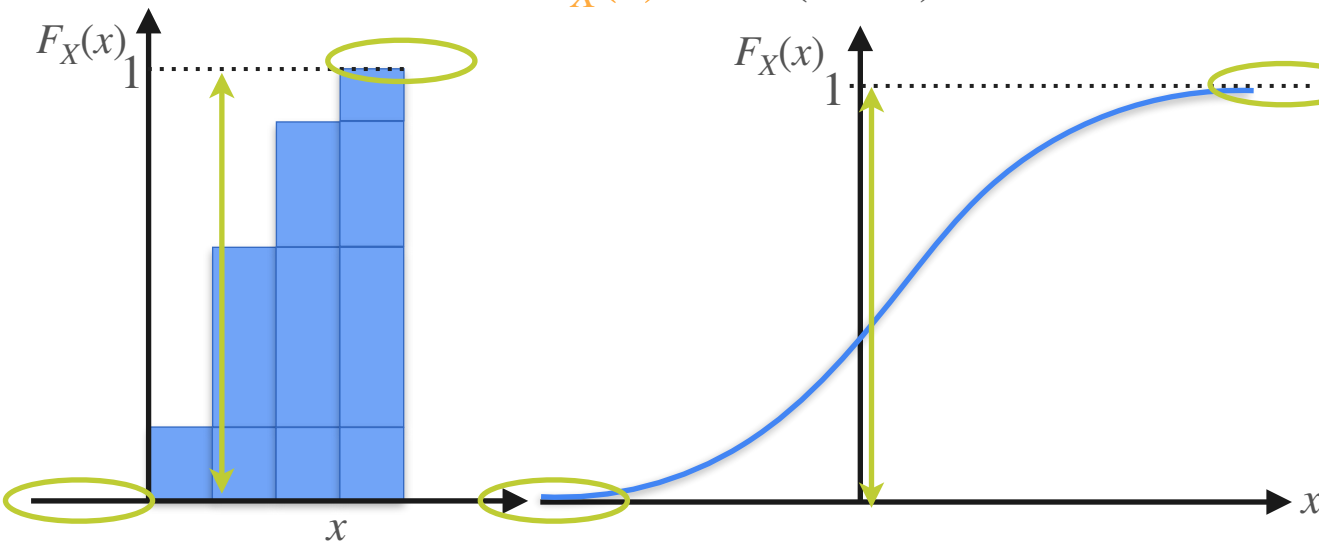


# Cumulative Distribution Function: Formal Definition

The CDF shows how much probability the variable has accumulated until a certain value

That means that

$$F_X(x) = \mathbf{P}(X \leq x) \leftarrow \text{It is defined for every real number}$$

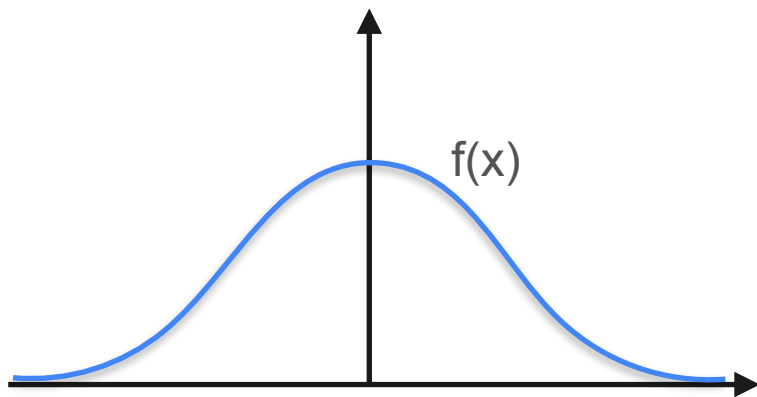


## Properties

- $0 \leq F_X(x) \leq 1$
- Left “endpoint” is 0
- Right “endpoint” is 1
- **Never decreases**

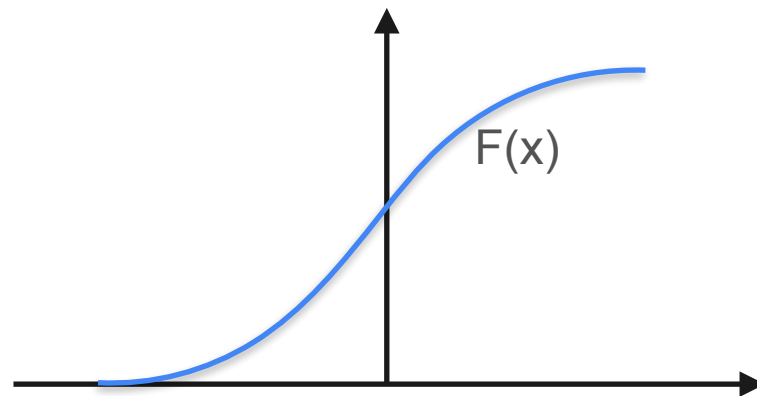
# PDF and CDF Summary

PDF



- area = 1
- Always positive

CDF



- left “endpoint” is 0
- right “endpoint” is 1
- (endpoints can be at infinity)
- Always positive and increasing



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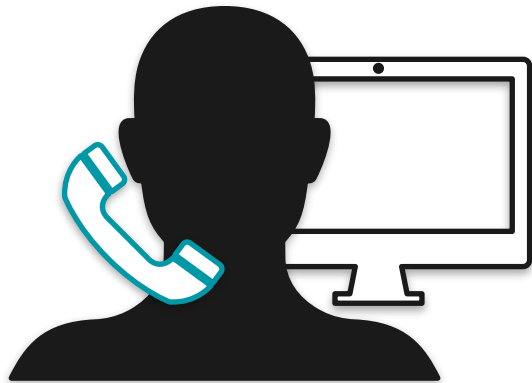
# Probability Distributions

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## Uniform Distribution

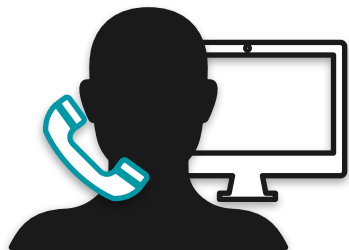


# Uniform Distribution: Motivation

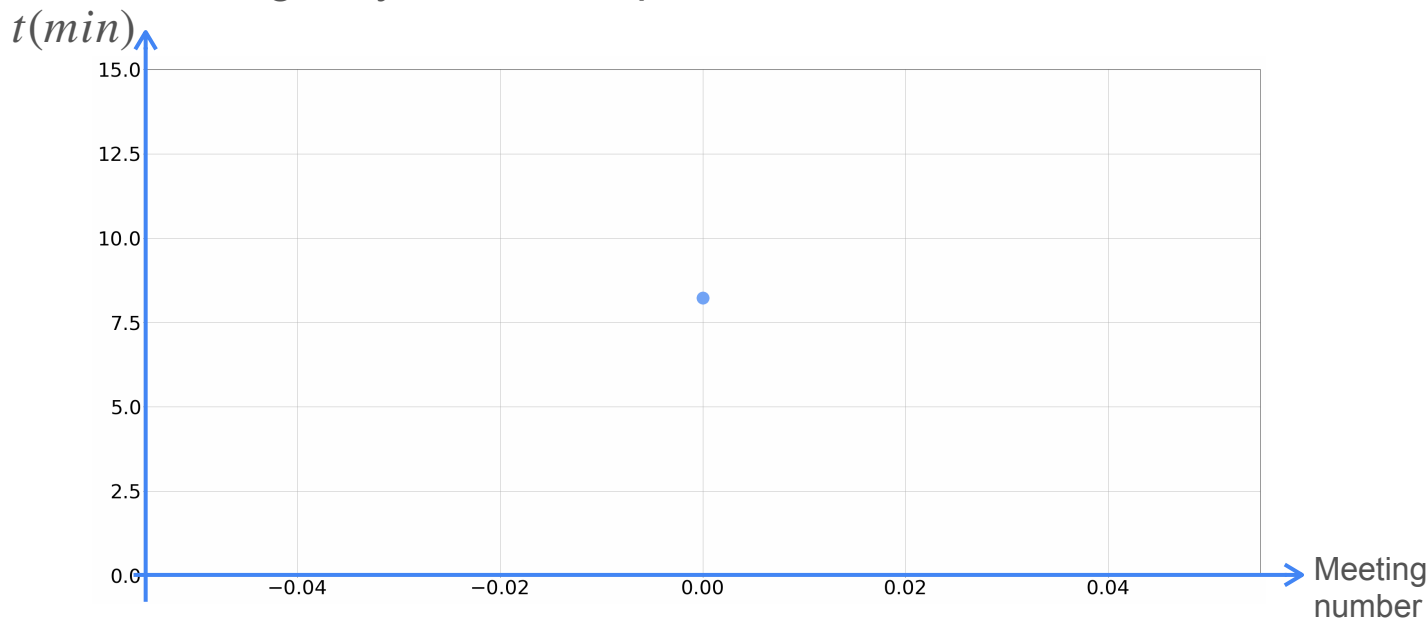


You're calling a tech support line.  
They can answer any time between zero and 15 minutes and if they don't answer in this time, the line is disconnected.

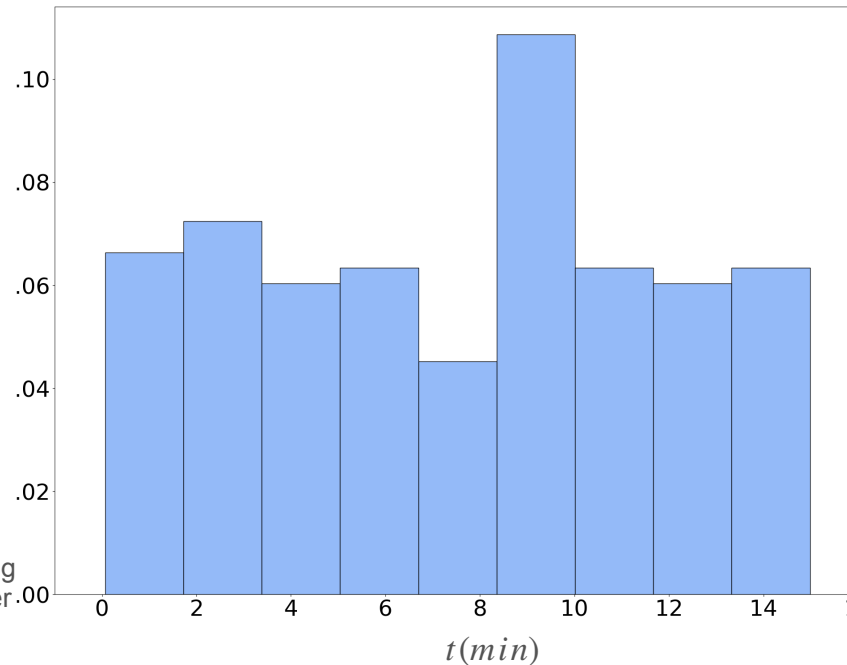
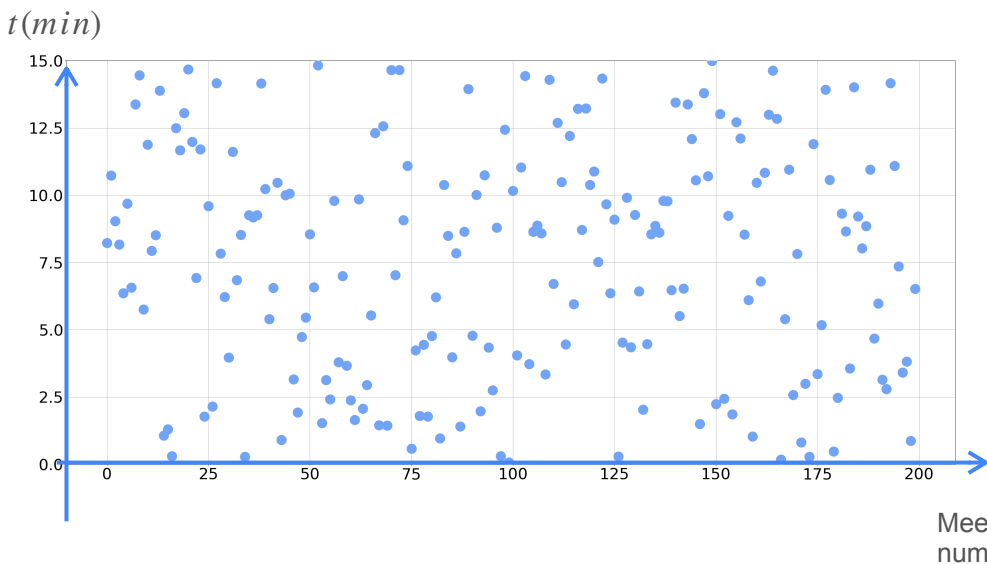
# Uniform Distribution: Motivation



Last 200 times you called them, you took down notes of how long they took to respond



# Uniform Distribution: Motivation



# Uniform Distribution: Motivation

T: time (in minutes) you have to wait

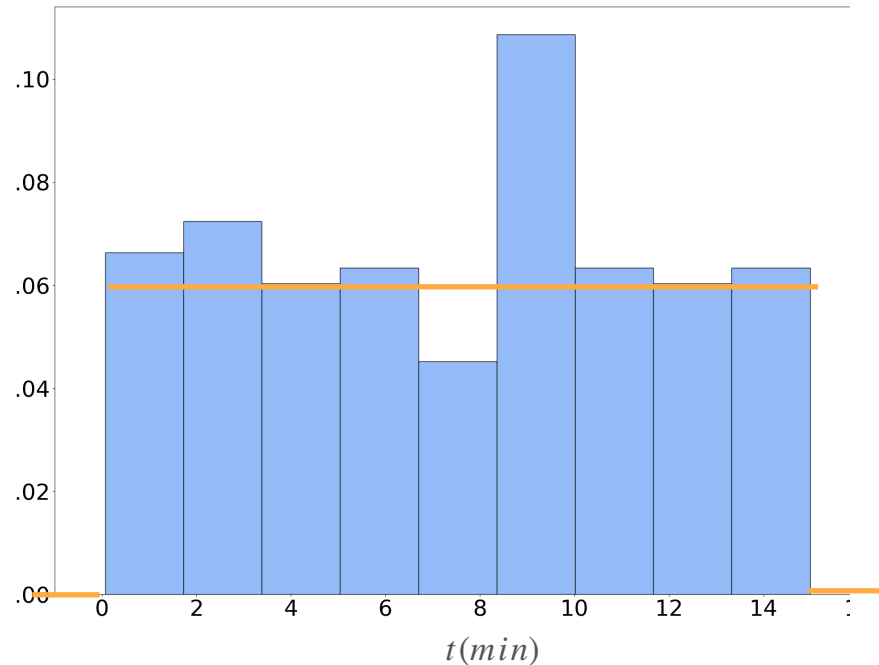
Any value between 0 and 15 minutes must have the same frequency of occurrence.



The pdf must be constant for all values in the interval (0,15)



Which constant?  $\rightarrow 15 \times h = 1 \rightarrow h = \frac{1}{15} = 0.06$



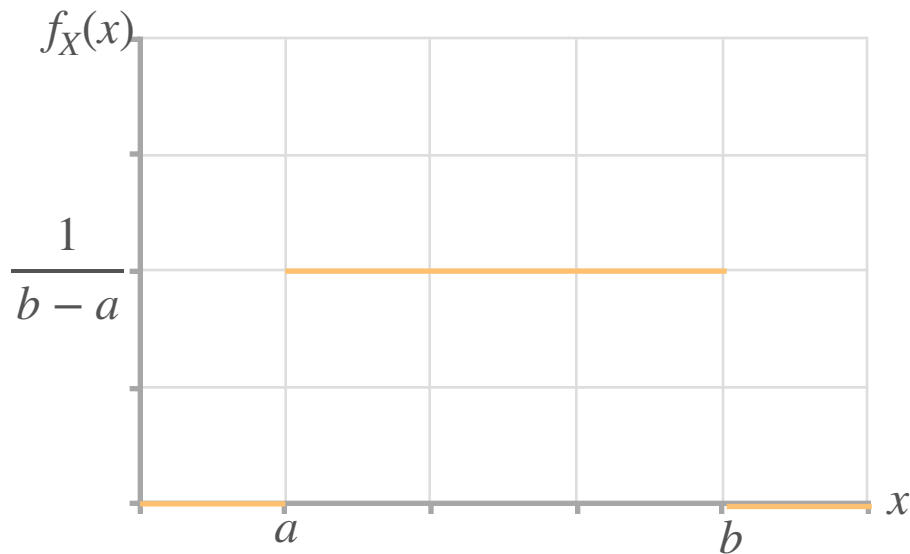
# Uniform Distribution: Model

A continuous random variable can be modeled with a **uniform** distribution if all possible values lie in an interval and have the **same frequency** of occurrence

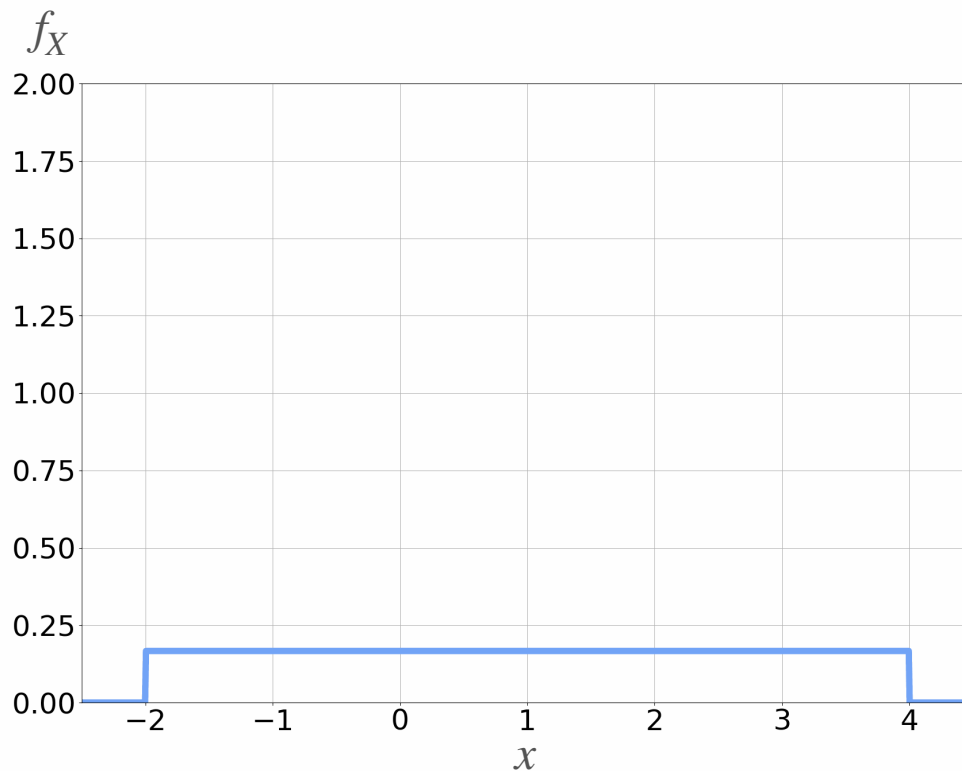
Parameters:

- $a$ : beginning of the interval
- $b$ : end of the interval

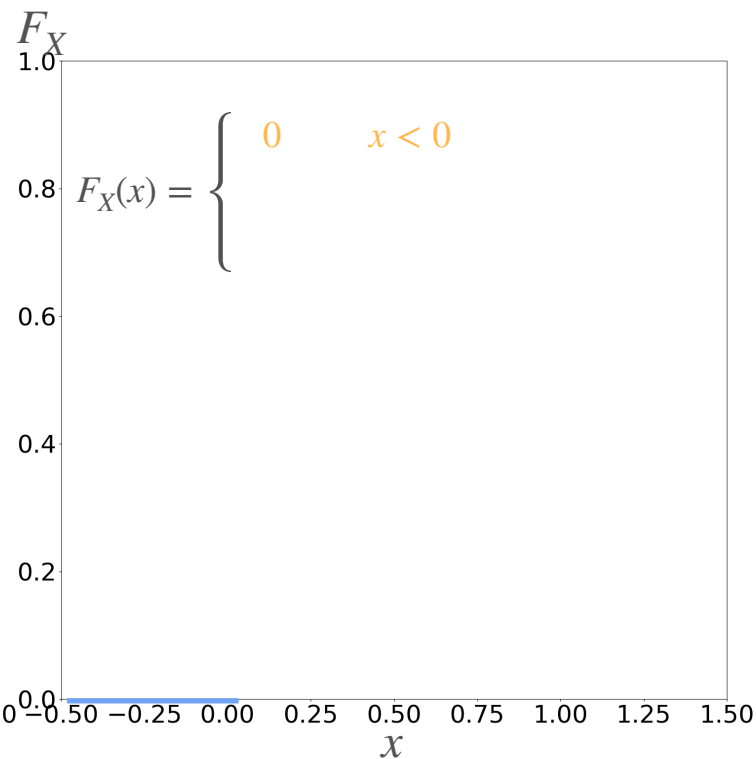
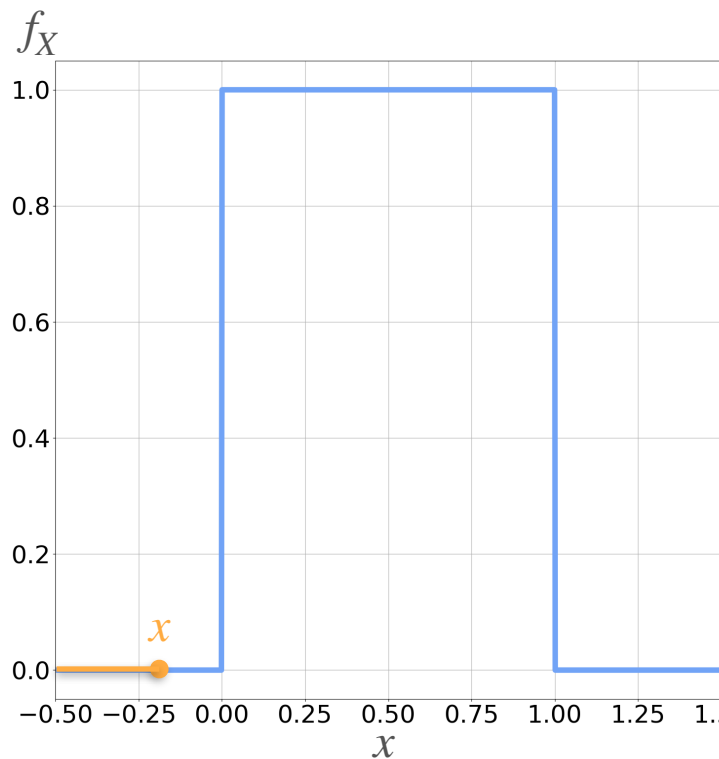
$$f_X(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & x \notin (a, b) \end{cases}$$



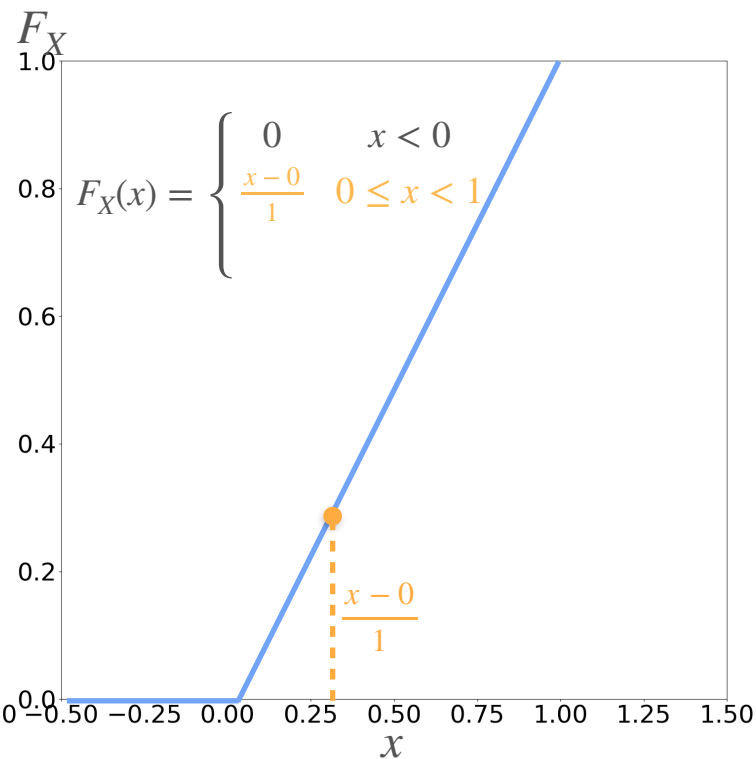
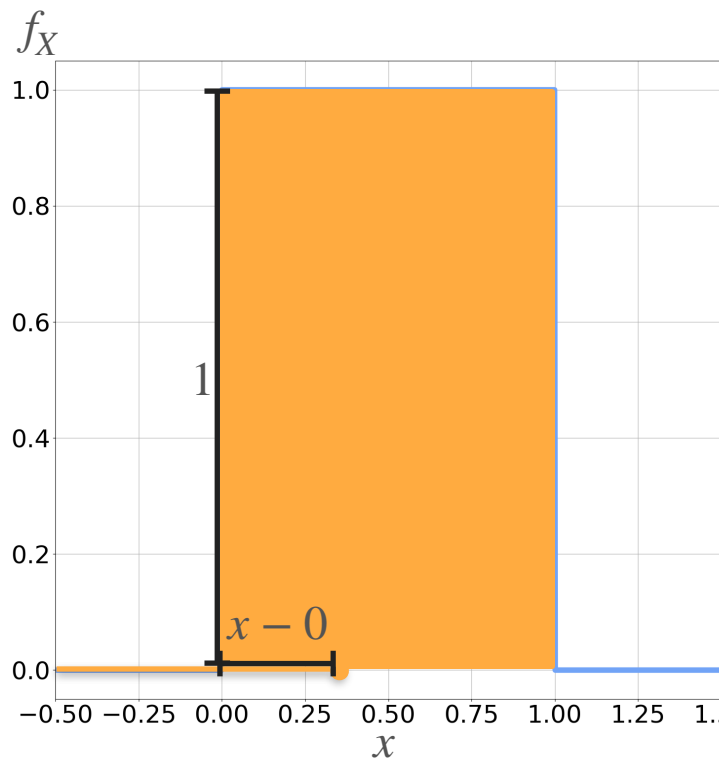
# Uniform Distribution: PDF



# Uniform Distribution: CDF

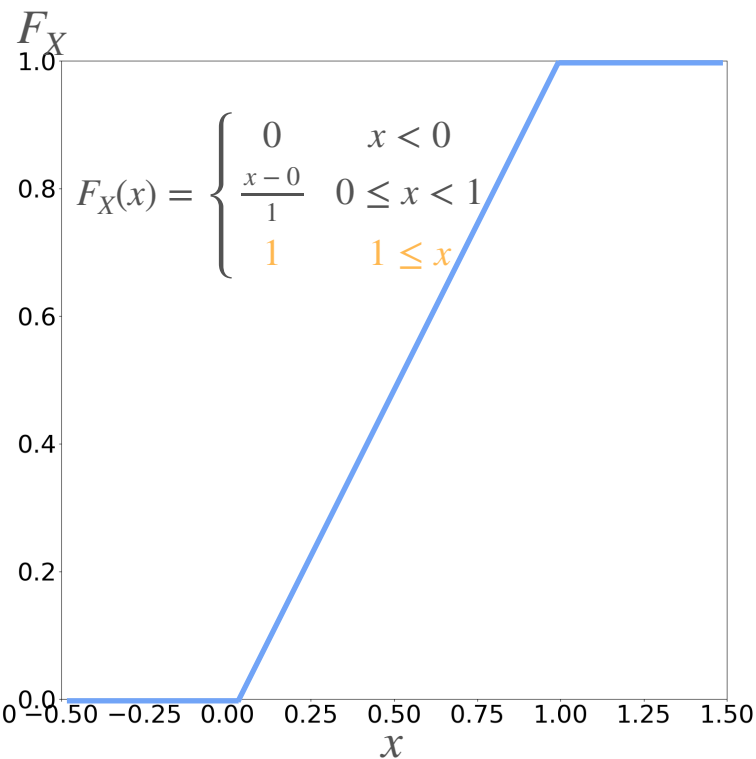
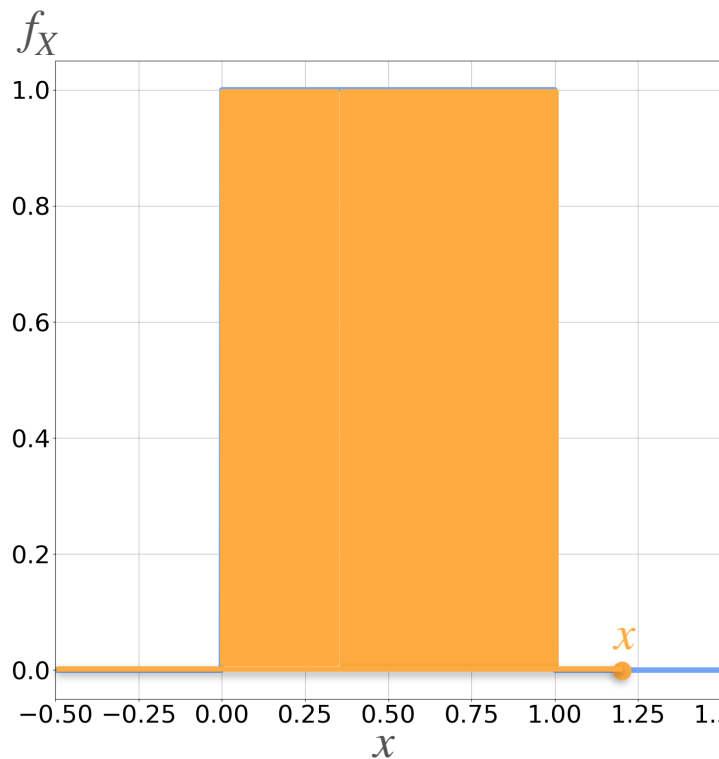


# Uniform Distribution: CDF

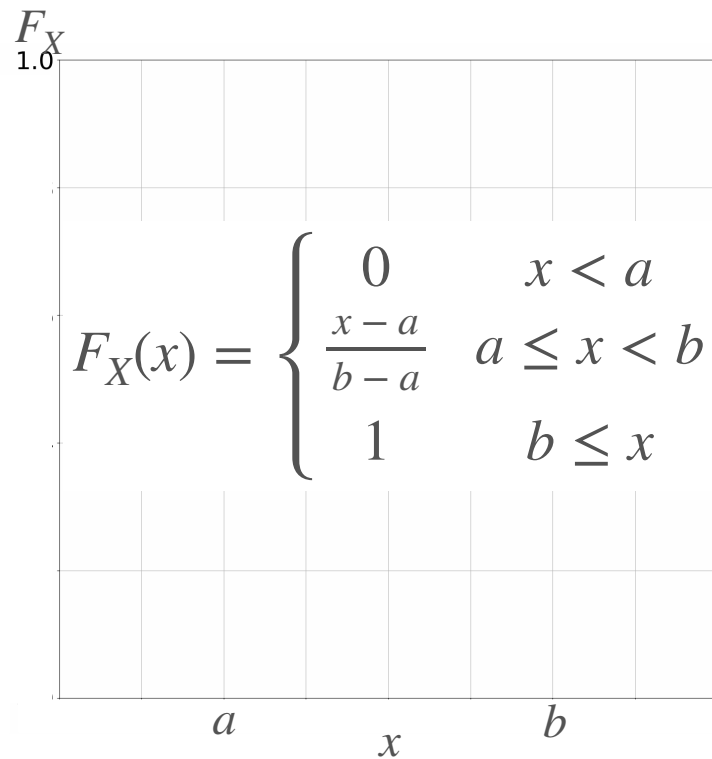
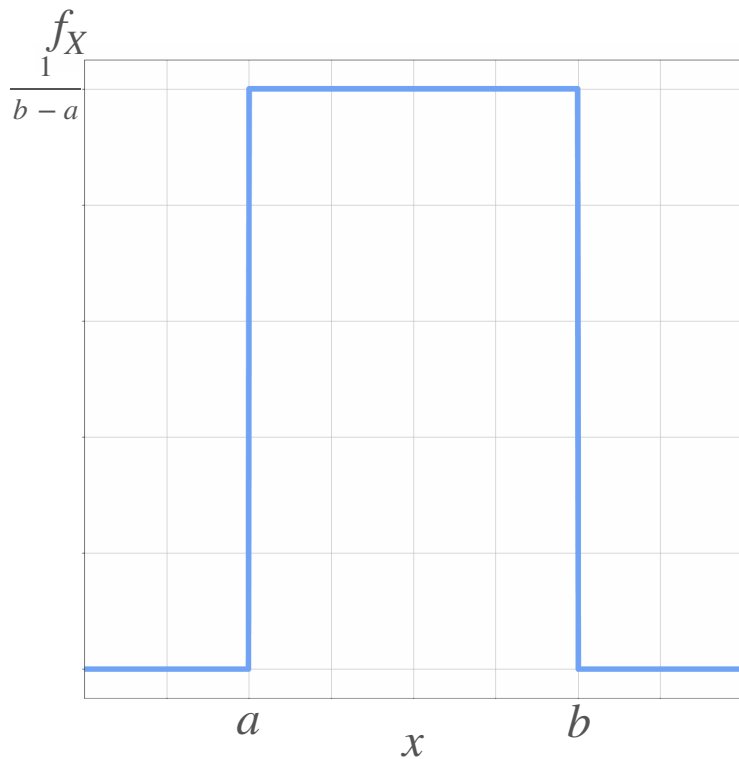




# Uniform Distribution: CDF



# Uniform Distribution: CDF





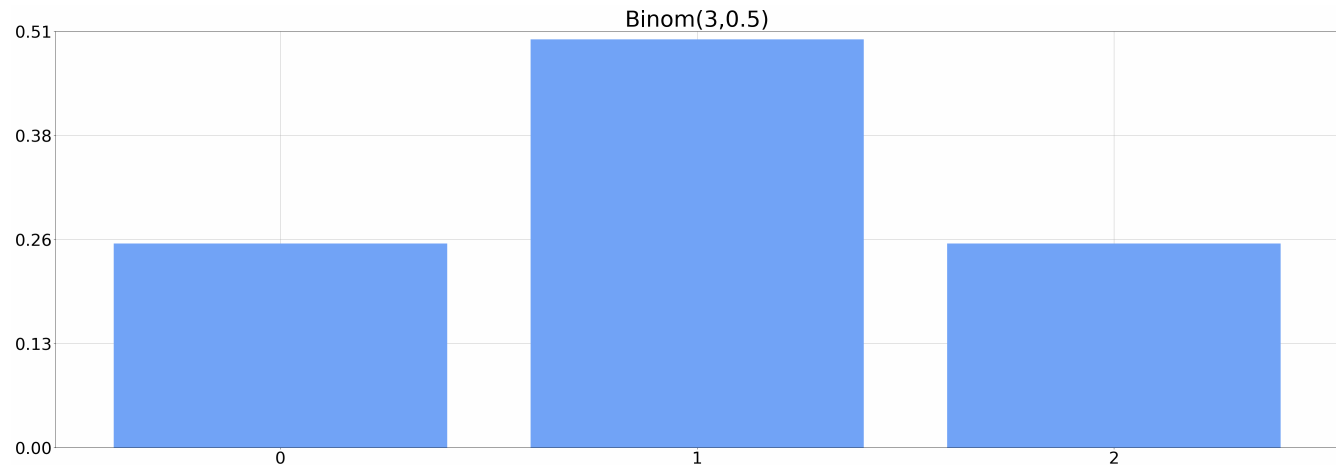
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# Probability Distributions

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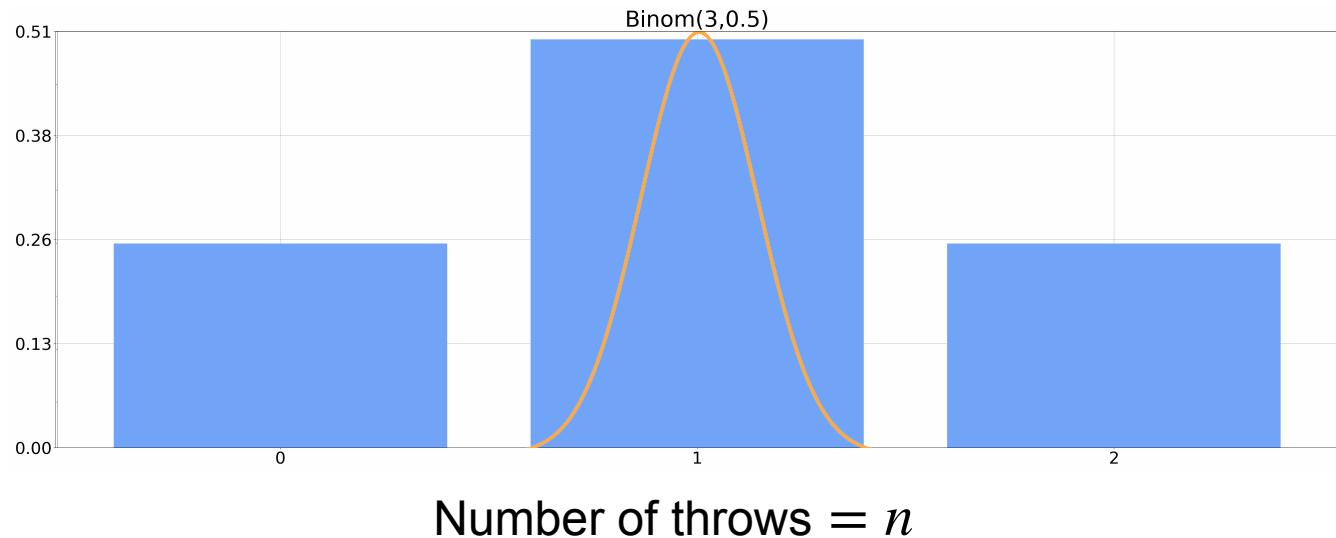
## **Normal distribution**

# Binomial Distribution With Very Large $n$

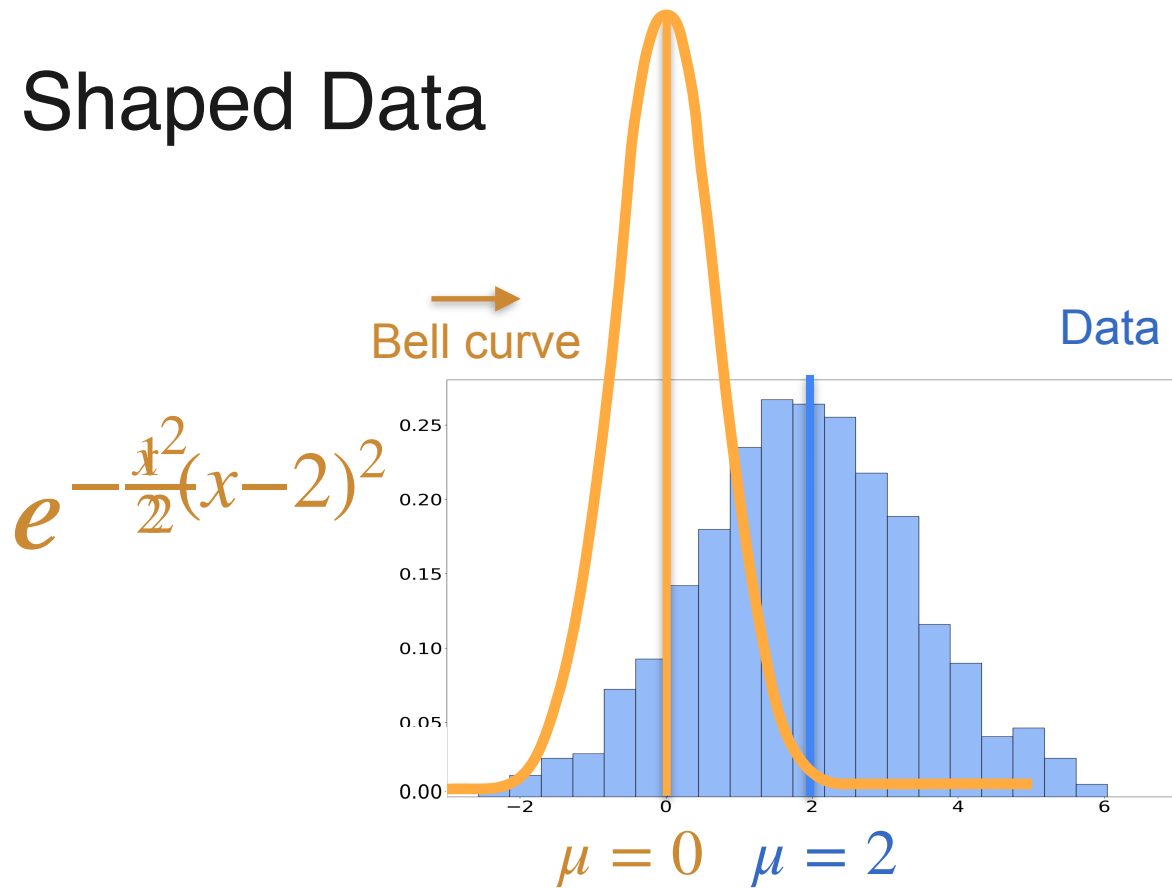


Number of throws = 2

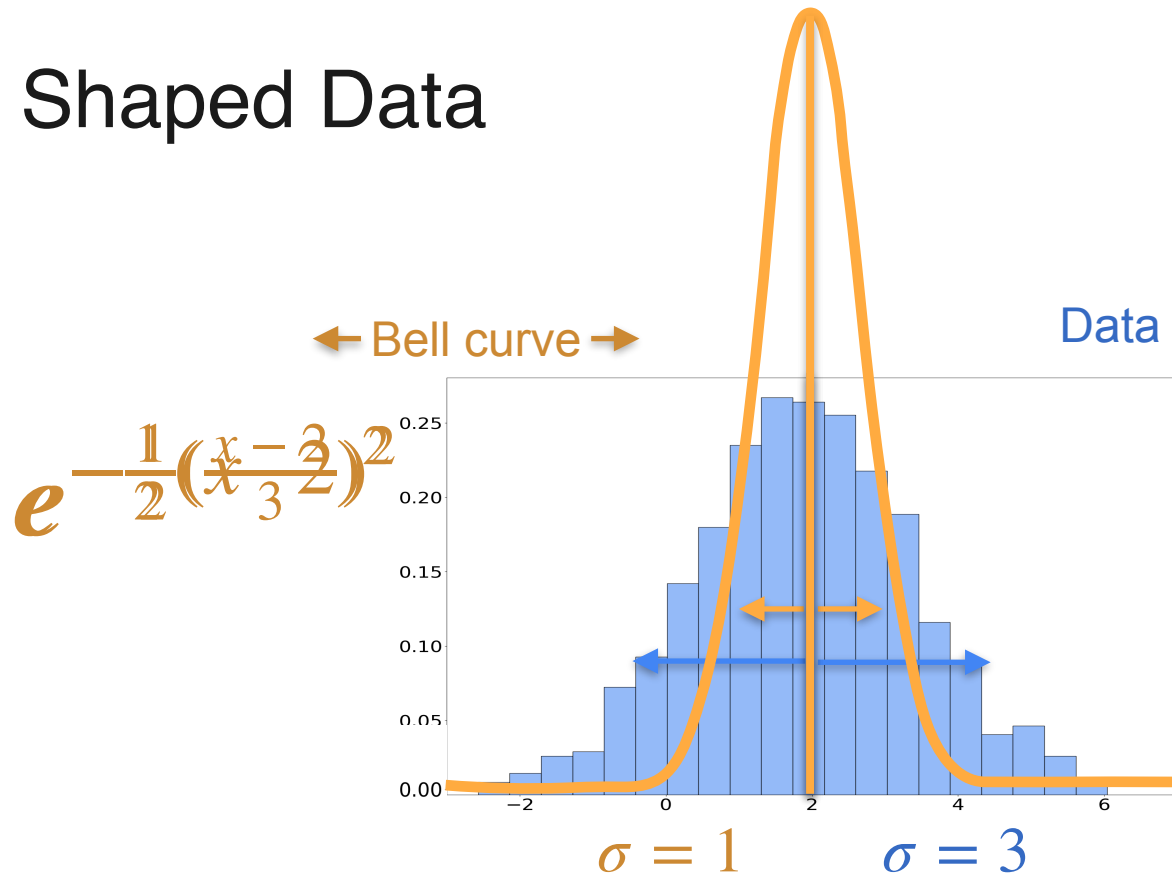
# Binomial Distribution With Very Large $n$



# Bell Shaped Data



# Bell Shaped Data

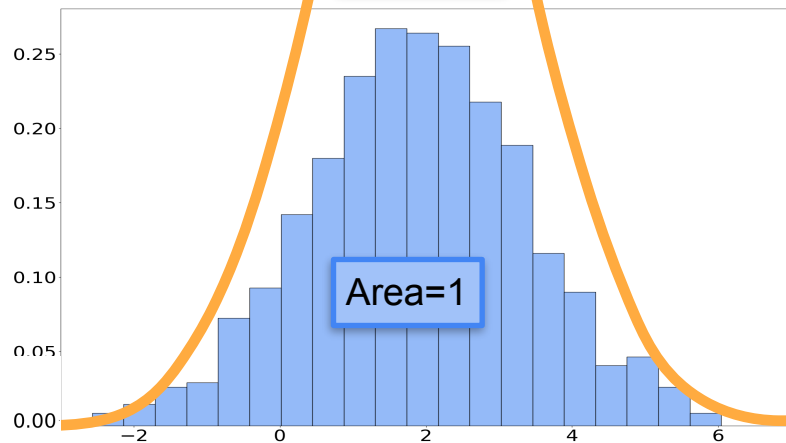


# Bell Shaped Data

$$\frac{1}{3\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-2}{3}\right)^2}$$

Bell curve

Data





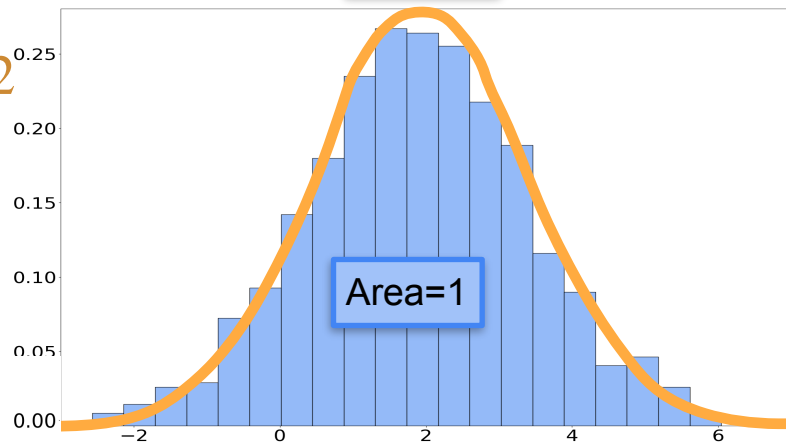
# Bell Shaped Data

$$\frac{1}{3\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-2}{3}\right)^2}$$

Bell curve

Area=1

Data



# Bell Shaped Data

Mean =  $\mu$

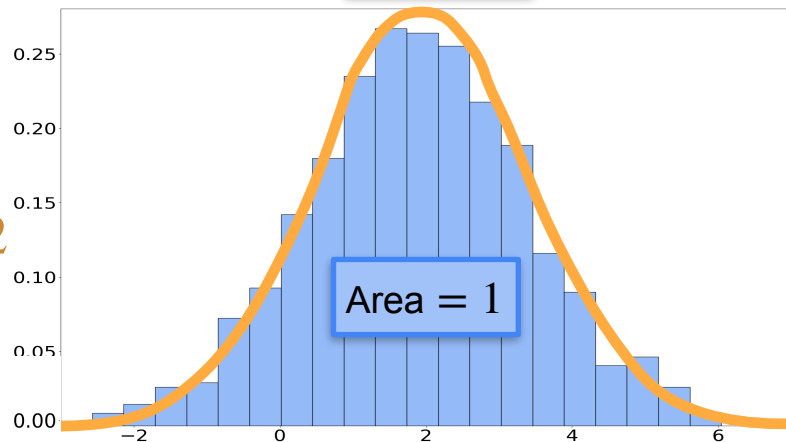
Standard deviation =  $\sigma$

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Bell curve

Area = 1

Data



# Bell Shaped Data

Mean =  $\mu$

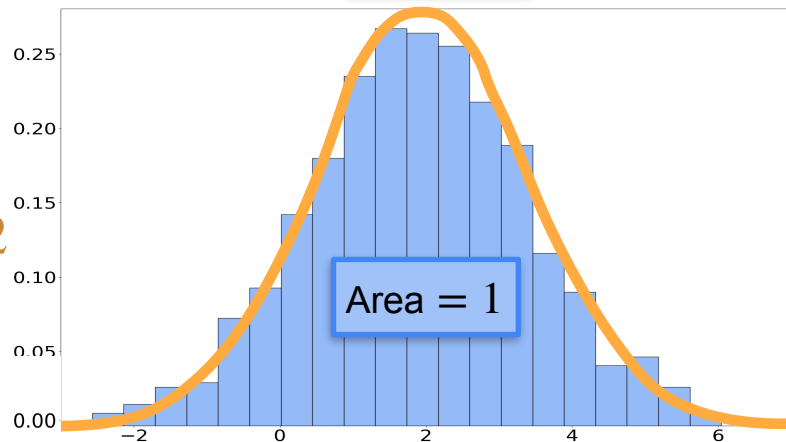
Standard deviation =  $\sigma$

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Bell curve

Area = 1

Data



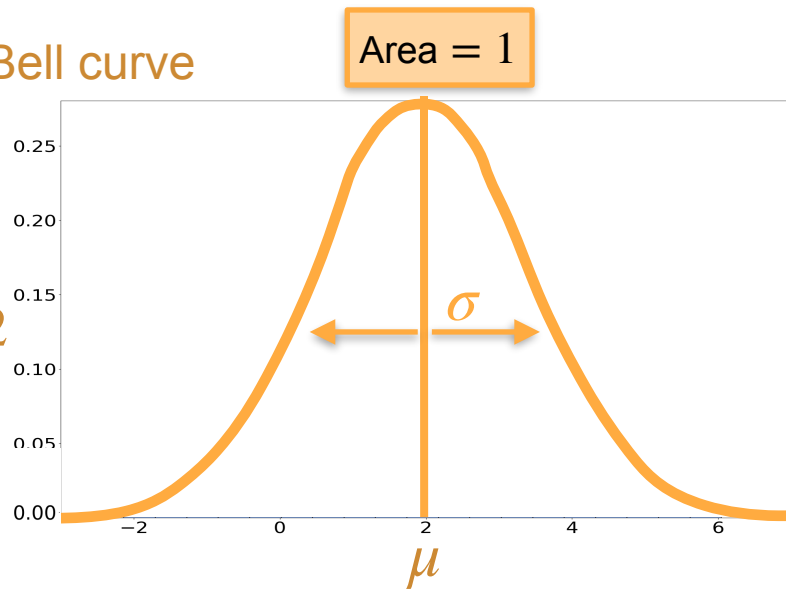
# Normal Distribution

Mean =  $\mu$

Standard deviation =  $\sigma$

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Bell curve



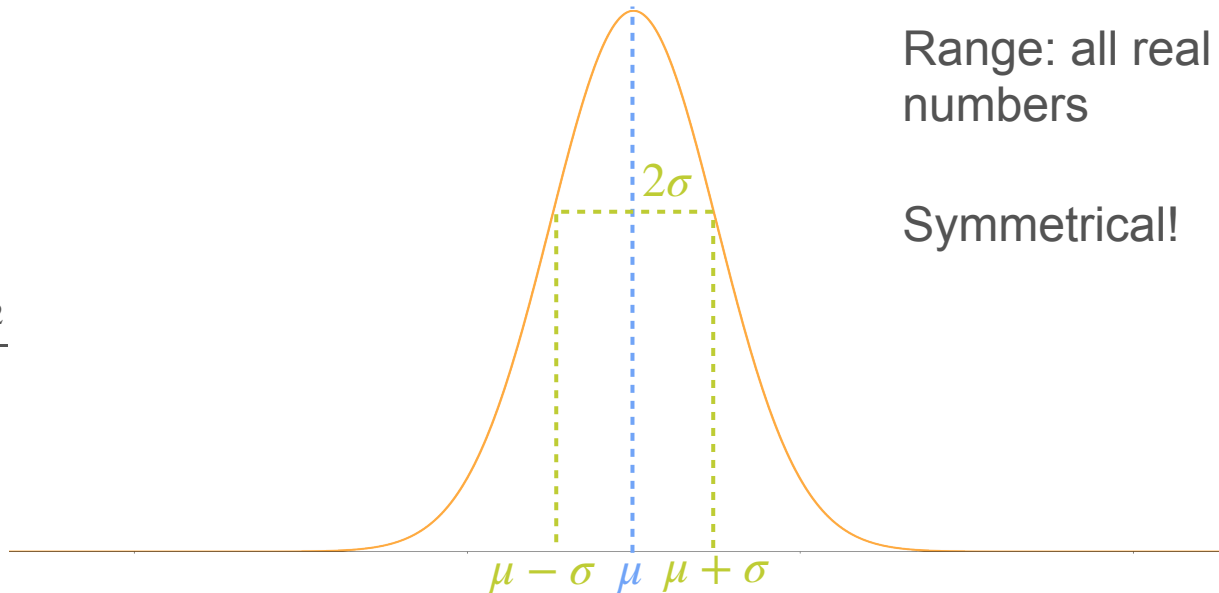
# Normal Distribution

Parameters:

- $\mu$ : center of the bell
- $\sigma$ : spread of the bell

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

Scaling constant

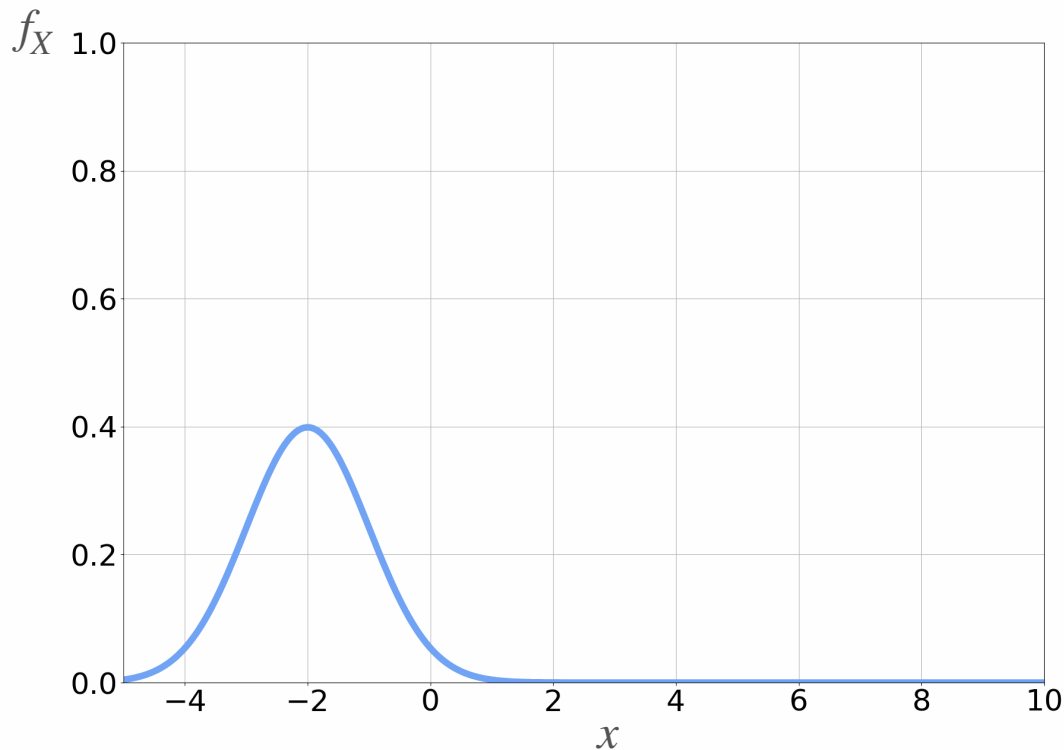


# Normal Distribution

Parameters:

- $\mu$ : center of the bell
- $\sigma$ : spread of the bell

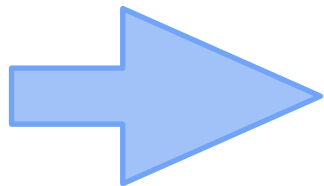
$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$



# Normal Distribution - Notation

Parameters:

- $\mu$ : center of the bell
- $\sigma$ : spread of the bell



$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

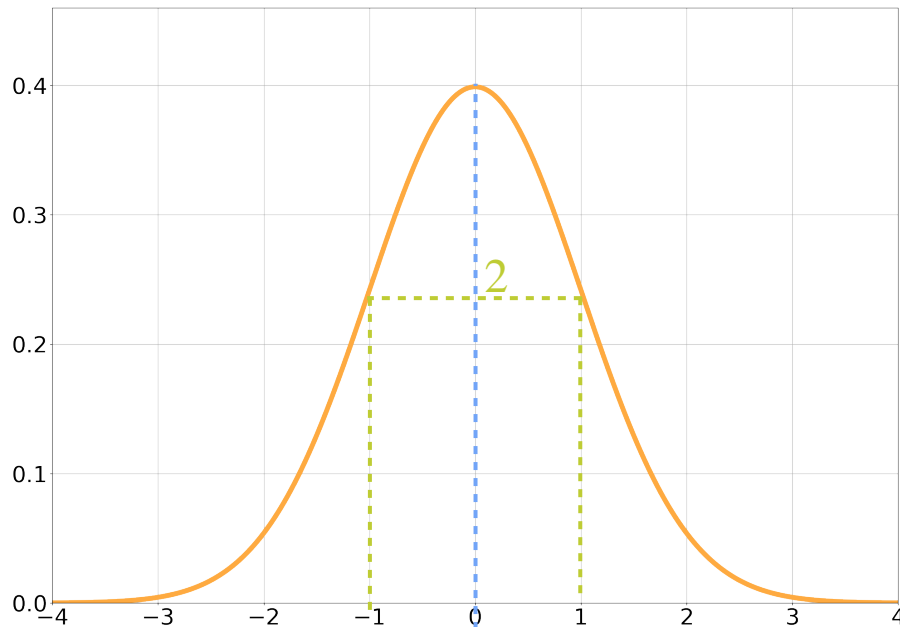
$$X \sim \mathcal{N}(\mu, \sigma^2)$$

# Standard Normal Distribution

Parameters:

- $\mu$ : 0
  - $\sigma$ : 1
- $$X \sim \mathcal{N}(0, 1^2)$$

$$\begin{aligned} f_X(x) &= \frac{1}{\sqrt{2\pi} \cdot 1} e^{-\frac{1}{2} \frac{(x-0)^2}{1^2}} \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} x^2} \end{aligned}$$





# Standardization

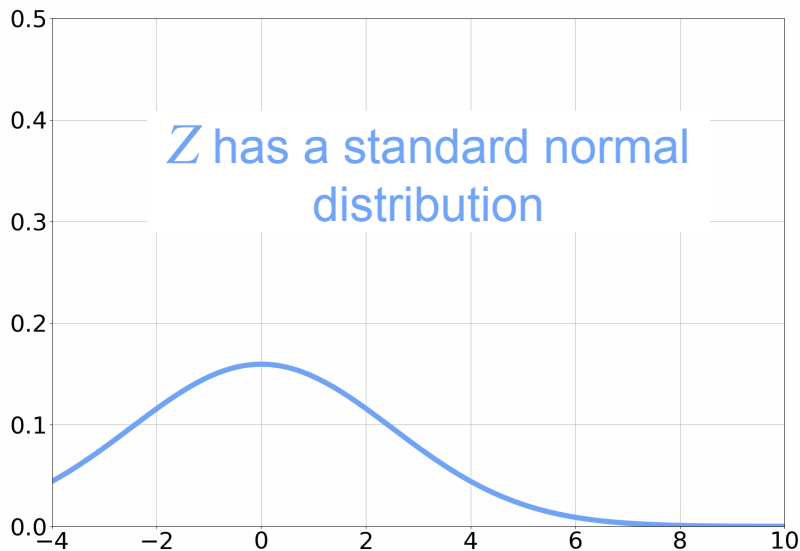
There's a really easy way to convert any normal distribution to the standard one!

$X$  distributes normally with

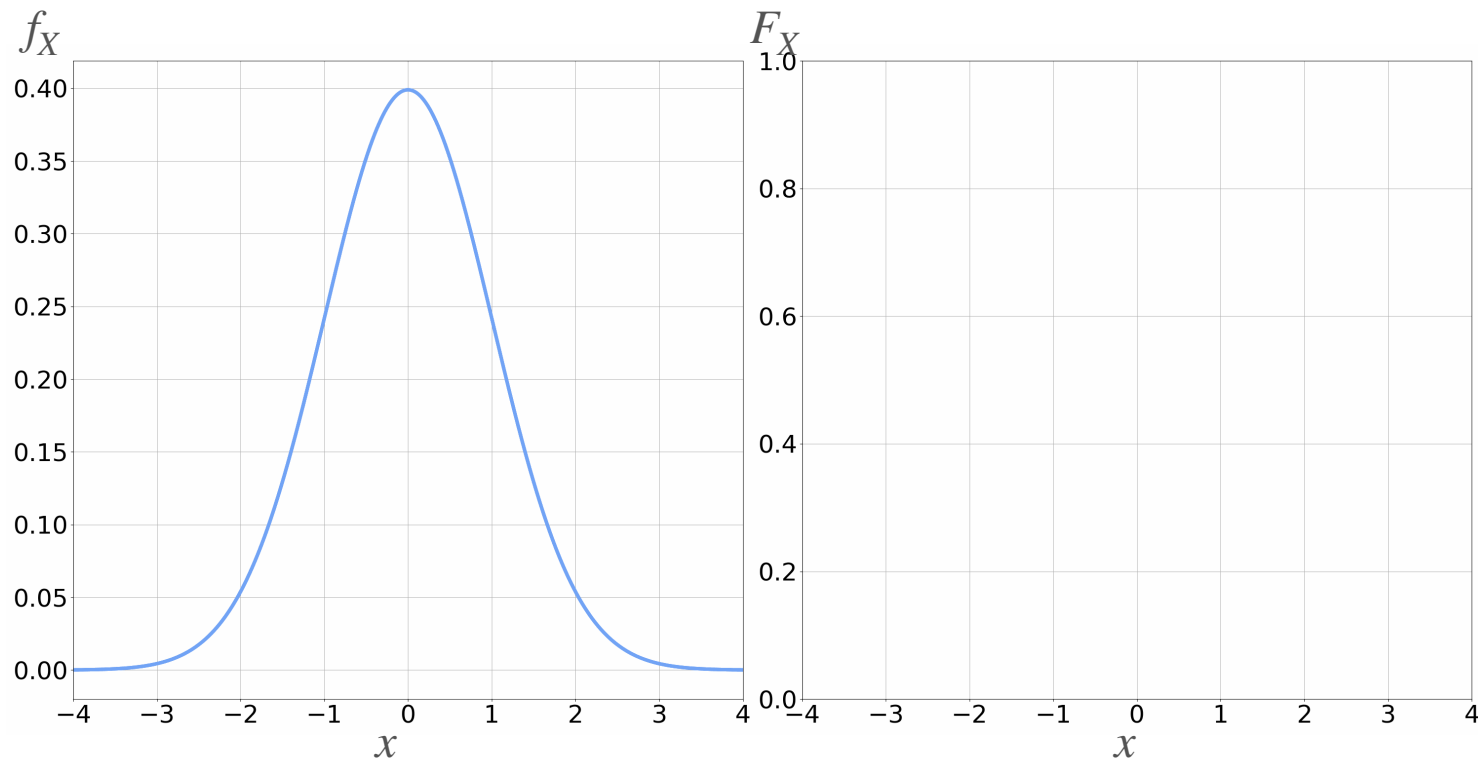
$$\mu = 2, \sigma = 2.5$$

$$Z = \frac{X - \mu}{\sigma}$$

Standardization is crucial to compare variables of different magnitudes!

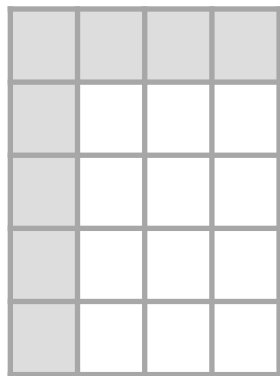


# What Does the CDF Look Like?



# Computing Probabilities From the PDF

This math can't be done by hand

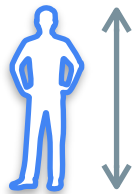


In the old days, people used tables of data



Now, you can use the help of some software to do the approximate area under the curve for you!

# Normal Distribution: Applications



Height



Weight



IQ



Noise in a  
communication channel

In general, characteristics that are the sum of many independent processes

Many models in ML are designed under the assumption that the variables follow a normal distribution



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# Probability Distributions

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**(Optional)**  
**Chi-squared distribution**

# Chi-Square Distribution: Motivation



Communication channel

Noise

Interference from other devices

Obstructions like walls, trees, etc.

Atmospheric conditions: rain, humidity, etc

Electrical interference, i.e. from power lines

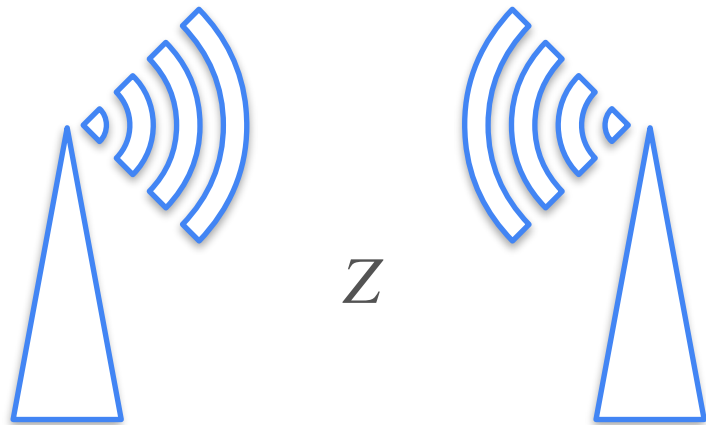
Others



Message sent: 10010

Message received:  $10010 + Z$

# Chi-Square Distribution: Motivation



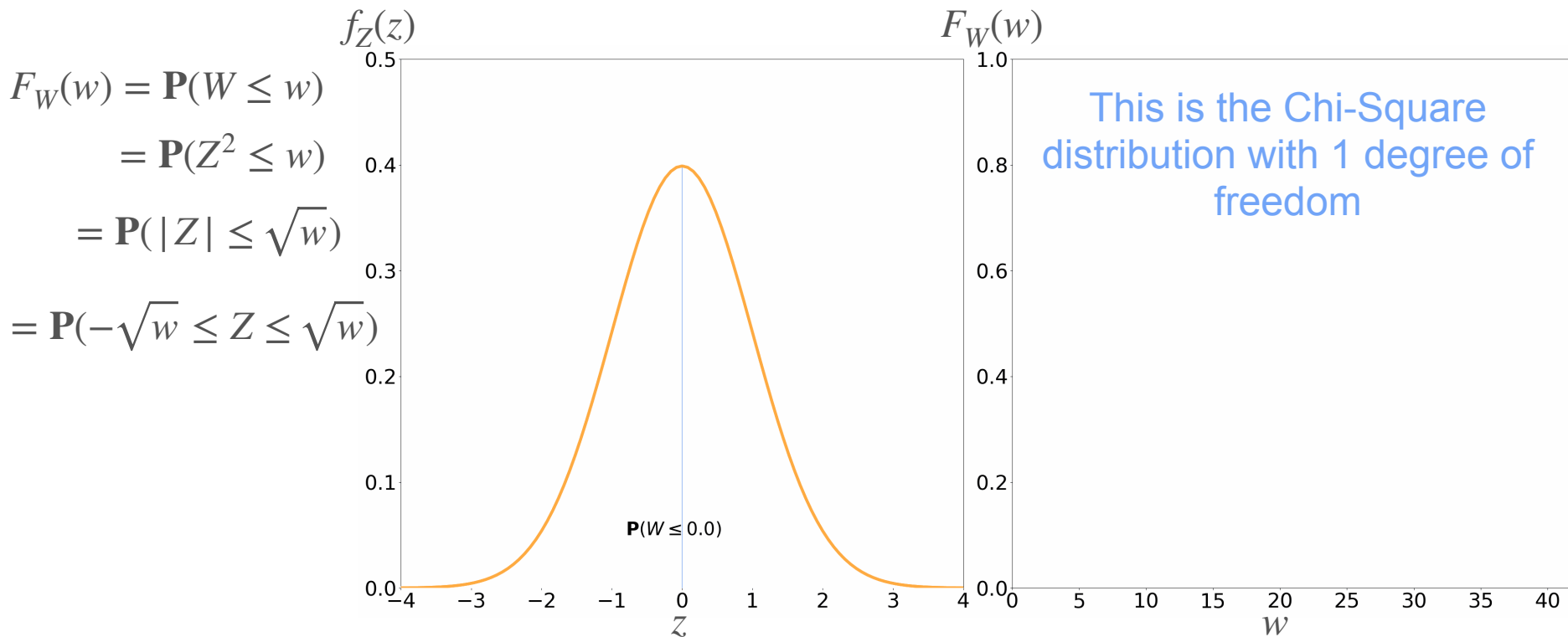
The communication channel has noise with a standard normal distribution

What is the **power** of the noise in the channel?

$$W = Z^2$$

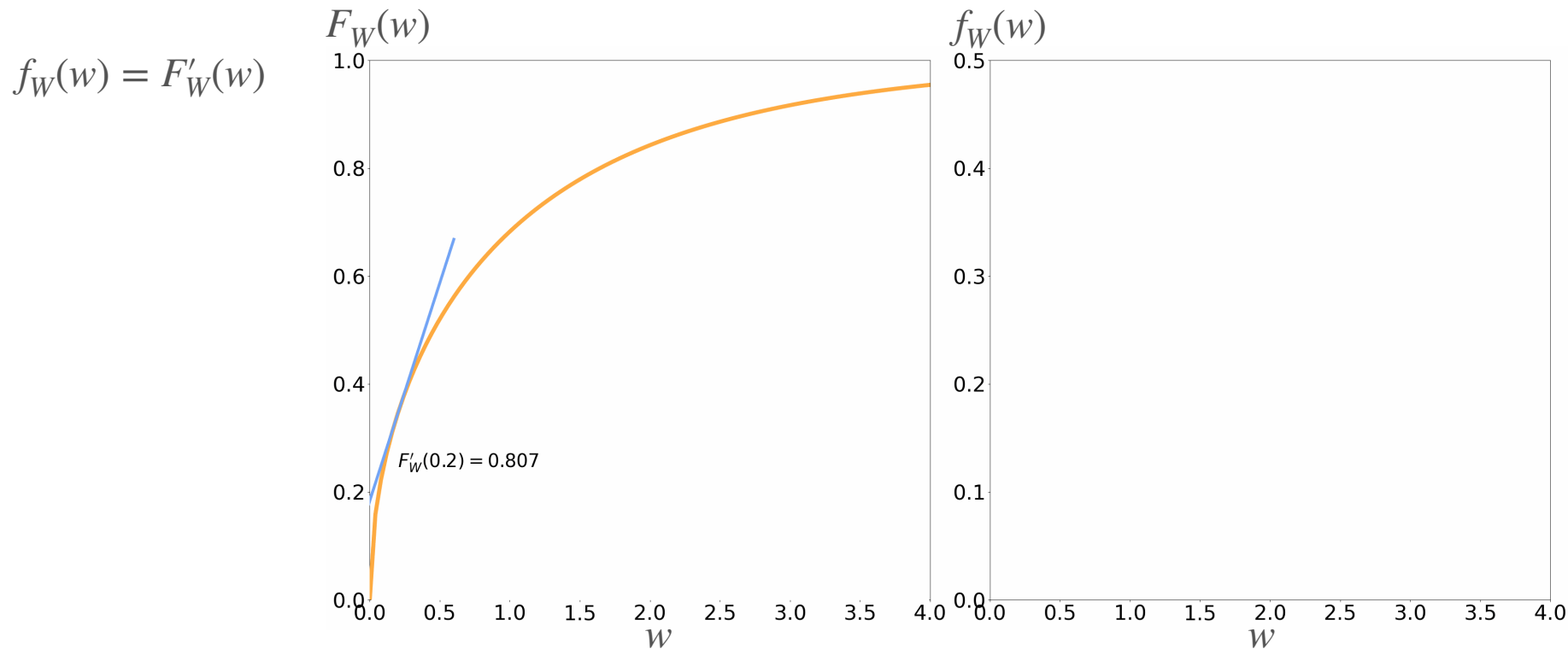
What is the distribution of  $W$ ?

# Chi Square Distribution





# Chi Square Distribution



# Chi-Square Distribution

Accumulated power over 2 transmissions?

$$W_2 = Z_1^2 + Z_2^2$$

Chi-Square with 2 df

Accumulated power over 5 transmissions?

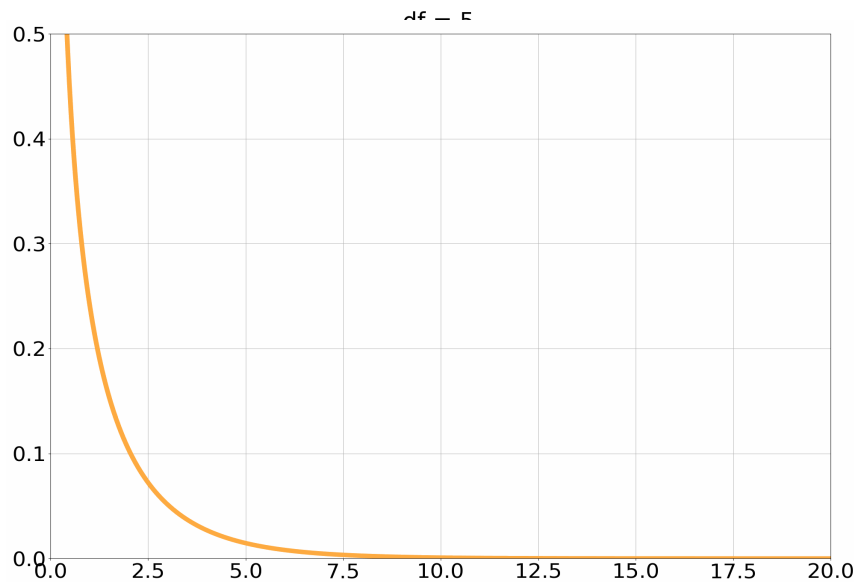
$$W_5 = Z_1^2 + Z_2^2 + Z_3^2 + Z_4^2 + Z_5^2$$

Chi-Square  
with 5 df

Accumulated power over  $k$  transmissions?

$$W_k = \sum_{i=1}^k Z_i^2$$

Chi-Square with  $k$  df





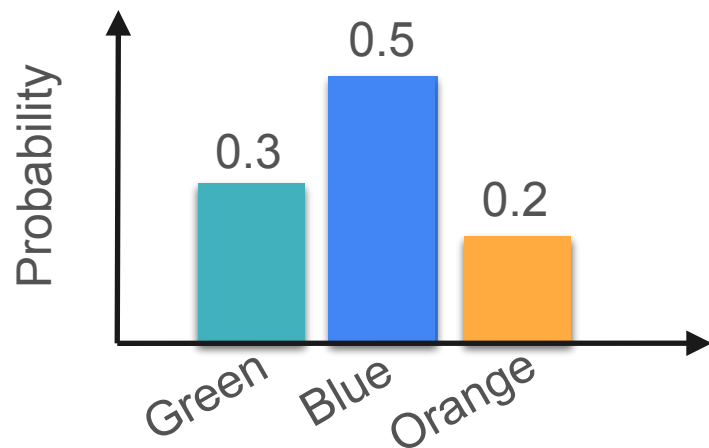
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# Probability Distributions

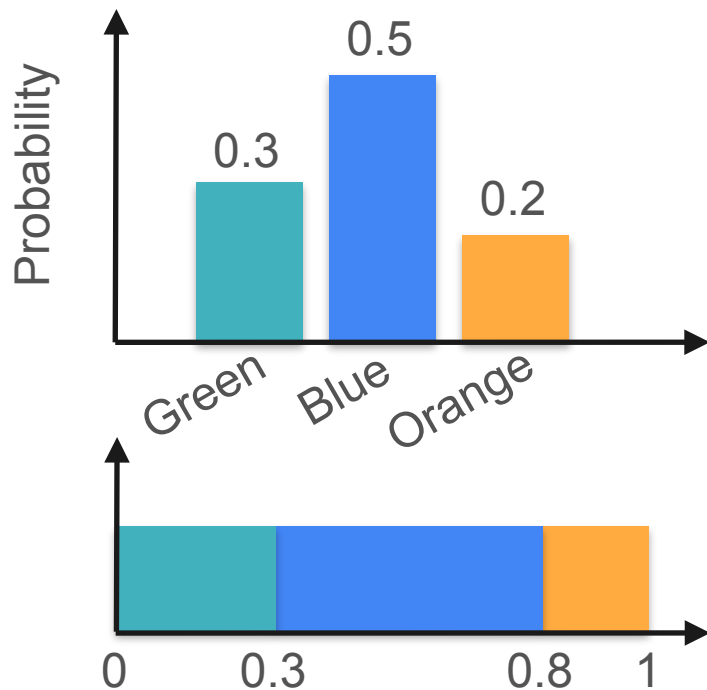
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## **Sampling from a Distribution**

# Sampling From a Distribution

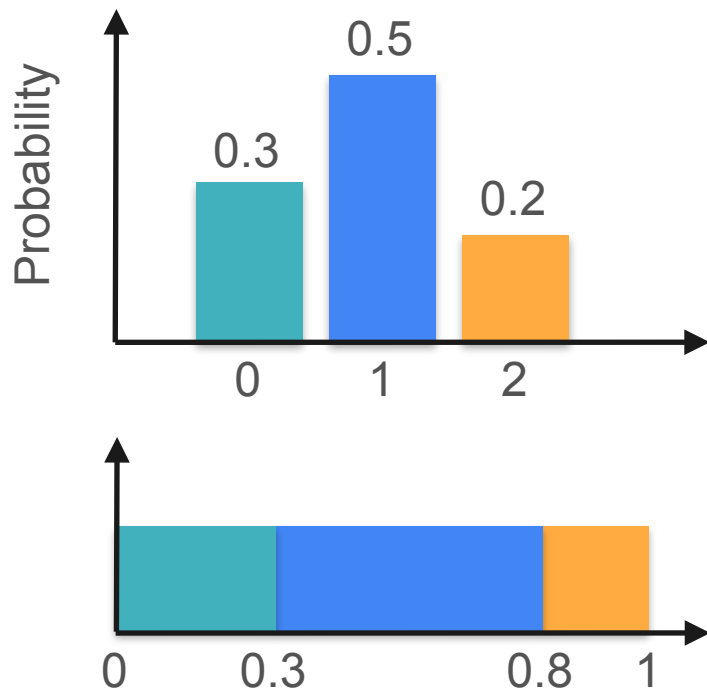


# Sampling From a Distribution



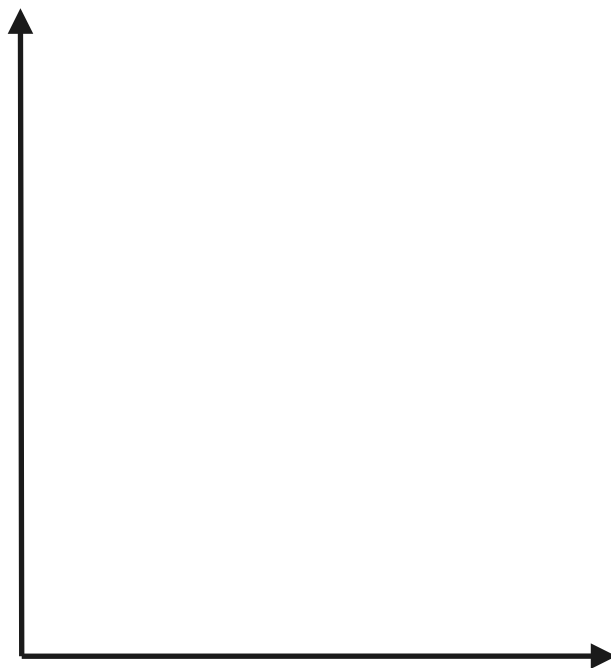
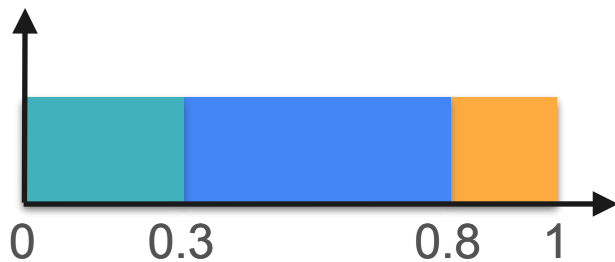
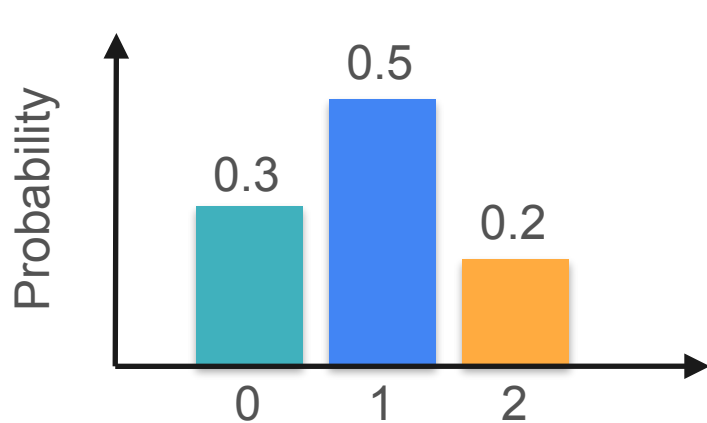
- **Step 1:** generate a random number between 0 and 1
- **Step 2:** find out which interval the number belongs to
  - [0, 0.3)
  - [0.3, 0.8)
  - [0.8, 1]
- **Step 3:** Assign an outcome based on the interval

# Sampling From a Distribution

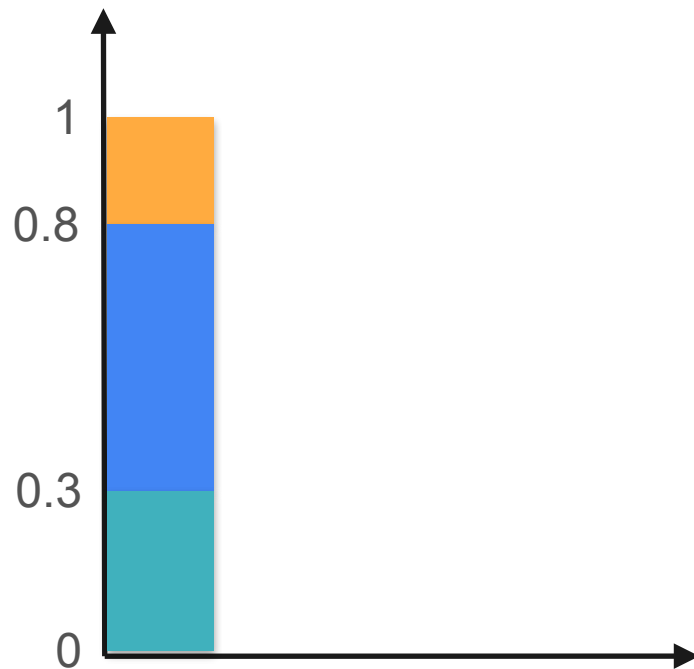
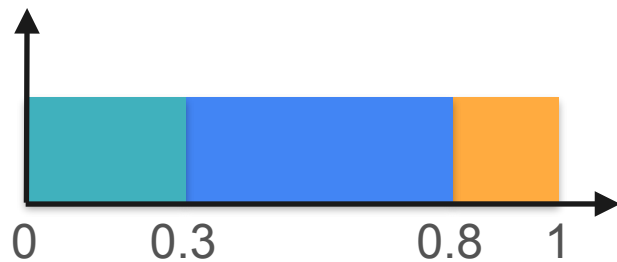
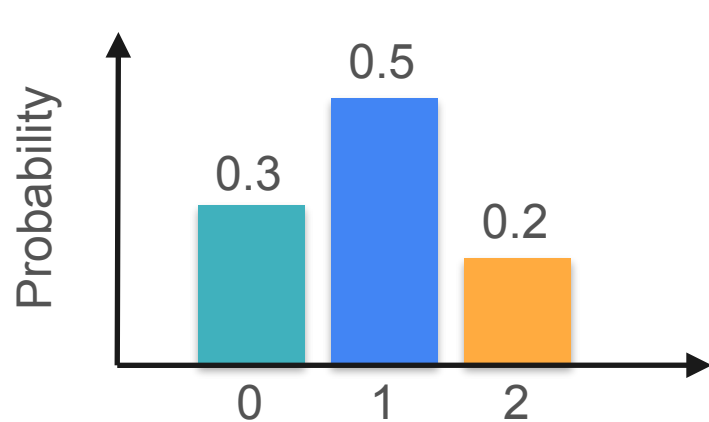


- **Step 1:** generate a random number between 0 and 1
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  - $[0, 0.3)$
  - $[0.3, 0.8)$
  - $[0.8, 1]$
- **Step 3:** Assign an outcome based on the interval

# Sampling From a Distribution

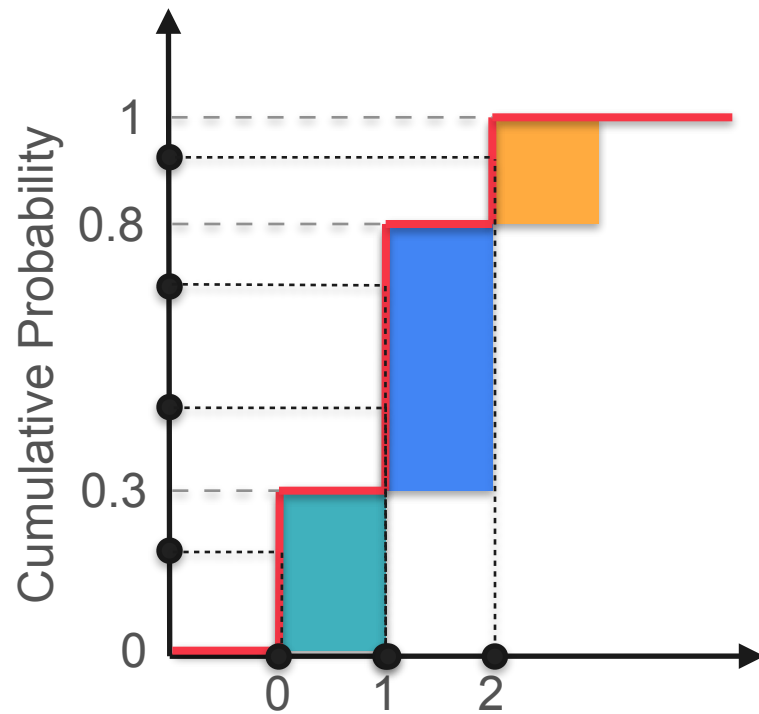
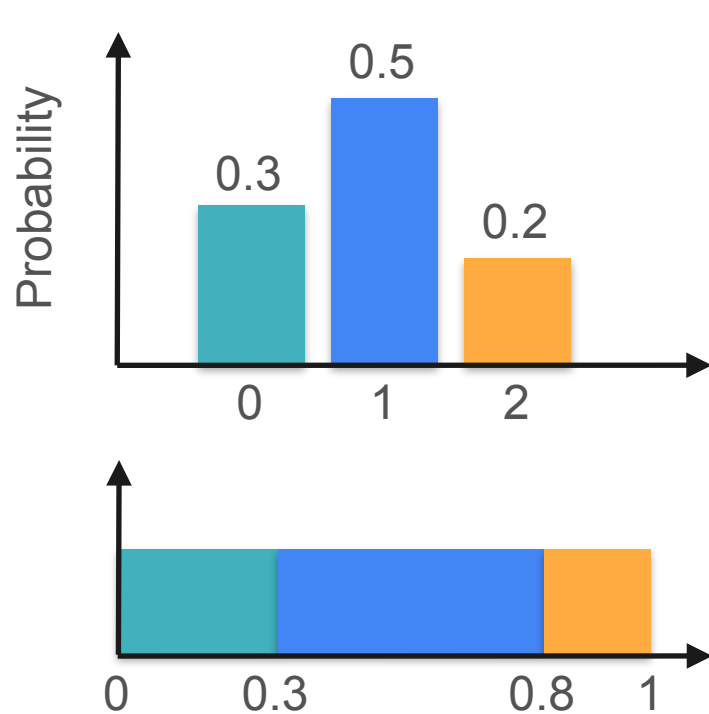


# Sampling From a Distribution

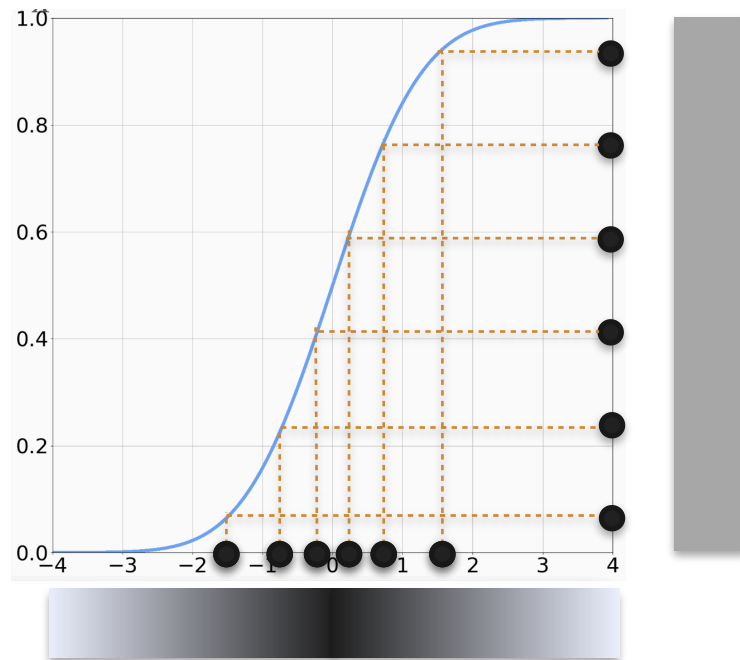
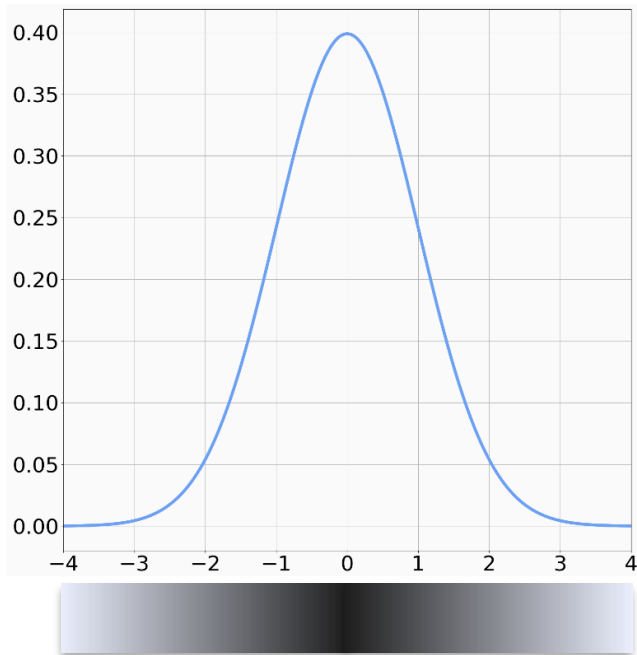




# Sampling From a Distribution



# Sampling From a Normal Distribution





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# Probability Distributions

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## **Conclusion**

# Week 1 - Conclusions

**Talking head**